Archival Research Edition · Scalar-Tensor & Disformal Action Mapping · V25 Audit Package

# Time Attraction Law — Scalar-Tensor Yukawa Fit, V25

## Derivation Steps 1–22, Coupling Dictionary, SPARC Galactic Consistency, and Reproducibility Manifest

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### Included Archival Files in This English Edition (24 Files)

This document compiles the 24 English files of the V25 audit package from `/workspace/ta-v27-work/scalar-tensor-yukawa-fit/` without modification:

- **Package Overview & Numerical Summary:** `README_en.md`
- **Derivation Chapters (Steps 1–22):** `step1_dilaton_form_en.md`, `step2_conformal_disformal_en.md`, `step3_yukawa_helmholtz_eom_en.md`, `step4_covariant_conservation_hydrostatic_en.md`, `step5_coupling_dictionary_ppn_screening_en.md`, `step6_sparc_galactic_ngc3198_consistency_en.md`, `step7_neutron_star_tov_stability_en.md`, `step8_microscopic_clock_coupling_siriusb_lab_v20_en.md`, `step9_vhe_photon_propagation_hess_dispersion_en.md`, `step10_bilateral_kernel_reciprocity_eft_synthesis_en.md`, `step11_chameleon_screening_multiscale_eft_synthesis_en.md`, `step12_second_variation_coupled_pulsation_hessian_en.md`, `step13_curved_window_metric_variation_synge_world_function_en.md`, `step14_gravitational_lensing_null_geodesics_cosmological_background_en.md`, `step15_electromagnetic_trace_anomaly_disformal_magnetic_source_v20_en.md`, `step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress_en.md`, `step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefiniteness_en.md`, `step18_nonlocal_tangent_bvp_shell_kernel_and_inertia_scaling_invariance_en.md`, `step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b_common_proxy_en.md`, `step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_quadrature_en.md`, `step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_lapse_en.md`, `step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability_en.md`
- **Reproducibility & SHA-256 Checksum Manifest:** `MANIFEST_en.txt`

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# 1 Scalar-Tensor and Yukawa/Disformal Action Mapping (V23 ↔ Jordan–Brans–Dicke / Dilaton Framework) — Steps 1–22 Derivation and Audit Reproduction Package

Source file: README_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Publication status and version delineation (V24 vs. V25):** The earlier Zenodo publication (**V24**, DOI: 10.5281/zenodo.23680715) refers exclusively to the earlier V24 release. The sign fixes in `curved_yukawa_adm_and_stability.py` (the exterior metric coupling term `a_m = (2.0 * psi0 / (r * r) + 2.0 * dpsi0 / r) / B0` at line 593 of `linearized_exterior_response`, and setting the exact left-hand side sign `scalar_m_sign = -1.0`, i.e., `a_m_sc = -(2.0 * ddphi_n / r_nodes + 2.0 * dphi_n / (r_nodes ** 2)) / B_n`, as the default in the $L_u u = R_h\,\delta h$ operator of `solve_yukawa_coupled_omega2_gl2`), the three new diagnostics in `curved_yukawa_audit_summary.json` (`full_jacobian_multi_eos_and_points`, `baryonic_mass_and_first_law_audit`, `eigenmode_xi_nodes_and_static_tangent_audit`), and the closure of the theoretical verification belong exclusively to the new **V25** version — the earlier V24 DOI must never be cited as if these fixes and closures were already contained in it.

**V25 diagnostics summary (exact data from `curved_yukawa_audit_summary.json` and `curved_yukawa_coupled_stability.csv`):**

- Derivative verification (SLy, $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km): `exterior_dphi_prime_dm_R` $= -1.683651 \times 10^{-4}$ km$^{-2}$ (rel. diff $3.98 \times 10^{-10}$); `interior_r6km_a_m_sc` $(\partial\phi_L'/\partial m) = -1.728334 \times 10^{-4}$ km$^{-3}$ (rel. diff $6.08 \times 10^{-9}$).
- Full 4×4 Jacobian: 4 configurations (SLy and APR4), 108 checks, max relative error $2.74 \times 10^{-5}$.
- Baryonic mass / First Law: `delta_dex_ADM_vs_baryon_turning` between $M_{\rm ADM}$ and the adaptive Gauss–Kronrod baryonic turning point ($M_B^{\rm quad}$) is $-4.58 \times 10^{-6}$ to $+8.43 \times 10^{-6}$ dex; non-conformal excess $8.23 \times 10^{-5}\,M_\odot$/dex.
- Main stability zero crossing and eigenmode audit (with exact LHS sign `scalar_m_sign = -1.0`): the main dynamical $\omega_0^2 = 0$ zero crossing in `curved_yukawa_coupled_stability.csv` for SLy $L = 10$ km, $\alpha_Y = 1$ is `15.316410` ($+5.80 \times 10^{-6}$ dex above the ADM peak `15.316404` and $+1.33 \times 10^{-6}$ dex above the $M_B^{\rm quad}$ peak `15.316409`; across all 5 sequences $\leq 6.9 \times 10^{-6}$ dex offset, nodeless $n = 0$ fundamental mode, and (0.6–5.0) $\times 10^{-4}$ relative $L^2$ agreement with the static equilibrium tangent; the diagnostic `scalar_m_sign = +1.0` root is `15.319129`, $+2.725 \times 10^{-3}$ dex — details: `step22`, Sections 4 and 7).

**Date:** 2026-10-27 **Working Directory:** `scalar-tensor-yukawa-fit` **Self-Contained Audit Archive:** `scalar-tensor-yukawa-fit_audit_package.zip` **License:** CC BY-NC 4.0 — Commercial use permitted only with the prior written consent of the author. **Official Archival DOIs:**

- **V14:** 10.5281/zenodo.23217189
- **V17 (SPARC 155 / 123 blind sample):** 10.5281/zenodo.19666220
- **V19 (3D spherical top-hat SPARC, 175 galaxies):** 10.5281/zenodo.23497128
- **V20 (Two-clock granite chamber and 8-coil magnetic test):** 10.5281/zenodo.23540741
- **V21 (Compact stars, TOV, and Sirius B audit):** 10.5281/zenodo.23602501
- **V22 Book Edition (Trilingual monograph, HU/EN/DE):** 10.5281/zenodo.23604423
- **V23 (Covariant action closure and coupled radial stability, `50aa2db0`):** 10.5281/zenodo.23639084

## 1.1 1. Research Question, Theoretical Framework & Branch Taxonomy

This audit package contains the exact mathematical correspondence and numerical verification mapping the covariant action candidates developed in the V19–V23 manuscripts and monograph edition of the **Time Attraction Law (TAL)** (10.5281/zenodo.23604423, 10.5281/zenodo.23639084) — **Formulation I** (nonlocal, positive-definite bilateral top-hat kernel action with auxiliary fields) and **Formulation II** (local, screened Helmholtz–Yukawa scalar-field action principle) — onto general **scalar–tensor theories of gravity** (Jordan–Brans–Dicke family, low-energy string-theory **dilaton** coupling, and Bekenstein-type **disformal** clock metrics).

### 1.1.1 1.1. Core Equations & Canonical Normalization

1. **Exponential Time Factor and Scalar/Dilaton Field (`step1_dilaton_form.md` / `step1_dilaton_form_en.md`):**

$$f_L(\mathbf{x}) \;=\; \exp\!\big(-\gamma_L\,\bar\rho_L(\mathbf{x})\big) \;\equiv\; e^{-\phi_L(\mathbf{x})}, \qquad \phi_L(\mathbf{x}) \;=\; \gamma_L\,\bar\rho_L(\mathbf{x}) \;\geq\; 0,$$

where the canonical coupling coefficient originates from the gravitational volumetric capacity ($C = \frac{4\pi G}{c^2} = 9.33306 \times 10^{-27}$ m/kg):

$$\gamma_L \;=\; \frac{GL^2}{c^2} \;=\; \frac{C}{4\pi}\,L^2, \qquad m_\phi \;=\; \frac{\hbar}{Lc}.$$

2. **Conformal Jordan Metric vs. Medium-Anchored Disformal Clock Metric (`step2`, `step5`, `step8`, `step14`):**

$$\tilde g^{\rm conf}_{\mu\nu} \;=\; A^2(\phi_L)\,g_{\mu\nu} \;=\; f_L^2\,g_{\mu\nu} \qquad \text{vs.} \qquad \tilde g^{\rm disf}_{\mu\nu} \;=\; g_{\mu\nu} + \frac{1-f_L^2}{c^2}\,u_\mu u_\nu \;\Longrightarrow\; \tilde g^{\rm disf}_{00} \;=\; f_L^2\,g_{00} \text{ (in the comoving frame).}$$

In a comoving ($u^\mu = (c/\sqrt{-g_{00}}, \mathbf{0})$, $u_\mu = (-\sqrt{-g_{00}}\,c, \mathbf{0}) \;\Longrightarrow\; u_0u_0/c^2 = -g_{00}$) baryonic medium and in atomic clocks, both metrics yield the same proper-time factor ($d\tilde\tau = f_L\,d\tau_g$). For ultrarelativistic light rays ($g_{\mu\nu}k^\mu k^\nu = 0$) and PPN tests, however, the two branches separate sharply:

- In the universal **conformal Jordan branch (`Branch II-Conf`)**, $\sqrt{-\tilde g^{\rm conf}} = e^{-4\phi_L}\sqrt{-g}$, the Maxwell action is conformally invariant, and the standard Brans–Dicke / PPN dictionary applies:

$$\alpha_{\rm std} \;=\; 2\beta^2 \;=\; 2\tilde\alpha_L \;=\; \frac{\alpha_Y}{4\pi}, \qquad \omega_{\rm BD} \;=\; \frac{1-3\alpha_{\rm std}}{2\alpha_{\rm std}} \;=\; \frac{4\pi}{\alpha_Y} - \frac{3}{2}, \qquad \gamma^{\rm conf}_{\rm PPN}(r) \;=\; \frac{1-\alpha_{\rm std}(1+r/L)e^{-r/L}}{1+\alpha_{\rm std}(1+r/L)e^{-r/L}}.$$

- In the medium-anchored **disformal branch (`Branch II-Disf`)**, $\sqrt{-\tilde g^{\rm disf}} = f_L\sqrt{-g}$, magnetic energy ($\rho_B = |\mathbf{B}|^2/(2\mu_0c^2)$) directly sources $\phi_L$ (`step15`), and light is deflected by the achromatic refractive index $n_t(s) = f_L(s)^{-1} = e^{+\phi_L(s)}$ (`step9`, `step14`), so the conformal Brans–Dicke $\gamma^{\rm conf}_{\rm PPN}$ formula does not apply to this branch.

3. **V23 Local Helmholtz–Yukawa Action Principle, Exact Normalization, and Curved Exterior Boundary-Value Problem (`step3`, `step5`, `step7`, `step22`):** The exact SI-unit coefficient of the scalar kinetic and mass terms (with $\alpha_Y \equiv 8\pi\tilde\alpha_L = 4\pi\alpha_{\rm std}$) is:

$$\boxed{S_{\rm II}[g_{\mu\nu},\phi_L,\Psi] \;=\; \int d^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}\left(R \;-\; \frac{1}{\tilde\alpha_L}\Big(g^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L + \frac{\phi_L^2}{L^2}\Big)\right) \;+\; e^{-\phi_L}\,\mathcal{L}_{\rm matter}\right] \;=\; \int d^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}R \;-\; \frac{c^4}{2G\,\alpha_Y}\Big(g^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L + \frac{\phi_L^2}{L^2}\Big) \;+\; e^{-\phi_L}\,\mathcal{L}_{\rm matter}\right],}$$

which in geometrized ($G = c = 1$) units yields the V23 code form $\mathcal{L}_\phi = -\frac{1}{2\alpha_Y}\big((\nabla\phi_L)^2 + \phi_L^2/L^2\big)$ and the exact curved-spacetime Euler–Lagrange equation (`yukawa_action_el`, $B(r) = 1 - 2m(r)/r$):

$$B\,\phi_L'' \;+\; \left[\frac{2B}{r} + B\nu' + \frac{B'}{2}\right]\phi_L' \;-\; \frac{\phi_L}{L^2} \;=\; -\frac{\alpha_Y G}{c^2}\,e^{-\phi_L}\,\varepsilon(r), \qquad \rho_\phi(r) \;=\; \frac{c^2}{2G\,\alpha_Y}\left[B\big(\phi_L'\big)^2 + \frac{\phi_L^2}{L^2}\right].$$

In the curved exterior vacuum outside the stellar surface ($r \geq R_*$, where $\varepsilon = P = 0$, $B\nu' + B'/2 = \frac{2m}{r^2} - \frac{4\pi r}{\alpha_Y L^2}\phi_L^2$), the coupled exterior Einstein–scalar system (`step22`, `curved_yukawa_adm_and_stability.py`) is:

$$\left(1 - \frac{2m(r)}{r}\right)\phi_L'' \;+\; \left[\frac{2}{r}\left(1 - \frac{m(r)}{r}\right) - \frac{4\pi r}{\alpha_Y L^2}\phi_L^2\right]\phi_L' \;-\; \frac{\phi_L}{L^2} \;=\; 0, \qquad \frac{dm(r)}{dr} \;=\; 4\pi r^2\rho_\phi(r),$$

which uniquely determines the genuine curved Dirichlet-to-Neumann slope $\phi'_{L,\rm curved}(R_*) = -k_{\rm curved}\phi_L(R_*)$ (for SLy $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$, $k_{\rm curved} = 0.233542$ km$^{-1}$, which is +24.74% steeper than the flat-space Robin approximation $k_{\rm flat} = 1/R_* + 1/L = 0.187216$ km$^{-1}$) and the asymptotic **ADM mass**:

$$M_{\rm ADM} \;\equiv\; \lim_{r\to\infty} m(r) \;=\; M_{\rm surf} \;+\; \Delta M^{\rm curved}_{\phi,\rm ext}, \qquad M_{\rm surf} \;\equiv\; m(R_*).$$

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### 1.1.2 1.2. Explicit Branch Taxonomy Table

The assumptions, kernel moments, and domain of validity of the six theoretical branches examined in this package are separated in the table below (`step2`, `step5`, `step10`, `step11`, `step21`, `step22`):

| Branch | Action Principle / Metric Definition | Kernel Moments (Expansion of $\tilde{\rho}_L(\mathbf{x})$) | Interior Hydrostatic Balance | Exterior ($r > R_*$) Vacuum and Mass | Domain of Validity & Steps |
|---|---|---|---|---|---|
| **Branch I-A** (V21 nonlocal 3D top-hat: `tophat_mass`, `tophat_gradient`) | Phenomenological 3D spherical top-hat average + algebraic Bianchi X-sector | $\left(1+\frac{L^2}{10}\nabla^2+\mathcal{O}(L^4)\right)\rho=$ | `mass`: $h'=-\nu'$; `gradient`: $h'=-\nu'=-f_L A_{\rm GR}$ | Compact support $[R_*, R_*+L]$; $m(r)=M_{\rm surf}$ constant for $r\geq R_*$ | V21 769 models, V23 Eqs. (8)–(9) static tangent BVP (`step7`, `step16`–`step18`, `step21`) |
| **Branch I-B** (V23 Formulation I: `bilateral_lapse_weighted`) | Auxiliary-field ($\phi_L, \lambda_L$) bilateral kernel action: $S_m=-\int\sqrt{-g}\,e^{-\phi_L}\varepsilon$ | $\left(1+\frac{L^2}{10}\nabla^2+\mathcal{O}(L^4)\right)\rho=$ | $h'=-\nu'-(\ln f_{\rm recip})'$ (reciprocal lapse-weighted force) | Compact support $[R_*, R_*+L]$ | V23 reciprocal bilateral potential $V_{{\rm eff},N}(r)$ and Synge boundary variation (`step10`, `step13`) |
| **Branch II-Fluid** (V23 Formulation II: `yukawa_action_el` / `yukawa_action_el_curved_adm`) | Local Helmholtz–Yukawa action with fluid-density coupling: $\mathcal{L}_m=-e^{-\phi_L}\varepsilon(n)$ | $(1-L^2\nabla^2)^{-1}\rho=\left(1+L^2\nabla^2+\mathcal{O}(L^4)\right)\rho$ | $h'(r)=\beta'(r)+\phi_L'(r)$ (`sym_yukawa_el_hydrostatic = 0`) | Curved Einstein–scalar tail $r\in[R_*,\infty)$; $M_{\rm ADM}=M_{\rm surf}+\Delta M^{\rm curved}_{\phi,\rm ext}$ | Curved exterior ADM mass, mass–radius curves, and coupled $(\delta h, \delta\phi_L, \delta m)$ stability (`step3`, `step4`, `step7`, `step12`, `step22`) |
| **Branch II-Conf** (Universal conformal Jordan branch) | $\tilde{g}^{\rm conf}_{\mu\nu}=e^{-2\phi_L}g_{\mu\nu}$, $\sqrt{-\tilde{g}^{\rm conf}}=e^{-4\phi_L}\sqrt{-g}$, $S_m[\tilde{g}^{\rm conf},\Psi]$ | $(1-L^2\nabla^2)^{-1}\rho_{\rm trace}$ | $h'=-\nu'+\phi_L'$ (trace-free EM does not source $\phi_L$) | Yukawa scalar tail; conformally invariant null geodesics | Standard Brans–Dicke / PPN dictionary ($\alpha_{\rm std}=\frac{\alpha_Y}{4\pi}$, $\omega_{\rm BD}=\frac{4\pi}{\alpha_Y}-\frac{3}{2}$, $\gamma^{\rm conf}_{\rm PPN}(r)$; `step1`, `step2`, `step5`, `step14`) |
| **Branch II-Disf** (Medium-anchored purely temporal disformal branch) | $\tilde{g}^{\rm disf}_{\mu\nu}=g_{\mu\nu}+\frac{1-f_L^2}{c^2}u_\mu u_\nu$, $\sqrt{-\tilde{g}^{\rm disf}}=f_L\sqrt{-g}$ | Top-hat or Yukawa kernel on comoving energy density | Comoving clocks $d\tilde{\tau}=f_L d\tau_g$; $u_\mu$ variation yields extra constraint | Magnetic energy ($\rho_B=B^2/(2\mu_0c^2)$) sources $\phi_L$; refractive index $n_t=e^{+\phi_L}$ | V20 8-coil magnetic test, V20 granite chamber H1, Sirius B H1, VHE gamma dispersion (`step2`, `step8`, `step9`, `step14`, `step15`, `step19`, `step20`) |
| **Branch II-Cham** (Nonlinear chameleon/ symmetron extension) | Hypothetical density-dependent $\beta_{\rm eff}(\rho)$ and self-interaction potential $V_0(\phi_L)$ | Density-dependent $L_{\rm eff}(\rho)=[\partial^2V_{\rm eff}/\partial\phi_L^2]^{-1/2}$ | Screened (thin shell) at high density, amplified at low density | Environment-dependent Yukawa range | Separate theoretical extension hypothesis beyond the constant-$\alpha_Y$ V23 action (`step6`, `step11`) |

*(Note on the top-hat ↔ Yukawa correspondence: the second-order Taylor moment of the 3D spherical top-hat kernel is $\frac{1}{10}L^2\nabla^2$, whereas that of the Yukawa Green's function is $L^2\nabla^2$; hence choosing $\alpha_Y=1$ matches the zeroth-order homogeneous [$\nabla^2\rho\to 0$] limit, while matching the gradient-sensitive second moment requires $L_{Yukawa}=L_{top\text{-}hat}/\sqrt{10}$ or rescaling $\alpha_Y$ — see `step10`.)*

---

## 1.2 2. Step-by-Step Summary of Hungarian Files (`step1`–`step22`, `README.md`, `MANIFEST.txt`) and English Files (`step1_*_en.md`–`step22_*_en.md`, `README_en.md`, `MANIFEST_en.txt`)

### 1.2.1 2.1. Hungarian Derivation Chapters (`step1_*.md`–`step22_*.md`)

| Step | File Name | Size (B) | SHA-256 (prefix) | One-Line Gist (HU / EN) |
|---|---|---|---|---|

| **1** | `step1_dilaton_form.md` | 11 20511 205 | ecd894436341cbdcecd894436341cbdc | **2** | `step2_conformal_disformal.md` | 18 40218 402 | 7a3761a4adcca6fd7a3761a4adcca6fd | **3** | `step3_yukawa_helmholtz_eom.md` | 12 64712 647 | 77fa5ad7646bad0b77fa5ad7646bad0b | **4** | `step4_covariant_conservation_hydrostatic.md` | 12 35612 356 | 52b1cab653da99d652b1cab653da99d6 | **5** | `step5_coupling_dictionary_ppn_screening.md` | 14 94914 949 | d9c9d69ff93e6625d9c9d69ff93e6625 | **6** | `step6_sparc_galactic_ngc3198_consistency.md` | 14 61514 615 | 8fbdd2157d0a4aa98fbdd2157d0a4aa9 | **7** | `step7_neutron_star_tov_stability.md` | 21 21521 215 | d656ac9249a495aed656ac9249a495ae | **8** | `step8_microscopic_clock_coupling_siriusb_lab_v20.md` | 16 30616 306 | f4d790f47b5ab8d4f4d790f47b5ab8d4 | **9** | `step9_vhe_photon_propagation_hess_dispersion.md` | 12 16412 164 | af5b8d0ab66ff02caf5b8d0ab66ff02c | **10** | `step10_bilateral_kernel_reciprocity_eft_synthesis.md` | 13 42013 420 | 3cfe4aaa90e66f3f3cfe4aaa90e66f3f | **11** | `step11_chameleon_screening_multiscale_eft_synthesis.md` | 13 29713 297 | 69c908b620c7562669c908b620c75626 | **12** | `step12_second_variation_coupled_pulsation_hessian.md` | 19 54519 545 | 66e0fdb1964263f166e0fdb1964263f1 | **13** | `step13_curved_window_metric_variation_synge_world_function.`

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md | 11 59511 595 261d16eadbf89cf4261d16eadbf89cf4 | **14** | step14_gravitational_lensing_null_geodesics_cosmological_background.md | 16 32416 324 d377321aba8ddfffd377321aba8ddfff | **15** | step15_electromagnetic_trace_anomaly_disformal_magnetic_source_v20.md | 13 34013 340 75332aacb1edc9f375332aacb1edc9f3 | **16** | step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress.md | 14 17714 177 54fd2d65267c463354fd2d65267c4633 | **17** | step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefiniteness.md | 14 38314 383 fce1edf9f49c8d62fce1edf9f49c8d62 | **18** | step18_nonlocal_tangent_bvp_shell_kernel_and_inertia_scaling_invariance.md | 21 17821 178 6ba1c66a313683626ba1c66a31368362 | **19** | step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b_common_proxy.md | 18 88218 882 bf6eea32f1a730c0bf6eea32f1a730c0 | **20** | step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_quadrature.md | 17 36717 367 bedb9d8511f3c287bedb9d8511f3c287 | **21** | step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_lapse.md | 18 43118 431 29eb46950ed70c5729eb46950ed70c57 | **22** | step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability.md | 35 77131 713 0c09a01ca8973d539013d9fd621f9235

#### 1.2.2 2.2. English Translation Files (step1_*_en.md–step22_*_en.md, README_en.md, MANIFEST_en.txt)

| Step | English File Name | Size (B) | SHA-256 (prefix) | One-Line Description |
|---|---|---|---|---|

| **1 (EN)** | step1_dilaton_form_en.md | 10 89010 890 83a2e8e88808373983a2e8e888083739 | **2 (EN)** | step2_conformal_disformal_en.md | 18 01418 014 abd9db8d104e3bcfabd9db8d104e3bcf | **3 (EN)** | step3_yukawa_helmholtz_eom_en.md | 12 25812 258 8ca1db6f110e26178ca1db6f110e2617 | **4 (EN)** | step4_covariant_conservation_hydrostatic_en.md | 11 98611 986 1deaba07182e7cfa1deaba07182e7cfa | **5 (EN)** | step5_coupling_dictionary_ppn_screening_en.md | 14 70114 701 e57d506cdd3ba02be57d506cdd3ba02b | **6 (EN)** | step6_sparc_galactic_ngc3198_consistency_en.md | 13 87013 870 6f75c1f0ccba91de6f75c1f0ccba91de | **7 (EN)** | step7_neutron_star_tov_stability_en.md | 20 54620 546 e219a8cfaf4c81a2e219a8cfaf4c81a2 | **8 (EN)** | step8_microscopic_clock_coupling_siriusb_lab_v20_en.md | 15 62015 620 b6caf38ba27692eab6caf38ba27692ea | **9 (EN)** | step9_vhe_photon_propagation_hess_dispersion_en.md | 11 45411 454 dabd6f01ef17584bdabd6f01ef17584b | **10 (EN)** | step10_bilateral_kernel_reciprocity_eft_synthesis_en.md | 12 89012 890 748620c6254121e1748620c6254121e1 | **11 (EN)** | step11_chameleon_screening_multiscale_eft_synthesis_en.md | 12 50812 508 375040d0e3dc34c7375040d0e3dc34c7 | **12 (EN)** | step12_second_variation_coupled_pulsation_hessian_en.md | 19 01019 010 798f21c32075f3a7798f21c32075f3a7 | **13 (EN)** | step13_curved_window_metric_variation_synge_world_function_en.md | 11 23311 233 dddf81914b76e505dddf81914b76e505 | **14 (EN)** | step14_gravitational_lensing_null_geodesics_cosmological_background_en.md | 15 51515 515 43051f7ff852c02543051f7ff852c025 | **15 (EN)** | step15_electromagnetic_trace_anomaly_disformal_magnetic_source_v20_en.md | 12 94712 947 8b537673b566d4cb8b537673b566d4cb | **16 (EN)** | step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress_en.md | 13 53013 530 f3886101a664595bf3886101a664595b | **17 (EN)** | step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefiniteness_en.md | 13 76513 765 672dc1adfa28d46f672dc1adfa28d46f | **18 (EN)** | step18_nonlocal_tangent_bvp_shell_kernel_and_inertia_scaling_invariance_en.md | 20 83220 832 ad7267dc00729932ad7267dc00729932 | **19 (EN)** | step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b_common_proxy_en.md | 18 06718 067 9fab3349db68588b9fab3349db68588b | **20 (EN)** | step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_quadrature_en.md | 16 82016 820 800ff0bb69664079800ff0bb69664079 | **21 (EN)** | step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_lapse_en.md | 17 54217 542 106b9ee27d97f4a4106b9ee27d97f4a4 | **22 (EN)** | step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability_en.md | 36 89631 863dc84080449d2fcbac89fccc6bcfe0ffe

| | | | | |
|---|---|---|---|---|
| **README (EN)** | README_en.md | — | — | **Full English edition of README.md listing all Hungarian and English chapters, numerical tables, and reproduction instructions.** |
| **MANIFEST (EN)** | MANIFEST_en.txt | — | — | Full English manifest listing all tracked files (including the 22 English files) with exact SHA-256 checksums and byte sizes. |

### 1.3 3. Summary of Key Numerical Results

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| Physical Domain | Quantity | Exact Numerical Value | Source Step & Script |
|---|---|---|---|
| **Scalar–Tensor / Yukawa Dictionary (`Branch II-Conf`)** | Canonical Yukawa strength ($\alpha_Y = 1$) and Brans–Dicke $\omega_{\rm BD}$ | $\alpha_{\rm std} = 2\beta^2 = 2\tilde{\alpha}_L = \frac{\alpha_Y}{4\pi} = \frac{1}{4\pi} \approx 0.0795775$; $\omega_{\rm BD} = \frac{4\pi}{\alpha_Y} - \frac{3}{2} \approx 11.06637 \approx 11.07$ ($\alpha_Y = 1$) or 36.1991 ($\alpha_Y = 1/3$) | `step5`, `covariant_closure_50aa2db0.py` |
| **SPARC 175 Galaxies ($\rho < 0$ positivity test)** | Number of negative-density points (across 3391 radial points) | **0 points** (positivity check of photometric + 3D top-hat + 3D Yukawa kernels) vs. **239 points** (140 non-holdout galaxies, $\Upsilon_d = 0.5$) / **301–303 points** (175 galaxies, spherical derivative formula) | `step6`, `SPARC_mode.py`, `sparc_tophat.py` |
| **NGC 3198 Negative Control ($R = 30.08$ kpc)** | Observed $C_{\rm req} = (V_{\rm obs}/V_{\rm bar})^2$ vs. isolated constant-coupling $C_{\rm pred}^{\rm isol}$ | $V_{\rm obs} = 146 \pm 2$ km/s; $\Upsilon_d = 1.0$: $V_{\rm bar} = 88.53$ km/s, $C_{\mathbf{req}} = 2.720$, whereas for the isolated constant-coupling model $C_{\mathbf{pred}}^{\mathbf{isol}} \leq 1 + \alpha_{\mathbf{std}} = 1.0796$ ($V_{\rm pred}^{\rm isol} = 88.53$–$91.99$ km/s; $\Upsilon_d = 0.5$: $V_{\rm bar} = 70.32$ km/s, $C_{\rm req} = 4.311$, $V_{\rm pred}^{\rm isol} = 70.32$–$73.06$ km/s) | `step6`, `SPARC_mode.py`, `data/NGC3198_table.csv` |
| **Cosmological Horizon Scales ($\Omega_b$ vs. $\Omega_m$)** | $L_{\rm hor} = \frac{c}{\sqrt{G\rho_{\rm bg}}} = \sqrt{\frac{8\pi}{3\Omega}}\,\frac{c}{H_0}$ ($H_0 = 67.4$ km/s/Mpc) | **Baryonic background ($\Omega_b = 0.0493$, $\rho_{b,0} = 4.2067 \times 10^{-28}$ kg/m$^3$):** $\sqrt{8\pi/(3\Omega_b)} = 13.031 \implies L_{\rm hor}^{(b)} = 1.7892 \times 10^{27}$ m $\approx$ **57.98** Gpc; **Matter background ($\Omega_m = 0.315$, $\rho_{m,0} = 2.686 \times 10^{-27}$ kg/m$^3$):** $\sqrt{8\pi/(3\Omega_m)} = 5.160 \implies L_{\rm hor}^{(m)} = 7.081 \times 10^{26}$ m $\approx$ **22.94** Gpc | `step14`, `sparc_tophat.py` |
| **Bianchi / Noether Conservation** | $\max_r | r\cdot\nabla_\mu(T^{\mu}_{\;r,\text{matter}} + T^{\mu}_{\;r,X}) | / $\varepsilon_c(s=r^2$ PCHIP) | |
| **Curved Exterior Yukawa BVP and ADM Mass (`Branch II-Fluid`)** | $R_{\rm max}$ convergence ($[R_*, R_* + N_{\rm mult}L]$, $N_{\rm mult} \in \{10, 12, 14, 16\}$), $k_{\rm curved}$ vs. $k_{\rm flat}$, and $M_{\rm ADM,max}$ | $R_{\rm max}$ spread: $\Delta k_{\rm curved}/k_{\rm curved} = 6.17 \times 10^{-12}$, $\Delta M_{\rm ADM}/M_{\rm ADM} = 5.54 \times 10^{-13}$; SLy 15.0, $L = 10$ km, $\alpha_Y = 1$: $k_{\rm curved} = 0.233542$ km$^{-1}$ vs. $k_{\rm flat} = 0.187216$ km$^{-1}$ (+24.74%); $M_{\rm ADM,max}^{\rm SLy} = \mathbf{2.03321\,M_\odot}$ ($M_{\rm surf,max} = 2.02911\,M_\odot$, $M_{B,\rm max}^{\rm quad} = \mathbf{2.42339\,M_\odot}$, $\Delta M_{\phi,\rm ext}^{\rm curved} = 0.00411\,M_\odot$), $R_{1.4}^{\rm ADM} = \mathbf{11.5359}$ km ($M_{\rm ADM,max}^{\rm APR4} = 2.18068\,M_\odot$, $M_{\rm ADM,max}^{\rm H4} = 1.97809\,M_\odot$) | `step7`, `step22`, `curved_yukawa_adm_and_stability.py` |
| **Coupled ($\delta h, \delta\phi_L, \delta m$) Stability on the Curved `yukawa_action_el` Branch** | $\omega_0^2 = 0$ zero crossing vs. $\frac{dM_{\rm ADM}}{d\rho_c} = 0$ and $\frac{dM_B}{d\rho_c} = 0$ (`scalar_m_sign = -1.0`) | SLy $L = 10$ km, $\alpha_Y = 1$: $\log_{10}\rho_{0c}(M_{\rm ADM,max}) = 15.316404$, $\log_{10}\rho_{0c}(M_{B,\rm max}^{\rm quad}) = 15.316409$, coupled $\omega_0^2 = 0$: 15.316410 ($+\mathbf{5.80 \times 10^{-6}}$ dex, 2097× reduction over frozen-$\phi_L$ shift $+1.22 \times 10^{-2}$ dex); $\alpha_Y = 1/3$: 15.307170 ($+4.20 \times 10^{-6}$ **dex**, 995× reduction); $L = 1$ km: 15.303080 ($-6.86 \times 10^{-6}$ **dex**, 165× reduction); APR4 ($L = 10$ km): 15.293025 ($+3.57 \times 10^{-6}$ **dex**, 2392× reduction); H4 ($L = 10$ km): 15.227783 ($-1.34 \times 10^{-6}$ **dex**, 17593× reduction) | `step12`, `step22`, `curved_yukawa_adm_and_stability.py` |
| **V21 Nonlocal 3D Top-Hat Tangent BVP (`Branch I-A`)** | SLy $L = 10$ km `mass` / `gradient` $\omega_0^2 = 0$ vs. $\partial M/\partial\rho_c = 0$ and $W \to 4W$ scaling | In `mass` mode: 15.29931 vs. 15.29924 ($+6.97 \times 10^{-5}$ dex, 58.3× improvement); in `gradient` mode: 15.30113 vs. 15.30106 ($+7.39 \times 10^{-5}$ dex, 59.3× improvement); inertia scaling: $\omega_{4W}^2/\omega_W^2 = 0.2500$ | `step7`, `step17`, `step18`, `covariant_closure_50aa2db0.py` |
| **Sirius B H0 vs. H1 Redshift** | $v_{\rm obs} = 80.65 \pm 0.77$ km/s ($M = 1.018\,M_\odot$, $R = 5846.6$ km) | H0: $ | \Delta v |
| **Granite Chamber (8558 kg) IN–OUT Test** | Conditional H1 differential clock signal $\Delta y_{\rm H1}$(IN − OUT) | At $L = 1.00$ m: Top-hat $-1.436 \times 10^{-24}$ ($\equiv 0$ for $L \geq 2.944$ m); Yukawa $-3.781 \times 10^{-25}$ ($-4.818 \times 10^{-25}$ at $L = 10$ m) | `step8`, `step19`, `covariant_closure_50aa2db0.py` |
| **8-Coil Solenoid ($N = 2000$, 14 mA)** | Magnetic peak signal ($L_p = 0.630957$ m), $r = L/t$ quadrature, and $L = 0.95$ m | $\Delta W(L_p) = 1.96114452$ J/A$^2$, $k_g = -6.13117767 \times 10^{-45}$ A$^{-2}$, $\Delta y(L_p) = -1.2017108 \times 10^{-48} \sim -10^{-48}$; Path A vs. B ($L = 3$ m): $7.44 \times 10^{-16}$; $L = 0.95$ m identity: $8.88 \times 10^{-16}$ J/A$^2$ | `step15`, `step20`, `make_revision3_checks.py` |

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### 1.3.1 3.1. V25 Fixes and Diagnostics: 4-Component Exterior BVP, Exact Exterior Linearization, Dynamical Scalar Wave, and GL2 N-Convergence

*(Source: `curved_yukawa_adm_and_stability.py`, `curved_yukawa_audit_summary.json`, `curved_yukawa_coupled_stability.csv`, `curved_yukawa_adm_sequences.csv`.)*

- **Exterior Lapse (Issue 5):** the exterior $\nu$ is co-integrated as the fourth ODE component $(\phi_L, \psi_L, m, \nu)$ with `solve_bvp tol=1e-12` and outer boundary condition $\nu(R_{\max}) = \frac{1}{2}\ln(1-2m(R_{\max})/R_{\max})$. Spreads for SLy, $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $N_{\rm mult} \in \{10, 12, 14, 16\}$: $\delta k/k = 6.17 \times 10^{-12}$, $\delta M_{\rm ADM}/M_{\rm ADM} = 5.54 \times 10^{-13}$, $\delta\nu_R = 1.63 \times 10^{-14}$.

| $N_{\rm mult}$ | $R_{\max}$ (km) | $k_{\rm curved}$ (km$^{-1}$) | $M_{\rm ADM}$ ($M_\odot$) | $\nu_R$ |
|---|---|---|---|---|
| 10 | 111.4657 | 0.233542329280 | 1.509998138706 | -0.246306687228 |
| 12 | 131.4657 | 0.233542329278 | 1.509998138707 | -0.246306687228 |
| 14 | 151.4657 | 0.233542329278 | 1.509998138707 | -0.246306687228 |
| 16 | 171.4657 | 0.233542329278 | 1.509998138707 | -0.246306687228 |

- **Exterior Linearization and Finite-Difference Derivative Tests (SLy, $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$):** `linearized_exterior_response` (line 593: `a_m = (2.0 * psi0 / (r * r) + 2.0 * dpsi0 / r) / B0`) and the left-hand side ($L_u u = R_h\,\delta h$) interior scalar operator coefficient `a_m_sc` (`scalar_m_sign = -1.0`: `a_m_sc = -(2.0 * ddphi_n / r_nodes + 2.0 * dphi_n / (r_nodes ** 2)) / B_n`) verified against central finite differences:
  - Exterior response $\partial\phi_L'(R_*)/\partial m(R_*)$: linearized BVP `-1.68365059e-04` km$^{-2}$ vs. central finite difference of the full 4-component exterior BVP `-1.68365059e-04` km$^{-2}$ (relative difference: $3.98 \times 10^{-10}$).
  - Interior $r = 6.0$ km scalar–mass derivative ($\partial\phi_L''/\partial m$): analytical `-1.72833429e-04` km$^{-3}$ vs. central finite difference of the interior scalar RHS `-1.72833428e-04` km$^{-3}$ (relative difference: $6.08 \times 10^{-9}$).
- **Dynamical Scalar Block:** the scalar block includes the $-\omega^2 e^{-2\nu}\delta\phi_L$ term and the exterior response is supplied by the $\omega$-dependent BVP (`dynamical_scalar_wave=True`); solver: GL2 piecewise-segment Volterra discretization, $N = 480$, with exact LHS sign `scalar_m_sign = -1.0`. The quasi-static and dynamical roots agree to all 6 reported decimal places.
- **Turning Points and Zero Crossings:** `minimize_scalar` + 7-point quadratic fit over $\pm 0.002$ dex; zero crossings from a 3-point quadratic fit over $\pm 0.0005$ dex around the root.

| Sequence | $\log_{10}\rho_{0c}(dM_{\rm ADM}/d\log_{10}\rho_{0c} = 0)$ | $\log_{10}\rho_{0c}(dM_B/d\log_{10}\rho_{0c} = 0)$ | Dynamical $\omega_0^2 = 0$ | $\Delta_{\rm ADM}$ Dynamical (dex) | $\Delta_{M_B}$ Dynamical (dex) | Quasi-Static $\omega_0^2 = 0$ | Frozen $\delta\phi_L = 0$ | $\Delta_{\rm ADM}$ Frozen (dex) |
|---|---|---|---|---|---|---|---|---|
| **SLy, $L = 10$ km, $\alpha_Y = 1$** | 15.316404 | 15.316409 | 15.316410 | +5.80e-06 | +1.33e-06 | 15.316410 | 15.328565 | +1.22e-02 |
| **SLy, $L = 10$ km, $\alpha_Y = 1/3$** | 15.307166 | 15.307169 | 15.307170 | +4.20e-06 | +6.41e-07 | 15.307170 | 15.311340 | +4.17e-03 |
| **SLy, $L = 1$ km, $\alpha_Y = 1$** | 15.303086 | 15.303082 | 15.303080 | -6.86e-06 | -2.61e-06 | 15.303080 | 15.304220 | +1.13e-03 |
| **APR4, $L = 10$ km, $\alpha_Y = 1$** | 15.293022 | 15.293025 | 15.293025 | +3.57e-06 | -1.40e-07 | 15.293025 | 15.301564 | +8.54e-03 |
| **H4, $L = 10$ km, $\alpha_Y = 1$** | 15.227785 | 15.227787 | 15.227783 | -1.34e-06 | -4.05e-06 | 15.227783 | 15.251384 | +2.36e-02 |

**GL2 Convergence on the $\omega_0^2 = 0$ Root (across $N$ and $R_{\max}$) and at $\log_{10}\rho_{0c} = 15.311404$** (SLy, $L = 10$ km, $\alpha_Y = 1$): on the root, $p_{\rm eff,root} = 4.248$, Richardson root $\log_{10}\rho_{0c,\rm Rich}(\omega_0^2 = 0) =$ @@TOK0@@ ($N = 480$ discretization error $6.72 \times 10^{-8}$ dex; spread across $R_{\max} \in [R_* + 10L, R_* + 20L]$ is $2.42 \times 10^{-13}$ dex); at $\log_{10}\rho_{0c} = 15.311404$, $p_{\rm eff} = 5.223$, Richardson value $\omega_0^2 = 6369417.611$ s$^{-2}$ ($N = 480$ relative error $1.29 \times 10^{-5}$).

| $N$ | $\log_{10}\rho_{0c}(\omega_0^2 = 0)$ | $ | \Delta\log_{10}\rho_{0c}$ | $ vs. Richardson Root (dex) | $\Delta_{\rm ADM}$ (dex) | $\Delta_{M_B}$ (dex) | $\omega_0^2(15.311404)$ (s$^{-2}$) | Rel. Diff. ($\omega_0^2$) |
|---|---|---|---|---|---|---|---|---|
| 80 | 15.31645808 | 4.83e-05 | +5.40e-05 | +4.95e-05 | 6432300.503336 | 9.87e-03 | | |
| 120 | 15.31642223 | 1.24e-05 | +1.82e-05 | +1.37e-05 | 6385729.603235 | 2.56e-03 | | |
| 160 | 15.31641426 | 4.46e-06 | +1.02e-05 | +5.72e-06 | 6375276.486537 | 9.20e-04 | | |
| 240 | 15.31641079 | 9.80e-07 | +6.71e-06 | +2.24e-06 | 6370703.105233 | 2.02e-04 | | |
| 360 | 15.31641001 | 2.06e-07 | +5.94e-06 | +1.47e-06 | 6369680.451299 | 4.13e-05 | | |
| 480 | 15.31640987 | 6.72e-08 | +5.80e-06 | +1.33e-06 | 6369499.539632 | 1.29e-05 | | |

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| $N$ | $\log_{10}\rho_{0c}(\omega_0^2=0)$ | $ | \Delta\log_{10}\rho_{0c} | $ vs. Richardson Root (dex) | $\Delta_{\mathrm{ADM}}$ (dex) | $\Delta_{M_B}$ (dex) | $\omega_0^2(15.311404)$ ($s^{-2}$) | Rel. Diff. ($\omega_0^2$) |
|---|---|---|---|---|---|---|---|---|
| 600 | 15.31640983 | 2.56e-08 | +5.76e-06 | +1.29e-06 | 6369446.532399 | 4.54e-06 | | |
| 800 | 15.31640982 | 7.55e-09 | +5.74e-06 | +1.27e-06 | 6369424.048161 | 1.01e-06 | | |
| 1200 | 15.31640981 | 1.35e-09 | +5.73e-06 | +1.26e-06 | 6369418.385568 | 1.22e-07 | | |

**Interpretive Note (Boundary of Theoretical Closure and Conditional Stability Closing Statement):** With the exact left-hand side sign (`scalar_m_sign = -1.0`), the root of the GL2 operator converges with respect to $N$ and $R_{\max}$ to the reported precision ($< 10^{-7}$ dex, `15.31640981`), and lies within $\leq 6.9\times10^{-6}$ dex of the ADM mass maximum across all 5 sequences (for SLy $L = 10$ km, $\alpha_Y = 1$: 15.316410 vs. 15.316404, $+5.80\times10^{-6}$ dex; the diagnostic comparison with `scalar_m_sign = +1.0` is documented in Section 7.4 of `step22`). For SLy $L = 10$ km, $\alpha_Y = 1$, the relative $L^2$ differences between the eigenfunction and the static tangent are $\delta h = 5.04\times10^{-4}$, $\delta m = 1.77\times10^{-4}$, $\delta\phi_L = 2.27\times10^{-4}$, $\delta\phi'_L = 3.09\times10^{-4}$, while the relative difference of the free-surface displacements is 3.54% (`rel_diff_xi_R_exact_minus_vs_static_tangent = 0.035406`). While the minus-sign root and grid convergence represent a major step forward, because of the 3.54% free-surface difference and the ODE vs. quadrature baryonic-mass turning-point spread they do not by themselves constitute an independent proof of the full radial fundamental spectrum and free-surface variation; therefore, the stability closing statement is formulated conditionally until the independent verification of the coupled variational derivation and boundary-condition matching is concluded.

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## 1.4 4. Local Archival Manuscripts Used (`manuscripts/`)

All referenced V19–V23 LaTeX (`.tex`) source files are included in the `manuscripts/` subdirectory:

1. **`manuscripts/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex`** and **`manuscripts/tal_covariant_closure_50aa2db0_EN_2026-10-24.tex`**
   - *Version & DOI:* V23 (`50aa2db0`), Zenodo DOI `10.5281/zenodo.23639084`.
   - *Role in the Derivation:* Primary source for the two covariant action candidates (Formulation I bilateral kernel action and Formulation II local Helmholtz–Yukawa scalar–tensor action), Bianchi residuals, coupled radial stability BVP (`eq:factored_diff`, `eq:coupled_bvp`), and H0/H1 clock coupling (Sirius B and 8558 kg granite chamber).
2. **`manuscripts/hu_p1.tex`**, **`manuscripts/hu_p2.tex`**, **`manuscripts/hu_p3.tex`**, **`manuscripts/en_part1.tex`**, **`manuscripts/en_part2.tex`**
   - *Version & DOI:* V22 Book Edition chapter sources, Zenodo DOI `10.5281/zenodo.23604423`.
   - *Role in the Derivation:* Source of verbatim citations for the Cauchy derivation (Chapter 2), 3D top-hat and SPARC 175-galaxy analysis (Chapters 4–5), V17 H.E.S.S./MAGIC gamma-ray timing audit (Chapter 6), 7-segment piecewise-polytrope TOV and $s = r^2$ regularization (Chapter 7), coupled radial pulsation (Chapter 8), Sirius B redshift (Chapter 9), 8558 kg granite chamber and 8-coil solenoid experiment (Chapter 10), and the numerical Appendix.
3. **`manuscripts/tal_book_HU_2026-10-24.tex`**, **`manuscripts/tal_book_EN_2026-10-24.tex`**, **`manuscripts/tal_book_DE_2026-10-24.tex`**
   - *Version & DOI:* V22 Book Edition unified Hungarian, English, and German LaTeX monograph files, Zenodo DOI `10.5281/zenodo.23604423`.
4. **`manuscripts/sparc_tophat_record_2026-10-22_v5.tex`**
   - *Version & DOI:* V19 SPARC 3D spherical top-hat manuscript, Zenodo DOI `10.5281/zenodo.23497128`.
5. **`manuscripts/tal_v20_fop.tex`**
   - *Version & DOI:* V20 laboratory two-clock and 8-coil magnetic manuscript (*Foundations of Physics* format), Zenodo DOI `10.5281/zenodo.23540741`.
6. **`manuscripts/tal_v21_fop.tex`**
   - *Version & DOI:* V21 compact stellar TOV and Sirius B manuscript, Zenodo DOI `10.5281/zenodo.23602501`.

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## 1.5 5. Python Programs & SPARC Data Files

### 1.5.1 5.1. Python Programs (in the Root Directory and code/ Subdirectory)

1. `curved_yukawa_adm_and_stability.py` — Module solving the coupled interior + curved exterior ($r \in [R_*, R_* + 12L]$) Einstein–scalar boundary-value problem of `yukawa_action_el` (**Branch II-Fluid**), verifying $R_{\max}$-independence, curved vs. flat Robin slope, recomputed mass–radius sequences in $M_{\rm ADM}$ and $M_B$ (`curved_yukawa_adm_sequences.csv`), and coupled $(\delta h, \delta\phi_L, \delta m)$ dynamical radial pulsation eigenvalues (`curved_yukawa_coupled_stability.csv`, `curved_yukawa_audit_summary.json`).
2. `covariant_closure_50aa2db0.py` — Main computational and symbolic verification module of V23 (`10.5281/zenodo.23639084`: SymPy Bianchi identities, Formulation I and II TOV comparison — including `yukawa_action_el_curved_adm` rows —, coupled `coupled_bvp` / `sym_galerkin` radial stability, $W \to 4W$ scaling invariance, Sirius B and 8558 kg granite chamber H0/H1 clock coupling).
3. `radial_modes.py` — Finite-element / banded-matrix eigenvalue solver for the Chandrasekhar Sturm–Liouville radial pulsation operator (`std`, `modA`, `modB`, and `fluid` vs. `metric` inertia weight).
4. `tal_profiles.py` — Profile module implementing $s = r^2$ even-parity central PCHIP interpolation (`Audit Fix 01`), the 3D top-hat shell integral, and the `gradient`-mode exterior vacuum-halo lapse correction $\Delta\nu_{\rm ext}$ on $r \in [R_*, R_* + L]$.
5. `stellar_tal_v21.py` (and `code/stellar_tal.py`, `code/diagnostics.py`) — V21/V23 module containing the Read et al. (2009) 7-segment piecewise-polytrope equations of state (SLy, APR4, H4), the log-enthalpy ($h$) TOV integrator (769 models, 33 causal endpoints), and the white-dwarf (Sirius B) integrator.
6. `make_revision3_checks.py` (along with `code/energy_integrator.py`, `code/fields.py`, `code/tophat_finite_source.py`, `code/v20_fixed_case_runner.py`, `code/build_pdf.py`) — Scripts running the V20 (`10.5281/zenodo.23540741`) 8-coil solenoid elliptic Biot–Savart check, $L_p = 0.630957$ m peak recalculation, $r = L/t$ compactified exterior integral (Paths A and B), and $L = 0.95$ m interior/exterior switching identity.
7. `sparkto_b_raw.py` and `sparc_tophat.py` — V19 program computing the 3D spherical top-hat mean density ($\bar\rho_L(\mathbf{x})$) and $f_L = \exp(-\gamma_L \bar\rho_L)$ factor of the 3D axisymmetric baryonic disk+gas+bulge model for all 175 SPARC galaxies.
8. `SPARC_mode.py`, `ta_pipeline.py`, `ta_foundation.py`, `sparc_full_runner.py`, `tal_factor_7th_model.py`, `ta_gammaL.py`, `ta_run.py`, `ta_run_l0.py`, `ta_run_l0_v2.py` — Programs running the SPARC 175 / 155 / 123 galaxy photometric vs. derivative-based (`foundation`) density comparison (0 vs. 239 / 301–303 negative-density points), the NGC 3198 ($R = 30.08$ kpc) control point, and V17/V19 7th-model rotation-curve fits.

### 1.5.2 5.2. SPARC and V19–V23 Data Files (data/, csv/, run_output/)

- `data/Rotmod_LTG.zip` (SHA-256: `0a80cc90714828cc28b7dd57923576714d209f2490328c087c4a4ad607faf588`) and `data/Rotmod_LTG/*_rotmod.dat` — Original Lelli, McGaugh & Schombert (2016) rotation-curve data files for all 175 SPARC galaxies (including `data/Rotmod_LTG/NGC3198_rotmod.dat`).
- `data/SPARC_Lelli2016c.mrt` (SHA-256: `5aa0501f6b0d881fa579030e315e7b5b6ef561a5bd3a07472f9929c7e5728243`) — SPARC Table 1 (distances, inclinations, $L_{3.6}$, $R_{\rm disk}$, $M_{\rm HI}$, $R_{\rm HI}$, quality flags).
- `data/blind_123.txt` (and `v17_package/github/data/blind_123.txt`, SHA-256: `3637d2f5bda6b2b97e645c5fba5cf39d310a1826bee388286c2c74b9171aa283`) — Hash-locked 123-galaxy blind sample list of V17/V19.
- `data/NGC3198_table.csv` and `data/NGC3198_table.md` — Detailed table of all 43 radial points of NGC 3198 (highlighting the $R = 30.08$ kpc control point).
- `data/sparc_tophat_per_galaxy_max.csv` and `data/sparc_tophat_summary.json` — V19 3D spherical top-hat peak values and Monte Carlo cross-checks across all 175 SPARC galaxies.
- `data/v19_7th_model/` and `run_output/` — Reference outputs of the V17/V19 baryonic baselines (`points_vbar.csv`, `summary_vbar.csv`) and the 7th model (`points_tal_fL_FITTED_forward.csv`, `fit_scan_L.csv`, `comparison.json`, `run_meta.json`).
- `csv/` and `data/v23/` — Reference files of the V20 laboratory granite chamber and 8-coil solenoid, together with the V23 (`50aa2db0`) and Step 22 curved exterior ADM mass and coupled stability output files (`curved_yukawa_adm_sequences.csv`, `curved_yukawa_coupled_stability.csv`, `curved_yukawa_audit_summary.json`, `conservation_Residuals.json`, `closure_tov_comparison.csv`, `coupled_stability_results.csv`, `clock_coupling_H0_H1.csv`).
- `data/v21/` and `observational_inputs.csv` — V21/V23 769 stellar models (`mr_sequences.csv`, `stellar_summary.csv`, `causal_endpoint_convergence.csv`, `profiles_L10km.csv`, `radius_shifts.csv`, `sirius_redshift.csv`, `numerical_diagnostics.json`, `wd_summary.json`) and NICER/LIGO/pulsar observational inputs.

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## 1.6 6. Exact Reproduction Instructions

### 1.6.1 6.1. Software Dependencies

- **Operating System:** Linux / macOS / POSIX-compatible environment
- **Python Version:** `Python >= 3.10` (tested under Python 3.11–3.13; on the local workstation, `/workspace/scratch/venv_openitems/bin/python3` has all scientific packages pre-installed)
- **Required Python Packages:** `numpy`, `scipy`, `pandas`, `sympy`, `matplotlib` Installation (if needed):

```
pip install numpy scipy pandas sympy matplotlib
```

### 1.6.2 6.2. Running Each Program & Expected Outputs

Every command below can be executed directly from the unpacked `scalar-tensor-yukawa-fit` working directory without network access:

1. **Curved Exterior `yukawa_action_el` Einstein–Scalar BVP, ADM Mass ($M_{\rm ADM}$), Mass–Radius Curves, and Coupled $(\delta h, \delta\phi_L, \delta m)$ Radial Stability (`curved_yukawa_adm_and_stability.py`):**

```
python3 curved_yukawa_adm_and_stability.py
```

   - *Expected Output (~2–3 minutes):*
   - Generates `curved_yukawa_adm_sequences.csv`, `curved_yukawa_coupled_stability.csv`, and `curved_yukawa_audit_summary.json` (and updates `data/v23/`).
   - Verifies $R_{\max}$-independence ($\Delta k_{\rm curved}/k_{\rm curved} = 6.17 \times 10^{-12}$, $\Delta M_{\rm ADM}/M_{\rm ADM} = 5.54 \times 10^{-13}$), curved vs. flat Robin slope (+24.74% for SLy $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$), ADM mass peaks ($M^{\rm SLy,10km}_{\rm ADM,max} = 2.03321\,M_\odot$, $R^{\rm ADM}_{1.4} = 11.5359$ km), finite-difference derivative tests ($\partial\phi'_L(R_*)/\partial m(R_*) = -1.68365 \times 10^{-4}$ km$^{-2}$, interior $r = 6$ km `a_m_sc` $= -1.72833 \times 10^{-4}$ km$^{-3}$), and coupled stability zero crossings (`scalar_m_sign = -1.0`: $\log_{10}\rho_{0c}(\omega_0^2 = 0) = 15.316410$, $\Delta_{\rm ADM} = +5.80 \times 10^{-6}$ dex, and $-6.86 \times 10^{-6}$ dex at $L = 1$ km).

2. **SPARC 175 Galaxies ($\rho < 0$ Audit), NGC 3198 ($R = 30.08$ kpc) Control Point, and V19 3D Top-Hat Peaks (`SPARC_mode.py`):**

```
python3 SPARC_mode.py --mode summary
```

   - *Expected Output (JSON on stdout, ~0.2 s):*
   - `"photometric_and_3D_tophat_yukawa_rho_le_0_points": 0` vs. `"derivative_formula_rho_lt_0_points_visible140_Ydisk_0p5": 239` (and 301 across all 175 galaxies).
   - `"NGC3198_control_point": "R_kpc": 30.08, "V_obs_km_s": 146.0, "V_bar_Ydisk_1p0_km_s": 88.53, "C_req_Ydisk_1p0": 2.72, "V_bar_Ydisk_0p5_km_s": 70.32, "C_req_Ydisk_0p5": 4.311`.

3. **SPARC 3D Spherical Top-Hat Mean Density and $f_L$ Factor Calculation (`sparkto_b_raw.py` / `sparc_tophat.py`):**

```
python3 sparkto_b_raw.py --galaxies NGC3198,UGC02487,UGC03546 --skip_checks --out ./output_tophat_check
```

4. **V20 Eight-Coil Magnetic-Field Experiment, Elliptic Biot–Savart, $r = L/t$ Exterior Quadrature, and $L = 0.95$ m Identity (`make_revision3_checks.py`):**

```
python3 make_revision3_checks.py . ./csv/revision3_checks_reproduced.json
```

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**5. V23 Covariant Action Closure, Bianchi Residuals, Coupled TOV Stability BVP, Sirius B, and 8558 kg Granite Chamber (`covariant_closure_50aa2db0.py`):**

```
python3 covariant_closure_50aa2db0.py
```

**6. SHA-256 Integrity Verification of Steps 1–22 and All Package Files:**

```
sha256sum -c SHA256SUMS.txt
```

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# 2 Step 1: Mathematical Equivalence of the Form of the TAL Factor $f_L = \exp(-\gamma_L \bar{\rho}_L)$ and the String-Theory Dilaton Coupling

Source file: step1_dilaton_form_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step1_dilaton_form_en.md (English translation of step1_dilaton_form.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V23 (50aa2db0): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_EN_2026-10-24.tex & tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (DOI: 10.5281/zenodo.23639084)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex & tal_book_HU_2026-10-24.tex (DOI: 10.5281/zenodo.23604423)
- V19 (model 7): 3D spherical top-hat law (DOI: 10.5281/zenodo.23497128)
- Local working note: docs/C_derivation_status.md (line 15: Time_Attraction.pdf (v1-v7))

## 2.1 1. Verbatim definitions of the Time Attraction Law (TAL) from the local manuscripts

In the local V19–V23 manuscripts, the positive-definite exponential time-flow / damping factor $f_L(\mathbf{x})$, the 3D spherical top-hat mean density $\bar{\rho}_L(\mathbf{x})$, and the scale-dependent coupling coefficient $\gamma_L$ (together with the coupling constant $C$) appear verbatim in the following form:

### 2.1.1 1.1. The exponential law and the definition of the constants (V22 hu_p1.tex, lines 141–144 and 245–253)

- **Exponential time-flow factor:**

$$f_L(\mathbf{x}) \;=\; \exp\!\big(-\gamma_L\,\bar{\rho}_L(\mathbf{x})\big), \qquad 0 < f_L(\mathbf{x}) \le 1$$

- **3D spherical top-hat mean density (with smoothing radius $L$ and volume $V_L = \frac{4}{3}\pi L^3$):**

$$\bar{\rho}_L(\mathbf{x}) \;=\; \frac{3}{4\pi L^3}\int_{|\mathbf{y}-\mathbf{x}|\le L} \rho(\mathbf{y})\, d^3y \;\ge\; 0$$

- **Fundamental gravitational–temporal constants (in SI units, CODATA $G = 6{,}67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$, $c = 2{,}99792458 \times 10^{8}\ \mathrm{m\,s^{-1}}$):**

$$C \;\equiv\; \frac{4\pi G}{c^2} \;=\; 9{,}33306 \times 10^{-27}\ \mathrm{m\,kg^{-1}}, \qquad \frac{G}{c^2} \;=\; \frac{C}{4\pi} \;=\; 7{,}425915 \times 10^{-28}\ \mathrm{m\,kg^{-1}}$$

$$\gamma_L \;=\; \frac{G\,L^2}{c^2} \;=\; \frac{C\,L^2}{4\pi} \;=\; 7{,}425915 \times 10^{-28}\, L^2\ \mathrm{m^3\,kg^{-1}}$$

### 2.1.2 1.2. The covariant auxiliary scalar and the matter-Lagrangian coupling in V23 (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex, lines 32–46 and 54; C_derivation_status.md, line 15)

- The early Time_Attraction.pdf (v1–v7, cited in line 15 of docs/C_derivation_status.md) and the V23 50aa2db0 manuscript (Section 1, Eq. 1–3 and line 54) weight the matter action by the exponential factor $f_L$:

$$S \;=\; \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R + \mathcal{L}_X(g_{\mu\nu},\phi_L,\lambda_L) + \mathcal{L}_{\mathrm{matter}}\right], \qquad \mathcal{L}_{\mathrm{fluid}}(n,\phi_L) \;=\; -f_L(\phi_L)\,\varepsilon(n)$$

or, in the formulation of versions 1–7:

$$S \;=\; \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R + f_L\,\mathcal{L}_m\right]$$

- In Formulation I of V23 (Eq. 2–3), the dimensionless scalar $\phi_L(x)$ is tied by the Lagrange multiplier $\lambda_L(x)$ to the 3D spherical top-hat average taken on the spatial hypersurface $\Sigma_u$ ($K_L(x,y) = \delta(\tau_u(x) - \tau_u(y))\,\Theta(L - d_\Sigma(x,y))/V_L(x)$):

$$\frac{\delta S}{\delta\lambda_L(x)} = 0 \quad\Longrightarrow\quad \phi_L(x) \;=\; \gamma_L\,\bar\rho_L(x), \qquad f_L(x) \;=\; e^{-\phi_L(x)} \;=\; \exp\!\big(-\gamma_L\,\bar\rho_L(x)\big).$$

---

## 2.2 2. The low-energy effective action of string theory and the conventions of the dilaton coupling

In the low-energy effective field theory of string theory (bosonic and superstring), reduced to 4 dimensions, the gravitational sector necessarily contains a scalar field, the **dilaton ($\phi$)** (see e.g. Polchinski: *String Theory*, Vol. I, Sec. 3.7; Gasperini & Veneziano, *Phys. Rept.* **373**, 1, 2003; Damour & Polyakov, *Nucl. Phys. B* **423**, 532, 1994).

1. **String frame:** At tree level (weighted by the square of the string coupling constant $g_s = e^{\phi}$, i.e. by $g_s^{-2} = e^{-2\phi}$), the effective action of the gravitational and closed/open string sector is:

$$S_{\rm string} \;=\; \int d^4x\sqrt{-g_{(s)}}\,e^{-2\phi}\left[\frac{c^4}{16\pi G_0}\Big(R_{(s)} + 4\,g^{\mu\nu}_{(s)}\partial_\mu\phi\,\partial_\nu\phi\Big) - \dots\right] + \int d^4x\sqrt{-g_{(s)}}\,e^{-\eta\phi}\,\mathcal{L}_m\big(g_{(s)\mu\nu},\Psi\big)\,,$$

where:

   - in the closed-string / tree-level sector $\eta = 2$, i.e. the matter Lagrangian is multiplied directly by the exponential factor $e^{-2\phi}\,\mathcal{L}_m$;
   - in the open-string / D-brane (Born–Infeld) sector $\eta = 1$, i.e. the coupling factor is $e^{-\phi}\,\mathcal{L}_m$.

2. **Einstein frame — exponential matter coupling alongside a canonical $R$ term:** In order for the factor in front of the curvature scalar $R$ (as in TAL V23 Eq. (1): $\frac{c^4}{16\pi G}R$) to take a canonical, $\phi$-independent form, we perform the conformal rescaling $g_{(s)\mu\nu} = e^{2\phi}g_{\mu\nu}$ (Damour & Polyakov 1994; Fujii & Maeda: *The Scalar-Tensor Theory of Gravitation*, Cambridge Univ. Press, 2003). The curvature term then becomes purely Einstein–Hilbert in form, while an exponential dilaton multiplier appears in the matter sector:

$$S_{\rm Einstein} \;=\; \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}\Big(R - 2\,g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi\Big) \;+\; e^{-2\beta_{\rm dil}\phi}\,\mathcal{L}_m(\Psi, g_{\mu\nu})\right],$$

(where, depending on the convention and the matter sector, the usual notation of the exponential multiplier is $e^{-2\phi}$, $e^{-\phi}$, or in general scalar–tensor notation $e^{-2\beta_{\rm dil}\phi}$).

---

## 2.3 3. The formal mathematical correspondence (Step 1 Mapping)

Comparing the TAL matter-coupling term ($f_L(\mathbf{x})\,\mathcal{L}_m$, where $f_L(\mathbf{x}) = e^{-\phi_L(\mathbf{x})} = \exp(-\gamma_L\bar\rho_L(\mathbf{x}))$) with the exponential matter coupling of the string-theory dilaton ($e^{-2\phi}\,\mathcal{L}_m$ in the $2\phi$ convention, or $e^{-\phi}\,\mathcal{L}_m$ in the $1\phi$ convention), the **mathematical functional form of the two expressions is exactly the same**:

$$\boxed{f_L(\mathbf{x}) \;=\; \exp\!\big(-\gamma_L\,\bar\rho_L(\mathbf{x})\big) \;=\; e^{-\phi_L(\mathbf{x})} \quad\Longleftrightarrow\quad e^{-2\phi(\mathbf{x})}\ \ (\text{or } e^{-\phi(\mathbf{x})})}$$

i.e. the direct correspondence of the dimensionless exponents:

$$\boxed{\phi_L(\mathbf{x}) \;\equiv\; \gamma_L\,\bar\rho_L(\mathbf{x}) \;=\; \frac{GL^2}{c^2}\,\bar\rho_L(\mathbf{x}) \quad\Longleftrightarrow\quad 2\phi(\mathbf{x})\ \ (\text{in the } e^{-2\phi}\text{ convention}),\ \text{or}\ \phi(\mathbf{x})\ \ (\text{in the } e^{-\phi}\text{ convention}).}$$

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#### 2.3.1 Open question recorded for later steps (not resolved in this step)

- **The similarity:** In both theories, a negative exponential function of a dimensionless scalar quantity ($e^{-\phi_L}$ and $e^{-2\phi}$, respectively) multiplies the matter Lagrangian.
- **The difference (open question for Step 2 and beyond):** In the basic definition of the Time Attraction Law (and in Formulation I of V23), the exponent $\phi_L(\mathbf{x}) = \gamma_L \bar{\rho}_L(\mathbf{x})$ is determined algebraically / via a nonlocal integral directly by the mass-energy density $\bar{\rho}_L$ averaged over the scale $L$, whereas in string theory the dilaton $\phi$ is an independent dynamical field with its own kinetic term ($\propto (\partial\phi)^2$). The connection between this dynamical kinetic term and the local Helmholtz–Yukawa operator of Formulation II of V23, $(-L^2\Box_g + 1)\phi_L = \alpha_Y \gamma_L \rho_{\rm src}$, is examined in later steps; we do not touch it here.

---

## 2.4 4. Checks

### 2.4.1 4.1. Dimensional analysis (inverse-density dimension of $\gamma_L$)

We check whether the exponent $\gamma_L \bar{\rho}_L$ is dimensionless in SI base units $(\mathrm{m}, \mathrm{kg}, \mathrm{s})$:

- Dimension of Newton's gravitational constant:

$$[G] \;=\; \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$$

- Dimension of the square of the speed of light:

$$[c^2] \;=\; \mathrm{m}^2\,\mathrm{s}^{-2}$$

- Dimension of the mass–length ratio ($G/c^2$, or $C = 4\pi G/c^2$):

$$\left[\frac{G}{c^2}\right] \;=\; \frac{\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}}{\mathrm{m}^2\,\mathrm{s}^{-2}} \;=\; \mathrm{m}\,\mathrm{kg}^{-1}$$

- Dimension of the square of the smoothing scale:

$$[L^2] \;=\; \mathrm{m}^2$$

- Consequently, the dimension of the coupling coefficient $\gamma_L = \frac{GL^2}{c^2}$:

$$[\gamma_L] \;=\; \left[\frac{G}{c^2}\right]\cdot[L^2] \;=\; \left(\mathrm{m}\,\mathrm{kg}^{-1}\right)\cdot \mathrm{m}^2 \;=\; \mathrm{m}^3\,\mathrm{kg}^{-1} \;=\; \frac{1}{\mathrm{kg}\,\mathrm{m}^{-3}} \;=\; \frac{1}{[\text{density}]} \quad \checkmark$$

- Since the dimension of the 3D volumetric mass density is $[\bar{\rho}_L] = \mathrm{kg}\,\mathrm{m}^{-3}$, the product is:

$$[\gamma_L\,\bar{\rho}_L] \;=\; \left(\mathrm{m}^3\,\mathrm{kg}^{-1}\right)\cdot\left(\mathrm{kg}\,\mathrm{m}^{-3}\right) \;=\; 1 \quad (\text{dimensionless}) \quad \checkmark$$

*(Note for the written-out check: the ratio $[G/c^2]$ by itself is $\mathrm{m\,kg}^{-1}$ [the reciprocal of a linear mass density], and multiplication by the factor $L^2$ [$\mathrm{m}^2$] makes the dimension of $\gamma_L = GL^2/c^2$ equal to $\mathrm{m}^3\,\mathrm{kg}^{-1}$, which is exactly the reciprocal of the 3D volumetric density [$\mathrm{kg\,m}^{-3}$], consistent with Eq. (3) and Eq. (5) of V22* `hu_p1.tex`*.)*

### 2.4.2 4.2. Checks of limits and signs

1. **Vacuum limit ($\bar{\rho}_L \to 0$):** When the averaged mass-energy density in the neighbourhood of radius $L$ vanishes ($\bar{\rho}_L \to 0$), the exponent $\phi_L = \gamma_L \bar{\rho}_L \to 0$, so:

$$\lim_{\bar{\rho}_L \to 0} f_L(\bar{\rho}_L) \;=\; \exp(0) \;=\; 1 \quad \checkmark$$

This exactly recovers the unmodified standard matter coupling ($\mathcal{L}_{\rm matter} \to 1 \cdot \mathcal{L}_m$), which in the dilaton picture corresponds to the asymptotic vacuum $\phi \to 0$ ($e^{-2\phi} \to 1$).

2. **Positive-density domain ($\bar{\rho}_L > 0$):** Since $G > 0$, $c > 0$, $L > 0 \implies \gamma_L > 0$, and the 3D spherical top-hat average is strictly positive ($\bar{\rho}_L > 0$) under the non-negative energy condition ($\rho(\mathbf{y}) \geq 0$), we have $\gamma_L \bar{\rho}_L > 0$, therefore:

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$$0 \;<\; f_L(\bar{\rho}_L) \;=\; \exp(-\gamma_L \bar{\rho}_L) \;<\; 1 \qquad (\forall\, \bar{\rho}_L > 0) \quad \checkmark$$

and the function is strictly monotonically decreasing:

$$\frac{df_L}{d\bar{\rho}_L} \;=\; -\gamma_L \exp(-\gamma_L \bar{\rho}_L) \;<\; 0 \quad \checkmark$$

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# 3 Step 2: Conformal, Disformal, and Effective Fluid-Coupled Jordan Metrics ($\tilde{g}_{\mu\nu}$) and the Taxonomy of V19–V23 Model Branches

Source file: step2_conformal_disformal_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step2_conformal_disformal_en.md **Previous Step:** step1_dilaton_form_en.md (step1_dilaton_form.md)
**License:** CC BY-NC 4.0 **Local Archival Sources:**

- V23 (50aa2db0 Hungarian manuscript): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Sections 1, 1.1, 1.2, and 3, Eqs. 1–12; DOI: 10.5281/zenodo.23639084)
- V23 (50aa2db0 English manuscript): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_EN_2026-10-24.tex
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex and hu_p2.tex (DOI: 10.5281/zenodo.23604423)
- V23 SymPy verification script: covariant_closure_50aa2db0.py (lines 105–138)

## 3.1 1. Verbatim Citations from the V23 Manuscript (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU)

### 3.1.1 1.1. Formulation of the Covariant Action (V23 EN/HU Section 1, Eq. (1), lines 33–35)

Section 1 of the V23 manuscript (*"Covariant Action Principle: Two Explicit Formulations"*) establishes the following covariant action using the $(-+++)$ metric signature convention:

$$S\big[g_{\mu\nu},\,\phi_L,\,\Psi_{\rm matter}\big] \;=\; \int d^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}\,R(g) \;+\; \mathcal{L}_{\rm matter}\big(\Psi_{\rm matter},\,\tilde{g}_{\mu\nu}(g,\phi_L)\big) \;+\; \mathcal{L}_{\phi}\big(g_{\mu\nu},\,\phi_L\big)\right]. \qquad \text{(V23 Eq. 1)}$$

### 3.1.2 1.2. Verbatim Definition of the Microscopic $H_0$ vs. $H_1$ Coupling and the Two Effective Jordan Metrics (V23 Section 3, Eqs. (11)–(12), lines 95–101)

Section 3 of the V23 manuscript (*"Microscopic Clock and Spectral-Line Coupling from the Action ($H_0$ vs. $H_1$)"*) defines the two coupling branches and the two Jordan metrics verbatim as follows:

> *"In the covariant action (1), the microscopic coupling of atomic clocks and electromagnetic spectral lines is uniquely determined by which metric contracts the indices of the Dirac + Maxwell Lagrangian $\mathcal{L}_{atom}(\psi_e, A_\mu)$:*
>
> – **Minimal physical-metric coupling ($H_0$ — default covariant closure):** *$\mathcal{L}_{atom}(\psi_e, A_\mu; g_{\mu\nu})$ couples directly to the physical spacetime metric $g_{\mu\nu}$, while the macroscopic baryonic energy–momentum source is weighted by $f_L$ through Formulation I or II. Then the proper time of a stationary atom is $d\tau_{H_0} = \sqrt{-g_{00}}\,dt = e^{\nu(r)}dt$, the local transition frequencies $\omega_{atom} = m_e c^2 \alpha_{EM}^2/(2\hbar)$ are spacetime constants, and atomic clocks measure precisely the metric lapse function arising from the $f_L$-modified Einstein equations. The macroscopic clock anomaly appears exclusively through the gravitational potential change $\Delta\Phi_N$ originating from the modified mass:*
>
> $$\left.\frac{\Delta\nu}{\nu}\right|_{H_0} = \frac{\Delta\Phi_{\rm metric}}{c^2}, \qquad \Delta z_{H_0}(\text{Sirius B}) \;=\; z_{\rm GR}\left(\frac{\Delta M_B}{M_B}\right)_{\rm TAL}. \qquad \text{(V23 Eq. 11)}$$
>
> – **Disformal / conformal proper-time coupling ($H_1$ — direct clock-coupling hypothesis):** *$\mathcal{L}_{atom}(\psi_e, A_\mu; \tilde{g}_{\mu\nu})$ couples to the effective metric*
>
> $$\tilde{g}^{\rm disf}_{\mu\nu} \;=\; g_{\mu\nu} \;+\; \frac{1-f_L(\phi_L)^2}{c^2}\,u_\mu u_\nu \qquad \text{or} \qquad \tilde{g}^{\rm conf}_{\mu\nu} \;=\; f_L(\phi_L)^2\,g_{\mu\nu} \qquad \text{(V23 Eq. 12)}$$

*(with fixed $m_e$), so that the proper-time element is $d\tilde{\tau}_{H_1} = \sqrt{-\tilde{g}_{00}}\,dt = f_L(\mathbf{x})\,e^{\nu(\mathbf{x})}dt$. This induces a direct non-metric frequency shift:*

$$\left.\frac{\Delta\nu}{\nu}\right|_{H_1}(\mathbf{x}) \;=\; \frac{\Delta\Phi_{\rm metric}(\mathbf{x})}{c^2} \;+\; \big(f_L(\mathbf{x})-1\big) \;\simeq\; \frac{\Delta\Phi_{\rm metric}(\mathbf{x})}{c^2} \;-\; \phi_L(\mathbf{x}).$$"

---

## 3.2 2. Derivation of the Conformal ($\tilde{g}^{\rm conf}_{\mu\nu}$) Transformation and the Brans–Dicke Relation

In Step 1 (`step1_dilaton_form.md`), we proved that $f_L(\phi_L) = e^{-\phi_L}$. Substituting this into the second equation of V23 Eq. (12):

$$\boxed{\tilde{g}^{\rm conf}_{\mu\nu}(x) \;=\; f_L(\phi_L(x))^2\,g_{\mu\nu}(x) \;=\; e^{-2\phi_L(x)}\,g_{\mu\nu}(x), \qquad \Omega(\phi_L) \;\equiv\; f_L(\phi_L) \;=\; e^{-\phi_L}.}$$

### 3.2.1 2.1. Inverse Metric, Determinant, and Proper-Time Element

1. **Inverse conformal metric:** Since $\tilde{g}^{\rm conf}_{\mu\alpha}\tilde{g}^{\alpha\nu}_{\rm conf} = \delta^\nu_\mu$,

$$\tilde{g}^{\mu\nu}_{\rm conf} \;=\; f_L(\phi_L)^{-2}\,g^{\mu\nu} \;=\; e^{+2\phi_L}\,g^{\mu\nu}.$$

2. **Four-volume element (determinant):** In $D = 4$ spacetime dimensions, $\det(e^{-2\phi_L}g_{\mu\nu}) = (e^{-2\phi_L})^4\det(g_{\mu\nu})$, therefore:

$$\sqrt{-\tilde{g}_{\rm conf}} \;=\; f_L(\phi_L)^4\,\sqrt{-g} \;=\; e^{-4\phi_L}\,\sqrt{-g}.$$

3. **Proper-time element of a static observer ($dx^i = 0$):** In a static spherically symmetric physical metric ($g_{00} = -e^{2\nu(r)}$):

$$d\tilde{\tau}^2_{\rm conf} \;=\; -\frac{1}{c^2}\tilde{g}^{\rm conf}_{00}\,dt^2 \;=\; -f_L^2\,\frac{g_{00}}{c^2}\,dt^2 \;=\; f_L(r)^2\,e^{2\nu(r)}\,dt^2 \quad\Longrightarrow\quad \boxed{d\tilde{\tau}_{\rm conf} \;=\; f_L(r)\,e^{\nu(r)}\,dt \;=\; e^{\nu(r)-\phi_L(r)}\,dt.}$$

### 3.2.2 2.2. Conformal Transformation of the Ricci Scalar (Jordan Frame ↔ Einstein Frame)

From standard $D = 4$ conformal geometry (e.g., Wald 1984, Appendix D; Fujii & Maeda 2003), if $g_{\mu\nu} = \Omega^{-2}\tilde{g}^{\rm conf}_{\mu\nu} = e^{+2\phi_L}\tilde{g}^{\rm conf}_{\mu\nu}$ (that is, $\tilde{g}^{\rm conf}_{\mu\nu} = \Omega^2 g_{\mu\nu}$ with $\Omega = e^{-\phi_L}$), the relation between the two Ricci scalars is:

$$R(g) \;=\; \Omega^2\left[\tilde{R}(\tilde{g}_{\rm conf}) \;+\; 6\,\tilde{\Box}\ln\Omega \;-\; 6\,\tilde{g}^{\mu\nu}_{\rm conf}\,\partial_\mu\ln\Omega\,\partial_\nu\ln\Omega\right].$$

Accordingly, rewriting the Einstein–Hilbert term in the Jordan frame ($\sqrt{-g} = \Omega^{-4}\sqrt{-\tilde{g}_{\rm conf}}$) yields:

$$\int d^4x\,\sqrt{-g}\,\frac{c^4}{16\pi G}\,R(g) \;=\; \int d^4x\,\sqrt{-\tilde{g}_{\rm conf}}\,\frac{c^4}{16\pi G}\left[\Omega^{-2}\tilde{R}(\tilde{g}_{\rm conf}) \;-\; 6\,\Omega^{-4}\tilde{g}^{\mu\nu}_{\rm conf}\partial_\mu\Omega\,\partial_\nu\Omega\right],$$

where the boundary divergence term $\nabla_\mu(\sqrt{-g}\,6\Omega\nabla^\mu\Omega)$ has been dropped via integration by parts. Introducing the **Brans–Dicke scalar field**:

$$\boxed{\Phi_{\rm BD}(x) \;\equiv\; \Omega(x)^{-2} \;=\; f_L(x)^{-2} \;=\; e^{+2\phi_L(x)},}$$

for which $\frac{\partial_\mu\Phi_{\rm BD}}{\Phi_{\rm BD}} = +2\partial_\mu\phi_L$, the local scalar kinetic term of V23 Formulation II ($\mathcal{L}_\phi = -\frac{c^4}{8\pi G\tilde{\alpha}_L}\left[\frac{1}{2}g^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L + \frac{1}{2L^2}\phi_L^2\right]$, V23 Eq. 8) in the Jordan frame becomes:

$$-\frac{1}{2}g^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L \;\longrightarrow\; -\frac{\omega_{\rm BD}}{\Phi_{\rm BD}}\,\tilde{g}^{\mu\nu}_{\rm conf}\partial_\mu\Phi_{\rm BD}\,\partial_\nu\Phi_{\rm BD}.$$

Matching the coefficients yields the standard scalar–tensor identity:

$$2\omega_{\rm BD}+3 \;=\; \frac{1}{\tilde{\alpha}_L} \;=\; \frac{1}{2\beta^2} \quad\Longleftrightarrow\quad \boxed{\omega_{\rm BD} \;=\; \frac{1}{2\tilde{\alpha}_L}-\frac{3}{2} \;=\; \frac{1}{4\beta^2}-\frac{3}{2}.}$$

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*(The exact numerical values of $\tilde{\alpha}_L$, $\alpha_Y = 8\pi\tilde{\alpha}_L$, $\alpha_{std} = 2\tilde{\alpha}_L = \frac{\alpha_Y}{4\pi}$, and $\omega_{BD}$, together with the Cassini PPN bound, are analyzed in detail in Step 5.)*

---

## 3.3 3. The Disformal ($\tilde{g}^{\mathbf{disf}}_{\mu\nu}$) Coupling and Algebraic Projector Decomposition

The first equation of V23 Eq. (12) defines a purely timelike **Bekenstein-type disformal metric** (Bekenstein 1993, *Phys. Rev. D* 48, 3641) using the four-velocity $u^\mu$ of the medium ($g_{\mu\nu}u^\mu u^\nu = -c^2$, $u_\mu = g_{\mu\nu}u^\nu$):

$$\boxed{\tilde{g}^{\rm disf}_{\mu\nu} \;=\; g_{\mu\nu} \;+\; \frac{1 - f_L(\phi_L)^2}{c^2}\, u_\mu u_\nu \;=\; g_{\mu\nu} \;+\; \frac{1 - e^{-2\phi_L}}{c^2}\, u_\mu u_\nu.}$$

### 3.3.1 3.1. Exact Projector Decomposition (SymPy-Verified Identity, `covariant_closure_50aa2db0.py` lines 108–125)

Let us introduce the spatial projector of the spacetime metric orthogonal to $u^\mu$:

$$h_{\mu\nu} \;\equiv\; g_{\mu\nu} + \frac{1}{c^2} u_\mu u_\nu \qquad \left(h_{\mu\nu}u^\nu = 0, \quad h^\mu_{\ \alpha} h^\alpha_{\ \nu} = h^\mu_{\ \nu}, \quad h^\mu_{\ \mu} = 3\right).$$

Then $g_{\mu\nu} = h_{\mu\nu} - \frac{1}{c^2}u_\mu u_\nu$, so the separated projector form of the disformal metric is:

$$\boxed{\tilde{g}^{\rm disf}_{\mu\nu} \;=\; h_{\mu\nu} \;-\; f_L(\phi_L)^2\, \frac{u_\mu u_\nu}{c^2} \;=\; \left(g_{\mu\nu} + \frac{u_\mu u_\nu}{c^2}\right) \;-\; e^{-2\phi_L}\, \frac{u_\mu u_\nu}{c^2}.}$$

From this, the physical interpretation follows immediately:

1. **The disformal coupling leaves the spatial geometry untouched:** for every spatial displacement $\delta x^\mu$ orthogonal to $u^\mu$ ($u_\mu \delta x^\mu = 0$), $\tilde{g}^{\rm disf}_{\mu\nu}\delta x^\mu \delta x^\nu = g_{\mu\nu}\delta x^\mu \delta x^\nu$.
2. **It scales the timelike (lapse) component by precisely the factor $f_L(\phi_L) = e^{-\phi_L}$:** the proper time of a clock comoving with the medium ($\delta x^\mu = \frac{u^\mu}{c} d\tau$) is:

$$d\tilde{\tau}^2_{\rm disf} \;=\; -\frac{1}{c^2}\tilde{g}^{\rm disf}_{\mu\nu}\, dx^\mu dx^\nu \;=\; f_L(\phi_L)^2 \left(-\frac{1}{c^2} g_{\mu\nu} dx^\mu dx^\nu\right) \;=\; f_L(\phi_L)^2\, d\tau^2.$$

3. **Inverse disformal metric and determinant:** Since $h^{\mu\alpha}u_\alpha = 0$ and $\left(-\frac{u_\mu u_\alpha}{c^2}\right)\left(-\frac{u^\alpha u^\nu}{c^2}\right) = -\frac{u_\mu u^\nu}{c^2}$, the inverse metric and volume element are exactly:

$$\boxed{\tilde{g}^{\mu\nu}_{\rm disf} \;=\; h^{\mu\nu} \;-\; f_L(\phi_L)^{-2}\, \frac{u^\mu u^\nu}{c^2} \;=\; g^{\mu\nu} \;-\; \frac{f_L(\phi_L)^{-2} - 1}{c^2}\, u^\mu u^\nu, \qquad \sqrt{-\tilde{g}_{\rm disf}} \;=\; f_L(\phi_L)\sqrt{-g} \;=\; e^{-\phi_L}\sqrt{-g}.}$$

### 3.3.2 3.2. General Static Spherically Symmetric Background and `SymPy` Verification

In a static spherically symmetric physical background metric:

$$ds^2 \;=\; g_{\mu\nu}dx^\mu dx^\nu \;=\; -e^{2\nu(r)}c^2dt^2 + e^{2\lambda(r)}dr^2 + r^2 d\Omega^2, \qquad g_{00} = -e^{2\nu(r)},$$

the four-velocity of a stationary (comoving) observer is $u^\mu = \left(c/\sqrt{-g_{00}},\, 0,\, 0,\, 0\right) = \left(c\, e^{-\nu(r)},\, 0,\, 0,\, 0\right)$, and its covariant form is $u_\mu = g_{\mu\nu}u^\nu = \left(-\sqrt{-g_{00}}\, c,\, 0,\, 0,\, 0\right) = \left(-c\, e^{\nu(r)},\, 0,\, 0,\, 0\right)$. Thus:

$$\frac{u_0 u_0}{c^2} \;=\; -g_{00} \;=\; +e^{2\nu(r)},$$

and substituting this yields the $(0,0)$-component of the disformal metric:

$$\tilde{g}^{\rm disf}_{00} \;=\; g_{00} \;+\; \left(1 - f_L(r)^2\right)\frac{u_0 u_0}{c^2} \;=\; g_{00} \;+\; \left(1 - f_L(r)^2\right)\left(-g_{00}\right) \;=\; f_L(r)^2\, g_{00} \;=\; -f_L(r)^2\, e^{2\nu(r)}.$$

In `covariant_closure_50aa2db0.py` (lines 108–125), within the local orthonormal tetrad ($g_{\hat{a}\hat{b}} = \eta_{\hat{a}\hat{b}} = \mathrm{diag}(-1,1,1,1)$, $u^{\hat{a}} = (c,0,0,0)$, $u_{\hat{a}} = (-c,0,0,0)$), the same identity is verified: $\tilde{g}^{\rm disf}_{\hat{0}\hat{0}} = -1 + (1 - f_L^2) = -f_L^2$, so the general coordinate expression is $\tilde{g}^{\rm disf}_{00} = e^{\hat{0}}_{\ 0} e^{\hat{0}}_{\ 0} \tilde{g}^{\rm disf}_{\hat{0}\hat{0}} = e^{2\nu}(-f_L^2) = f_L^2 g_{00}$.

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## 3.4 4. Explicit Taxonomy of V19–V23 Model Branches (Branch Taxonomy Table)

For peer-review transparency, the table below unambiguously separates the model branches appearing across the V19–V23 manuscripts and the present derivation chain, including their action principle/definition, volume element, hydrostatic equation, clock/light coupling, and domain of validity:

| Branch Code | Name and Manuscript Source | Scalar / Kernel Definition and 2nd Moment | Matter Coupling / Volume Element | Internal Hydrostatic Equilibrium | Electromagnetic / Clock / PPN Behavior |
|---|---|---|---|---|---|
| **Branch I-A** (`tophat_mass`, `tophat_gradient`) | V21 phenomenological 3D spherical top-hat TOV (`stellar_tal_v21.py`, V23 Eqs. 4–5) | Nonlocal 3D spherical window: $\phi_L = \gamma_L(K_L * \varepsilon)$, $K_L = \frac{3}{4\pi L^3}\Theta(L-r)$, $1 + \frac{L^2}{10}\nabla^2 + \dots$ | Phenomenological $f_L$ multiplier in $m'$ (`mass`) or $\nu'$ (`gradient`), with algebraic $T^\mu_\nu(X)$ closure | $h'(r) = -\nu'(r)$ (`mass`: $\nu' = A_{\rm geo}$; `gradient`: $\nu' = f_L A_{\rm geo}$) | $H_0$: minimal physical metric; $H_1$: direct $f_L$ clock shift (excluded by Sirius B for $L \geq 1$ km) |
| **Branch I-B** (`bilateral_eom`, `bilateral_lapse_weighted`) | V23 Formulation I: bilateral nonlocal kernel action (V23 Eqs. 2–3, 6–7) | Symmetric two-point kernel $K_L(x,y) = K_L(y,x)$, $1 + \frac{L^2}{10}\nabla^2 + \dots$ | $S_{\mathrm{I},m} = -\int d^4x\sqrt{-g}\, e^{-\phi_L}\varepsilon$ (in fixed-metric Euclidean kernel approximation) | $V_{\rm eff}(r) = f_L(r) - \gamma_L \int K_L f_L \varepsilon$ with reciprocity force | Macroscopic hydrostatic test branch; metric variation inside the kernel is not closed |
| **Branch II-Fluid** (`yukawa_action_el`) | V23 Formulation II: local Yukawa-screened Einstein–scalar–fluid action (V23 Eqs. 8–10) | Local Helmholtz–Yukawa PDE: $(-\Box_g + L^{-2})\phi_L = \frac{\alpha_Y G}{c^2} e^{-\phi_L}\varepsilon$, $1 + L^2\nabla^2 + \dots$ | $S_m = -\int d^4x\sqrt{-g}\, e^{-\phi_L}\varepsilon(n)$, $P_{\rm matter} = e^{-\phi_L}P$, $\rho_\phi = \frac{B(\phi_L')^2 + \phi_L^2/L^2}{2\alpha_Y}$ | $h'(r) = -\nu'(r) + \phi_L'(r)$ (fifth force + curved exterior scalar tail $M_{\rm ADM}$) | Under $H_0$, atomic clocks couple to physical $g_{\mu\nu}$; turning points of $M_{\rm ADM}$ and $M_B$ coincide |
| **Branch II-Conf** ($\tilde g^{\rm conf}_{\mu\nu} = e^{-2\phi_L} g_{\mu\nu}$) | Universal conformal Jordan–Brans–Dicke / dilaton metric (V23 Eq. 12b, `step1`, `step5`) | Same local Yukawa PDE; $\Phi_{\rm BD} = e^{+2\phi_L}$ | $\sqrt{-\tilde g_{\rm conf}} = e^{-4\phi_L}\sqrt{-g}$; conformally invariant Maxwell action ($\tilde F_{\mu\nu}\tilde F^{\mu\nu}\sqrt{-\tilde g} = F_{\mu\nu}F^{\mu\nu}\sqrt{-g}$) | In the non-relativistic limit ($\varepsilon \simeq \rho_0 c^2 \gg 3P$), coincides with the $\beta = \sqrt{\alpha_{\rm std}/2}$ fifth force | Standard conformal Brans–Dicke / PPN dictionary applies to this branch: $\alpha_{\rm std} = \frac{\alpha_Y}{4\pi}$, $\omega_{\rm BD} = \frac{4\pi}{\alpha_Y} - \frac{3}{2} \approx 11.07$ ($\alpha_Y = 1$), light deflection is conformally invariant |
| **Branch II-Disf** ($\tilde g^{\rm disf}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2}{c^2} u_\mu u_\nu$) | Comoving timelike disformal clock and EM metric (V23 Eq. 12a, V20/V22 solenoid, `step14`–`step15`) | Same local Yukawa PDE, augmented with magnetic source $\rho_B = \frac{ | \mathbf{B} | ^2}{2\mu_0 c^2}$ | $\sqrt{-\tilde g_{\rm disf}} = f_L\sqrt{-g} = e^{-\phi_L}\sqrt{-g}$ (coincides with the $e^{-\phi_L}$ weight of Branch II-Fluid!) |
| **Branch II-Cham** ($\beta_{\rm eff}(\rho)$ extension) | Density-dependent chameleon / symmetron nonlinear EFT hypothesis (`step11`) | Density-dependent effective mass $m^2_{\rm eff}(\rho)$ and coupling $\beta_{\rm eff}(\rho)$ | Nonlinear $V(\phi) + A(\phi)\rho$ extension beyond the quadratic constant-$\alpha_Y$ V23 action | Environment-dependent fifth force (strong screening in laboratory/NS, enhancement in dilute galactic medium) | **Standalone extension hypothesis**, not a consequence of the constant-$\alpha_Y$ V23 `yukawa_action_el` equations |

## 3.5 5. Checks

1. **Limit Check ($\phi_L \to 0$, i.e., $f_L \to 1$):**
   - When $\phi_L = 0$ ($f_L = 1$), $\tilde g^{\rm conf}_{\mu\nu} = g_{\mu\nu}$ and $\tilde g^{\rm disf}_{\mu\nu} = g_{\mu\nu} + 0 = g_{\mu\nu}$, so both Jordan metrics reduce to the physical Einstein metric $g_{\mu\nu}$, and $d\tilde\tau_{H_1} = d\tau_{H_0}$ ✓.
2. **Proper-Time Agreement for Stationary Clocks (along $u^\mu$):**
   - For a clock comoving with the baryonic medium ($dx^\mu \parallel u^\mu$), the conformal and disformal metrics yield the **identical** proper-time element and frequency shift:

$$d\tilde\tau_{\rm conf} \;=\; d\tilde\tau_{\rm disf} \;=\; f_L(x)\, d\tau \;=\; e^{-\phi_L(x)}\sqrt{-g_{00}}\, dt,$$

   which is also verified symbolically by `covariant_closure_50aa2db0.py` (`sym_res["proper_time_ratio_H1_over_H0"] = "f_L"`, `sym_res["weak_field_H1_clock_shift"] = "Phi_N/c^2 - phi_L"`) ✓.
3. **Difference in the Spatial Sector, Volume Element, and Light Propagation:**
   - Whereas $\sqrt{-\tilde g_{\rm conf}} = e^{-4\phi_L}\sqrt{-g}$ and the conformal metric leaves null geodesics (light rays) invariant ($\tilde g^{\rm conf}_{\mu\nu}k^\mu k^\nu = 0 \iff g_{\mu\nu}k^\mu k^\nu = 0$), the disformal metric satisfies $\sqrt{-\tilde g_{\rm disf}} = e^{-\phi_L}\sqrt{-g}$ and induces an effective refractive index ($n = f_L^{-1} = e^{+\phi_L}$) for light while leaving spatial distances unscaled ✓.

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# 4 Step 3: Mapping the V23 Local Screened Helmholtz–Yukawa Action and Field Equation to the Massive Scalar-Tensor / Dilaton Klein–Gordon Equation

Source file: step3_yukawa_helmholtz_eom_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step3_yukawa_helmholtz_eom_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md), step2_conformal_disformal_en.md (step2_conformal_disformal.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V23 (50aa2db0 Hungarian manuscript): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Subsection 1.2, lines 56–65, Eqs. 6–7; DOI: 10.5281/zenodo.23639084)
- V23 (50aa2db0 English manuscript): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_EN_2026-10-24.tex (Sec. 1.2, lines 56–65, Eqs. 6–7)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex (lines 200–206 and 245–254; DOI: 10.5281/zenodo.23604423)
- V23 SymPy verification script: covariant_closure_50aa2db0.py (lines 126–149) and conservation_Residuals.json

## 4.1 1. Verbatim Citations from the V23 (50aa2db0) and V22 Manuscripts

In Step 1 (step1_dilaton_form.md), we recorded as an open question how the nonlocal exponent $\phi_L(\mathbf{x}) = \gamma_L \bar{\rho}_L(\mathbf{x})$ becomes a local dynamical scalar field endowed with its own kinetic term. **Subsection 1.2** of the V23 manuscript (*Formulation II: Local Screened Scalar-Tensor (Helmholtz–Yukawa) Action*, lines 56–65, **Eqs. 6–7**) defines this verbatim as follows:

- **Lagrangian and Field Equation of the Scalar Field (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, lines 57–60, Eq. 6):**

  *"In Formulation II, I replace the nonlocal integral kernel with a local canonical screened scalar action:"*

$$\mathcal{L}_\phi = -\frac{c^4}{8\pi G\tilde{\alpha}_L}\left[\frac{1}{2}g^{\mu\nu}\nabla_\mu\phi_L\nabla_\nu\phi_L + \frac{1}{2L^2}\phi_L^2\right], \quad \left(-\Box_g + \frac{1}{L^2}\right)\phi_L = \frac{4\pi G}{c^2}\alpha_{\rm std}\,\rho_{\rm src} \iff (-L^2\nabla^2+1)\phi_L = \alpha_Y\gamma_L\rho_{\rm src}. \qquad \text{(V23 Eq. 6)}$$

- **Static Spherically Symmetric Green's Function Solution and Scalar Stress–Energy Tensor (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, lines 61–65, Eq. 7):**

  *"In a static spherical system, the exact positive-definite Green's function solution of the radial Helmholtz operator $-\frac{1}{r^2}\frac{d}{dr}(r^2\phi_L') + L^{-2}\phi_L = \frac{4\pi G}{c^2}\alpha_{std}\rho_{src}$ ($\alpha_Y = 4\pi\alpha_{std}$) is"*

$$\phi_L^{\rm Yuk}(r) = \alpha_Y\,\frac{G}{c^2}\int_0^{R_*} dr'\,\frac{r'L}{r}\,\sinh\frac{r_<}{L}\,e^{-r_>/L}\rho_{\rm src}(r'), \qquad \phi_L^{\rm Yuk}(0) = \alpha_Y\,\frac{G}{c^2}\int_0^{R_*} r' e^{-r'/L}\rho_{\rm src}(r')\,dr', \qquad \text{(V23 Eq. 7)}$$

  *"and the variation $\delta S_\phi/\delta g^{\mu\nu}$ provides the local covariant stress–energy tensor $T^{(\phi)}_{\mu\nu} = \frac{c^4}{8\pi G\tilde{\alpha}_L}\left[\nabla_\mu\phi_L\nabla_\nu\phi_L - \frac{1}{2}g_{\mu\nu}\left((\nabla\phi_L)^2 + L^{-2}\phi_L^2\right)\right]$. (SymPy: sym_yukawa_center_limit_residual = 0, no extra L-factor). The unit-normalized kernel corresponds to $\alpha_Y = 1$, i.e., $\alpha_{std} = 1/(4\pi) \approx 0.0796$, which recovers the homogeneous top-hat limit ($\phi_L = \gamma_L\rho_{src}$); $\alpha_Y = 1/3$ ($\alpha_{std} = 1/(12\pi)$) is the volume-matched variant."*

- **Action-Consistent (yukawa_action_el) Euler–Lagrange Branch (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, line 65):**

*"The `yukawa_action_el` rows follow from the action $\mathcal{L} = -[(\nabla\phi)^2 + \phi^2/L^2]/(2\alpha_Y) - f_L\varepsilon$ ($G = c = 1$; `SymPy: sym_yukawa_el_source_factor` and `_hydrostatic` residual $= 0$): $B\phi'' + (2B/r + B\nu'/2 + B'/2)\phi' - \phi/L^2 = -\alpha_Y f_L \varepsilon$, $B = 1 - 2m/r$, $P_{matter} = f_L P$, $h' = -\nu' + \phi'$, $\rho_\phi = (B\phi'^2 + \phi^2/L^2)/(2\alpha_Y)$, exterior matching $\phi'(R) = -(1/R + 1/L)\phi(R)$."*

---

## 4.2 2. Derivation of the Euler–Lagrange Field Equation and Mapping to a Massive Dilaton / Scalar-Tensor Field

### 4.2.1 2.1. Variation with Respect to $\phi_L$ (with Matter-Sector Coupling $f_L(\phi_L) = e^{-\phi_L}$)

Consider the local action of V23 Formulation II, where the matter energy density couples with the factor $f_L(\phi_L) = e^{-\phi_L}$ derived in Steps 1 and 2 ($\mathcal{L}_m = -f_L(\phi_L)\,\rho c^2 = -e^{-\phi_L}\varepsilon$):

$$S_{\mathrm{II}}[g_{\mu\nu}, \phi_L, \Psi] \;=\; \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R \;-\; \frac{c^4}{8\pi G\tilde{\alpha}_L}\left(\frac{1}{2}g^{\mu\nu}\nabla_\mu\phi_L\nabla_\nu\phi_L + \frac{1}{2L^2}\phi_L^2\right) \;-\; e^{-\phi_L}\,\rho c^2\right].$$

Varying the action with respect to $\phi_L$ ($\delta S_{\mathrm{II}}/\delta\phi_L = 0$):

$$\frac{1}{\sqrt{-g}}\frac{\delta S_{\mathrm{II}}}{\delta\phi_L} \;=\; \frac{c^4}{8\pi G\tilde{\alpha}_L}\left(\Box_g\phi_L - \frac{1}{L^2}\phi_L\right) \;+\; e^{-\phi_L}\,\rho c^2 \;=\; 0,$$

where $\Box_g\phi_L = \frac{1}{\sqrt{-g}}\partial_\mu\big(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\phi_L\big)$. Rearranging and multiplying by $-\frac{8\pi G\tilde{\alpha}_L}{c^4}$ yields:

$$\left(-\Box_g + \frac{1}{L^2}\right)\phi_L \;=\; \frac{8\pi G}{c^2}\,\tilde{\alpha}_L\,\rho_{\mathrm{src}}, \qquad \rho_{\mathrm{src}} \;=\; \begin{cases}\rho & \text{(weak-field / linear source, } f_L \simeq 1\text{)},\\ f_L(\phi_L)\,\rho = e^{-\phi_L}\rho & \text{(exact } \texttt{yukawa_action_el} \text{ source)}.\end{cases}$$

### 4.2.2 2.2. Identification of the Constants ($\tilde{\alpha}_L$, $\alpha_{\mathbf{std}}$, $\alpha_Y$, $\gamma_L$, and $C$)

Comparing term by term with V23 Eq. (6):

1. **Relation between the Poisson and Yukawa normalizations:**

$$\frac{8\pi G}{c^2}\,\tilde{\alpha}_L \;=\; \frac{4\pi G}{c^2}\,\alpha_{\mathrm{std}} \;=\; C\,\alpha_{\mathrm{std}} \quad\Longrightarrow\quad \tilde{\alpha}_L \;=\; \frac{\alpha_{\mathrm{std}}}{2},$$

where $C = \frac{4\pi G}{c^2}$ is the fundamental coupling constant $C$ of the V22 monograph (`hu_p1.tex`, Eq. 4).

2. **Factoring out $\gamma_L = \frac{GL^2}{c^2}$ ($-L^2\Box_g + 1$ form):** Multiplying both sides of the field equation by $L^2$ (in the quasi-static limit $\Box_g \to \nabla^2$):

$$(-L^2\nabla^2 + 1)\phi_L \;=\; 4\pi\alpha_{\mathrm{std}}\left(\frac{GL^2}{c^2}\right)\rho_{\mathrm{src}} \;=\; \alpha_Y\,\gamma_L\,\rho_{\mathrm{src}}, \qquad \boxed{\alpha_Y \;\equiv\; 4\pi\,\alpha_{\mathrm{std}} \;=\; 8\pi\,\tilde{\alpha}_L.}$$

Or in an equivalent notation (as written in Section 1.1 of `README.md`):

$$(\nabla^2 - L^{-2})\phi_L \;=\; -\frac{4\pi G}{c^2}\alpha_{\mathrm{std}}\,\rho_{\mathrm{src}} \qquad \left(\text{or in the } \phi_L \to -\phi_L \text{ sign convention, } +\frac{\alpha_{\mathrm{std}}G}{c^2}\rho_{\mathrm{src}} \text{ with the } 4\pi \text{ factor absorbed}\right).$$

### 4.2.3 2.3. Mapping to the Massive Scalar-Tensor / Dilaton Klein–Gordon Equation

In string-theoretic dilaton gravity or general scalar–tensor gravity (e.g., Damour & Polyakov 1994; Fujii & Maeda 2003; Khoury & Weltman 2004), the field equation of a canonical scalar field of mass $m_\phi$ with a quadratic self-interaction potential ($V(\varphi) = \frac{1}{2}\mu_\phi^2\varphi^2$, where $\mu_\phi = \frac{m_\phi c}{\hbar}$) is:

$$\left(-\Box_g + \mu_\phi^2\right)\phi_L \;=\; \frac{4\pi G}{c^2}\,\alpha_{\mathrm{std}}\,\rho_{\mathrm{src}}.$$

From this, the exact dictionary correspondence between the TAL smoothing scale ($L$) and the effective Compton wavelength and mass of the scalar/dilaton field is:

$$\boxed{\mu_\phi \;\equiv\; \frac{1}{L}, \qquad m_\phi \;=\; \frac{\hbar}{L\,c}}$$

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- **Physical interpretation:** What the nonlocal formulation of V19/V21 describes as a 3D spherical top-hat windowing of radius $L$ corresponds in V23 Formulation II to the exchange of a **Yukawa-screened scalar/dilaton field with range $\lambda_{\text{Compton}} = L$ (i.e., mass $m_\phi = \hbar/(Lc)$)**.

---

## 4.3 3. How Does the Yukawa Solution Recover the 3D Top-Hat Law $\phi_L = \gamma_L \bar{\rho}_L$?

1. **The 3D Yukawa Green's Function:** The fundamental solution of $(-L^2\nabla^2 + 1)G_L(\mathbf{x}-\mathbf{y}) = \delta^{(3)}(\mathbf{x}-\mathbf{y})$ in three-dimensional Euclidean space is:

$$G_L(\mathbf{x}-\mathbf{y}) \;=\; \frac{\exp\!\left(-\frac{|\mathbf{x}-\mathbf{y}|}{L}\right)}{4\pi L^2\,|\mathbf{x}-\mathbf{y}|}, \qquad \int_{\mathbb{R}^3} G_L(\mathbf{r})\,d^3r \;=\; \frac{4\pi}{4\pi L^2}\int_0^\infty r\,e^{-r/L}\,dr \;=\; \frac{L^2}{L^2} \;=\; 1.$$

Thus, the solution of V23 Eq. (6) for an arbitrary distribution $\rho_{\text{src}}(\mathbf{y})$ is a **positive-definite Yukawa smoothing**:

$$\phi_L^{\text{Yuk}}(\mathbf{x}) \;=\; \alpha_Y\,\gamma_L \int_{\mathbb{R}^3} G_L(\mathbf{x}-\mathbf{y})\,\rho_{\text{src}}(\mathbf{y})\,d^3y \;=\; \alpha_Y\,\frac{G}{c^2}\int_{\mathbb{R}^3}\frac{e^{-|\mathbf{x}-\mathbf{y}|/L}}{4\pi\,|\mathbf{x}-\mathbf{y}|}\,\rho_{\text{src}}(\mathbf{y})\,d^3y.$$

2. **Homogeneous / Slowly Varying Density Limit ($\rho_{\text{src}} = \text{const.}$):** Since $\int_{\mathbb{R}^3} G_L(\mathbf{r})\,d^3r = 1$, in a homogeneous medium:

$$\phi_L^{\text{Yuk}} \;=\; \alpha_Y\,\gamma_L\,\rho_{\text{src}}.$$

Choosing the **unit-normalized kernel** ($\alpha_Y = 1 \iff \alpha_{\text{std}} = \frac{1}{4\pi} \approx 0.079577$):

$$\phi_L^{\text{Yuk}} \;=\; \gamma_L\,\rho_{\text{src}} \quad\Longrightarrow\quad f_L \;=\; e^{-\phi_L^{\text{Yuk}}} \;=\; \exp(-\gamma_L\rho_{\text{src}}),$$

which recovers precisely the TAL exponential law examined in Step 1 (`step1_dilaton_form.md`)!

3. **Point-Mass / Compact-Source Limit ($R_* \ll r$):** Outside a compact source of mass $M = \int \rho_{\text{src}}\,d^3y$ ($r > R_*$), the solution V23 Eq. (7) simplifies to a classical fifth-force Yukawa potential:

$$\phi_L^{\text{Yuk}}(r) \;\propto\; \alpha_Y\,\frac{GM}{c^2 r}\,e^{-r/L}.$$

---

## 4.4 4. Checks

### 4.4.1 4.1. Dimensional Analysis of V23 Eqs. (6) and (7)

- **Left-hand side of V23 Eq. (6):** $\phi_L$ is dimensionless ($[\phi_L] = 1$), the dimension of the d'Alembert/Laplace operator is $[\Box_g] = [\nabla^2] = \text{m}^{-2}$, and $[L^{-2}] = \text{m}^{-2}$, therefore:

$$\left[\left(-\Box_g + \frac{1}{L^2}\right)\phi_L\right] \;=\; \text{m}^{-2}.$$

- **Right-hand side of V23 Eq. (6):** $[G/c^2] = \text{m}\,\text{kg}^{-1}$, $[\alpha_{\text{std}}] = 1$, $[\rho_{\text{src}}] = \text{kg}\,\text{m}^{-3}$, therefore:

$$\left[\frac{4\pi G}{c^2}\alpha_{\text{std}}\,\rho_{\text{src}}\right] \;=\; \left(\text{m}\,\text{kg}^{-1}\right)\cdot\left(\text{kg}\,\text{m}^{-3}\right) \;=\; \text{m}^{-2} \quad \checkmark$$

- **Central integral of V23 Eq. (7) ($\phi_L^{\text{Yuk}}(0)$):**

$$\left[\alpha_Y\,\frac{G}{c^2}\int_0^{R_*} r'\,e^{-r'/L}\rho_{\text{src}}(r')\,dr'\right] \;=\; 1\cdot\left(\text{m}\,\text{kg}^{-1}\right)\cdot\text{m}\cdot 1\cdot\left(\text{kg}\,\text{m}^{-3}\right)\cdot\text{m} \;=\; \text{m}^3\,\text{kg}^{-1}\cdot\text{kg}\,\text{m}^{-3} \;=\; 1 \quad \text{(dimensionless)} \quad \checkmark$$

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#### 4.4.2 4.2. Agreement of SymPy Verifications with the V23 Package (`conservation_Residuals.json`)

In V23, `covariant_closure_50aa2db0.py` (lines 126–148) runs three symbolic checks on this sector, each yielding an exact residual of 0:

1. `sym_yukawa_center_limit_residual = 0`: $\lim_{r\to 0} \frac{L}{r}\sinh(r/L) = 1$, so $\phi_L^{\mathrm{Yuk}}(0)$ contains no extra $L$-multiplier ✓.
2. `sym_yukawa_alphaY_eq_4pi_alphastd_residual = 0`: $\alpha_{\mathrm{std}} = \frac{1}{4\pi} \implies \alpha_Y = 4\pi\alpha_{\mathrm{std}} = 1 \implies \phi_L = \gamma_L\rho$ in a homogeneous medium ✓.
3. `sym_yukawa_el_source_factor_residual = 0` and `sym_yukawa_el_matter_pressure_residual = 0`: variation of the matter term $\mathcal{L} \supset -e^{-\phi_L}\varepsilon(n)$ yields the source term $-\alpha_Y f_L \varepsilon$, the effective pressure $P_{\mathrm{matter}} = f_L P$, and the modified hydrostatic equilibrium $h' = -\nu' + \phi_L'$ ✓.

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# 5 Step 4: Covariant Stress-Energy Conservation, Scalar-Matter Momentum Exchange, and Modified Hydrostatic Equilibrium ($h' = -\nu' + \phi_L'$)

Source file: step4_covariant_conservation_hydrostatic_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step4_covariant_conservation_hydrostatic_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md), step2_conformal_disformal_en.md (step2_conformal_disformal.md), step3_yukawa_helmholtz_eom_en.md (step3_yukawa_helmholtz_eom.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V23 (50aa2db0 Hungarian manuscript): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Subsections 1.1 and 1.2, lines 47–65, Eqs. 4–5 and Eqs. 6–7; DOI: 10.5281/zenodo.23639084)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex (Chapter 2, lines 205–207; DOI: 10.5281/zenodo.23604423)
- V23 SymPy verification script: covariant_closure_50aa2db0.py (lines 72–98 and 143–160) and conservation_Residuals.json

## 5.1 1. Verbatim Citations from the V22 and V23 Manuscripts

In Step 3 (step3_yukawa_helmholtz_eom.md), we derived the Euler–Lagrange equation of the scalar/dilaton field $\phi_L$. The next necessary theoretical step is the variation with respect to the metric $g^{\mu\nu}$ and the derivation of **covariant energy–momentum conservation following from the Bianchi identity** ($\nabla_\mu G^{\mu\nu} \equiv 0$) ($\nabla_\mu (T^{\mu\nu}_{\rm matter} + T^{\mu\nu}_X) = 0$), together with the resulting **modified hydrostatic equilibrium (fifth force / scalar-gradient force)**.

1. **Formulation of the Physical Problem in the V22 Monograph (hu_p1.tex, line 206):**

   *"At the same time, inserting $f_L(\mathbf{x})$ as an external multiplier into the Tolman–Oppenheimer–Volkoff equations or the clock-frequency formulas leaves behind a residual $\nabla_\mu T^{\mu\nu}_{matter} \neq 0$ unless balanced by a dynamical field sector. Therefore, I formulated the dynamical 'why' as task **50aa2db0**: elevating the nonlocal kernel into a covariant scalar–tensor or nonlocal action principle $S = \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R + \mathcal{L}_{matter}(\psi, \tilde{g}_{\mu\nu}) + \mathcal{L}_\phi(\phi, \nabla\phi)\right]$ whose Euler–Lagrange equation recovers the profile $\phi(\mathbf{x}) \to \gamma_L \bar{\rho}_L(\mathbf{x})$ in the quasi-static limit while guaranteeing the conservation law $\nabla_\mu(T^{\mu\nu}_{matter} + T^{\mu\nu}_X) = 0$."*

2. **Algebraic $X$-Sector Reconstruction in Formulation I of V23 (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, lines 47–52, Eqs. 4–5):**

   *"In a static spherically symmetric interior, if the auxiliary sector $T^\mu_\nu(X) = diag(-\varepsilon_X, P_{r,X}, P_{t,X}, P_{t,X})$ absorbs the window stress on the right-hand side of $G^\mu_\nu = \frac{8\pi G}{c^4}(T^\mu_{\nu,matter} + T^\mu_{\nu,X})$ while the fluid elements satisfy $\nabla_\mu T^{\mu\nu}_{matter} = 0$ ($P' = -(\varepsilon + P)\nu'$ in geometrized units, $G = c = 1$, $\nu = \Phi_{eff}/c^2$, $g_{tt} = -e^{2\nu} = -e^{2\Phi_{eff}/c^2}$), then the Bianchi identity $\nabla_\mu G^\mu_r \equiv 0$ uniquely fixes $T^\mu_\nu(X)$ in both V21 modes:"*

$$\textbf{Mass mode } (m' = 4\pi r^2 f_L \varepsilon,\ \nu' = A_{\rm geo}): \quad \varepsilon_X = -(1-f_L)\varepsilon, \quad P_{r,X} = 0, \quad P_{t,X} = -\frac{r}{2}(1-f_L)\varepsilon\nu',$$

$$\textbf{Gradient mode } (m' = 4\pi r^2 \varepsilon,\ \nu' = f_L A_{\rm geo}): \quad \varepsilon_X = 0, \quad P_{r,X} = -(1-f_L)\Big(P + \frac{m}{4\pi r^3}\Big),\ P_{t,X} = P_{r,X} + \frac{r}{2}(P'_{r,X} + P_{r,X}\nu'),$$

(V23 Eq. 4, V23 Eq. 5)

   *"where $A_{geo}(r) = (m + 4\pi r^3 P)/[r(r-2m)]$. Symbolic verification performed in **SymPy** (**conservation_Residuals.json**) confirms that $\nabla_\mu T^\mu_r(matter) \equiv 0$ and $\nabla_\mu T^\mu_r(X) \equiv 0$ hold identically."*

3. **Local Scalar–Tensor (Yukawa) Stress–Energy Tensor and Action-Consistent Hydrostatic Equilibrium in Formulation II (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, line 65):**

   *"and the variation $\delta S_\phi/\delta g^{\mu\nu}$ provides the local covariant stress–energy tensor $T^{(\phi)}_{\mu\nu} = \frac{c^4}{8\pi G\bar{\alpha}_L}\left[\nabla_\mu\phi_L\nabla_\nu\phi_L - \frac{1}{2}g_{\mu\nu}\left((\nabla\phi_L)^2 + L^{-2}\phi_L^2\right)\right]$. [...] The **yukawa_action_el** rows follow from the action $\mathcal{L} = -[(\nabla\phi)^2 + \phi^2/L^2]/(2\alpha_Y) - f_L\varepsilon$ ($G = c = 1$; **SymPy: sym_yukawa_el_source_factor** and **_hydrostatic** residual $= 0$): $B\phi'' + (2B/r + B\nu' + B'/2)\phi' - \phi/L^2 = -\alpha_Y f_L \varepsilon$, $B = 1 - 2m/r$, $P_{matter} = f_L P$, $h' = -\nu' + \phi'$, $\rho_\phi = (B\phi'^2 + \phi^2/L^2)/(2\alpha_Y)$, exterior matching $\phi'(R) = -(1/R + 1/L)\phi(R)$."*

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## 5.2 2. Derivation of Scalar–Matter Momentum Exchange and Modified Hydrostatic Equilibrium

### 5.2.1 2.1. Derivation of the Effective Matter Pressure ($P_{\mathbf{matter}} = f_L P$)

In the V23 `yukawa_action_el` branch (and in general Einstein-frame scalar–tensor theory where the matter Lagrangian is $\mathcal{L}_{\rm fluid}(n,\phi_L) = -f_L(\phi_L)\,\varepsilon(n) = -e^{-\phi_L}\varepsilon(n)$, with $n$ denoting the baryon number density), the thermodynamic pressure is defined via the Legendre transformation under baryon-number conservation:

$$P_{\rm matter} \;=\; n\,\frac{\partial\big(f_L(\phi_L)\,\varepsilon(n)\big)}{\partial n} \;-\; f_L(\phi_L)\,\varepsilon(n) \;=\; f_L(\phi_L)\Big(n\,\frac{d\varepsilon}{dn} - \varepsilon(n)\Big) \;=\; f_L(\phi_L)\,P(n).$$

Thus the stress–energy tensor of the $f_L$-weighted ideal fluid is:

$$T^{\mu\nu}_{\rm matter} \;=\; f_L(\phi_L)\left[\big(\varepsilon + P\big)\frac{u^\mu u^\nu}{c^2} + P\,g^{\mu\nu}\right] \quad\Longrightarrow\quad \varepsilon_{\rm matter} = f_L\varepsilon, \qquad P_{\rm matter} = f_L P.$$

### 5.2.2 2.2. Divergence of the Scalar-Field Stress–Energy Tensor ($\nabla^\mu T^{(\phi)}_{\mu\nu}$)

In geometrized units ($G = c = 1$, where $\alpha_Y = 8\pi\tilde{\alpha}_L$, so that $\frac{c^4}{8\pi G\tilde{\alpha}_L} = \frac{1}{\alpha_Y}$), the stress–energy tensor of the scalar field is:

$$T^{(\phi)}_{\mu\nu} \;=\; \frac{1}{\alpha_Y}\left[\nabla_\mu\phi_L\nabla_\nu\phi_L - \frac{1}{2}g_{\mu\nu}\left(g^{\alpha\beta}\nabla_\alpha\phi_L\nabla_\beta\phi_L + \frac{1}{L^2}\phi_L^2\right)\right].$$

Let us compute its covariant divergence ($\nabla^\mu T^{(\phi)}_{\mu\nu}$):

$$\nabla^\mu T^{(\phi)}_{\mu\nu} \;=\; \frac{1}{\alpha_Y}\left[(\Box_g\phi_L)\,\nabla_\nu\phi_L + (\nabla^\mu\phi_L)\,\nabla_\mu\nabla_\nu\phi_L - (\nabla^\alpha\phi_L)\,\nabla_\nu\nabla_\alpha\phi_L - \frac{1}{L^2}\phi_L\nabla_\nu\phi_L\right].$$

Since second covariant derivatives commute on a scalar field ($\nabla_\mu\nabla_\nu\phi_L = \nabla_\nu\nabla_\mu\phi_L$), the second and third terms cancel:

$$\nabla^\mu T^{(\phi)}_{\mu\nu} \;=\; \frac{1}{\alpha_Y}\left(\Box_g\phi_L - \frac{1}{L^2}\phi_L\right)\nabla_\nu\phi_L.$$

Now substitute the exact `yukawa_action_el` Euler–Lagrange field equation derived in Step 3 (`step3_yukawa_helmholtz_eom.md`):

$$\left(-\Box_g + \frac{1}{L^2}\right)\phi_L \;=\; \alpha_Y\,f_L(\phi_L)\,\varepsilon \quad\Longleftrightarrow\quad \Box_g\phi_L - \frac{1}{L^2}\phi_L \;=\; -\alpha_Y\,f_L\,\varepsilon.$$

With this, the divergence of the scalar sector in closed form is:

$$\boxed{\nabla^\mu T^{(\phi)}_{\mu\nu} \;=\; -f_L(\phi_L)\,\varepsilon\,\nabla_\nu\phi_L \;=\; \varepsilon\,\nabla_\nu f_L(\phi_L).}$$

### 5.2.3 2.3. Joint Conservation ($\nabla^\mu(T^{\mathbf{matter}}_{\mu\nu} + T^{(\phi)}_{\mu\nu}) = 0$) and the Modified Euler/TOV Equation

Because of the contracted Bianchi identity ($\nabla^\mu G_{\mu\nu} \equiv 0$) on the left-hand side of the Einstein equations ($G_{\mu\nu} = 8\pi(T^{\rm matter}_{\mu\nu} + T^{(\phi)}_{\mu\nu})$), the joint divergence of the right-hand side vanishes identically:

$$\nabla^\mu T^{\rm matter}_{\mu\nu} \;+\; \nabla^\mu T^{(\phi)}_{\mu\nu} \;=\; 0 \quad\Longrightarrow\quad \nabla^\mu T^{\rm matter}_{\mu\nu} \;=\; +f_L(\phi_L)\,\varepsilon\,\nabla_\nu\phi_L.$$

In a static spherically symmetric spacetime ($ds^2 = -e^{2\nu(r)}dt^2 + B(r)^{-1}dr^2 + r^2d\Omega^2$, $B(r) = 1 - 2m(r)/r$), the radial ($\nu = r$) covariant divergence of the matter tensor is:

$$\nabla_\mu T^\mu_r(\text{matter}) \;=\; \frac{d}{dr}\big(f_L P\big) \;+\; f_L(\varepsilon + P)\,\nu' \;=\; f_L\,P' \;-\; f_L\,P\,\phi_L' \;+\; f_L(\varepsilon + P)\,\nu',$$

where we used $f_L' = \frac{d}{dr}(e^{-\phi_L}) = -f_L\,\phi_L'$. Adding the divergence of the scalar sector $\nabla_\mu T^\mu_r(\phi) = -f_L\,\varepsilon\,\phi_L'$ gives:

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$$\nabla_\mu T^\mu_r(\text{matter}) + \nabla_\mu T^\mu_r(\phi) \;=\; f_L\Big[P' + (\varepsilon+P)\nu' - (\varepsilon+P)\phi_L'\Big] \;=\; 0.$$

Since $f_L = e^{-\phi_L} > 0$, we can divide by $f_L(\varepsilon+P)$, and introducing the specific enthalpy ($dh = \frac{dP}{\varepsilon+P}$) yields:

$$\boxed{h'(r) \;\equiv\; \frac{P'(r)}{\varepsilon(r)+P(r)} \;=\; -\nu'(r) \;+\; \phi_L'(r) \;=\; -\nu'(r) \;-\; \frac{d\ln f_L(r)}{dr}.}$$

---

## 5.3 3. Physical Interpretation in the Scalar–Tensor / Dilaton Framework

The relation $h' = -\nu' + \phi_L'$ derived above is the direct dynamical bridge between scalar–tensor gravity (Damour & Esposito-Farèse 1992; Fujii & Maeda 2003) and the Time Attraction Law:

1. **Decomposition of the Effective Potential:** The net equilibrium force acting on a baryonic fluid element consists of two parts:
   - the standard metric gravitational attraction $-\nu'(r) = -\frac{1}{c^2}\frac{d\Phi_{\rm eff}}{dr}$, and
   - the **scalar / time-coherence gradient force** $+\phi_L'(r) = -\frac{d\ln f_L}{dr}$ (referred to in the scalar–tensor literature as the *fifth force* / dilaton-gradient force).
2. **Weak-Field / Galactic Limit:** In a non-relativistic medium ($P \ll \varepsilon \simeq \rho c^2$), the effective acceleration of a test body or fluid element is:

$$\mathbf{a}_{\rm eff} \;=\; -\nabla\Phi_N \;+\; c^2\,\nabla\phi_L \;=\; -\nabla\left(\Phi_N - c^2\phi_L\right) \;=\; -\nabla\left(\Phi_N + c^2\ln f_L\right).$$

Because the density around a mass concentration increases inward ($\frac{d\bar\rho_L}{dr} < 0$), $\phi_L(r)$ increases inward ($\phi_L'(r) < 0$, or $\frac{d\ln f_L}{dr} > 0$), so the $+\phi_L'(r)$ term provides a density-gradient-driven contribution in both hydrostatic equilibrium and orbital dynamics.

---

## 5.4 4. Checks

1. **SymPy Symbolic Identities in the V23 Package (`covariant_closure_50aa2db0.py`, lines 72–98 and 150–160; `conservation_Residuals.json`):**
   - `sym_yukawa_el_matter_pressure_residual = 0`: verifies the identity $P_{\rm matter} = n\frac{\partial(f_L\varepsilon)}{\partial n} - f_L\varepsilon = f_L P$ ✓.
   - `sym_yukawa_el_hydrostatic_residual = 0`: verifies that substituting the Yukawa EL equation into $\nabla_\mu T^\mu_r(\text{matter}) + \nabla_\mu T^\mu_r(\phi) = 0$ yields $h' = -\nu' + \phi_L'$ exactly ✓.
   - `div_matter_mass_symbolic = 0`, `div_X_mass_symbolic = 0`, `div_matter_grad_symbolic = 0`, `div_X_grad_symbolic = 0`: verifies conservation of the algebraic $X$-sector in V23 Formulation I Eqs. (4)–(5) ✓.
2. **General Relativistic (GR) Limit ($\phi_L \to 0$, $f_L \to 1$):** When $\phi_L \equiv 0$ ($\phi_L' = 0$, $f_L = 1$), the scalar stress–energy tensor vanishes ($T^{(\phi)}_{\mu\nu} = 0$), the effective pressure reduces to $P_{\rm matter} \to P$, and hydrostatic equilibrium returns to the standard GR Tolman–Oppenheimer–Volkoff equation ($h' = -\nu'$, i.e., $P' = -(\varepsilon+P)\nu'$) ✓.

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# 6 Step 5: Dictionary of Coupling Constants, Weak-Field Yukawa Fifth Force, and Conformal vs. Disformal PPN Light Deflection

Source file: step5_coupling_dictionary_ppn_screening_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step5_coupling_dictionary_ppn_screening_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md), step2_conformal_disformal_en.md (step2_conformal_disformal.md), step3_yukawa_helmholtz_eom_en.md (step3_yukawa_helmholtz_eom.md), step4_covariant_conservation_hydrostatic_en.md (step4_covariant_conservation_hydrostatic.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V23 (50aa2db0 Hungarian manuscript): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Section 1.2, Eqs. 6–7, lines 56–65; Section 3, lines 161–167; DOI: 10.5281/zenodo.23639084)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex (Chapter 3, lines 245–255, Eqs. 4–5) and hu_p2.tex (Chapter 4, lines 23–38; DOI: 10.5281/zenodo.23604423)

## 6.1 1. Verbatim Citations from the V22 and V23 Manuscripts

In Step 2 (step2_conformal_disformal.md), we established the taxonomy of the V19–V23 model branches and showed that one must distinguish between the universal conformal Jordan metric (**Branch II-Conf**), the comoving timelike disformal metric (**Branch II-Disf**), and the effective fluid-density coupling (**Branch II-Fluid**).

1. **The Coupling Constant $C$, $\gamma_L$, and the Compactness Scale in the V22 Monograph (hu_p1.tex, lines 247–254, Eqs. 4–5):**

$$C \equiv \frac{4\pi G}{c^2} = 9.33306 \times 10^{-27}\,\mathrm{m\,kg^{-1}}, \qquad \gamma_L = \frac{G\,L^2}{c^2} = \frac{C\,L^2}{4\pi} = 7.425915 \times 10^{-28}\,L^2\,\mathrm{m^3\,kg^{-1}}. \quad \text{(V22 Eqs. 4–5)}$$

*"When a point mass $M$ is averaged over a sphere of radius $L$, its mean density is $\bar{\rho}_L = \frac{3M}{4\pi L^3}$, so the exponent is $\gamma_L \bar{\rho}_L = \frac{3GM}{4\pi c^2 L} = \frac{3\,C\,M}{(4\pi)^2 L}$, which up to the geometric factor $3/(4\pi)$ recovers the dimensionless compactness scale $GM/(c^2 L)$."*

2. **Solar System Screening in Chapter 4 of V22 (hu_p2.tex, lines 32 and 38):**

*"For subgalactic $L \gtrsim 1\,pc$, $\gamma_L \bar{\rho}_L^{\odot} = \frac{3GM_\odot}{4\pi c^2 L} \lesssim 10^{-14}$; planetary orbits remain untouched! [...] when $L$ exceeds the size of planetary orbits, the solar mass $M_\odot$ is diluted over the volume $\frac{4}{3}\pi L^3$, so gradients of $f_L$ along planetary orbits are suppressed below the Cassini and ephemeris bounds."*

3. **The V23 Yukawa Normalization Constants (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, line 65):**

*"The unit-normalized kernel corresponds to $\alpha_Y = 1$, i.e., $\alpha_{std} = 1/(4\pi) \approx 0.0796$, which recovers the homogeneous top-hat limit ($\phi_L = \gamma_L \rho_{src}$); $\alpha_Y = 1/3$ ($\alpha_{std} = 1/(12\pi)$) is the volume-matched variant."*

## 6.2 2. Weak-Field Effective Potential, Yukawa Fifth Force, and the Effective Gravitational Constant ($G_{\mathrm{eff}}(r)$)

In Step 4 (step4_covariant_conservation_hydrostatic.md), we derived from the covariant conservation law ($\nabla_\mu(T^{\mu\nu}_{\mathrm{matter}} + T^{\mu\nu}_{(\phi)}) = 0$) the modified hydrostatic / Euler equation:

$$h'(r) = -\nu'(r) + \phi_L'(r), \qquad \mathbf{a}_{\mathrm{eff}}(\mathbf{r}) = -\nabla\Phi_{\mathrm{eff}}(\mathbf{r}) = -\nabla\Big(\Phi_N(\mathbf{r}) - c^2\,\phi_L(\mathbf{r})\Big).$$

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### 6.2.1 2.1. Exterior Yukawa Potential Around a Compact Source ($r > R_*$)

In the exterior of a spherically symmetric body of mass $M$ and radius $R_*$ ($r > R_*$, in the weak-field approximation), the flat-space Green's function solution written in Step 3 (`step3_yukawa_helmholtz_eom.md`, V23 Eq. 7) takes the form:

$$\phi_L^{\rm Yuk}(r) \;=\; \alpha_Y \,\frac{G}{c^2}\,\frac{L\,e^{-r/L}}{r}\int_0^{R_*} r'\,\sinh\!\left(\frac{r'}{L}\right)\rho_{\rm src}(r')\,dr' \;=\; \frac{\alpha_{\rm std}\,GM_{\rm eff}(L)}{c^2\,r}\,e^{-r/L},$$

where $\alpha_{\rm std} = \frac{\alpha_Y}{4\pi}$, and in the point-source limit ($R_* \ll L$, where $\sinh(r'/L) \simeq r'/L$):

$$M_{\rm eff}(L) \;=\; 4\pi\int_0^{R_*} r'^2\left[\frac{L}{r'}\sinh\!\left(\frac{r'}{L}\right)\right]\rho_{\rm src}(r')\,dr' \;\xrightarrow{R_*\ll L}\; \int_0^{R_*} 4\pi r'^2\rho_{\rm src}(r')\,dr' \;=\; M.$$

Substituting this alongside the Newtonian potential $\Phi_N(r) = -\frac{GM}{r}$ into the effective potential gives:

$$\Phi_{\rm eff}(r) \;=\; \Phi_N(r) \;-\; c^2\,\phi_L^{\rm Yuk}(r) \;=\; -\frac{GM}{r}\Big(1 \;+\; \alpha_{\rm std}\,e^{-r/L}\Big).$$

Differentiating with respect to $r$ yields the distance-dependent effective gravitational constant:

$$\boxed{a_{\rm eff}(r) \;=\; -\frac{G_{\rm eff}(r)\,M}{r^2}, \qquad G_{\rm eff}(r) \;=\; G\left[1 \;+\; \alpha_{\rm std}\left(1+\frac{r}{L}\right)e^{-r/L}\right].}$$

### 6.2.2 2.2. Canonical Scalar–Tensor Normalization and the Relation $\alpha_{\rm std} = 2\beta^2$

Let us rewrite the V23 Eq. (8) scalar action:

$$S_\phi \;=\; -\int d^4x\,\sqrt{-g}\,\frac{c^4}{16\pi G}\,\frac{1}{\tilde\alpha_L}\left(g^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L + \frac{\phi_L^2}{L^2}\right) \;=\; -\int d^4x\,\sqrt{-g}\,\frac{c^4}{2G\,\alpha_Y}\left(g^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L + \frac{\phi_L^2}{L^2}\right)$$

(where $\alpha_Y = 8\pi\tilde\alpha_L = 4\pi\alpha_{\rm std}$, so $\tilde\alpha_L = \frac{\alpha_{\rm std}}{2} = \frac{\alpha_Y}{8\pi}$, and in units $G = c = 1$ the scalar Lagrangian is precisely $-\frac{1}{2\alpha_Y}[(\nabla\phi_L)^2 + \phi_L^2/L^2]$) in terms of the canonical Einstein-frame scalar field $\varphi$ of Damour–Esposito-Farèse (1992) / Damour–Polyakov (1994), whose kinetic term is $-\frac{c^4}{4\pi G}\,\frac{1}{2}(\nabla\varphi)^2 = -\frac{c^4}{8\pi G}(\nabla\varphi)^2$:

$$\frac{c^4}{2G\,\alpha_Y}(\nabla\phi_L)^2 \;=\; \frac{c^4}{8\pi G\,\alpha_{\rm std}}(\nabla\phi_L)^2 \;\equiv\; \frac{c^4}{8\pi G}\,2(\nabla\varphi)^2 \qquad\Longrightarrow\qquad \phi_L \;=\; \sqrt{2\alpha_{\rm std}}\,\varphi \;=\; 2\beta\,\varphi.$$

Since the matter-coupling factor is $f_L = e^{-\phi_L} = e^{-2\beta\varphi}$, the exact relation between the dimensionless scalar–matter coupling constant ($\beta$) and the V23 parameters $\alpha_{\rm std}, \alpha_Y, \tilde\alpha_L$ is:

$$\boxed{\alpha_{\rm std} \;=\; 2\beta^2 \;=\; 2\tilde\alpha_L \;=\; \frac{\alpha_Y}{4\pi}, \qquad \beta \;=\; \sqrt{\frac{\alpha_{\rm std}}{2}} \;=\; \sqrt{\frac{\alpha_Y}{8\pi}}.}$$

- For the **unit-normalized top-hat counterpart** ($\alpha_Y = 1$):

$$\alpha_{\rm std} \;=\; \frac{1}{4\pi} \;\approx\; 0.079577, \qquad \beta \;=\; \frac{1}{\sqrt{8\pi}} \;\approx\; 0.199471, \qquad 2\omega_{\rm BD}+3 \;=\; \frac{1}{\beta^2} \;=\; 8\pi \;\Longrightarrow\; \omega_{\rm BD} \;=\; 4\pi - \frac{3}{2} \;\approx\; 11.066.$$

- For the **volume-matched variant** ($\alpha_Y = 1/3$):

$$\alpha_{\rm std} \;=\; \frac{1}{12\pi} \;\approx\; 0.026526, \qquad \beta \;=\; \frac{1}{\sqrt{24\pi}} \;\approx\; 0.115165, \qquad \omega_{\rm BD} \;=\; 12\pi - \frac{3}{2} \;\approx\; 36.199.$$

---

## 6.3 3. Strict Separation of the Three Coupling Branches (Branch II-Conf, Branch II-Fluid, Branch II-Disf) in the PPN and Light-Deflection Sector

In accordance with peer-review requirements, we record that the conformal coupling $\tilde g_{\mu\nu}^{\rm conf}$, the disformal coupling $\tilde g_{\mu\nu}^{\rm disf}$, and the effective fluid coupling $S_m = -\int d^4x\sqrt{-g}\,e^{-\phi_L}\varepsilon(n)$ represent **three distinct physical hypotheses** regarding the electromagnetic/null-geodesic sector, and consequently their PPN behaviors also differ:

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1. **Branch II-Conf — Universal Conformal Jordan Coupling ($\tilde{g}^{\mathbf{conf}}_{\mu\nu} = e^{-2\phi_L} g_{\mu\nu}$, $\sqrt{-\tilde{g}_{\mathbf{conf}}} = e^{-4\phi_L}\sqrt{-g}$):**
   - In this branch, every matter and electromagnetic field couples universally to the metric $\tilde{g}^{\rm conf}_{\mu\nu} = e^{-2\phi_L} g_{\mu\nu}$.
   - Because the Maxwell action is conformally invariant in $D = 4$ dimensions ($\sqrt{-\tilde{g}_{\rm conf}}\,\tilde{g}^{\mu\alpha}_{\rm conf}\tilde{g}^{\nu\beta}_{\rm conf}F_{\mu\nu}F_{\alpha\beta} = \sqrt{-g}\,g^{\mu\alpha}g^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}$), photon null geodesics ($\tilde{g}^{\rm conf}_{\mu\nu}dx^\mu dx^\nu = 0 \iff g_{\mu\nu}dx^\mu dx^\nu = 0$) sense only the Einstein-frame metric $g_{\mu\nu}$, and the electromagnetic field does not source the scalar $\phi_L$ ($T^\mu_{\ \mu}(\mathrm{EM}) = 0$).
   - The standard scalar–tensor / Brans–Dicke PPN parameter (Will, *Living Rev. Relativ.* **17**, 4, 2014) holds **exclusively for this Branch II-Conf**:

$$\boxed{\gamma^{\rm conf}_{\rm PPN}(r) \;=\; \frac{1-\alpha_{\rm std}\,e^{-r/L}}{1+\alpha_{\rm std}\,e^{-r/L}}, \qquad \gamma^{\rm conf}_{\rm PPN}(r\ll L) \;=\; \frac{2\omega_{\rm BD}+1}{2\omega_{\rm BD}+3} \;=\; \frac{4\pi-1}{4\pi+1} \;\approx\; 0.8526\ (\alpha_Y = 1), \qquad \gamma^{\rm conf}_{\rm PPN}(r\gg L) \;\to\; 1.}$$

2. **Branch II-Fluid + $H_0$ — Effective Baryonic Fluid Coupling with Minimal Photon/Atomic Metric ($S_m = -\int d^4x\sqrt{-g}\,e^{-\phi_L}\varepsilon(n)$, $\Psi_{\mathbf{atomic/EM}}$ Coupled to $g_{\mu\nu}$):**
   - Macroscopic baryonic fluid experiences the fifth force ($h' = -\nu' + \phi_L'$), so the effective gravitational constant between two macroscopic bodies is $G_{\rm eff}(r) = G\big[1+\alpha_{\rm std}(1+r/L)e^{-r/L}\big]$.
   - However, in the regime $r \gg L$ (for example when $L$ is submillimeter or when on astronomical scales $r_{\rm AU} \gg L$), the Yukawa term decays exponentially ($e^{-r/L} \to 0$), the exterior spacetime is asymptotically Schwarzschild geometry (with mass $M_{\rm ADM}$), and $\gamma_{\rm PPN}(r\gg L) = 1$, $\beta_{\rm PPN}(r\gg L) = 1$.

3. **Branch II-Disf + $H_1$ — Comoving Timelike Disformal Coupling ($\tilde{g}^{\mathbf{disf}}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2}{c^2}u_\mu u_\nu$, $\sqrt{-\tilde{g}_{\mathbf{disf}}} = f_L\sqrt{-g}$):**
   - In this branch, the volume element $\sqrt{-\tilde{g}_{\rm disf}} = e^{-\phi_L}\sqrt{-g}$ coincides exactly with the $e^{-\phi_L}$ weight factor of Branch II-Fluid, but the four-velocity $u^\mu$ enters the metric explicitly (which either requires a comoving khronon/aether fixing or yields an additional constraint equation under free $u^\mu$ variation).
   - Because the disformal transformation is non-conformal, it breaks conformal invariance in the Maxwell sector: on the one hand, a static magnetic field sources $\phi_L$ with effective mass density $\rho_B = \frac{|\mathbf{B}|^2}{2\mu_0 c^2}$ (Steps 15 and 20), and on the other hand, light rays propagate on the disformal light cone $\tilde{g}^{\rm disf}_{\mu\nu}k^\mu k^\nu = 0$ with an effective refractive index $n_{\rm refr}(r) = f_L(r)^{-1} = e^{+\phi_L(r)}$ (Step 14).
   - Consequently, light deflection in Branch II-Disf is **not** described by the conformal Brans–Dicke formula $\gamma^{\rm conf}_{\rm PPN} = (1-\alpha_{\rm std}e^{-r/L})/(1+\alpha_{\rm std}e^{-r/L})$, but rather by the refractive deflection angle $\hat{\alpha}_{\rm disf}(b) = \frac{4GM}{c^2 b} - \int_{-\infty}^{\infty}\frac{\partial_\perp \phi_L}{c^2}dz$ derived in Step 14.

## 6.4 4. Summary Coupling Dictionary (TAL ↔ Scalar–Tensor / Dilaton)

| Physical Quantity | Time Attraction Law (V19–V23) | Scalar–Tensor / Dilaton Counterpart | Exact Conversion and Branch-Specific Note |
|---|---|---|---|
| **Dimensionless scalar field** | $\phi_L(\mathbf{x})$ | $2\beta\varphi(\mathbf{x})$ (or $2\phi_{\rm dil}(\mathbf{x})$) | $f_L(\mathbf{x}) = e^{-\phi_L} \leftrightarrow e^{-2\beta\varphi} \leftrightarrow e^{-2\phi_{\rm dil}}$ |
| **Fundamental gravitational–temporal constant** | $C \equiv \frac{4\pi G}{c^2} = 9.33306\times10^{-27}$ m/kg | $8\pi G/c^4$ ($\times c^2/2$) | $\frac{G}{c^2} = \frac{C}{4\pi} = 7.425915\times10^{-28}$ m/kg |
| **Scale-dependent density coupling** | $\gamma_L = \frac{GL^2}{c^2} = \frac{CL^2}{4\pi}$ (m$^3$/kg) | $\frac{G}{c^2\mu_\phi^2}$ | $[\gamma_L] = \mathrm{m}^3\,\mathrm{kg}^{-1} = [\text{density}]^{-1}$ |
| **Smoothing radius / Range** | $L$ (3D top-hat / Yukawa scale) | Compton wavelength $\lambda_\phi = \mu_\phi^{-1}$ | $m_\phi = \frac{\hbar}{Lc}$ (scalar/dilaton mass) |
| **Action normalization constants** | $\alpha_Y = 4\pi\alpha_{\rm std} = 8\pi\tilde{\alpha}_L$ | $8\pi\beta^2 = \frac{4\pi}{2\omega_{\rm BD}+3}$ | Scalar prefactor in action: $-\frac{c^4}{16\pi G}\frac{1}{\tilde{\alpha}_L} = -\frac{c^4}{2G\alpha_Y}$ ($-\frac{1}{2\alpha_Y}$ in $G=c=1$) |
| **Conformal branch (Branch II-Conf)** | $\tilde{g}^{\rm conf}_{\mu\nu} = e^{-2\phi_L}g_{\mu\nu}$ | Jordan–Brans–Dicke metric | $\sqrt{-\tilde{g}_{\rm conf}} = e^{-4\phi_L}\sqrt{-g}$; $\alpha_{\rm std} = \frac{\alpha_Y}{4\pi}$, $\omega_{\rm BD} = \frac{4\pi}{\alpha_Y} - \frac{3}{2}$ |
| **Disformal branch (Branch II-Disf)** | $\tilde{g}^{\rm disf}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2}{c^2}u_\mu u_\nu$ | Bekenstein disformal metric | $\tilde{g}_{00} = f_L^2 g_{00}$, $\tilde{g}_{ij} = g_{ij}$, $\sqrt{-\tilde{g}_{\rm disf}} = f_L\sqrt{-g}$; magnetic source $\rho_B$ and refractive index $n = e^{\phi_L}$ |

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## 6.5 5. Checks

1. **Yukawa Limits ($r \ll L$ and $r \gg L$):**
   - $r \gg L$ (strongly screened regime, $e^{-r/L} \to 0$): $G_{\rm eff}(r) \to G$ and $\gamma_{\rm PPN}^{\rm conf}(r) \to 1$, so the model recovers general relativity at all distances $r \gg L$ ✓.
   - $r \ll L$ (unscreened short distance, $e^{-r/L} \to 1$): $G_{\rm eff}(r) \to G(1+\alpha_{\rm std}) = G(1+2\beta^2)$, which matches the standard scalar–tensor gravity two-body interaction formula (Damour & Esposito-Farèse 1992, Eq. 2.12) ✓.
2. **Solar System Dilution Bound of the 3D Top-Hat (V22 `hu_p2.tex`, line 32):** For the solar mass ($M_\odot = 1.98847 \times 10^{30}$ kg) at $L = 1$ pc $= 3.0857 \times 10^{16}$ m:

$$\gamma_L \bar{\rho}_L^{\odot} \;=\; \frac{3GM_\odot}{4\pi c^2 L} \;=\; \frac{3 \times 1.4766 \times 10^3 \text{ m}}{4\pi \times 3.0857 \times 10^{16} \text{ m}} \;\approx\; 1.14 \times 10^{-14} \;\ll\; 10^{-5} \quad \text{(Cassini PPN bound)} \quad \checkmark$$

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# 7 Step 6: Galactic Rotation Curves (155 / 175 SPARC Galaxies), Positive-Definite 3D Top-Hat / Yukawa Kernel Positivity, and the NGC 3198 ($R = 30.08$ kpc) Negative Control Check

Source file: step6_sparc_galactic_ngc3198_consistency_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step6_sparc_galactic_ngc3198_consistency_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step5_coupling_dictionary_ppn_screening_en.md (step5_coupling_dictionary_ppn_screening.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex (Chapter 3, lines 256–265, Eq. 6) and hu_p2.tex (Chapters 4–5, lines 1–61, Eqs. 7–8; DOI: 10.5281/zenodo.23604423)
- V17 and V19 Zenodo records: DOI 10.5281/zenodo.19666220 (V17 SPARC 155 / 123 blind sample) and DOI 10.5281/zenodo.23497128 (V19 3D spherical top-hat, 175 SPARC galaxies)
- Local SPARC and NGC 3198 working documents: docs/README_draft.md (lines 6–24), negative_density.md (lines 1–12 and 55–63), foundation_mode.md (lines 13–15 and 50–63)

## 7.1 1. Verbatim Citations from the V22 Monograph and Local SPARC Documents

In Steps 1–5, we constructed the V23 covariant scalar–tensor / Yukawa framework and verified its weak-field / Solar System (PPN) limit. In this step, we examine **the mathematical positivity of the 3D kernel on the galactic scale (exclusion of negative-density points across all 175 SPARC galaxies)** and the **NGC 3198 ($R = 30.08$ kpc) negative control point**, strictly separating the boost required by observation ($C_{\rm req}$) from the boost actually predicted by the isolated model with fixed coupling $\alpha_{\rm std} = 1/(4\pi)$ ($C_{\rm pred}^{\rm isol}$).

1. **Definition of the 3D Spherical Top-Hat and Exclusion of Negative Density ($\rho < 0$) (hu_p2.tex, lines 12–21, Eqs. 7–8):**

   *"Instead of pointwise derivatives or linear approximations, I define the source of temporal damping at any point $\mathbf{x} \in \mathbb{R}^3$ as the unweighted three-dimensional top-hat average of the physical mass–energy density $\rho(\mathbf{y}) = T_{00}(\mathbf{y})/c^2 \geq 0$ inside a ball $B(\mathbf{x}, L)$ of radius $L > 0$:"*

   $$\bar{\rho}_L(\mathbf{x}) \;\equiv\; \frac{3}{4\pi L^3}\int_{|\mathbf{y}-\mathbf{x}|\leq L} \rho(\mathbf{y})\, d^3y \;=\; \frac{M_{\rm enc}(\mathbf{x}; L)}{\frac{4}{3}\pi L^3}, \qquad f_L(\mathbf{x}) \;=\; \exp\!\big(-\gamma_L\, \bar{\rho}_L(\mathbf{x})\big). \tag{V22 Eq. 7}$$

   *"Consequently, the exponential damping factor satisfies the exact analytical bounds: $0 < f_L(\mathbf{x}) \leq 1 \quad \forall\, \mathbf{x} \in \mathbb{R}^3$, $\forall\, L > 0$. Neither negative density, nor sign reversal, nor a metric horizon inversion arising from $f_L$ can occur."*

2. **The SPARC 175 / 155 / 123 Galaxy Sample and the 3D Axisymmetric Disk Model (hu_p2.tex, lines 47–55):**

   *"The main SPARC catalog contains **175 late-type and dwarf galaxies** with 3.6 $\mu$m surface photometry and resolved HI/H$\alpha$ rotation curves. During my V17 processing, after filtering out 20 systems exhibiting severe kinematic asymmetry or inclination quality flags ($i < 30^\circ$ or $Q = 3$), the **processed rotation-curve sample of 155 galaxies** was obtained. Strict data-hygiene cuts (at least 5 radial measurement points and uninhibited $V_{gas}, V_{disk}, V_{bulge}$ baryonic decomposition) defined the **123-galaxy blind test sample**, encompassing **2596 radial velocity measurements**."*

   *"In Version 19, I replaced the thin-disk midplane approximation $\rho_{mid}(r) = \Sigma_{bary}(r)/(2h_z)$ with the exact 3D axisymmetric mass distribution $\rho_{bary}(R, z) = \frac{\Sigma_{bary}(R)}{2h_z}\exp(-|z|/h_z)$ (with scale height $h_z = 0.196\, R_d^{0.633}$) [...]. Across all 175 SPARC galaxies and over the range $L \in [10^{16}, 10^{23}]$ m, the maximum isolated galactic damping is $\max|1-f_L| = 2.284 \times 10^{-7}$ on the logarithmic grid (and $2.300 \times 10^{-7}$ under continuous optimization at $L \approx 23.09$ kpc in the massive spiral galaxy **UGC02487**)."*

3. **Exact Clarification of the NGC 3198 ($R = 30.08$ kpc) Velocity Values (docs/README_draft.md, lines 6–19 and foundation_mode.md, lines 13–14):**

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*"At $R = 30.08$ kpc (`NGC3198_rotmod.dat`, $D = 13.8$ Mpc):*

- **`Vobs = 146 ± 2 km/s`**: *Observed rotation speed. '145' in earlier text is consistent with this within $1\sigma$. Neighbouring points: 148 at 28.16 kpc, 147 at 32.14 kpc.*
- **`Vbar` ($\Upsilon_{\mathbf{disk}} = 0.5, \Upsilon_{\mathbf{bul}} = 0.7$, signed gas) = `70.32 km/s`**: *Baryonic speed with the documented v16 mass-to-light ratio.*
- **`Vbar` ($\Upsilon_{\mathbf{disk}} = 1.0$) = `88.53 km/s`**: *Baryonic speed computed from the raw SPARC columns $\sqrt{V_{gas}^2 + V_{disk}^2}$.* ***'88.5' is a baryonic value with $\Upsilon_{disk} = 1$, not an observed speed.***

*The required boost is $C_{req} = (V_{obs}/V_{bar})^2$, which is 4.311 for $\Upsilon = 0.5$ and 2.720 for $\Upsilon = 1.0$."*

---

## 7.2 2. Derivation of the Galactic Circular Velocity and Positivity Check of the 3D Kernel

### 7.2.1 2.1. Circular Velocity from the Modified Euler Equation ($h' = -\nu' + \phi_L'$)

In Steps 4 and 5 (`step4_covariant_conservation_hydrostatic.md`, `step5_coupling_dictionary_ppn_screening.md`), we derived from the covariant conservation law ($\nabla_\mu(T^{\mu\nu}_{\rm matter} + T^{\mu\nu}_{(\phi)}) = 0$) that the radial effective acceleration acting on a baryonic element orbiting in the midplane ($z = 0$) of a galactic disk is:

$$a_{r,\rm eff}(r) \;=\; -\frac{d\Phi_N(r)}{dr} \;+\; c^2\,\frac{d\phi_L(r)}{dr} \;=\; -\frac{V_{\rm bar}^2(r)}{r} \;-\; c^2\,\frac{d\ln f_L(r)}{dr}.$$

The equilibrium circular velocity ($V_{\rm rot}^2(r) = -r\,a_{r,\rm eff}(r)$) can therefore be written in two ways:

1. **Additive scalar-gradient (V19–V23 canonical scalar–tensor / Yukawa) form:**

$$\boxed{V_{\rm rot}^2(r) \;=\; V_{\rm bar}^2(r) \;-\; c^2\,r\,\frac{d\phi_L(r)}{dr} \;=\; V_{\rm bar}^2(r) \;+\; c^2\,r\,\gamma_L\left|\frac{d\bar\rho_L(r)}{dr}\right|}$$

(since $\frac{d\bar\rho_L}{dr} < 0 \implies \frac{d\phi_L}{dr} < 0$, the scalar gradient provides an inward additional attraction).

2. **Multiplicative $C(\rho)$ force-multiplier (V14–V16 phenomenological) form (`hu_p1.tex`, Eq. 6):**

$$V_{\rm rot}^2(r) \;=\; C(\rho(r))\,V_{\rm bar}^2(r), \qquad C(\rho) \;=\; 1 + C_0\exp\!\left(-\frac{\rho}{\rho_c}\right).$$

The exact algebraic relation between the two formulations at any radius $r$ is:

$$C_{\rm eff}(r) \;\equiv\; \frac{V_{\rm rot}^2(r)}{V_{\rm bar}^2(r)} \;=\; 1 \;-\; \frac{c^2\,r}{V_{\rm bar}^2(r)}\,\frac{d\phi_L(r)}{dr}.$$

### 7.2.2 2.2. What Does the Result "0 Negative-Density Points Across 175 SPARC Galaxies" Prove, and What Does It Not Prove?

For peer-review clarity, we strictly state the exact meaning of the V19 (`sparc_tophat.py`) test performed across the 175 SPARC galaxies:

- **Failure of the Earlier Spherical Derivative Formula (`negative_density.md`, line 11; `foundation_mode.md`, lines 16–18 and 55–63):** When density is estimated from the numerical derivative of the rotation curve ($\rho_{\rm deriv}(r) = \frac{1}{4\pi G r^2}\frac{d(rV_{\rm bar}^2)}{dr}$), non-monotonic gas and disk profiles cause $\rho_{\rm deriv}(r) < 0$ at **239** ($\Upsilon_{\rm disk} = 0.5$) and **303** ($\Upsilon_{\rm disk} = 1.0$) interior points across the 140 non-holdout SPARC galaxies (of which 22 points fall in the 29.5–31.5 kpc band, closest to 30.37 kpc in **NGC 4013** at $R = 30.31$ kpc with $\rho = -1.56 \times 10^{-24}$ kg/m$^3$).
- **Mathematical Positivity of the V19 3D Top-Hat and V23 Yukawa Kernels:** Both the 3D spherical top-hat kernel ($K_L(\mathbf{x},\mathbf{y}) = \frac{\Theta(L-|\mathbf{x}-\mathbf{y}|)}{\frac{4}{3}\pi L^3} \geq 0$) and the 3D Yukawa Green's function derived in Step 3 ($G_L(\mathbf{x}-\mathbf{y}) = \frac{e^{-|\mathbf{x}-\mathbf{y}|/L}}{4\pi L^2|\mathbf{x}-\mathbf{y}|} > 0$) **integrate the photometric surface-brightness density $\rho_{\mathbf{bary}}(R,z) \geq 0$ with a strictly positive kernel**. Thus, at every radial point of every galaxy:

$$\bar\rho_L(r) > 0, \qquad \phi_L^{\rm Yuk}(r) > 0, \qquad 0 < f_L(r) \leq 1 \quad \checkmark$$

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- **Critical Methodological Distinction:** This result is a **proof of the mathematical positivity of the 3D integral kernel** on the positive photometric mass model of all 175 SPARC galaxies (`n_negative_density_points = 0`), **not** a rotation-curve fitting success of the constant-$\alpha_Y = 1$ ($\alpha_{\rm std} = 1/(4\pi)$) Yukawa action! As demonstrated numerically in Section 3 below, the isolated model with fixed coupling $\alpha_{\rm std} = 1/(4\pi)$ by itself does not reproduce flat galactic rotation curves.

---

## 7.3 3. Exact Numerical Negative Control: Isolated Fixed-Coupling ($\alpha_{\rm std} = 1/(4\pi)$) Model vs. NGC 3198 ($R = 30.08$ kpc)

In accordance with peer-review requirements, the table below lists in separate columns for the control point `NGC3198_rotmod.dat` ($R = 30.08$ kpc) the observed velocity ($V_{\rm obs}$), the baryonic velocity ($V_{\rm bar}$), the boost **required** by observation ($C_{\rm req} = (V_{\rm obs}/V_{\rm bar})^2$), and the boost ($C^{\rm isol}_{\rm pred}$) and rotation velocity ($V^{\rm isol}_{\rm pred} = \sqrt{C^{\rm isol}_{\rm pred}}\, V_{\rm bar}$) **actually predicted** by the isolated model with fixed $\gamma_L = GL^2/c^2$ ($\alpha_Y = 1$, $\alpha_{\rm std} = 1/(4\pi) \approx 0.07958$):

| $\Upsilon_{\rm disk}$, $\Upsilon_{\rm bul}$ **Choice** | **Observed** $V_{\rm obs}$ **(km/s)** | **Baryonic** $V_{\rm bar}$ **(km/s)** | **Required Boost** $C_{\rm req} = (V_{\rm obs}/V_{\rm bar})^2$ | **Predicted Boost** $C^{\rm isol}_{\rm pred}$ **(Isolated 3D Top-Hat,** $\gamma_L = GL^2/c^2$**)** | **Predicted Boost** $C^{\rm isol,max}_{\rm pred}$ **(Point-Source Yukawa Upper Bound,** $1+\alpha_{\rm std}$**)** | **Predicted Velocity** $V^{\rm isol}_{\rm pred}$ **(Isolated** $\alpha_{\rm std} = 1/(4\pi)$**, km/s)** | **Observed Minus Predicted Residual** $\Delta V = V_{\rm obs} - V^{\rm isol}_{\rm pred}$ |
|---|---|---|---|---|---|---|---|
| $\Upsilon_{\rm disk} = 1.0$, $\Upsilon_{\rm bul} = 0$ (raw SPARC columns) | $146.0 \pm 2.0$ | 88.53 | 2.720 | $1 + \mathcal{O}(10^{-7}) \approx 1.0000002$ | $\leq 1 + \frac{1}{4\pi} = \mathbf{1.07958}$ | 88.53–91.99 | $-54.0$ **to** $-57.5$ **km/s** (does not fit the flat curve!) |
| $\Upsilon_{\rm disk} = 0.5$, $\Upsilon_{\rm bul} = 0.7$ (V16/V17 mass-to-light ratio) | $146.0 \pm 2.0$ | 70.32 | 4.311 | $1 + \mathcal{O}(10^{-7}) \approx 1.0000002$ | $\leq 1 + \frac{1}{4\pi} = \mathbf{1.07958}$ | 70.32–73.06 | $-72.9$ **to** $-75.7$ **km/s** (does not fit the flat curve!) |

### 7.3.1 Physical Conclusion (Negative Control):

1. In an isolated galaxy, the depth of the gravitational potential is $\frac{GM_{\rm gal}}{c^2 R} \sim \frac{V^2_{\rm bar}}{c^2} \sim 10^{-7}$, and the 3D top-hat damping computed with the canonical coupling $\gamma_L = \frac{GL^2}{c^2}$ across all 175 SPARC galaxies satisfies $\max|1 - f_L| \leq 2.30 \times 10^{-7}$ (`sparc_tophat_summary.json`).
2. Even in the optimal transition zone $r \sim L$, the Yukawa fifth force with fixed $\alpha_Y = 1$ ($\alpha_{\rm std} = \frac{1}{4\pi} \approx 0.07958$) increases $V^2_{\rm bar}$ by at most a factor of $1 + \alpha_{\rm std} \approx 1.0796$ (yielding 91.99 km/s for $\Upsilon_{\rm disk} = 1.0$ and 73.06 km/s for $\Upsilon_{\rm disk} = 0.5$, instead of the observed 146 km/s).
3. Consequently, the NGC 3198 ($R = 30.08$ kpc) control point proves as an **exact negative control** that the isolated V23 Yukawa model with constant coupling $\alpha_{\rm std} = 1/(4\pi)$ by itself **does not** explain flat galactic rotation curves; doing so requires either the V16/V17 phenomenological midplane boost $C(\rho) = 1 + C_0 e^{-\rho/\rho_c}$ ($C_0 \approx 3.34$–$5.09$) or a chameleon/symmetron extension with density-dependent coupling ($\beta_{\rm eff}(\rho)$, $\alpha^{\rm gal}_{\rm eff} = 2\beta^2_{\rm eff} \approx 1.72$–$3.31$) going beyond the constant-$\alpha_Y$ V23 action (see Step 11, **Branch II-Cham**).

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## 7.4 4. Checks

1. **NGC 3198** $R = 30.08$ **kpc Numerical Identities (**$C_{\rm req}$ **vs.** $C^{\rm isol}_{\rm pred}$**):**
   - $V_{\rm bar}(\Upsilon_{\rm disk} = 1.0) = 88.53$ km/s $\implies$ $C_{\rm req} = (146/88.53)^2 = 2.7199 \approx 2.720$, whereas $C^{\rm isol}_{\rm pred} \leq 1 + \frac{1}{4\pi} = 1.07958 \implies V^{\rm isol}_{\rm pred} \leq 91.99$ km/s $< 146$ km/s ✓.
   - $V_{\rm bar}(\Upsilon_{\rm disk} = 0.5) = 70.32$ km/s $\implies$ $C_{\rm req} = (146/70.32)^2 = 4.3107 \approx 4.311$, whereas $C^{\rm isol}_{\rm pred} \leq 1.07958 \implies V^{\rm isol}_{\rm pred} \leq 73.06$ km/s $< 146$ km/s ✓.
2. **Positivity Check Across All (175) SPARC Galaxies:**
   - Photometric 3D disk model ($\rho_{\rm bary}(R, z) \geq 0$) + 3D spherical top-hat or 3D Yukawa Green's function: number of points with $\rho \leq 0$ is **0** across all 175 galaxies (`negative_density.md`, lines 5–7 and `hu_p2.tex`, lines 17–21) ✓.

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# 8 Step 7: Compact Objects, Curved-Spacetime Scalar-Tensor / Yukawa TOV Equations (Flat-Robin Approximation vs. Curved Exterior ADM Mass), and Static Nonlocal Tangent Stability Consistency (V21 / V23)

Source file: step7_neutron_star_tov_stability_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step7_neutron_star_tov_stability_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step6_sparc_galactic_ngc3198_consistency_en.md (step6_sparc_galactic_ngc3198_consistency.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p2.tex (Chapter 7, lines 102–124, Eq. 10) and hu_p3.tex (Chapter 8, lines 1–28, Eqs. 11–12; DOI: 10.5281/zenodo.23604423)
- V21 Zenodo record: DOI 10.5281/zenodo.23602501 (Neutron Star Conditional TOV Sensitivity and Radial Pulsation Study)
- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Sections 1.1–1.2 and 2, lines 37–159, Eqs. 4–9, Tables 1–2; DOI: 10.5281/zenodo.23639084)
- Curved exterior yukawa_action_el ADM mass and coupled stability: curved_yukawa_adm_and_stability.py and step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability.md

## 8.1 1. Verbatim Citations from the V22 Monograph and the V23 (50aa2db0) Manuscript

In this step, we derive the equations of **compact objects and neutron-star TOV stability (V21 / V23)**, strictly separating:

1. the nonlocal static tangent operator V23 Eq. (9) formulated on the **Branch I-A** (V21 nonlocal 3D top-hat tophat_mass and tophat_gradient) background, and
2. the **Branch II-Fluid** (V23 local yukawa_action_el) coupled Einstein–scalar–fluid system, for which we separately compare the **flat-space Robin approximation and surface mass** ($M_{\rm surf} = m(R_*)$) used in Table 1 of the V23 manuscript with the **exact Dirichlet–Neumann slope and asymptotic ADM mass determined from the coupled curved exterior vacuum solution** ($r \in [R_*, \infty)$) ($M_{\rm ADM} = \lim_{r\to\infty} m(r) = M_{\rm surf} + \Delta M_{\phi,\rm ext}^{\rm curved}$, detailed derivation in Step 22, step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability.md).

1. **The Two Conditional TOV Modes in V21, the $1.214 \times 10^{-6}$ Kernel Accuracy, the 769 Stellar Models (Including the 33 Causality Endpoints), and the Methodological Stipulation on $e^{\nu}F$ (hu_p2.tex, Chapter 7, lines 107–124):**

*"In our audited code, the measured maximum relative discrepancy between the analytical shell kernel and independent 3D quadrature across all profiles is $1.214 \times 10^{-6}$. I then integrated two conditional Tolman–Oppenheimer–Volkoff (TOV) closures:"*

$$\textbf{(A) mass mode:}\quad \frac{dm_{\rm eff}}{dr} = 4\pi r^2 \rho(r)\, f_L(r), \qquad \textbf{(B) gradient mode:}\quad \frac{d\Phi_{\rm eff}}{dr} = f_L(r)\, \frac{G\big(m(r) + 4\pi r^3 P(r)/c^2\big)}{c^2 r\big(r - 2Gm(r)/c^2\big)}, \qquad \text{(V22 Eq. 10)}$$

*"I integrated 33 equilibrium sequences across three nuclear equations of state — **SLy** (nucleonic), **APR4** (variational nucleonic; reaching the causal sound-speed limit $c_s = c$ at rest-mass density $\rho_0 \approx 1.4458 \times 10^{15}$ **g/cm$^3$**), and **H4** (stiff hyperonic) — over 11 logarithmically spaced smoothing scales $L \in [1\,m, 100\,km]$. The V21 archival baseline table contains **182 equilibrium models**, while the dense diagnostic stability grid consists of **769 stellar models in total** (including all **33 exact causality endpoints** where $c_s/c = 1$)."*

*"When evaluating the TOV equations, I explicitly state: **the appearance of the modified hydrostatic factor $e^{\nu}F$ (or $\Phi'_{\rm eff}$) in the conditional pressure-gradient equation is an algebraic equilibrium assumption, and by itself does not derive how atomic clocks, nuclear transitions, or surface photon redshifts couple to the factor $f_L$.**"*

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**2. Algebraic $X$-Sector Stress Reconstruction and the Limitations of Frozen-$f_L$ Chandrasekhar Operators (`hu_p3.tex`, Chapter 8, lines 8–22 and `tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, lines 48–52):**

*"Across the 769 audited neutron-star models, the relative tangential stress remains bounded, with a maximum of* $\max|p_{tX}/P_c| = 0.140$ *(14.0% of the central pressure). I emphasize that this $X$-sector reconstruction is an algebraic bookkeeping closure of the background Bianchi identity, not yet a dynamical field equation derived from an action principle."*

$$\begin{aligned}
&\textbf{Mass mode } (m' = 4\pi r^2 f_L\varepsilon,\ \nu' = A_{\rm geo}): \quad \varepsilon_X = -(1-f_L)\varepsilon, \quad P_{r,X} = 0, \quad P_{t,X} = -\frac{r}{2}(1-f_L)\varepsilon\nu',\\
&\textbf{Gradient mode } (m' = 4\pi r^2\varepsilon,\ \nu' = f_L A_{\rm geo}): \quad \varepsilon_X = 0, \quad P_{r,X} = -(1-f_L)\Big(P + \frac{m}{4\pi r^3}\Big),\ P_{t,X} = P_{r,X} + \frac{r}{2}(P'_{r,X} + P_{r,X}\nu').
\end{aligned}$$

(V23 Eq. 4, V23 Eq. 5)

**3. Curved-Spacetime Yukawa Action-EL (`yukawa_action_el`) Equations of V23 (`tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, line 65):**

*"The `yukawa_action_el` rows follow from the action* $\mathcal{L} = -[(\nabla\phi)^2 + \phi^2/L^2]/(2\alpha_Y) - f_L\varepsilon$ *($G = c = 1$; `SymPy: sym_yukawa_el_source_factor` and `_hydrostatic` residual $= 0$): $B\phi'' + (2B/r + B\nu' + B'/2)\phi' - \phi/L^2 = -\alpha_Y f_L\varepsilon$, $B = 1 - 2m/r$, $P_{matter} = f_L P$, $h' = -\nu' + \phi'$, $\rho_\phi = (B\phi'^2 + \phi^2/L^2)/(2\alpha_Y)$, exterior matching $\phi'(R) = -(1/R + 1/L)\phi(R)$; the exterior tail $\Delta M_{\phi,ext}$ is listed separately (SLy, $L = 10$ km: $3.4\times10^{-3}\,M_\odot$ at $\log_{10}\rho_{0c} = 15$, $9.0\times10^{-3}\,M_\odot$ at $M_{\max}$)."*

**4. The Frozen-$f_L$ Sturm–Liouville Residual and the Nonlocal Static Tangent Operator (`tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, Section 2, lines 122–135, Eqs. 8–9):**

$$\mathcal{L}_{\rm SL}^{\rm matter}[\xi] - \sqrt{\frac{r}{r-2m}}\,e^{2\nu}\,\mathcal{H}_{\rm lin}[\xi]\Big|_{\delta f_L = 0} = \frac{4\pi r^{5/2}(f_L - 1)\Gamma P\varepsilon e^{2\nu}}{(r-2m)^{5/2}}\Big[(4\pi r^3 P - 3r + 7m)\xi - r(r-2m)\xi'\Big]. \qquad \text{(V23 Eq. 8)}$$

$$\delta f_L(r_i) = -\gamma_L f_L(r_i)\sum_j K_{ij}\,\frac{\varepsilon(r_j) + P(r_j)}{c_s^2(r_j)}\,\delta h(r_j), \qquad \begin{cases} \dfrac{d(\delta h)}{dr} = -\delta\nu'[\delta h, \delta m, \delta f_L[\delta h]] + \omega_0^2\, e^{\lambda - 2\nu}\xi, \\ \dfrac{d(\delta m)}{dr} = 4\pi r^2\Big[f_{\rm mass}\frac{\varepsilon + P}{c_s^2}\delta h + \varepsilon\,\delta f_{\rm mass}[\delta h]\Big]. \end{cases} \qquad \text{(V23 Eq. 9)}$$

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## 8.2 2. Derivation of the Curved-Spacetime Scalar-Tensor / Yukawa TOV System (`yukawa_action_el`)

### 8.2.1 2.1. The Curved-Spacetime d'Alembertian Operator and the Yukawa Field Equation

In a static, spherically symmetric spacetime (in geometrized units $G = c = 1$):

$$ds^2 \;=\; -e^{2\nu(r)}dt^2 + B(r)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta\, d\varphi^2), \qquad B(r) \;\equiv\; e^{-2\lambda(r)} \;=\; 1 - \frac{2m(r)}{r},$$

the metric determinant is $\sqrt{-g} = e^{\nu(r)}B(r)^{-1/2}r^2\sin\theta$. For a scalar field $\phi_L(r)$ depending only on $r$, the covariant Laplace–Beltrami / d'Alembert operator is:

$$\Box_g\phi_L \;=\; \frac{1}{\sqrt{-g}}\partial_r\big(\sqrt{-g}\,g^{rr}\,\partial_r\phi_L\big) \;=\; \frac{B^{1/2}}{e^\nu r^2}\frac{d}{dr}\Big(e^\nu B^{1/2}r^2\phi'_L\Big) \;=\; B\phi''_L + \left(\frac{2B}{r} + B\nu' + \frac{B'}{2}\right)\phi'_L.$$

For the Lagrangian $\mathcal{L}_{\phi+m} = -\frac{1}{2\alpha_Y}\left[g^{\mu\nu}\nabla_\mu\phi_L\nabla_\nu\phi_L + \frac{\phi_L^2}{L^2}\right] - f_L(\phi_L)\varepsilon(n)$ ($f_L = e^{-\phi_L}$, $\alpha_Y = 4\pi\alpha_{\rm std} = 8\pi\tilde\alpha_L = 2\beta^2\cdot 4\pi$), the Euler–Lagrange equation with respect to $\phi_L$ ($\frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_L)}) - \frac{\partial\mathcal{L}}{\partial\phi_L} = 0$, taking into account $\frac{\partial(-e^{-\phi_L}\varepsilon)}{\partial\phi_L} = +f_L\varepsilon$) is:

$$\boxed{B\phi''_L \;+\; \left(\frac{2B}{r} + B\nu' + \frac{B'}{2}\right)\phi'_L \;-\; \frac{\phi_L}{L^2} \;=\; -\alpha_Y\, f_L(\phi_L)\,\varepsilon(r).}$$

This is precisely the `yukawa_action_el` equation appearing in Section 1.2 of V23 (`tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, line 65), where the source term is multiplied by the dilaton-like self-consistent weight factor $f_L = e^{-\phi_L}$ (`sym_yukawa_el_source_factor = 0`).

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### 8.2.2 2.2. The Coupled Einstein–Yukawa–Euler (TOV) System, the Curved Exterior Boundary Condition, and the ADM Mass ($M_{\mathrm{ADM}}$ vs. $M_{\mathrm{surf}}$)

From the tensors $T^{\mu}_{\nu}(\mathrm{matter}) = \mathrm{diag}(-f_L\varepsilon, f_L P, f_L P, f_L P)$ and $T^{\mu}_{\nu}(\phi)$ derived in Step 4 (`step4_covariant_conservation_hydrostatic.md`), the $(tt)$ and $(rr)$ Einstein equations together with the conservation law $\nabla_\mu(T^{\mu\nu}_{\mathrm{matter}} + T^{\mu\nu}_{(\phi)}) = 0$ yield the coupled, action-closed TOV system:

$$\rho_\phi(r) \;=\; -T^t_t(\phi) \;=\; \frac{1}{2\alpha_Y}\left(B(r)\big[\phi_L'(r)\big]^2 + \frac{\phi_L(r)^2}{L^2}\right), \qquad p_{r,\phi}(r) \;=\; T^r_r(\phi) \;=\; \frac{1}{2\alpha_Y}\left(B(r)\big[\phi_L'(r)\big]^2 - \frac{\phi_L(r)^2}{L^2}\right),$$

$$\frac{dm(r)}{dr} \;=\; 4\pi r^2\big[f_L(r)\varepsilon(r) + \rho_\phi(r)\big],$$

$$\frac{d\nu(r)}{dr} \;=\; \frac{m(r) + 4\pi r^3\big[f_L(r)P(r) + p_{r,\phi}(r)\big]}{r\big(r - 2m(r)\big)},$$

$$\frac{dh(r)}{dr} \;=\; \frac{P'(r)}{\varepsilon(r) + P(r)} \;=\; -\frac{d\nu(r)}{dr} \;+\; \frac{d\phi_L(r)}{dr}.$$

**Critical Methodological Clarification (Point 1 and Point 2 — detailed derivation in Step 22):**

1. **Difference Between the Flat-Space Robin Approximation and the True Curved Exterior Dirichlet–Neumann Slope:**
   - The V23 manuscript (the original `yukawa_action_el` row of `covariant_closure_50aa2db0.py`) used as boundary condition at the stellar surface $r = R_*$ the derivative of the flat-space Helmholtz solution ($\phi_L^{\mathrm{flat}}(r) = \phi_L(R_*)\frac{R_*}{r}e^{-(r-R_*)/L}$):

$$\phi_{L,\mathrm{flat}}'(R_*) \;=\; -k_{\mathrm{flat}}\,\phi_L(R_*), \qquad k_{\mathrm{flat}} \;=\; \frac{1}{R_*} + \frac{1}{L}.$$

   - In reality, however, in the exterior vacuum outside the stellar surface ($r \geq R_*$, where $\varepsilon = P = 0$), spacetime is strongly curved ($B(R_*) = 1 - 2M_{\mathrm{surf}}/R_* \approx 0.50$–$0.62$). In the exterior vacuum, the exact identity is $B\nu' + B'/2 = \frac{2m(r)}{r^2} - \frac{4\pi r}{\alpha_Y L^2}\phi_L(r)^2$, so the exterior coupled Einstein–scalar system is:

$$\left(1 - \frac{2m(r)}{r}\right)\phi_L''(r) + \left[\frac{2}{r}\left(1 - \frac{m(r)}{r}\right) - \frac{4\pi r}{\alpha_Y L^2}\phi_L(r)^2\right]\phi_L'(r) - \frac{\phi_L(r)}{L^2} \;=\; 0, \qquad m'(r) \;=\; 4\pi r^2\rho_\phi(r).$$

Because $B(R_*) < 1$, the effective local decay rate of the mass term $\phi_L/L^2$ at the surface is $\sim B(R_*)^{-1/2}/L > 1/L$. As proven in Step 22 (`curved_yukawa_adm_and_stability.py`), for SLy at $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$, the true curved Dirichlet–Neumann slope is $k_{\mathrm{curved}} = -\phi_{L,\mathrm{curved}}'(R_*)/\phi_L(R_*) = 0.233542\ \mathrm{km}^{-1}$, which is +24.74% **larger** than the flat Robin approximation $k_{\mathrm{flat}} = 0.187216\ \mathrm{km}^{-1}$!

2. **Surface Mass ($M_{\mathrm{surf}}$) vs. Asymptotic ADM Mass ($M_{\mathrm{ADM}}$):**
   - In Table 1 of the V23 manuscript, the $M_{\max}$ and $R_{1.4}$ columns of the `yukawa_action_el` row were optimized for the surface mass $m(R_*) = M_{\mathrm{surf}}$, while the exterior scalar mass $\Delta M_{\phi,\mathrm{ext}}$ was listed in a separate column.
   - For a distant observer (and for the Bardeen–Carter–Hawking / Harrison–Wheeler turning-point theorem), the physical gravitational mass is the asymptotic **ADM mass**:

$$\boxed{M_{\mathrm{ADM}} \;\equiv\; \lim_{r\to\infty} m(r) \;=\; M_{\mathrm{surf}} \;+\; \Delta M^{\mathrm{curved}}_{\phi,\mathrm{ext}}, \qquad \Delta M^{\mathrm{curved}}_{\phi,\mathrm{ext}} \;=\; \int_{R_*}^{\infty} 4\pi r^2\rho_\phi(r)\,dr.}$$

   - On the `yukawa_action_el_curved_adm` sequences recomputed with the curved exterior solution (SLy, $L = 10$ km, $\alpha_Y = 1$):
   - $M_{\mathrm{ADM,max}} = \mathbf{2.03321\,M_\odot}$ ($\log_{10}\rho_{0c}(M_{\mathrm{ADM,max}}) = \mathbf{15.316404}$, $R(M_{\mathrm{ADM,max}}) = 9.8853$ km, $\Delta M^{\mathrm{curved}}_{\phi,\mathrm{ext}} = 0.00411\,M_\odot$, $M_{\mathrm{surf}} = 2.02911\,M_\odot$),
   - the maximum of the conserved baryonic rest mass ($M_B = \int_0^{R_*} 4\pi r^2 B(r)^{-1/2}\rho_0(r)\,dr$) occurs at $\log_{10}\rho_{0c}(M_{B,\max}^{\mathrm{quad}}) = \mathbf{15.316409}$ ($|\Delta\log_{10}\rho_{0c}| = 4.47\times10^{-6}$ dex agreement with the ADM-mass turning point!),
   - $R(M_{\mathrm{ADM}} = 1.4\,M_\odot) = \mathbf{11.5359}$ km (whereas $R(M_{\mathrm{surf}} = 1.4\,M_\odot) = 11.5348$ km),
   - and the zero crossing of the coupled $(\delta h, \delta\phi_L)$ dynamical radial pulsation eigenvalue solved on the `yukawa_action_el` background (with the `scalar_m_sign = -1.0` main operator) is $\log_{10}\rho_{0c}(\omega_0^2 = 0) = \mathbf{15.316410}$

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($+\mathbf{5.80 \times 10^{-6}}$ dex from the turning point $dM_{\rm ADM}/d\rho_c = 0$ and $+1.33 \times 10^{-6}$ dex from $dM_B/d\rho_c = 0$; at $L = 1$ km, **15.303080**, $-6.86 \times 10^{-6}$ dex; see Steps 12 and 22).

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## 8.3 3. Stability Analysis of the V21 Nonlocal 3D Top-Hat Background (Branch I-A): Frozen-$f_L$ Residual vs. $\delta f_L[\delta h]$ Tangent Operator

*(Note on separating model branches — Point 4: Subsections 3.1–3.2 below apply specifically to the V21 nonlocal 3D spherical top-hat* `tophat_mass` *and* `tophat_gradient` *[**Branch I-A**] backgrounds, whereas the coupled stability of* `yukawa_action_el` *[**Branch II-Fluid**] is presented in Steps 12 and 22.)*

### 8.3.1 3.1. Algebraic Residual Term of the Frozen-$f_L$ Chandrasekhar Operator (V23 Eq. 8)

In general relativity ($f_L \equiv 1$), taking the derivative $\partial/\partial\rho_c$ along the static equilibrium sequence generates a zero-eigenfrequency ($\omega_0^2 = 0$) radial trial perturbation at the extremal point $dM/d\rho_c = 0$. However, the derivation of the classical second-order Chandrasekhar Sturm–Liouville operator ($\mathcal{L}_{\rm SL}^{\rm matter}[\xi]$) substitutes during integration by parts the GR background identity

$$\lambda' + \nu' \;=\; 4\pi r e^{2\lambda}(\varepsilon + P) \qquad \text{(only when } f_L \equiv 1\text{!)}.$$

When the background is modified according to the V21 `mass` mode, the difference between the Chandrasekhar operator and the linearized hydrostatic operator evaluated at $\delta f_L = 0$ ($\mathcal{H}_{\rm lin}[\xi]\big|_{\delta f_L=0}$) does not vanish, but produces precisely the residual proportional to $(f_L - 1)$ given in V23 Eq. (8):

$$\mathcal{L}_{\rm SL}^{\rm matter}[\xi] - \sqrt{\frac{r}{r-2m}}\, e^{2\nu}\, \mathcal{H}_{\rm lin}[\xi]\Big|_{\delta f_L=0} \;=\; \frac{4\pi r^{5/2}(f_L-1)\Gamma P \varepsilon e^{2\nu}}{(r-2m)^{5/2}} \Big[(4\pi r^3 P - 3r + 7m)\xi - r(r-2m)\xi'\Big].$$

### 8.3.2 3.2. Incorporating the Nonlocal $\delta f_L[\delta h]$ Tangent Perturbation on the V21 Background (V23 Eq. 9 and Table 2)

When the central density is varied by $\rho_{0c} \to \rho_{0c} + \delta\rho_{0c}$ on the V21 3D top-hat background, the Eulerian perturbation of the energy density is $\delta\varepsilon(r) = \frac{\varepsilon(r)+P(r)}{c_s^2(r)}\,\delta h(r)$, which also perturbs the factor $f_L$ through the nonlocal kernel:

$$\delta f_L(r_i) \;=\; -\gamma_L\, f_L(r_i) \sum_j K_{ij}\, \frac{\varepsilon(r_j) + P(r_j)}{c_s^2(r_j)}\, \delta h(r_j).$$

Substituting this operator $\delta f_L[\delta h]$ back into the linearized $(\delta h, \delta m)$ boundary-value problem around the V21 background (V23 Eq. 9), at $\omega_0^2 = 0$ the system coincides exactly with the tangent operator of the V21 static equilibrium equations with respect to $\log_{10}\rho_{0c}$. As a result, on the SLy ($L = 10$ km) V21 sequence, the discrepancy between the zero crossing $\omega_0^2 = 0$ and the static turning point $dM/d\rho_c = 0$ (V23 Table 2) is:

- **GR baseline ($L = 0$ km):** $\Delta \log_{10}\rho_{0c} = +7.12 \times 10^{-5}$ dex (discretization baseline),
- **V21 SLy `mass` mode ($L = 10$ km, turning point $\log_{10}\rho_{0c} = 15.29924$):** improves from the frozen-$f_L$ BVP discrepancy of $-4.06 \times 10^{-3}$ dex (and `matter` SL $-1.34 \times 10^{-3}$ dex) to $+\mathbf{6.97 \times 10^{-5}}$ **dex** (a 58.3× and 19.2× improvement, respectively),
- **V21 SLy `gradient` mode ($L = 10$ km, turning point $\log_{10}\rho_{0c} = 15.30106$):** improves from the frozen-$f_L$ BVP discrepancy of $-4.39 \times 10^{-3}$ dex (and `matter` SL $-3.57 \times 10^{-3}$ dex) to $+\mathbf{7.39 \times 10^{-5}}$ **dex** (a 59.3× and 48.3× improvement, respectively).

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## 8.4 4. Checks

1. $f_L \to 1$ **($L \to 0$ or $\alpha_Y \to 0$) GR Limit:**
   - $\phi_L \to 0 \implies \rho_\phi \to 0,\ p_{r,\phi} \to 0,\ f_L \to 1$, the `yukawa_action_el` TOV system recovers the standard GR TOV equations ($M_{\rm max}^{\rm SLy} = 2.04841\,M_\odot$, $M_{\rm max}^{\rm APR4} = 2.18800\,M_\odot$, $M_{\rm max}^{\rm H4} = 2.01385\,M_\odot$), and the V23 Eq. (8) residual term vanishes identically because $(f_L - 1) = 0$ ✓.

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2. **Agreement Between Curved Exterior `yukawa_action_el` ADM Mass and Coupled Stability Turning Point (see `step22`):**
   - SLy $L = 10$ km, $\alpha_Y = 1$: $\log_{10}\rho_{0c}(dM_{\rm ADM}/d\rho_c = 0) = 15.316404$, $\log_{10}\rho_{0c}(dM_B/d\rho_c = 0) = 15.316409$, coupled $(\delta h, \delta\phi_L)$ zero crossing $\log_{10}\rho_{0c}(\omega_0^2 = 0) = 15.316410$ ($\Delta_{\rm ADM} = +5.80 \times 10^{-6}$ dex, a 2097× reduction compared to the frozen-$\phi_L$ shift of $+1.22 \times 10^{-2}$ dex; at $L = 1$ km, 15.303080 vs. 15.303086, $-6.86 \times 10^{-6}$ dex, a 165× reduction) ✓.
3. **Inertia-Scaling Invariance at the V21 Zero Crossing ($\omega_0^2 = 0$):**
   - Scaling $W \to 4W$ at $\log_{10}\rho_{0c} = 15.30427$ yields the exact ratio $\omega_{4W}^2/\omega_W^2 = 0.2500$ ($\omega_W^2 = 4.897 \times 10^6$ s$^{-2}$), whereas at the zero crossing $\omega_0^2 = 0$, $F(0)$ is invariant, confirming that the $+6.97 \times 10^{-5}$ dex and $+7.39 \times 10^{-5}$ dex agreement in V21 is a consequence of the exact consistency of the static nonlocal tangent operator ✓.

**V25 Update (curved `yukawa_action_el` branch):** 4-component exterior BVP ($\delta\nu_R = 1.63 \times 10^{-14}$), 108-entry multi-EOS / multi-point $4 \times 4$ Jacobian verification (max rel. err $= 2.74 \times 10^{-5}$; SLy 15.0, $L = 10$ km: $\partial\phi_L'(R_*)/\partial m(R_*) = -1.68365 \times 10^{-4}$ km$^{-2}$, rel. diff. $3.98 \times 10^{-10}$; interior $r = 6$ km `a_m_sc` $= -1.72833 \times 10^{-4}$ km$^{-3}$, rel. diff. $6.08 \times 10^{-9}$), three-method baryonic mass ($M_B^{\rm quad}, M_B^{\rm ODE}, M_B^{\rm trapz}$) and First-Law audit, and nodeless ($n = 0$) fundamental mode and static tangent projection ($|\langle \mathbf{\Psi}_0, \mathbf{\Psi}_{\rm tan}\rangle| \in [0.9963,\ 0.99999993]$). SLy $L = 10$ km, $\alpha_Y = 1$: $M_{\rm ADM}$ turning point 15.316404, $M_B^{\rm quad}$ turning point 15.316409 ($M_B^{\rm trapz}$: 15.316389, $M_B^{\rm ODE}$: 15.316522), main dynamical $\omega_0^2 = 0$ root (`scalar_m_sign=-1.0`) 15.316410 ($+5.80 \times 10^{-6}$ dex; with diagnostic `scalar_m_sign=+1.0`, 15.319129, $+2.72 \times 10^{-3}$ dex). Detailed tables: `step22`, Sections 4 and 7.

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# 9 Step 8: Microscopic Dirac–Maxwell Clock and Photon Coupling from the Action (H0 vs. Conditional H1), Sirius B Redshift Constraints, and the V20 Laboratory Tests (8558 kg Granite Chamber and 8-Coil Magnetic-Field Experiment)

Source file: step8_microscopic_clock_coupling_siriusb_lab_v20_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step8_microscopic_clock_coupling_siriusb_lab_v20_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step7_neutron_star_tov_stability_en.md (step7_neutron_star_tov_stability.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Section 3, lines 161–190, Table 3; DOI: 10.5281/zenodo.23639084)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p3.tex (Chapters 9 and 10, lines 30–91, Eq. 13; DOI: 10.5281/zenodo.23604423)
- V20 and V21 Zenodo records: DOI 10.5281/zenodo.23540741 (V20 8-coil magnetic field and 8558 kg granite chamber) and DOI 10.5281/zenodo.23602501 (V21 Sirius B and NICER conditional H0/H1 analysis)
- Local solenoid working report: /workspace/artifacts/solenoid_update/report_hu.md (lines 17–60)

## 9.1 1. Verbatim Citations from the V23 (50aa2db0) Manuscript and the V22 Monograph

The last two test items of the test plan in README.md (the second half of Section 3.4 and Section 3.5) concern the derivation of **microscopic clock and photon coupling (H0 vs. conditional H1)** from the covariant scalar–tensor / disformal action, together with its quantitative evaluation on the **Sirius B white dwarf** and on the **two branches of the V20 laboratory program** (8558 kg granite chamber and 8-coil solenoid).

1. **The Two Branches of Microscopic Dirac–Maxwell Clock Coupling in the Action Principle (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, Section 3, lines 162–166):**

   *"Let $\Psi_{atomic} = (\psi_e, A_\mu)$ denote the Dirac electron and Maxwell electromagnetic fields of an atomic clock comoving with the four-velocity $u^\mu$ of the baryonic medium, coupled to the effective Jordan-frame metric $\tilde{g}_{\mu\nu}[\phi_L]$:*

   - **Minimal / purely auxiliary-field coupling ($\tilde{g}_{\mu\nu} = g_{\mu\nu}$ on the $\Psi_{\text{atomic}}$ sector $\Longrightarrow$ H0 hypothesis):** *If $\phi_L$ couples exclusively to the stress–energy tensor of the macroscopic baryonic fluid while atomic electrons and photons propagate on the physical spacetime metric $g_{\mu\nu}$, then the proper time of an atomic clock stationary at the surface $r = R_*$ is $d\tau_{clock} = \sqrt{-g_{00}(R_*)}\, dt = e^{\Phi_{eff}(R_*)/c^2} dt$. The surface redshift is purely metric ($1 + z_{H0} = e^{-\Phi_{eff}(R_*)/c^2}$), with no direct $f_L(R_*)$ multiplier.*

   - **Universal disformal lapse or conformal Jordan coupling ($\tilde{g}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2(\phi_L)}{c^2} u_\mu u_\nu$ $\Longrightarrow$ H1 hypothesis):** *If atomic clocks sense the disformal metric $\tilde{g}_{\mu\nu}$, whose comoving time–time component for $g_{00} = -1$, $u_0 = -c$ is $\tilde{g}_{00} = -1 + (1 - f_L^2) = -f_L^2$ (**SymPy: sym_disformal_g00_residual** $= 0$), then the proper-time element of an atomic transition is $d\tilde{\tau}_{clock} = f_L(R_*)\, e^{\Phi_{eff}(R_*)/c^2} dt$. The coordinate frequency of photons propagating to infinity is conserved, so the direct surface clock multiplier appears: $1 + z_{H1} = e^{-\Phi_{eff}(R_*)/c^2}\, f_L(R_*)^{-1}$."*

2. **Conditional H0 and Conditional H1 Redshift Results for the Sirius B White Dwarf (hu_p3.tex, Chapter 9, lines 53–57):**

   *"I tested all six hypotheses against the high-precision HST/STIS Balmer-line measurement of the **Sirius B** white dwarf ($M = 1.018 \pm 0.011\, M_\odot$, $R = 0.008098 \pm 0.000046\, R_\odot$, observed redshift velocity $v_{obs} = 80.65 \pm 0.77$ km/s, GR baseline $v_{GR} \approx 80.42$ km/s), across both non-relativistic ($n = 1.5$) and relativistic ($n = 3$) polytropic interior profiles:*

   - **Under H0-gradient lapse (Hydrostatic coupling only):** *[The] maximum velocity shift across all $L$ is $|\Delta v| \leq 1.27$ **m/s** (0.00127 km/s), three orders of magnitude smaller than the $\pm 770$ m/s measurement uncertainty. Thus Sirius B places no constraint on $L$ under H0.*

- **Under conditional H1 (Direct surface clock coupling, $\nu \propto f_L(R_*)$):** *When atomic transitions in the photosphere are directly multiplied by $f_L(R_*)$, a smoothing sphere of radius $L \sim R_* \approx 5.63 \times 10^6$ m encloses a substantial fraction of the white dwarf's mass, inducing up to $\Delta v \approx +5.56$ km/s of excess redshift and reaching $\max \Delta\chi^2 = 52.3$ (single-factor H1) and $\max \Delta\chi^2 = 72.0$ (H1+gradient lapse). At the $2\sigma$ ($\Delta\chi^2 = 4$) level, conditional H1 excludes the intermediate band $L \in [4.08 \times 10^6, 3.37 \times 10^7]$ **m** ($n = 1.5$) and $L \in [4.95 \times 10^6, 3.37 \times 10^7]$ **m** ($n = 3$), while leaving both laboratory ($L \sim 1$ m) and sub-kilometer neutron-star scales unconstrained!"*

3. **The Two Branches of the V20 Laboratory Program (`hu_p3.tex`, Chapter 10, lines 71–85):**
   - **Branch "A" (8558 kg granite chamber):** *"Inner bore radius $R_{in} = 0.25$ m, outer radius $R_{out} = 0.85$ m, vertical height $H = 1.20$ m ($z \in [-0.60, +0.60]$ m), volume $V = \pi(R_{out}^2 - R_{in}^2)H = 3.1102\,m^3$, and total granite mass $M_{granite} = 8558$ **kg**."*
   - **Branch "B" (8-coil magnetic-field measurement):** *"8 identical solenoids, each of length $l_{coil} = 400$ mm, outer structural **flange diameter** ∅300 **mm**, effective **copper winding diameter** ∅291.7 **mm** ($R_{coil} = 0.14585$ m), $N = 2000$ turns of 0.15 mm enameled copper wire, with DC resistance $R_{DC} = 1789\,\Omega$ per coil, driven by current $I = 14.0$ mA (0.351 W/coil, 2.805 W in total)."*

$$\Delta y_{\rm peak}(L = 0.630957\,\mathrm{m}) \;=\; -1.2017108 \times 10^{-48}. \qquad \text{(V22 Eq. 13)}$$

---

## 9.2 2. Mathematical Derivation: From Dirac–Maxwell Proper Time to the Clock-Shift Formulas

### 9.2.1 2.1. How Does the Dirac–Maxwell Action Distinguish Between H0 and the Conditional H1 Hypothesis?

Consider the local Dirac electron ($\psi_e$, mass $m_e$) and Coulomb/Maxwell ($A_\mu$) action of a stationary atomic clock ($\mathbf{x} = \text{const.}$, $u^\mu = (c/\sqrt{-g_{00}}, \mathbf{0})$) on the effective clock metric $\tilde{g}_{\mu\nu}$:

$$S_{\rm atomic}[\tilde{g}_{\mu\nu}, \psi_e, A_\mu] \;=\; \int d^4x\,\sqrt{-\tilde{g}}\left[\bar{\psi}_e\left(i\hbar c\,\tilde{e}^\mu_a\gamma^a D_\mu - m_e c^2\right)\psi_e - \frac{1}{4\mu_0}\tilde{g}^{\mu\alpha}\tilde{g}^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}\right].$$

By the tetrad transformation derived in Step 2 (`step2_conformal_disformal.md`), in the local comoving frame $\tilde{e}^0_0 = \sqrt{-\tilde{g}_{00}}$, so the energy (Bohr/hyperfine transition frequency $\nu_0 = \Delta E_{\rm atomic}/h$) of stationary Dirac eigenstates ($\psi_e \propto e^{-i\omega_{\rm coord}t}$) is constant in the local clock proper time ($d\tilde{\tau}_{\rm clock} = \sqrt{-\tilde{g}_{00}}\,dt/c$):

$$\nu_{\rm coord}(\mathbf{x}) \;=\; \nu_0\,\sqrt{-\tilde{g}_{00}(\mathbf{x})}.$$

Because in a static spacetime ($\partial_t g_{\mu\nu} = 0$, $\partial_t \phi_L = 0$) the coordinate frequency $\omega_{\rm coord}$ of photons is conserved along the Killing vector $\xi^\mu = (\partial_t)^\mu$ between the emitter ($\mathbf{x}_A$) and a distant observer ($\mathbf{x}_B \to \infty$, where $g_{00} \to -1$, $f_L \to 1$), the observed frequency ratio is:

1. **H0 branch ($\tilde{g}_{\mu\nu} = g_{\mu\nu}$ on the $\Psi_{\rm atomic}$ sector — minimal clock coupling):**

$$\sqrt{-\tilde{g}_{00}(\mathbf{x})} \;=\; \sqrt{-g_{00}(\mathbf{x})} \;=\; e^{\Phi_{\rm eff}(\mathbf{x})/c^2} \quad\Longrightarrow\quad \left.\frac{\nu_{\rm obs}}{\nu_0}\right|_{\rm H0} \;=\; e^{\Phi_{\rm eff}(\mathbf{x})/c^2}, \qquad 1 + z_{\rm H0} \;=\; e^{-\Phi_{\rm eff}(\mathbf{x})/c^2}.$$

In a weak field ($\Phi_{\rm eff} = \Phi_N + \Delta\Phi_{\rm TAL}$), the fractional frequency shift relative to standard Newtonian GR is:

$$\Delta y_{\rm H0}(\mathbf{x}) \;\equiv\; \left(\frac{\nu_{\rm obs}}{\nu_0}\right)_{\rm H0} - \left(\frac{\nu_{\rm obs}}{\nu_0}\right)_{\rm GR} \;\simeq\; \frac{\Delta\Phi_{\rm TAL}(\mathbf{x})}{c^2} \;=\; \mathcal{O}\!\left(\frac{GM}{c^2R}\cdot\phi_L\right) \;=\; \mathcal{O}\!\left(\frac{G^2}{c^4}\right).$$

Therefore, the H0 signal is suppressed at second order ($\propto G^2/c^4$): in the V20 granite chamber $\Delta y_{\rm H0} \sim +10^{-48}$ to $+10^{-47}$, and for Sirius B $|\Delta v_{\rm H0}| \le 1.27$ m/s.

2. **Conditional H1 branch ($\tilde{g}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2}{c^2}u_\mu u_\nu \;\Longrightarrow\; \tilde{g}_{00} = f_L^2 g_{00}$ — disformal lapse / Jordan clock coupling):**

$$\sqrt{-\tilde{g}_{00}(\mathbf{x})} \;=\; f_L(\mathbf{x})\,e^{\Phi_{\rm eff}(\mathbf{x})/c^2} \quad\Longrightarrow\quad \left.\frac{\nu_{\rm obs}}{\nu_0}\right|_{\rm H1} \;=\; f_L(\mathbf{x})\,e^{\Phi_{\rm eff}(\mathbf{x})/c^2}, \qquad 1 + z_{\rm H1} \;=\; \frac{e^{-\Phi_{\rm eff}(\mathbf{x})/c^2}}{f_L(\mathbf{x})}.$$

In a weak field ($\phi_L \ll 1 \;\Longrightarrow\; f_L = e^{-\phi_L} \simeq 1 - \phi_L$), the excess frequency shift beyond Newtonian GR is measurable at first order ($\propto G/c^2$):

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$$\Delta y_{\mathrm{H1}}(\mathbf{x}) \;\simeq\; f_L(\mathbf{x}) - 1 \;\simeq\; -\phi_L(\mathbf{x}) \;=\; \begin{cases} -\gamma_L\,\bar{\rho}_L(\mathbf{x}) & \text{(Formulation I: 3D top-hat),} \\ -\phi_L^{\mathrm{Yuk}}(\mathbf{x}) & \text{(Formulation II: screened Yukawa).} \end{cases}$$

## 9.3 3. Quantitative Verification on Sirius B and on the Two-Arm V20 Laboratory Program

### 9.3.1 3.1. Sirius B ($M = 1.018\,M_\odot$, $R_* = 5586.5$ km, $n = 3$ Polytrope) and the V20 8558 kg Granite Chamber (V23 Table 3)

The exact numerical values from V23 `clock_coupling_H0_H1.csv` (Table 3) across both formulations are:

| System | $L$ [m] | $f_L(R_*)$ Top-Hat | $f_L(R_*)$ Yukawa | H0 Prediction | Conditional H1 Top-Hat | Conditional H1 Yukawa ($\alpha_Y = 1$) | Unit |
|---|---|---|---|---|---|---|---|
| **Sirius B** ($1.018\,M_\odot$) | $1.0 \times 10^3$ | 1.000000 | 1.000000 | 80.70055 | 80.70055 | 80.70055 | km/s |
| **Sirius B** ($1.018\,M_\odot$) | $1.0 \times 10^4$ | 1.000000 | 1.000000 | 80.70055 | 80.70055 | 80.70055 | km/s |
| **Sirius B** ($1.018\,M_\odot$) | $1.0 \times 10^6$ | 1.000000 | 1.000000 | 80.69839 | 80.70094 | 80.74476 | km/s |
| **Sirius B** ($1.018\,M_\odot$) | $5.0 \times 10^6$ | 0.999984 | 0.999993 | 80.69487 | 85.38911 | 82.85137 | km/s |
| **V20 granite** (8558 kg) | 0.50 | 1.000000 | 1.000000 | $+3.075\times10^{-48}$ | $-3.318 \times 10^{-25}$ $(-3.318 \times 10^{-25})$ | $-2.008 \times 10^{-25}$ $(-1.943 \times 10^{-25})$ | $\Delta y$ |
| **V20 granite** (8558 kg) | 0.85 | 1.000000 | 1.000000 | $+1.208\times10^{-47}$ | $-1.303 \times 10^{-24}$ $(-1.303 \times 10^{-24})$ | $-3.373 \times 10^{-25}$ $(-3.101 \times 10^{-25})$ | $\Delta y$ |
| **V20 granite** (8558 kg) | 1.00 | 1.000000 | 1.000000 | $+1.331\times10^{-47}$ | $-1.436 \times 10^{-24}$ $(-1.436 \times 10^{-24})$ | $-3.781 \times 10^{-25}$ $(-3.407 \times 10^{-25})$ | $\Delta y$ |

*(Note on the V20 rows: in the conditional H1 columns, the first value is the IN shift at $\mathbf{r} = \mathbf{0}$, and the value in parentheses is the IN−OUT differential signal at $D_{out} = 2.0$ m; the Newtonian GR shift is $\Phi_N(IN)/c^2 = -9.269 \times 10^{-24}$, IN−OUT $= -6.094 \times 10^{-24}$. The upper bound of the 103 kg static granite offset under canonical $\gamma_L$ is $\leq 5.8 \times 10^{-26}$).*

### 9.3.2 3.2. The V20 8-Coil Magnetic-Field Energy Experiment in the Scalar–Tensor / Disformal Interpretation

If the source term of the scalar field also couples to the energy density $T_{00}^{(\mathrm{EM})} = u_B = \frac{|\mathbf{B}(\mathbf{x})|^2}{2\mu_0}$ of the electromagnetic energy–momentum tensor ($\rho_B(\mathbf{x}) = \frac{|\mathbf{B}(\mathbf{x})|^2}{2\mu_0 c^2}$; although the trace of conformally invariant 4D Maxwell theory vanishes, $T^\mu_\mu(\mathrm{EM}) = 0$, in the medium-anchored disformal coupling $\tilde{g}_{00} = -f_L^2$ the electromagnetic energy component is weighted directly by $f_L$), then on the conditional H1 branch the ON/OFF modulation of the magnetic field gives:

$$\Delta y_B(\mathbf{0}; L) \;=\; -\gamma_L\,\bar{\rho}_{B,L}(\mathbf{0}) \;=\; -\frac{GL^2}{c^2} \cdot \frac{3}{4\pi L^3} \int_{|\mathbf{y}|\leq L} \frac{|\mathbf{B}(\mathbf{y})|^2}{2\mu_0 c^2}\, d^3y \;=\; -\frac{3G}{8\pi\mu_0 c^4 L} \int_{|\mathbf{y}|\leq L} |\mathbf{B}(\mathbf{y})|^2\, d^3y.$$

- **Geometric peak at $L = 0.630957$ m:** Integrating the Biot–Savart field of the 8 coils (⌀300 mm flange boundary, ⌀291.7 mm effective copper winding diameter, $l_{\mathrm{coil}} = 400$ mm, $N = 2000$, $I = 14.0$ mA), the polarity-even signal $\Delta y_{\mathrm{even}}(L) = \frac{1}{2}[\Delta y(+I) + \Delta y(-I)]$ reaches its extremum precisely at $L = 0.630957$ m ($\Delta y_{\mathrm{peak}} = -1.2017108 \times 10^{-48}$).
- **Zeeman masking and subtraction of the background cross-term:** Polarity reversal $+I \leftrightarrow -I$ cancels the linear Zeeman term and the $2\mathbf{B}_{\mathrm{bg}} \cdot \mathbf{B}_{\mathrm{coil}}$ cross-term; the co-directional quadratic Zeeman shift (at 0.86–43 $\mu$T: $-4 \times 10^{-20}$ to $-9.9 \times 10^{-17}$ for Sr, $3.4 \times 10^{-12}$ to $8.6 \times 10^{-9}$ for Cs) can be symmetrically reduced to $\mathbf{B}(\mathbf{0}) = \mathbf{0}$ at the center by alternating coil polarities.

## 9.4 4. Checks

1. **Analytical 3D Top-Hat Value of the V20 Granite Chamber at $L = 1.00$ m:**

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– Because the outer radius of the hollow granite cylinder is $R_{\rm out} = 0.85$ m and its half-height is $H/2 = 0.60$ m, the corner distance from the origin is $r_{\rm corner} = \sqrt{0.85^2 + 0.60^2} = \sqrt{1.0825} \approx 1.0404$ m. At $L = 1.00$ m, the sphere $B(\mathbf{0}, 1\text{ m})$ encloses nearly the entirety of the 8558 kg granite mass:

$$\Delta y_{\rm H1}^{\text{top-hat}}(L = 1.00\text{ m}) \;=\; -\frac{GL^2}{c^2}\,\frac{M_{\rm enc}(<1\text{ m})}{\frac{4}{3}\pi L^3} \;=\; -1.436 \times 10^{-24} \quad \checkmark$$

**2. Order-of-Magnitude Verification of the Sirius B H1 Velocity Shift at $L = 5.0 \times 10^6$ m:**

– $1 - f_L(R_*) = 1 - 0.999984 = 1.6 \times 10^{-5} \implies \Delta v_{\rm H1} \simeq c\,\big(1 - f_L(R_*)\big) \approx 3 \times 10^5$ km/s $\times\, 1.56 \times 10^{-5} \approx +4.69$ km/s, so $v_{\rm H1} = 80.69487 + 4.69424 = 85.38911$ km/s (Top-Hat) and 82.85137 km/s (Yukawa), matching Table 3 of V23 ✓.

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# 10 Step 9: High-Energy Photon Propagation in the Disformal/Conformal Scalar-Tensor Metric, the Effective Refractive Index $n_t(s) = f_L(s)^{-1}$, and the Multi-Source V17 H.E.S.S./MAGIC Gamma-Ray Timing Audit

Source file: step9_vhe_photon_propagation_hess_dispersion_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step9_vhe_photon_propagation_hess_dispersion_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step8_microscopic_clock_coupling_siriusb_lab_v20_en.md (step8_microscopic_clock_coupling_siriusb_lab_v20.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p2.tex (Chapter 6, lines 63–99, Eq. 9 and the 5-source H.E.S.S./MAGIC control table; DOI: 10.5281/zenodo.23604423)
- V17 Zenodo record: DOI 10.5281/zenodo.19666220 (SPARC 155 galaxies and multi-source H.E.S.S. VHE gamma-ray timing audit)
- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Sections 1 and 3, Eq. 1 and lines 161–166; DOI: 10.5281/zenodo.23639084)

## 10.1 1. Verbatim Citations from Chapter 6 of the V22 Monograph (hu_p2.tex)

In Step 8 (step8_microscopic_clock_coupling_siriusb_lab_v20.md), we derived the stationary proper-time and redshift coupling of a comoving atomic clock ($\Psi_{\rm atomic} = (\psi_e, A_\mu)$) on the effective metric $\tilde{g}_{\mu\nu}$. The next logical step is to derive the **propagation of the Maxwell photon field ($F_{\mu\nu}$) across astronomical and cosmological baseline distances ($s \in [0, D]$)** from the scalar–tensor / disformal action principle, comparing it with the multi-source V17 H.E.S.S./MAGIC gamma-ray timing audit documented in Chapter 6 of the V22 monograph:

1. **Local Refractive Index of the Temporal Carrier ($n_t(s) = f_L(s)^{-1}$), Achromatic Delay, and Dispersion Parameterization (hu_p2.tex, Chapter 6, lines 68–74, Eq. 9):**

   *"When a VHE gamma-ray photon traverses cosmic voids, filaments, host-galaxy halos, and galactic interstellar structures along a line of sight $s \in [0, D]$, the local refractive index of the temporal carrier is $n_t(s) = f_L(s)^{-1} = \exp(+\gamma_L \bar{\rho}_L(s))$. In the purely metric (geometric) limit, where all photon energies couple identically to the effective lapse factor $f_L(s)$, the arrival time experiences an achromatic Shapiro-like excess delay: $\Delta t_{geom} = c^{-1} \int_0^D \left[\exp(\gamma_L \bar{\rho}_L(s)) - 1\right] ds$. If, however, high-energy ($E$) photon wave packets undergo finite-wavelength dispersion by scattering across spatial gradients $\nabla\bar{\rho}_L$ on scale $L$, the parameterized form of the energy-dependent time lag between energy channels $E_1$ and $E_2$ is:"*

$$\Delta t(E_1, E_2) \;=\; \tau_n \, \frac{E_2^n - E_1^n}{E_{\rm ref}^n} \int_0^D \gamma_L \, \bar{\rho}_L(s) \, (1 + z(s))^n \, ds, \qquad n \in \{1, 2\}, \tag{V22 Eq. 9}$$

   *"where $\tau_n$ is the dispersion coupling coefficient and $D(z)$ is the redshift-weighted comoving path length."*

2. **The Five-Source V17 H.E.S.S. / MAGIC Control Table (hu_p2.tex, Chapter 6, lines 78–91):**

| VHE Source | Redshift / Dist. | Energy Band (TeV) | Observed Lag / Bound | V17 Audit Result |
|---|---|---|---|---|
| **PKS 2155-304 (H.E.S.S.)** | $z = 0.116$ | 0.2–4.0 TeV | $\Delta t \approx +28 \pm 12$ s | Marginal $\sim 2\sigma$ lag; dominated by intrinsic SSC emission profile. |
| **Mrk 501 (1997 / H.E.S.S.)** | $z = 0.034$ | 0.15–10 TeV | $\Delta t \approx +240 \pm 60$ s | Larger lag at 3.4× lower $z$ violates monotonic distance scaling! |
| **GRB 190114C (MAGIC)** | $z = 0.4245$ | 0.2–1.0 TeV | $ | \Delta t / \Delta E |

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| VHE Source | Redshift / Dist. | Energy Band (TeV) | Observed Lag / Bound | V17 Audit Result |
|---|---|---|---|---|
| **Crab Pulsar (H.E.S.S./MAGIC)** | $d \approx 2.0\,\mathrm{kpc}$ | 0.1–1.5 TeV | $ | \Delta t |
| **Vela Pulsar (H.E.S.S. II)** | $d \approx 0.287\,\mathrm{kpc}$ | 0.02–0.1 TeV | $ | \Delta t |

3. **Two Main Conclusions of the V17 Multi-Source Audit (`hu_p2.tex`, Chapter 6, lines 95–98):**
   - **Rejection of Naive Distance-Proportional Dispersion:** *"Because Mrk 501 ($z = 0.034$) exhibits a larger apparent flare lag than PKS 2155-304 ($z = 0.116$) while GRB 190114C ($z = 0.4245$) and the Crab/Vela pulsars yield tight null bounds, explaining the 28 s lag of PKS 2155-304 via a uniform intergalactic dispersion constant is refuted. Intrinsic blazar synchrotron self-Compton (SSC) acceleration timescales must be modeled first."*
   - **Unique Environmental Overdensity Prediction vs. ΛCDM:** *"Unlike Lorentz-invariance-violating (LIV) models that predict a purely redshift-dependent delay $\Delta t \propto \int_0^z \frac{(1+z')\,dz'}{H(z')}$, the Time Attraction Law predicts that any genuine non-geometric timing residual must correlate with the baryonic overdensity integral ($\int \bar{\rho}_L(s)\,ds$) along lines of sight passing through rich galaxy clusters (such as Coma or Perseus) compared to cosmic voids at the same redshift."*

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## 10.2 2. Mathematical Derivation: Light Cone, Effective Refractive Index, and EFT Dispersion in the Scalar-Tensor Framework

### 10.2.1 2.1. Conformal Invariance vs. Disformal Light Cone: Derivation of the Refractive Index $n_t(s) = f_L(s)^{-1}$

Let us examine the free Maxwell action ($S_{\mathrm{EM}} = -\frac{1}{4\mu_0}\int d^4x\sqrt{-\tilde{g}}\,\tilde{g}^{\mu\alpha}\tilde{g}^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}$) under the two metric transformations derived in Step 2 (`step2_conformal_disformal.md`):

1. **Pure Conformal Transformation ($\tilde{g}^{\mathbf{conf}}_{\mu\nu} = f_L^2 g_{\mu\nu}$):** In $D = 4$ spacetime dimensions, $\sqrt{-\tilde{g}^{\mathrm{conf}}} = f_L^4\sqrt{-g}$ and $\tilde{g}^{\mu\alpha}_{\mathrm{conf}}\tilde{g}^{\nu\beta}_{\mathrm{conf}} = f_L^{-4}g^{\mu\alpha}g^{\nu\beta}$, so the factor $f_L^4 f_L^{-4} = 1$ cancels:

$$S_{\mathrm{EM}}\big[f_L^2 g_{\mu\nu}, A_\mu\big] \;=\; S_{\mathrm{EM}}[g_{\mu\nu}, A_\mu]\,.$$

Therefore, in a purely conformal scalar–tensor coupling, the light cone ($\tilde{g}^{\mathrm{conf}}_{\mu\nu}k^\mu k^\nu = 0 \iff g_{\mu\nu}k^\mu k^\nu = 0$) is identical to that of the background metric.

2. **Medium-Anchored Disformal Clock Metric of V23 ($\tilde{g}^{\mathbf{disf}}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2}{c^2}u_\mu u_\nu$):** If electromagnetic waves couple to the disformal metric $\tilde{g}^{\mathrm{disf}}_{\mu\nu}$ defined by the four-velocity $u^\mu$ of the baryonic medium, then in the local rest frame ($g_{\mu\nu} \simeq \eta_{\mu\nu}$, $u^\mu = (c, \mathbf{0})$) the line element is:

$$d\tilde{s}^2_{\mathrm{disf}} \;=\; -f_L(s)^2c^2dt^2 + ds^2, \qquad ds^2 = dx^2 + dy^2 + dz^2.$$

In the geometric-optics (eikonal / null-geodesic) limit, $d\tilde{s}^2_{\mathrm{disf}} = 0$, from which the coordinate speed of light of the photon and the effective temporal refractive index of the medium follow:

$$v_{\mathrm{eff}}(s) \;=\; \frac{ds}{dt} \;=\; c\,f_L(s) \;=\; c\,\exp\!\big(-\phi_L(s)\big) \quad\Longrightarrow\quad \boxed{n_t(s) \;\equiv\; \frac{c}{v_{\mathrm{eff}}(s)} \;=\; f_L(s)^{-1} \;=\; \exp\!\big(+\phi_L(s)\big) \;=\; \exp\!\big(+\gamma_L\,\bar{\rho}_L(s)\big).}$$

Integrating over the path $s \in [0, D]$ directly yields the achromatic Shapiro-like travel-time excess given in line 69 of V22 `hu_p2.tex`:

$$\Delta t_{\mathrm{geom}} \;=\; \int_0^D \left(\frac{1}{v_{\mathrm{eff}}(s)} - \frac{1}{c}\right) ds \;=\; \frac{1}{c}\int_0^D \Big[\exp\!\big(\gamma_L\bar{\rho}_L(s)\big) - 1\Big]\,ds \;\simeq\; \frac{1}{c}\int_0^D \phi_L(s)\,ds \qquad (\phi_L \ll 1).$$

Because $\Delta t_{\mathrm{geom}}$ does not depend on the photon energy $E = \hbar\omega$, in the pure eikonal limit the relative delay between photons of energies $E_1$ and $E_2$ emitted simultaneously vanishes: $\Delta t_{\mathrm{geom}}(E_1, E_2) = 0$.

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#### 10.2.2 2.2. Finite-Wavelength Scalar–Photon EFT Operators and Derivation of the Dispersion Formula V22 Eq. (9)

When the wavelength of high-energy (VHE, $E \sim 0.1$–$10$ TeV) gamma-ray photons interacts with gradients of the scalar field $\phi_L = \gamma_L \bar{\rho}_L$ (or $\phi_L^{\rm Yuk}$), the leading dimension-$4+n$ ($n \in \{1,2\}$) scalar–Maxwell operator in the effective field theory (EFT) expansion with the medium four-velocity $u^\mu$ is:

$$\Delta\mathcal{L}^{(n)}_{\rm disp} \;=\; -\frac{c\,\tau_n}{4\mu_0}\,\frac{\phi_L(x)}{E^n_{\rm ref}}\,F_{\mu\nu}\left(i\hbar\,u^\alpha\nabla_\alpha\right)^n F^{\mu\nu}.$$

In the plane-wave approximation ($A_\mu \propto e^{ik_\alpha x^\alpha}$, where the local comoving photon energy is $E_{\rm loc}(s) = -\hbar u^\alpha k_\alpha = E_{\rm obs}\,(1+z(s))$), the modified dispersion relation is:

$$\omega^2 \;=\; c^2k^2\left[1 - \frac{2c\,\tau_n}{n+1}\,\phi_L(s)\,\left(\frac{E_{\rm obs}(1+z(s))}{E_{\rm ref}}\right)^n\right].$$

Integrating the reciprocal of the group velocity ($v_g = \partial\omega/\partial k$) between two distinct observed energy channels ($E_1 < E_2$) yields V22 Eq. (9):

$$\boxed{\Delta t(E_1,E_2) \;=\; t_{\rm arr}(E_2) - t_{\rm arr}(E_1) \;=\; \tau_n\,\frac{E_2^n - E_1^n}{E^n_{\rm ref}}\int_0^D \phi_L(s)\,\big(1+z(s)\big)^n\,ds \;=\; \tau_n\,\frac{E_2^n - E_1^n}{E^n_{\rm ref}}\int_0^D \gamma_L\,\bar{\rho}_L(s)\,\big(1+z(s)\big)^n\,ds.}$$

---

## 10.3 3. Checks and Physical Implications

1. **Why Does the 5-Source V17 Audit Exclude Universal Vacuum Dispersion ($\phi_L = $ const.) and Support Local $\phi_L(\mathbf{x})$ Coupling?**
   - If dispersion were caused by density-independent vacuum LIV ($\phi_L(s) \equiv$ const.), then $\Delta t/\Delta E$ would increase strictly monotonically with redshift $z$. Instead, **Mrk 501** ($z = 0.034$, $\Delta t \approx +240 \pm 60$ s) shows an $\sim 8.6\times$ larger lag at $3.4\times$ lower redshift than **PKS 2155-304** ($z = 0.116$, $\Delta t \approx +28 \pm 12$ s), while **GRB 190114C** at $z = 0.4245$ gives $|\Delta t/\Delta E| \leq 19$ s/TeV and the Galactic **Crab** ($|\Delta t| < 0.4$ ms) / **Vela** ($|\Delta t| < 0.2$ ms) pulsars impose tight upper bounds ✓.
2. **Discriminating Prediction of Scalar–Tensor / Yukawa TAL Against LIV Models:**
   - Because the scalar field $\phi_L(s)$ is proportional to the baryonic mass density along the line of sight ($\phi_L = \gamma_L\bar{\rho}_L$ or $\phi_L^{\rm Yuk}$), for sources at identical redshift $z$ the line-of-sight integral $\mathcal{I}_L \equiv \int_0^D \phi_L(s)(1+z(s))^n\,ds$ is proportional to the baryonic overdensity of intervening galaxy clusters (e.g., Coma, Perseus) relative to cosmic voids ✓.
3. **Vacuum Limit ($\bar{\rho}_L \to 0 \implies \phi_L \to 0$):**
   - $\phi_L(s) \to 0 \implies n_t(s) = e^0 = 1$, $\Delta t_{\rm geom} = 0$, and $\Delta t(E_1,E_2) = 0$: light propagation recovers the special/general relativistic vacuum light cone ✓.

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# 11 Step 10: The Nonlocal Bilateral Kernel Action (Formulation I), Lapse-Weighted Reciprocity Potential ($V_{\mathrm{eff},N}$), and the Moment/EFT Bridge Between Formulations I and II (Yukawa Scalar-Tensor)

Source file: step10_bilateral_kernel_reciprocity_eft_synthesis_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step10_bilateral_kernel_reciprocity_eft_synthesis_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step9_vhe_photon_propagation_hess_dispersion_en.md (step9_vhe_photon_propagation_hess_dispersion.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Sections 1.1 and 1.2, lines 37–66, Eqs. 2–3, Table 1; DOI: 10.5281/zenodo.23639084)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p3.tex (Chapters 11 and 12, lines 94–120; DOI: 10.5281/zenodo.23604423)

## 11.1 1. Verbatim Citations from the V23 (50aa2db0) Manuscript and the V22 Monograph

In Steps 1–9, we developed Formulation II (the local screened Helmholtz–Yukawa scalar–tensor action) together with its observational and laboratory consequences. In this step, we derive **Formulation I (the nonlocal bilateral kernel action with auxiliary fields $\phi_L, \lambda_L$, unweighted bilateral_eom, and lapse-weighted bilateral_lapse_weighted reciprocal equilibrium)** as well as the **exact moment expansion connecting Formulation I (3D spherical top-hat, Branch I-A/I-B) and Formulation II (Yukawa scalar–tensor, Branch II-Fluid, $\alpha_Y = 1$ vs. $\alpha_Y = 1/3$)**, strictly recording that the second-order gradient moments of the two kernels ($\frac{1}{10}L^2\nabla^2$ vs. $L^2\nabla^2$) differ from each other.

1. **Formulation I of V23: Nonlocal Positive-Definite Kernel Action and Lagrange Multiplier (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, Section 1.1, lines 38–46, Eqs. 2–3):**

   *"In Formulation I, a Lagrange multiplier field $\lambda_L(x)$ couples the auxiliary scalar $\phi_L(x)$ to the spatial top-hat average taken on the hypersurface $\Sigma_u$ orthogonal to the fluid four-velocity $u^\mu$ ($h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu/c^2$):"*

   $$S_{\mathrm{I},X}[\phi_L, \lambda_L, g_{\mu\nu}] = -\frac{c^4}{8\pi G\gamma_L}\int d^4x\sqrt{-g}\,\lambda_L(x)\left[\phi_L(x) - \gamma_L\int_{\Sigma_u} d^4y\sqrt{-g(y)}\,K_L(x,y)\,\rho_{\mathrm{src}}(y)\right], \qquad \text{(V23 Eq. 2)}$$

   *"where $K_L(x,y) = \delta(\tau_u(x) - \tau_u(y))\,\Theta(L - d_\Sigma(x,y))/V_L(x)$ is the positive-definite 3D spherical kernel of radius $L$ and proper volume $V_L = \frac{4}{3}\pi L^3$. Variation with respect to $\lambda_L(x)$, $\phi_L(x)$, and $\rho_{src}(x)$ yields the following Euler–Lagrange closure:"*

   $$\frac{\delta S}{\delta\lambda_L(x)} = 0 \implies \phi_L(x) = \gamma_L\bar{\rho}_L(x), \quad f_L(x) = e^{-\phi_L(x)}; \qquad \frac{\delta S}{\delta\phi_L(x)} = 0 \implies \lambda_L(x) = \frac{8\pi G\gamma_L}{c^4}\frac{\partial\mathcal{L}_{\mathrm{eff}}}{\partial\phi_L}(x). \qquad \text{(V23 Eq. 3)}$$

2. **Bilateral Kernel Reciprocity ($x \leftrightarrow y$), Lapse-Weighted $V_{\mathrm{eff},N}$, and the Two-Cell Variational Test (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, line 54):**

   *"If $\lambda_L$ is absorbed into the fluid in the form $\mathcal{L}_{fluid}(n,\phi_L) = -f_L(\phi_L)\,\varepsilon(n)$ and $\varepsilon$ is varied both explicitly and inside the kernel, then $V_{eff}(x) = f_L(x) - \gamma_L\int K_L(y,x)f_L(y)\varepsilon(y)\,d^3y$ and the modified equilibrium is $dh/dr = -A(r) - dV_{recip}/dr$, $V_{recip} = \ln V_{eff}$. I solve this as a self-consistent fixed point (**bilateral_eom** rows in Table 1); it differs from the algebraically closed V21 mass mode. Varying the lapse-weighted integral $\int dV\,N f_L\varepsilon$ with $N = e^\nu$ (at fixed kernel and metric) gives $V_{eff,N}(x) = f_L - \frac{\gamma_L}{N(x)}\int dV_y\,N(y)K(y,x)f_L\varepsilon$, which I solve in the lapse equation with $P_{matter} = f_L P$ (**bilateral_lapse_weighted** rows; two-cell test $N = (1; 0.8)$, $\varepsilon = (1; 2)$, $\gamma = 0.1$: $(0.74882; 0.72084)$, unweighted $0.73160$). Covariant closure still requires metric variation of the kernel and volume element; neither row is the EL solution of action (2)."*

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3. **The $\alpha_Y = 1$ (Unit-Normalized) and $\alpha_Y = 1/3$ (Volume-Matched) Yukawa Normalizations (`tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, line 65):**

   *"The unit-normalized kernel corresponds to $\alpha_Y = 1$, i.e., $\alpha_{std} = 1/(4\pi) \approx 0.0796$, which recovers the homogeneous top-hat limit ($\phi_L = \gamma_L \rho_{src}$); $\alpha_Y = 1/3$ ($\alpha_{std} = 1/(12\pi)$) is the volume-matched variant."*

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## 11.2 2. Mathematical Derivation

### 11.2.1 2.1. Derivation of Bilateral Kernel Reciprocity ($x \leftrightarrow y$) and Lapse-Weighted $V_{\mathbf{eff},N}(x)$

Consider the static energy functional in which the matter Lagrangian $\mathcal{L}_{\rm fluid}(n, \phi_L) = -f_L(\phi_L)\,\varepsilon(n)$ ($f_L(x) = \exp[-\gamma_L \int dV_y\, K_L(x,y)\varepsilon(y)]$) appears weighted by the lapse factor $N(x) = e^{\nu(x)}$ (with fixed background metric and fixed kernel $K_L(x,y)$):

$$E_{\rm matter}[\varepsilon] \;=\; \int dV_x\, N(x)\, f_L(x)\, \varepsilon(x), \qquad \phi_L(x) \;=\; \gamma_L \int dV_y\, K_L(x,y)\, \varepsilon(y).$$

Let us vary $E_{\rm matter}[\varepsilon]$ with respect to the energy density $\varepsilon(x)$! Because $\varepsilon$ appears in two places — once as an explicit factor at point $x$ and once in the exponent of $f_L(y) = \exp[-\gamma_L \int dV_x\, K_L(y,x)\varepsilon(x)]$ at every point $y$ whose smoothing ball contains $x$ — the functional derivative by the chain rule and by exchanging the dummy integration variables $x \leftrightarrow y$ (*bilateral kernel reciprocity*) is:

$$\delta E_{\rm matter} \;=\; \int dV_x\, N(x)\, f_L(x)\, \delta\varepsilon(x) \;+\; \int dV_y\, N(y)\, \varepsilon(y)\, \frac{\partial f_L(y)}{\partial \phi_L(y)}\, \gamma_L \int dV_x\, K_L(y,x)\, \delta\varepsilon(x).$$

Since $\frac{\partial f_L(y)}{\partial \phi_L(y)} = -f_L(y)$, exchanging the integration order of $dV_y$ and $dV_x$ in the second integral and factoring out $N(x)\,\delta\varepsilon(x)$ yields:

$$\delta E_{\rm matter} \;=\; \int dV_x\, N(x) \underbrace{\left[ f_L(x) \;-\; \frac{\gamma_L}{N(x)} \int dV_y\, N(y)\, K_L(y,x)\, f_L(y)\, \varepsilon(y) \right]}_{\equiv\, V_{{\rm eff},N}(x)} \delta\varepsilon(x).$$

This derives the two bilateral potentials appearing in Section 1.1 (line 54) of V23:

1. **Lapse-weighted bilateral potential (`bilateral_lapse_weighted`):**

$$\boxed{V_{{\rm eff},N}(x) \;=\; f_L(x) \;-\; \frac{\gamma_L}{N(x)} \int dV_y\, N(y)\, K_L(y,x)\, f_L(y)\, \varepsilon(y).}$$

2. **Unweighted bilateral potential (`bilateral_eom`, $N(y)/N(x) \to 1$):**

$$\boxed{V_{\rm eff}(x) \;=\; f_L(x) \;-\; \gamma_L \int dV_y\, K_L(y,x)\, f_L(y)\, \varepsilon(y).}$$

In the hydrostatic Euler equation ($\delta\varepsilon/\delta n = (\varepsilon + P)/n = \mu e^h$), the condition of constant chemical potential ($N(r)\, V_{\rm eff}(r)\, e^{h(r)} = \text{const.}$) after logarithmic differentiation directly gives the modified equilibrium equation:

$$\frac{dh}{dr} \;=\; -\frac{d\nu}{dr} \;-\; \frac{dV_{\rm recip}}{dr}, \qquad V_{\rm recip}(r) \;\equiv\; \ln V_{\rm eff}(r).$$

Since $\gamma_L \int K_L f_L \varepsilon\, dV_y > 0$, the bilateral reciprocal term **reduces** $V_{\rm eff}$ in the dense stellar core ($V_{\rm eff}(0) < f_L(0)$), producing an inward additional compressive force that explains why the `bilateral_eom` and `bilateral_lapse_weighted` branches yield a lower maximum mass than GR ($M_{\rm max}^{\rm SLy} = 1.93214\, M_\odot$ and $1.90217\, M_\odot$ at $L = 10$ km, compared to $2.09822\, M_\odot$ in the V21 `mass` mode; V23 Table 1).

### 11.2.2 2.2. Exact Moment-Expansion Comparison: Where Do the 3D Top-Hat (Branch I) and Yukawa (Branch II) Formulations Agree and Where Do They Differ?

Expanding the source density $\rho_{\rm src}(\mathbf{x} + \mathbf{u})$ in a Taylor series around point $\mathbf{x}$ inside the 3D spherical top-hat average ($|\mathbf{u}| \le L$) gives:

$$\bar{\rho}_L(\mathbf{x}) \;=\; \frac{3}{4\pi L^3}\int_{|\mathbf{u}|\le L}\left[\rho_{\rm src}(\mathbf{x}) + u_i\partial_i\rho_{\rm src}(\mathbf{x}) + \frac{1}{2}u_iu_j\partial_i\partial_j\rho_{\rm src}(\mathbf{x}) + \dots\right] d^3u.$$

By spherical symmetry, $\int u_i\, d^3u = 0$ and $\int u_iu_j\, d^3u = \frac{1}{3}\delta_{ij}\int_0^L 4\pi u^4\, du = \frac{4\pi}{15}L^5\,\delta_{ij}$, so the gradient expansion of the 3D top-hat operator is:

$$\boxed{\phi_L^{\text{top-hat}}(\mathbf{x}) \;=\; \gamma_L\,\bar{\rho}_L(\mathbf{x}) \;=\; \gamma_L\left(1 \;+\; \frac{L^2}{10}\nabla^2 \;+\; \mathcal{O}(L^4\nabla^4)\right)\rho_{\rm src}(\mathbf{x}).}$$

On the other hand, the formal inverse of the Helmholtz–Yukawa equation of Formulation II ($(-L^2\nabla^2+1)\phi_L^{\rm Yuk} = \alpha_Y\gamma_L\rho_{\rm src}$) for small gradients is:

$$\boxed{\phi_L^{\rm Yuk}(\mathbf{x}) \;=\; \alpha_Y\,\gamma_L\left(1 - L^2\nabla^2\right)^{-1}\rho_{\rm src}(\mathbf{x}) \;=\; \alpha_Y\,\gamma_L\left(1 \;+\; L^2\nabla^2 \;+\; \mathcal{O}(L^4\nabla^4)\right)\rho_{\rm src}(\mathbf{x}).}$$

**Critical Theoretical Distinction (Point 5):**

1. **Zeroth-Moment (Homogeneous $\nabla^2\rho_{\rm src} \to 0$) Limit:** Under the normalization $\alpha_Y = 1$ ($\alpha_{\rm std} = \frac{1}{4\pi} \approx 0.07958$), the 3D top-hat and the Yukawa Green's function agree **exclusively in the zeroth moment** ($\int K_L(\mathbf{u})\, d^3u = \int G_L(\mathbf{u})\, d^3u = 1$, i.e., in a homogeneous medium: $\phi_L^{\rm Yuk} = \gamma_L\rho_{\rm src} = \phi_L^{\text{top-hat}}$)!
2. **Difference in the Second-Order Gradient Moment ($\frac{1}{10}L^2\nabla^2$ vs. $L^2\nabla^2$) and Compact Support:** In an inhomogeneous medium ($\nabla^2\rho_{\rm src} \neq 0$), the two models are **not** the same effective field theory:
   - at fixed $L$ and $\alpha_Y = 1$, the second-order $\nabla^2\rho_{\rm src}$ coefficient differs by a factor of 10 ($\frac{1}{10}L^2$ for the 3D top-hat kernel vs. $1\cdot L^2$ for the Yukawa Green's function; matching the second moment would require $\ell_Y = L/\sqrt{10}$);
   - the 3D top-hat kernel has compact support ($r \le L$, so the exterior stellar halo terminates sharply at $r = R_* + L$), whereas the Yukawa Green's function generates an exponentially decaying ($\propto e^{-r/L}/r$) curved exterior scalar tail over an infinite domain and an ADM mass contribution $\Delta M_{\phi,\rm ext}^{\rm curved}$;
   - the $\alpha_Y = \frac{1}{3}$ ($\alpha_{\rm std} = \frac{1}{12\pi} \approx 0.02653$) variant matches the ratio of the spherical volume $\frac{4}{3}\pi L^3$ to the Yukawa normalization volume $4\pi L^3$ ($\frac{4\pi/3}{4\pi} = \frac{1}{3}$) at the stellar center ($r \to 0$).

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## 11.3 3. Checks

1. **Exact Reproduction of the V23 Two-Cell Variational Test (`bilateral_lapse_weighted` vs. `bilateral_eom`):**
   - Consider two cells of equal volume ($\Delta V_1 = \Delta V_2 = 1/2$, symmetric overlapping kernel $K_{ij} = 1$, so $\bar{\varepsilon} = \frac{1}{2}(\varepsilon_1 + \varepsilon_2)$), where according to line 54 of V23, $N = (1.0;\ 0.8)$, $\varepsilon = (1.0;\ 2.0)$, and $\gamma_L = 0.1$:
   - Mean density: $\bar{\varepsilon} = \frac{1+2}{2} = 1.5 \implies \phi_L = \gamma_L\bar{\varepsilon} = 0.15 \implies f_L(1) = f_L(2) = e^{-0.15} = 0.86070798$.
   - **Unweighted (`bilateral_eom`):**

$$V_{\rm eff}(1) = V_{\rm eff}(2) = f_L\big(1 - \gamma_L\bar{\varepsilon}\big) = e^{-0.15}\times(1 - 0.15) = 0.85\, e^{-0.15} = \mathbf{0.73160178} \approx \mathbf{0.73160} \quad \checkmark$$

   - **Lapse-weighted (`bilateral_lapse_weighted`):** The lapse-weighted source average is $\overline{N\varepsilon} = \frac{1}{2}(N_1\varepsilon_1 + N_2\varepsilon_2) = \frac{1}{2}(1.0\times 1.0 + 0.8\times 2.0) = \frac{2.6}{2} = 1.3$:

$$V_{\text{eff},N}(1) = f_L\left(1 - \frac{\gamma_L\,\overline{N\varepsilon}}{N_1}\right) = e^{-0.15}\left(1 - \frac{0.13}{1.0}\right) = 0.87\, e^{-0.15} = \mathbf{0.74881594} \approx \mathbf{0.74882} \quad \checkmark$$

$$V_{\text{eff},N}(2) = f_L\left(1 - \frac{\gamma_L\,\overline{N\varepsilon}}{N_2}\right) = e^{-0.15}\left(1 - \frac{0.13}{0.8}\right) = 0.8375\, e^{-0.15} = \mathbf{0.72084293} \approx \mathbf{0.72084} \quad \checkmark$$

   - All three values agree digit for digit with the text of the V23 manuscript (`tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, line 54).

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# 12 Step 11: Environment-Dependent Scalar-Tensor Screening Extension Hypothesis (Branch II-Cham: $V_{\text{eff}}(\phi_L;\rho)$, Chameleon/Symmetron $\beta_{\text{eff}}(\rho)$), Correlation Length $L_{\text{eff}}(\rho)$, and Multi-Scale EFT Synthesis

Source file: step11_chameleon_screening_multiscale_eft_synthesis_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step11_chameleon_screening_multiscale_eft_synthesis_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step10_bilateral_kernel_reciprocity_eft_synthesis_en.md (step10_bilateral_kernel_reciprocity_eft_synthesis.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex (Chapters 1 and 3, lines 171–180 and 256–265, Eq. 6), hu_p2.tex (Chapters 4–5, lines 24–61), and hu_p3.tex (Chapters 11–12, lines 99–120; DOI: 10.5281/zenodo.23604423)
- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Sections 1–3; DOI: 10.5281/zenodo.23639084)

## 12.1 1. Verbatim Citations and Methodological Delimitation of the Branch II-Cham Extension

In Steps 1–10, under a fixed correlation/Yukawa length $L$ and fixed coupling $\alpha_Y \in \{1, 1/3\}$ ($\alpha_{\text{std}} = \frac{\alpha_Y}{4\pi}$), we derived the equations of V23 Formulation I (3D spherical top-hat, **Branch I-A/I-B**) and Formulation II (local quadratic Helmholtz–Yukawa scalar–tensor, **Branch II-Fluid / II-Conf / II-Disf**). As shown in Step 6 (step6_sparc_galactic_ngc3198_consistency.md) via the NGC 3198 ($R = 30.08$ kpc) negative control check, the isolated V23 model with constant coupling $\alpha_{\text{std}} = 1/(4\pi) \approx 0.07958$ yields at most a squared-velocity boost of $C_{\text{pred}}^{\text{isol}} \leq 1 + \alpha_{\text{std}} = 1.0796$, compared to the observed $C_{\text{req}} = 2.720$–$4.311$.

**Methodological Delimitation (Point 5):** The density-dependent $\beta_{\text{eff}}(\rho)$ and $V(\phi_L)$ chameleon/symmetron mechanism examined in this step (**Branch II-Cham** in the taxonomy of Step 2) is a **standalone nonlinear extension hypothesis** that goes beyond the quadratic, constant-$\alpha_Y$ Yukawa action of V23 (50aa2db0), providing a candidate 4D effective field theory (EFT) framework for open task **(d)** of Chapter 11 of the V22 monograph (hu_p3.tex, line 104: *"together with a physical scale-selection mechanism for $L$"*).

1. **The V5–V16 Chameleon and Horndeski Precedents, and the Force Multiplier $C(\rho)$ (hu_p1.tex, lines 172–174 and 256–265, Eq. 6):**

   *"V5 – V7: Weak-Field and Chameleon Explorations — I investigated scalar–tensor and chameleon-like screening analogies to connect local density gradients with low-density spatial effective acceleration."*

   *"In Versions 14–16 (DOI 10.5281/zenodo.23217189), during my first phenomenological studies of galactic rotation curves, I parameterized the effective gravitational boost using the following density-dependent force multiplier:"*

$$C(\rho) \;=\; 1 + C_0 \exp\!\left(-\frac{\rho}{\rho_c}\right), \qquad V_{\text{rot}}^2(r) \;=\; C(\rho(r))\, V_{\text{bar}}^2(r), \tag{V22 Eq. 6}$$

   *"where $\rho_c$ is the critical transition density, and in the early works I set $C_0 = 5$ (so that for $\rho \ll \rho_c$, $C(\rho) \to 6$, whereas at high density $\rho \gg \rho_c \implies C(\rho) \to 1$, recovering the Newtonian limit in the Solar System)."*

2. **Open Theoretical Task (d) of Chapter 11 of the V22 Monograph (hu_p3.tex, line 104):**

   *"**(d) Self-Consistent Multi-Channel Observational Evaluation:** Once the action principle and matter coupling in (a)–(c) are fixed, galactic rotation curves (including cosmic-background embedding $\rho_{bg}$), VHE gamma-ray timing across cluster overdensities, Sirius B, and NICER posteriors must be jointly fitted **together with a physical scale-selection mechanism for** $L$."*

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## 12.2 2. Mathematical Derivation: The Environment-Dependent Effective Potential $V_{\text{eff}}(\phi_L;\rho)$ and Scale Selection $L_{\text{eff}}(\rho)$

### 12.2.1 2.1. General Self-Interacting Scalar-Tensor Action and the Effective Potential

Let us generalize the quadratic mass term ($\frac{1}{2L_0^2}\phi_L^2$) of V23 Formulation II (Eq. 8) to a general self-interaction potential $V(\phi_L)$ in the 4-dimensional scalar–tensor action:

$$S = \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R - \frac{\kappa_\phi}{2}g^{\mu\nu}\nabla_\mu\phi_L\nabla_\nu\phi_L - \kappa_\phi V(\phi_L) + f_L(\phi_L)\mathcal{L}_m\right], \qquad \kappa_\phi = \frac{c^4}{4\pi G\alpha_{\text{std}}} = \frac{c^4}{G\alpha_Y}, \quad f_L(\phi_L) = e^{-\phi_L}.$$

*(Note on normalization: $\kappa_\phi/2 = \frac{c^4}{2G\alpha_Y} = \frac{c^4}{16\pi G}\frac{1}{\tilde{\alpha}_L}$, agreeing with the V23 Eq. 8 normalization used in Steps 3 and 7!)* In a non-relativistic baryonic medium ($\mathcal{L}_m = -\rho c^2$), the Euler–Lagrange equation for the field $\phi_L$ is:

$$\Box_g\phi_L = \frac{\partial V_{\text{eff}}(\phi_L;\rho)}{\partial\phi_L}, \qquad \boxed{V_{\text{eff}}(\phi_L;\rho) \equiv V(\phi_L) + \frac{\rho c^2}{\kappa_\phi}e^{-\phi_L} = V(\phi_L) + \frac{\alpha_Y G}{c^2}\rho\, e^{-\phi_L}}$$

(where $\alpha_Y = 4\pi\alpha_{\text{std}} = 8\pi\tilde{\alpha}_L$).

### 12.2.2 2.2. Equilibrium Minimum $\phi_{\min}(\rho)$ and the Density-Dependent Yukawa Length $L_{\text{eff}}(\rho)$

In a locally quasi-homogeneous region of density $\rho$, the scalar field takes the minimum $\frac{\partial V_{\text{eff}}}{\partial\phi_L}\big|_{\phi_{\min}} = 0$, and small perturbations $\delta\phi_L = \phi_L - \phi_{\min}(\rho)$ propagate with an **environment-dependent Yukawa length** ($L_{\text{eff}}(\rho)$) determined by the local curvature:

$$\left.\frac{\partial V_{\text{eff}}}{\partial\phi_L}\right|_{\phi_{\min}} = V'\big(\phi_{\min}(\rho)\big) - \frac{\alpha_Y G}{c^2}\rho\, e^{-\phi_{\min}(\rho)} = 0,$$

$$\boxed{L_{\text{eff}}^{-2}(\rho) \equiv \mu_{\text{eff}}^2(\rho) \equiv \left.\frac{\partial^2 V_{\text{eff}}(\phi_L;\rho)}{\partial\phi_L^2}\right|_{\phi_L=\phi_{\min}(\rho)} = V''\big(\phi_{\min}(\rho)\big) + \frac{\alpha_Y G}{c^2}\rho\, e^{-\phi_{\min}(\rho)}.}$$

Let us examine the two fundamental physical cases of this formula:

1. **Quadratic (Constant Bare Mass) Limit of V23 (**Branch II-Fluid**,** $V(\phi_L) = \frac{\phi_L^2}{2L_0^2}$**):** Here $V'(\phi_L) = \phi_L/L_0^2$ and $V''(\phi_L) = L_0^{-2}$. The minimum condition can be written in terms of the Lambert $W$ function ($\phi_{\min}e^{\phi_{\min}} = \alpha_Y\gamma_{L_0}\rho$, so $\phi_{\min}(\rho) \simeq \alpha_Y\gamma_{L_0}\rho$ for small $\phi_L$), and the local effective Yukawa length **decreases with increasing density** even from the bare $e^{-\phi_L}\rho c^2$ coupling alone:

$$L_{\text{eff}}^{-2}(\rho) = L_0^{-2}\Big(1+\phi_{\min}(\rho)\Big) \simeq L_0^{-2}\Big(1+\alpha_Y\,\gamma_{L_0}\,\rho\Big) \quad\Longrightarrow\quad L_{\text{eff}}(\rho) = \frac{L_0}{\sqrt{1+\phi_{\min}(\rho)}}.$$

2. **Nonlinear Chameleon / Symmetron Extension Hypothesis (**Branch II-Cham**,** `hu_p1.tex` **V5–V7):** If, going beyond the V23 quadratic model, the bare potential is $V(\phi_L) = \Lambda_0^4\exp(\sigma\phi_L)$ or $V(\phi_L) = M_0^4\phi_L^{-n}$ ($n>0$) and the coupling function has a general nonlinear form $f_{\text{eff}}(\phi_L) = \exp[-A(\phi_L)]$ whose effective slope at the local minimum is $\beta_{\text{eff}}(\rho) = \beta_0\exp\Big(-\frac{\rho}{2\rho_c}\Big)$, then the effective gravitational multiplier ($G_{\text{eff}}/G = 1+2\beta_{\text{eff}}^2(\rho)$) reproduces the V14–V16 phenomenological force multiplier $C(\rho)$ (`hu_p1.tex`, Eq. 6):

$$\boxed{\frac{G_{\text{eff}}(\rho)}{G} = 1+2\beta_{\text{eff}}^2(\rho) = 1+2\beta_0^2\exp\left(-\frac{\rho}{\rho_c}\right) \equiv C(\rho), \qquad C_0 \equiv 2\beta_0^2 = \alpha_{\text{std},0}.}$$

   - **Galactic outer regime ($\rho \ll \rho_c \sim 10^{-23}$–$10^{-21}$ kg/m$^3$):** in this nonlinear extension hypothesis, screening is lifted and $2\beta_{\text{eff}}^2(\rho) \to C_0$ (at the NGC 3198 $R = 30.08$ kpc point, $C_{\text{req}} - 1 = 1.720$ [$\Upsilon = 1.0$] and 3.311 [$\Upsilon = 0.5$]).
   - **High-density Solar System and terrestrial regime ($\rho \gg \rho_c$):** the exponential factor suppresses the long-range scalar force multiplier ($C(\rho) - 1 = C_0 e^{-\rho/\rho_c} \to 0$), leaving only the short-scale ($L$) canonical $\gamma_L = GL^2/c^2$ ($\alpha_{\text{std}} = 1/(4\pi)$) V23 mode.

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## 12.3 3. Multi-Scale EFT Overview Across the V17–V23 Investigated Regimes (with Branch Labels)

The table below specifies for every physical system examined which model branch (**Branch I-A**, **Branch II-Fluid**, **Branch II-Disf**, **Branch II-Cham**) governs the corresponding result:

| Physical Scale and System | Model Branch (step2 Taxonomy) | Characteristic Density $\rho$ [kg/m$^3$] | Investigated $L$ Scale | $\phi_L = \gamma_L \bar{\rho}_L$ (or $\phi_L^{\text{Yuk}}$) | Primary Observable and V17–V23 Audit Outcome |
|---|---|---|---|---|---|
| **V20 8-coil magnetic field** (step8, step20) | **Branch II-Disf / I-A** ($H_1$) | $\rho_B = \frac{B^2}{2\mu_0 c^2} \sim 10^{-22}$ | $L_{\text{peak}} = 0.630957$ m | $1.2017108 \times 10^{-48}$ | Polarity-even $\Delta y_{\text{peak}} = -1.2017108 \times 10^{-48}$; Zeeman + cross-term canceled. |
| **V20 8558 kg granite chamber** (step8, step19) | **Branch I-A / II-Fluid** ($H_0$ vs. $H_1$) | $2.75 \times 10^3$ | $L =$ 0.50–1.00 m | $1.436 \times 10^{-24}$ (1 m) | $H_0$: $+1.331 \times 10^{-47}$; conditional $H_1$: $-1.436 \times 10^{-24}$ (top-hat), $-3.781 \times 10^{-25}$ (Yukawa). |
| **Sirius B white dwarf** (step8, step19) | **Branch I-A / II-Fluid** ($H_0$ vs. $H_1$) | $\bar{\rho} \sim 3 \times 10^9$ | $L =$ 1–5000 km | $1.6 \times 10^{-5}$ (5000 km) | $H_0$: $ |
| **Neutron stars** (step7, step18, step22) | **Branch I-A** vs. **Branch II-Fluid** | $10^{17}$–$1.45 \times 10^{18}$ | $L =$ 1–10 km | 0.016–0.050 (10 km) | Curved exterior `yukawa_action_el` $M_{\text{ADM}}$ and coupled $(\delta h, \delta\phi_L)$ stability (step22); V21 nonlocal tangent BVP (step18). |
| **Solar System (PPN / Cassini)** (step5) | **Branch II-Conf** vs. **II-Disf / I-A** | $\bar{\rho}_L^{\odot} \propto L^{-3}$ | $L =$ 1 AU–1 pc | $1.14 \times 10^{-14}$ (1 pc) | 3D top-hat dilution: $ |
| **SPARC (175/155) and NGC 3198** (step6) | **Branch I-A / II-Fluid** (negative control) vs. **II-Cham** | $10^{-24}$–$10^{-20}$ | $L \sim$ 1–23.09 kpc | $\leq 2.300 \times 10^{-7}$ (isolated) | 0 points with $\rho \leq 0$ (kernel positivity); NGC 3198 (30.08 kpc): $C_{\text{req}} = 2.720/4.311$, whereas constant $\alpha_{\text{std}}$: $C_{\text{pred}}^{\text{isol}} \leq 1.0796$. |
| **V17 H.E.S.S./ MAGIC gamma timing** (step9) | **Branch II-Disf** ($H_1$ EM) | $\rho_{\text{IGM/ISM}} \sim 10^{-27}$–$10^{-21}$ | $z =$ 0.034–0.4245 | $\int_0^D \phi_L(s)\, ds$ | Naive $z$-proportional LIV excluded; prediction correlates with cluster overdensity ($\int \bar{\rho}_L\, ds$). |

## 12.4 4. Checks

1. **Limiting Values of the Correspondence $C(\rho) = 1 + 2\beta_{\text{eff}}^2(\rho)$ (Branch II-Cham):**
   - $\rho \to 0$ (deep galactic outer regime): $2\beta_{\text{eff}}^2(0) = 2\beta_0^2 = C_0 \implies C(0) = 1 + C_0$ ✓.
   - $\rho \gg \rho_c$ (stellar interior, Solar System, laboratory): $2\beta_{\text{eff}}^2(\rho) = C_0 e^{-\rho/\rho_c} \to 0 \implies C(\rho) \to 1$ ✓.
2. **Verification of the Quadratic Yukawa Limit $L_{\text{eff}}(\rho) = L_0/\sqrt{1 + \phi_{\min}(\rho)}$ (Branch II-Fluid):**
   - At low density ($\phi_{\min}(\rho) \to 0$), $L_{\text{eff}}(\rho) \to L_0$, recovering the constant Yukawa length $L_0$ of V23 Formulation II ✓.

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# 13 Step 12: Second Variation ($\delta^2 S$) of the Scalar-Tensor / Yukawa Action, Explicit Coefficients of the Coupled $(\xi, \delta\phi_L)$ Radial Pulsation Operator, and Derivation of the Positive-Definite Physical Kinetic Hessian

Source file: step12_second_variation_coupled_pulsation_hessian_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step12_second_variation_coupled_pulsation_hessian_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step11_chameleon_screening_multiscale_eft_synthesis_en.md (step11_chameleon_screening_multiscale_eft_synthesis.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p3.tex (Chapter 8, lines 13–23, Eq. 12, and Chapter 11 task **(b)**, line 102; DOI: 10.5281/zenodo.23604423)
- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Section 1.2, line 65 and Section 2, lines 120–153, Eqs. 8–9, Table 2; DOI: 10.5281/zenodo.23639084)
- V21 Zenodo record: DOI 10.5281/zenodo.23602501
- Numerical implementation and verification on the yukawa_action_el sequence: curved_yukawa_adm_and_stability.py, curved_yukawa_coupled_stability.csv, and step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability.md

## 13.1 1. Verbatim Citations from the V22 Monograph and the V23 (50aa2db0) Manuscript

In Step 7 (step7_neutron_star_tov_stability.md), we derived the curved-spacetime yukawa_action_el TOV background equations as well as the **static ($\omega_0^2 = 0$) nonlocal tangent model** ($\delta f_L[\delta h]$) around the V21 background. Both open task **(b)** of Chapter 11 of the V22 monograph (hu_p3.tex, line 102) and Section 2 of the V23 manuscript (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, lines 134–136) explicitly distinguish the static tangent model (and its Gram matrix) from the **genuine coupled dynamical pulsation system and physical kinetic Hessian derived from the second variation $\delta^2 S$ of the action**:

1. **Theoretical Task (b) of Chapter 11 of the V22 Monograph (hu_p3.tex, line 102):**

   *"**(b) Genuine Modified Radial Stability Analysis Derived from the Coupling:** Once task (a) provides the dynamical field equations of the perturbations $\delta\phi$ and $\delta T_X^{\mu\nu}$, the Chandrasekhar radial pulsation operator must be re-derived variationally from the second variation $\delta^2 S$ of the action, eliminating the non-variational metric operator pathology flagged above $L \geq 367\,m$ and lifting the frozen-$f$ ($\delta f_L = 0$) approximation."*

2. **Non-Variational Character of the V21 gradient/metric Operator Above $L \geq 367$ m (hu_p3.tex, Chapter 8, line 21):**

   *"**gradient/metric operator (flagged as non-variational above $L \geq 367\,m$):** Substitutes the metric lapse derivative $\nu' = 2\Phi'_{eff}$ into the standard GR curvature terms of $\mathcal{Q}(r)$ without accounting for the perturbation of the $X$-sector tangential stress $\delta p_{tX}$. My audit openly flags this operator as **non-positive-definite and non-variational above $L \geq 367\,m$ across 12 sequences (609 models)**, where uncompensated residual terms produce unphysical eigenvalues $\omega_0^2 < 0$ even along the rising branch $\partial M/\partial\rho_c > 0$."*

3. **Inertia Invariance of the V23 Static Tangent Model and the Difference Between the Gram Matrix and the Physical Kinetic Hessian (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, Section 2, lines 134–136):**

   *"Because at $\omega_0^2 = 0$ the problem reduces to $\mathcal{H}_{lin}[\delta h] = 0$, this agreement verifies static nonlocal linearization rather than dynamical inertia: scaling $W \to 4W$ at $\log_{10}\rho_{0c} = 15.30427$ yields a ratio $\omega_{4W}^2/\omega_W^2 = 0.2500$ ($\omega_W^2 = 4.897 \times 10^6\ s^{-2}$) with the identical shooting function $F(0) = 0.023252$, so the zero crossing is inertia-invariant. The raw BVP $W$ matrix is not symmetric; the symmetrized Galerkin form $\mathbf{M}_{sym} = \mathbf{D}^T \mathrm{diag}(w\,\Delta r)\mathbf{D}$ is symmetric with $\min \mathrm{diag}\,\mathbf{M}_{sym} = 5.68 \times 10^{-9} > 0$; this is a Gram-matrix property, not a proof of positive-definiteness of the physical kinetic Hessian. The omega0_sq_sym_tangent_s2 column gives the first-order tangent estimate ($\omega^2 \simeq -F(0)/F'(0)$)."*

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## 13.2 2. Mathematical Derivation: The Second Variation $\delta^2 S$, Elimination of Metric Constraints, and Explicit Coefficients

### 13.2.1 2.1. Time-Dependent Spherically Symmetric Perturbations and the Metric Constraints $(\delta\lambda, \delta m, \delta\nu')$

Consider the covariant action of V23 Formulation II (`yukawa_action_el`, **Branch II-Fluid**, $G = c = 1$) in a time-dependent, spherically symmetric spacetime ($ds^2 = -e^{2\nu(r,t)}dt^2 + e^{2\lambda(r,t)}dr^2 + r^2 d\Omega^2$, where on the static background $B_0(r) = e^{-2\lambda_0(r)} = 1 - 2m_0(r)/r$):

$$S[g_{\mu\nu}, \phi_L, \Psi_{\rm fluid}] \;=\; \int dt \int_0^\infty dr\, 4\pi r^2 e^{\nu+\lambda} \left[ \frac{R}{16\pi} - \frac{1}{2\alpha_Y} \left( -e^{-2\nu}(\dot{\phi}_L)^2 + e^{-2\lambda}(\phi_L')^2 + \frac{\phi_L^2}{L^2} \right) - e^{-\phi_L}\varepsilon(n) \right].$$

Let us introduce small radial Lagrangian fluid displacements $\xi(r,t)$ (for which the enthalpy perturbation is $\delta h(r,t) = \frac{\delta P(r,t)}{\varepsilon_0(r)+P_0(r)}$) and the Eulerian scalar-field perturbation $\delta\phi_L(r,t)$ around the static equilibrium background $(\nu_0(r), \lambda_0(r), \phi_{L,0}(r), \varepsilon_0(r), P_0(r))$:

$$r_{\rm fluid}(r,t) \;=\; r + \xi(r,t), \qquad \phi_L(r,t) \;=\; \phi_{L,0}(r) + \delta\phi_L(r,t).$$

1. **Metric Constraints from the $(tr)$ and $(tt)$ Einstein Equations ($\delta\lambda$ and $\delta m$):** Integrating the spherically symmetric $(tr)$ Einstein equation ($\delta G_t^{\,r} = 8\pi\, \delta T_t^{\,r}$) with respect to time in the interior ($0 < r < R_*$) gives:

$$\frac{\partial(\delta\lambda)}{\partial t} \;=\; -4\pi r e^{2\lambda_0} \left[ f_{L,0}(\varepsilon_0 + P_0)\, \dot{\xi} - \frac{1}{\alpha_Y} \phi_{L,0}'\, \delta\dot{\phi}_L \right],$$

whereas from the $(tt)$ Einstein equation ($\delta m'(r,t) = 4\pi r^2\, \delta(-T_t^{\,t})$, where $\delta B = -2\delta m/r = -2B_0\, \delta\lambda$) at fixed baryon number ($\delta N_B = 0$), the radial derivative of the mass perturbation in variables $(\delta h, \delta\phi_L, \delta\phi_L', \delta m)$ is:

$$\boxed{\frac{\partial(\delta m)}{\partial r} \;=\; 4\pi r^2 \left[ f_{L,0} \frac{\varepsilon_0 + P_0}{c_s^2}\, \delta h - f_{L,0}\varepsilon_0\, \delta\phi_L + \frac{1}{\alpha_Y} \left( B_0 \phi_{L,0}'\, \delta\phi_L' + \frac{\phi_{L,0}}{L^2}\delta\phi_L - \frac{(\phi_{L,0}')^2}{r}\delta m \right) \right].}$$

Equivalently, expressed in terms of the Lagrangian displacement $\xi(r)$ (by substituting the baryon-number conservation constraint $\delta n = -\frac{1}{r^2 e^{\lambda_0}}(r^2 e^{\lambda_0} n_0 \xi)' - n_0 \delta\lambda$ and integrating by parts), the mass perturbation is:

$$\delta m(r) \;=\; -4\pi r^2 f_{L,0}(\varepsilon_0 + P_0)\, \xi(r) \;+\; \int_0^r 4\pi\tilde{r}^2 \left[ \frac{B_0 \phi_{L,0}'}{\alpha_Y}\delta\phi_L' + \left( \frac{\phi_{L,0}}{\alpha_Y L^2} - f_{L,0}\varepsilon_0 \right) \delta\phi_L + f_{L,0}(\varepsilon_0 + P_0)\phi_{L,0}'\, \xi - \frac{(\phi_{L,0}')^2}{\alpha_Y \tilde{r}}\delta m \right] d\tilde{r}.$$

2. **Lapse-Gradient Constraint from the $(rr)$ Einstein Equation ($\delta\nu'$):** Because $\nu_0'(r) = \frac{m_0 + 4\pi r^3 P_{\rm tot,0}}{r(r-2m_0)}$, where $P_{\rm tot,0} = f_{L,0}P_0 + p_{r,\phi,0}$ and $p_{r,\phi,0} = \frac{1}{2\alpha_Y}[B_0(\phi_{L,0}')^2 - \phi_{L,0}^2/L^2]$, the total pressure perturbation is:

$$\delta P_{\rm tot} \;=\; f_{L,0}(\varepsilon_0 + P_0)\, \delta h - f_{L,0}P_0\, \delta\phi_L + \frac{1}{\alpha_Y} \left( B_0 \phi_{L,0}'\, \delta\phi_L' - \frac{\phi_{L,0}}{L^2}\delta\phi_L - \frac{(\phi_{L,0}')^2}{r}\delta m \right),$$

from which the perturbation of the lapse gradient follows:

$$\boxed{\delta\nu'(r) \;=\; \frac{1 + 2r\nu_0'(r)}{r\big(r - 2m_0(r)\big)}\, \delta m(r) \;+\; \frac{4\pi r^3}{r\big(r - 2m_0(r)\big)}\, \delta P_{\rm tot}(r).}$$

### 13.2.2 2.2. Second-Order Action $\delta^2 S$ and the Positive-Definite Physical Kinetic Hessian ($\mathbf{T_{kin}}$)

After substituting the metric constraints back into the quadratic second-order action ($\delta^2 S = \int dt\, (T_{\rm kin}[\dot{\xi}, \delta\dot{\phi}_L] - U_{\rm pot}[\xi, \delta\phi_L])$), the kinetic part under harmonic time dependence ($\xi(r,t) = \xi(r)e^{i\omega t}$, $\delta\phi_L(r,t) = \delta\phi_L(r)e^{i\omega t}$) is:

$$\boxed{T_{\rm kin}\big[\xi, \delta\phi_L\big] \;=\; \frac{\omega^2}{2} \int_0^{R_*} dr\, \mathcal{W}_\xi(r)\, \xi(r)^2 \;+\; \frac{\omega^2}{2} \int_0^\infty dr\, \mathcal{W}_\phi(r) \left[\delta\phi_L(r)\right]^2,}$$

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where the two physical inertia (kinetic) weight functions are:

$$\boxed{\mathcal{W}_\xi(r) \;=\; 4\pi r^2\, e^{3\lambda_0(r)-\nu_0(r)}\, f_{L,0}(r)\,\big(\varepsilon_0(r)+P_0(r)\big) \;>\; 0 \quad (0<r<R_*), \qquad \mathcal{W}_\phi(r) \;=\; \frac{4\pi r^2}{\alpha_Y}\, e^{\lambda_0(r)-\nu_0(r)} \;>\; 0 \quad (r>0).}$$

### 13.2.3 2.3. Explicit Coefficients of the Coupled $(\xi, \delta\phi_L)$ and $(\delta h, \delta\phi_L, \delta m)$ Eigenvalue System (Point 3)

The coupled radial pulsation system can be written in two strictly equivalent forms — the second-order Lagrangian $(\xi, \delta\phi_L)$ form and the first-/second-order canonical $(\delta h, \delta\phi_L, \delta m)$ form:

**(A) Explicit Second-Order Coupled $(\xi, \delta\phi_L)$ Sturm–Liouville / Klein–Gordon Form:**

$$\begin{cases} -\dfrac{d}{dr}\left(\mathcal{P}_\xi(r)\dfrac{d\xi}{dr}\right) + \mathcal{Q}_{\xi\xi}(r)\,\xi(r) + \mathcal{C}_{\xi\phi}[\delta\phi_L]\,(r) \;=\; \omega^2\,\mathcal{W}_\xi(r)\,\xi(r), \\ -\dfrac{d}{dr}\left(\mathcal{P}_\phi(r)\dfrac{d(\delta\phi_L)}{dr}\right) + \mathcal{Q}_{\phi\phi}(r)\,\delta\phi_L(r) + \mathcal{C}_{\phi\xi}[\xi]\,(r) \;=\; \omega^2\,\mathcal{W}_\phi(r)\,\delta\phi_L(r), \end{cases}$$

where the principal-part coefficients are:

$$\mathcal{P}_\xi(r) \;=\; 4\pi r^2\, e^{\nu_0(r)+\lambda_0(r)}\, f_{L,0}(r)\,\Gamma_1(r)\,P_0(r) \;>\; 0, \qquad \mathcal{P}_\phi(r) \;=\; \frac{4\pi r^2}{\alpha_Y}\, e^{\nu_0(r)-\lambda_0(r)} \;>\; 0,$$

the fluid–fluid potential coefficient ($\mathcal{Q}_{\xi\xi}$, with the modified hydrostatic gradient $P_0' = -(\varepsilon_0+P_0)(\nu_0' - \phi_{L,0}')$ and the combined background pressure/energy $\rho_{\mathrm{tot},0} = f_{L,0}\varepsilon_0 + \rho_{\phi,0}$, $P_{\mathrm{tot},0} = f_{L,0}P_0 + p_{r,\phi,0}$) is:

$$\boxed{\begin{aligned} \mathcal{Q}_{\xi\xi}(r) \;=\; 4\pi r^2 e^{\nu_0+\lambda_0} f_{L,0}\Bigg\{ &\frac{4}{r}P_0' + \frac{(P_0')^2}{\varepsilon_0+P_0} - 8\pi e^{2\lambda_0}(\varepsilon_0+P_0)\big(f_{L,0}P_0 + p_{r,\phi,0}\big) \\ &- \frac{d}{dr}\left(\frac{2\Gamma_1 P_0}{r}\right) + (\varepsilon_0+P_0)\left[\phi_{L,0}'' + \big(\nu_0' + \lambda_0' - \phi_{L,0}'\big)\phi_{L,0}'\right]\Bigg\}, \end{aligned}}$$

the scalar–scalar potential coefficient ($\mathcal{Q}_{\phi\phi}$, including the metric back-reaction from $\delta\nu'$ and $\delta B$) is:

$$\boxed{\mathcal{Q}_{\phi\phi}(r) \;=\; \frac{4\pi r^2}{\alpha_Y}\, e^{\nu_0+\lambda_0}\left[\frac{1}{L^2} + \alpha_Y\, f_{L,0}(r)\,\varepsilon_0(r) + \frac{8\pi r\, e^{2\lambda_0}}{\alpha_Y L^2}\phi_{L,0}(r)\phi_{L,0}'(r)\right] \;+\; \mathcal{Q}_{\phi\phi}^{\mathrm{metric}}[\delta m],}$$

and the mutually adjoint fluid–scalar cross-coupling operators ($\langle \eta, \mathcal{C}_{\xi\phi}[\delta\phi_L]\rangle = \langle \mathcal{C}_{\phi\xi}[\eta], \delta\phi_L\rangle$) are:

$$\boxed{\begin{aligned} \mathcal{C}_{\xi\phi}[\delta\phi_L](r) \;&=\; 4\pi r^2 e^{\nu_0+\lambda_0} f_{L,0}(\varepsilon_0+P_0)\left[-\delta\phi_L'(r) + \delta\nu'\big|_{\delta\phi_L,\delta m[\delta\phi_L]}(r)\right], \\ \mathcal{C}_{\phi\xi}[\xi](r) \;&=\; -\frac{d}{dr}\Big(4\pi r^2 e^{\nu_0+\lambda_0} f_{L,0}(\varepsilon_0+P_0)\,\xi(r)\Big) \;+\; 4\pi r^2 e^{\nu_0+\lambda_0} f_{L,0}\frac{\varepsilon_0+P_0}{c_s^2}\,\delta h[\xi](r) \;+\; \mathcal{C}_{\phi\xi}^{\mathrm{metric}}[\delta m[\xi]](r). \end{aligned}}$$

**(B) Exact Canonical $(\delta h, \delta\phi_L, \delta m)$ System and Moving-Surface Matching Conditions (Implemented in `curved_yukawa_adm_and_stability.py`):** For numerical integration and exact handling of polytrope transitions ($r_{\mathrm{trans}}$), the canonical variables $(\delta h, \delta\phi_L, \delta m)$ are optimal (since $\delta h$ and $\delta m$ remain continuous even across jump points of $c_s^2$!):

$$\boxed{\begin{aligned} \frac{d(\delta h)}{dr} \;&=\; -\delta\nu'(r) \;+\; \delta\phi_L'(r) \;+\; \omega^2\, e^{2(\lambda_0-\nu_0)}\,\xi(r), \\ \frac{d(\delta m)}{dr} \;&=\; 4\pi r^2\left[f_{L,0}\frac{\varepsilon_0+P_0}{c_s^2}\,\delta h - f_{L,0}\varepsilon_0\,\delta\phi_L + \frac{1}{\alpha_Y}\left(B_0\phi_{L,0}'\,\delta\phi_L' + \frac{\phi_{L,0}}{L^2}\delta\phi_L - \frac{(\phi_{L,0}')^2}{r}\delta m\right)\right], \\ B_0\,\delta\phi_L'' \;+\; &\left(\frac{2B_0}{r} + B_0\nu_0' + \frac{B_0'}{2}\right)\delta\phi_L' - \left(\frac{1}{L^2} + \alpha_Y f_{L,0}\varepsilon_0 - \omega^2 e^{-2\nu_0}\right)\delta\phi_L \\ &=\; -\alpha_Y f_{L,0}\frac{\varepsilon_0+P_0}{c_s^2}\,\delta h + \frac{2\,\delta m}{r}\phi_{L,0}'' - \left[\delta(B\nu') + \frac{\delta B'}{2} - \frac{2\,\delta m}{r^2}\right]\phi_{L,0}'. \end{aligned}}$$

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- **Moving-Surface Boundary and Matching Conditions ($r = R_*$):**
  1. At the free surface, the Lagrangian pressure perturbation vanishes: $\Delta P(R_*) = \delta P(R_*) + P_0'(R_*)\xi(R_*) = 0 \implies \delta h(R_*) = -h_0'(R_*)\xi(R_*)$, which together with integrated mass conservation $(\delta m(R_*^-) = -4\pi R_*^2 f_{L,0}(R_*)(\varepsilon_0 + P_0)\big|_{R_*^-}\xi(R_*) = 0$, since for a polytropic crust $\varepsilon_0(R_*^-) = P_0(R_*^-) = 0$) guarantees $\delta m(R_*) \in C^0$.
  2. Across the moving surface $r = R_* + \xi(R_*)$, the scalar field and its normal derivative are continuous:

$$[\![\delta\phi_L + \xi\phi_{L,0}']\!]_{R_*} = 0, \qquad [\![\delta\phi_L' + \xi\phi_{L,0}'']\!]_{R_*} = 0.$$

Since $\phi_{L,0} \in C^1([0,\infty))$, $[\![\phi_{L,0}']\!]_{R_*} = 0 \implies [\![\delta\phi_L]\!]_{R_*} = 0$; furthermore, $[\![\phi_{L,0}'']\!]_{R_*} = \frac{\alpha_Y}{B_0(R_*)} f_{L,0}(R_*)\varepsilon_0(R_*^-) = 0$ at the surface of the polytropic crust ($\varepsilon_0(R_*^-) = 0$), so $[\![\delta\phi_L']\!]_{R_*} = 0$ holds as well!

  3. In the curved exterior vacuum ($r \in [R_*, R_{\max}]$), $\varepsilon_0 = P_0 = 0$, so the $(\delta\phi_L, \delta m)$ system is integrated in coupled form out to $R_{\max} = R_* + 12L$, where the asymptotic Yukawa decay condition closes the boundary-value problem, and the asymptotic mass perturbation $\delta M_{\rm ADM}(\omega^2) = \delta m(R_{\max}; \omega^2) = 0$ determines the eigenvalue!

---

## 13.3 3. Numerical Verification on the Curved Exterior `yukawa_action_el` Sequence (Point 4)

The script `curved_yukawa_adm_and_stability.py` solves the coupled $(\delta h, \delta\phi_L, \delta m)$ eigenvalue problem ($r \in [0, R_* + 14L]$, with the exact left-hand side sign `scalar_m_sign = -1.0`) directly on the `yukawa_action_el` equilibrium sequences equipped with the curved exterior solution. The results (`curved_yukawa_coupled_stability.csv` and `curved_yukawa_audit_summary.json`) are:

| Sequence | $\log_{10}\rho_{0c}(dM_{\rm ADM}=0)$ | $\log_{10}\rho_{0c}(dM_B=0)$ | Dynamical $\omega_0^2=0$ | $\Delta_{\rm ADM}$ Dynamical (dex) | $\Delta_{M_B}$ Dynamical (dex) | Quasi-Static $\omega_0^2=0$ | Frozen $\delta\phi_L=0$ | $\Delta_{\rm ADM}$ Frozen (dex) |
|---|---|---|---|---|---|---|---|---|
| **SLy,** $L = 10$ **km,** $\alpha_Y = 1$ | 15.316404 | 15.316409 | 15.316410 | +5.80e-06 | +1.33e-06 | 15.316410 | 15.328565 | +1.22e-02 |
| **SLy,** $L = 10$ **km,** $\alpha_Y = 1/3$ | 15.307166 | 15.307169 | 15.307170 | +4.20e-06 | +6.41e-07 | 15.307170 | 15.311340 | +4.17e-03 |
| **SLy,** $L = 1$ **km,** $\alpha_Y = 1$ | 15.303086 | 15.303082 | 15.303080 | -6.86e-06 | -2.61e-06 | 15.303080 | 15.304220 | +1.13e-03 |
| **APR4,** $L = 10$ **km,** $\alpha_Y = 1$ | 15.293022 | 15.293025 | 15.293025 | +3.57e-06 | -1.40e-07 | 15.293025 | 15.301564 | +8.54e-03 |
| **H4,** $L = 10$ **km,** $\alpha_Y = 1$ | 15.227785 | 15.227787 | 15.227783 | -1.34e-06 | -4.05e-06 | 15.227783 | 15.251384 | +2.36e-02 |

*(GL2 Volterra, $N = 480$, `scalar_m_sign = -1.0`, dynamical scalar wave, finite-difference verified exterior $\partial\phi_L'(R_*)/\partial m(R_*) = -1.68365 \times 10^{-4}$ $km^{-2}$ and interior $r = 6$ km $\partial\phi_L''/\partial m = -1.72833 \times 10^{-4}$ $km^{-3}$ derivatives; $N$- and $R_{\max}$-convergence on the root: $p_{eff,root} = 4.248$, $\log_{10}\rho_{0c,Rich}(\omega_0^2 = 0) = 15.31640981$, see `step22`, Sections 4 and 7.)*

---

## 13.4 4. Checks

1. $f_{L,0} \to 1$, $\alpha_Y \to 0$ **(GR Limit):**
   - For $\phi_{L,0} \to 0$ and $\delta\phi_L \to 0$, $\mathcal{W}_\xi(r) \to 4\pi r^2 e^{3\lambda_0 - \nu_0}(\varepsilon_0 + P_0)$, $\mathcal{P}_\xi(r) \to 4\pi r^2 e^{\nu_0 + \lambda_0}\Gamma_1 P_0$, and $\mathcal{Q}_{\xi\xi}(r)$ recovers the classical Chandrasekhar (1964) general-relativistic radial pulsation Sturm–Liouville operator ✓.
2. **Comparison Between the Coupled $(\delta h, \delta\phi_L, \delta m)$ Eigenvalue Zero Crossing and the ADM/Baryonic Mass Peaks on the `yukawa_action_el` Branch:**
   - For SLy $L = 10$ km, $\alpha_Y = 1$, the frozen-$\phi_L$ ($\delta\phi_L = 0$) approximation shifts the zero crossing by $+1.22 \times 10^{-2}$ dex from the $dM_{\rm ADM}/d\rho_c = 0$ turning point (15.328565 vs. 15.316404), whereas the coupled $(\delta h, \delta\phi_L, \delta m)$ operator (`scalar_m_sign = -1.0`) yields $\log_{10}\rho_{0c}(\omega_0^2 = 0) = 15.316410$ ($+5.80 \times 10^{-6}$ dex above the ADM mass peak and $+1.33 \times 10^{-6}$ dex above the baryonic mass peak, a 2097× reduction relative to the frozen-$\phi_L$ shift; across all 5 sequences the agreement is within $\leq 6.9 \times 10^{-6}$ dex, a 165 × –17593× reduction) ✓.
3. **Positivity of the Kinetic Weights Above $L \geq 367$ m:**
   - Because $f_{L,0}(r) = e^{-\phi_{L,0}(r)} > 0$ and $\alpha_Y > 0$, $\mathcal{W}_\xi(r) > 0$ and $\mathcal{W}_\phi(r) > 0$ hold across every scale $L \in (0,\infty)$ without sign reversal ✓.

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# 14 Step 13: Metric Variation of the Curved-Spacetime Bilateral Top-Hat Window and Volume Element ($\delta[\sqrt{h}\,K_L(x,y)]/\delta g^{\mu\nu}$), Synge's World Function, and the $\delta_D(L-d_\Sigma)$ Boundary Surface Stress

Source file: step13_curved_window_metric_variation_synge_world_function_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step13_curved_window_metric_variation_synge_world_function_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step12_second_variation_coupled_pulsation_hessian_en.md (step12_second_variation_coupled_pulsation_hessian.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Abstract, line 28 and Section 1.1, lines 38–55, Eqs. 2–3; DOI: 10.5281/zenodo.23639084)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p2.tex (Chapters 4 and 7) and hu_p3.tex (Chapters 8 and 11; DOI: 10.5281/zenodo.23604423)

## 14.1 1. Verbatim Citations from the V23 (50aa2db0) Manuscript

In Step 10 (step10_bilateral_kernel_reciprocity_eft_synthesis.md), we derived the unweighted (bilateral_eom) and lapse-weighted (bilateral_lapse_weighted) reciprocal equilibrium of V23 Formulation I **at fixed kernel and metric**. However, the abstract (line 28) and Section 1.1 (lines 38–54) of the V23 manuscript explicitly state that one open mathematical step remains for the covariant Euler–Lagrange closure of Formulation I (Eq. 2): **the variation of the curved-space window function $K_L(x,y)$ and the volume element with respect to the metric $g_{\mu\nu}$**:

1. **Covariant Action of V23 Formulation I and the $\Sigma_u$ Hypersurface Kernel (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, Section 1.1, lines 38–46, Eqs. 2–3):**

   *"In Formulation I, a Lagrange multiplier field $\lambda_L(x)$ couples the auxiliary scalar $\phi_L(x)$ to the spatial top-hat average taken on the hypersurface $\Sigma_u$ orthogonal to the fluid four-velocity $u^\mu$ ($h_{\mu\nu}=g_{\mu\nu}+u_\mu u_\nu/c^2$):"*

   $$S_{\mathrm{I},X}[\phi_L,\lambda_L,g_{\mu\nu}]=-\frac{c^4}{8\pi G\gamma_L}\int d^4x\sqrt{-g}\,\lambda_L(x)\left[\phi_L(x)-\gamma_L\int_{\Sigma_u}d^4y\sqrt{-g(y)}\,K_L(x,y)\,\rho_{\mathrm{src}}(y)\right], \qquad \text{(V23 Eq. 2)}$$

   *"where $K_L(x,y)=\delta(\tau_u(x)-\tau_u(y))\,\Theta(L-d_\Sigma(x,y))/V_L(x)$ is the positive-definite 3D spherical kernel of radius $L$ and proper volume $V_L=\frac{4}{3}\pi L^3$."*

2. **Curved-Space Window and Volume-Element Variation Flagged as Open in the V23 Manuscript (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, lines 28 and 54):**
   - **Abstract (line 28):** *"[T]his is not a dynamical proof of **curved-space window variation**, which remains an open question."*
   - **Section 1.1 (line 54):** *"Varying the lapse-weighted integral $\int dV\,N f_L\varepsilon$ with $N=e^\nu$ (at fixed kernel and metric) gives $V_{\mathrm{eff},N}(x)=f_L-\frac{\gamma_L}{N(x)}\int dV_y\,N(y)K(y,x)f_L\varepsilon$, which I solve in the lapse equation with $P_{matter}=f_L P$ (**bilateral_lapse_weighted** rows; two-cell test $N=(1;0.8)$, $\varepsilon=(1;2)$, $\gamma=0.1$: $(0.74882;0.72084)$, unweighted $0.73160$). **Covariant closure still requires metric variation of the kernel and volume element; neither row is the EL solution of action (2).**"*

## 14.2 2. Mathematical Derivation: Metric Variation of the Curved-Space Volume Element and the Geodesic Window $d_\Sigma(x,y)$

### 14.2.1 2.1. Variation of the Hypersurface Integration Measure on $\Sigma_u$ and the Proper Ball Volume $V_L(x)$

In a static spacetime ($ds^2=-N(x)^2c^2dt^2+h_{ij}(x)dx^idx^j$, where $N(x)=e^{\nu(x)}$ is the lapse function and $h_{ij}(x)$ is the induced 3-metric on the spatial hypersurface $\Sigma_u$, with $\sqrt{-g(x)}=N(x)\sqrt{h(x)}$), the kernel integral restricted to the

simultaneous hypersurface $\tau_u(x) = \tau_u(y)$ is:

$$\bar{\rho}_L(x) \;=\; \int_{\Sigma_u} d^3y\,\sqrt{h(y)}\,\mathcal{K}_L(x,y)\,\varepsilon(y), \qquad \mathcal{K}_L(x,y) \;\equiv\; \frac{\Theta\big(L - d_\Sigma(x,y)\big)}{V_L(x)},$$

where $d_\Sigma(x,y)$ is the minimal spatial geodesic arc length connecting points $x$ and $y$ on $\Sigma_u$, and the proper volume of the geodesic ball $B_\Sigma(x,L) = \{y \in \Sigma_u : d_\Sigma(x,y) \le L\}$ is:

$$V_L(x) \;=\; \int_{\Sigma_u} d^3y\,\sqrt{h(y)}\,\Theta\big(L - d_\Sigma(x,y)\big).$$

By V23 Eq. (3), the Lagrange multiplier evaluated on the solution ($\phi_L = \gamma_L\bar{\rho}_L$, $\mathcal{L}_{\rm eff} = -f_L(\phi_L)\varepsilon$) is:

$$\lambda_L(x) \;=\; \frac{8\pi G\gamma_L}{c^4}\,\frac{\partial\big(-e^{-\phi_L(x)}\varepsilon(x)\big)}{\partial\phi_L(x)} \;=\; \frac{8\pi G\gamma_L}{c^4}\,f_L(x)\,\varepsilon(x).$$

When the action $S_{\rm I} = \int dt \int d^3x\, N(x)\sqrt{h(x)}\,\big[-f_L(x)\varepsilon(x)\big]$ is varied with respect to the spatial metric $h_{ij}(z)$ (at fixed $N(x)$ and $\varepsilon(x)$), three distinct metric contributions appear in the variation $\delta\phi_L(x) = \gamma_L\,\delta\bar{\rho}_L(x)$:

$$\delta\bar{\rho}_L(x) \;=\; \int_{\Sigma_u} d^3y\,\delta\Big[\sqrt{h(y)}\,\mathcal{K}_L(x,y)\Big]\varepsilon(y) \;=\; \delta\bar{\rho}_L^{(\rm vol)}(x) \;+\; \delta\bar{\rho}_L^{(\rm norm)}(x) \;+\; \delta\bar{\rho}_L^{(\rm bdry)}(x).$$

1. **Direct Variation of the Volume Element $\sqrt{h(y)}$ ($\delta\bar{\rho}_L^{(\mathbf{vol})}$):** Since $\delta\sqrt{h(y)} = \frac{1}{2}\sqrt{h(y)}\,h^{ij}(y)\,\delta h_{ij}(y)$:

$$\delta\bar{\rho}_L^{(\rm vol)}(x) \;=\; \frac{1}{2}\int_{\Sigma_u} d^3y\,\sqrt{h(y)}\,\mathcal{K}_L(x,y)\,\varepsilon(y)\,h^{ij}(y)\,\delta h_{ij}(y).$$

2. **Variation of the Normalization Volume $V_L(x)$ ($\delta\bar{\rho}_L^{(\mathbf{norm})}$):** Since $\delta\big(V_L(x)^{-1}\big) = -V_L(x)^{-2}\,\delta V_L(x)$, the volume-element variation from the normalization contributes a subtraction weighted by the local mean $\bar{\rho}_L(x)$:

$$\delta\bar{\rho}_L^{(\rm norm)}\big|_{\rm vol}(x) \;=\; -\bar{\rho}_L(x)\,\frac{1}{2}\int_{\Sigma_u} d^3y\,\sqrt{h(y)}\,\mathcal{K}_L(x,y)\,h^{ij}(y)\,\delta h_{ij}(y).$$

The sum of the two volume-element terms is remarkable in its own right: **in a homogeneous medium ($\varepsilon(y) = \bar{\rho}_L(x)$), the two volume-element variations ($\delta\bar{\rho}_L^{(\mathbf{vol})} + \delta\bar{\rho}_L^{(\mathbf{norm})}\big|_{\mathbf{vol}}$) cancel each other identically!**

### 14.2.2 2.2. Variation of the Geodesic Distance $\delta d_\Sigma(x,y)$ (Synge's World Function) and the Boundary Surface Stress $\delta_D(L - d_\Sigma)$

Let $\gamma_{xy} : [0, d_\Sigma(x,y)] \to \Sigma_u$, $l \mapsto z^k(l)$ denote the minimal spatial geodesic curve connecting points $x$ and $y$, parameterized by proper arc length $l$, with unit tangent vector $\hat{t}^k(l) = \frac{dz^k}{dl}$ ($h_{ij}\hat{t}^i\hat{t}^j = 1$). Because of the geodesic equation, the first variation of Synge's spatial world function ($\sigma_\Sigma(x,y) = \frac{1}{2}d_\Sigma(x,y)^2$) at fixed endpoints $(x,y)$ is independent of the path displacement $\delta z^k(l)$ of the geodesic, and reduces purely to the line integral of the metric perturbation $\delta h_{ij}$ along the geodesic:

$$\boxed{\delta d_\Sigma(x,y) \;=\; \frac{1}{2}\int_0^{d_\Sigma(x,y)} dl\,\hat{t}^i(l)\,\hat{t}^j(l)\,\delta h_{ij}\big(\gamma_{xy}(l)\big) \;=\; \frac{1}{2}\int_{\Sigma_u} d^3z\,\delta^{(3)}\big(z - \gamma_{xy}(l)\big)\,\hat{t}^i(l)\hat{t}^j(l)\,\delta h_{ij}(z)\,dl.}$$

Because the derivative of the Heaviside step function is the Dirac delta ($\frac{\partial}{\partial d_\Sigma}\Theta(L - d_\Sigma) = -\delta_D(L - d_\Sigma)$), the displacement of the window boundary generates a **spherical surface ($d_\Sigma(x,y) = L$) boundary stress**:

$$\boxed{\delta\bar{\rho}_L^{(\rm bdry)}(x) \;=\; -\frac{1}{2V_L(x)}\oint_{d_\Sigma(x,y)=L} dA_y\,\big[\varepsilon(y) - \bar{\rho}_L(x)\big]\int_0^L dl\,\hat{t}^i(l)\hat{t}^j(l)\,\delta h_{ij}\big(\gamma_{xy}(l)\big),}$$

where $dA_y$ is the proper surface element of the geodesic sphere $d_\Sigma(x,y) = L$, and the factor $\varepsilon(y) - \bar{\rho}_L(x)$ arises from combining the boundary variation of $\Theta(L - d_\Sigma)$ in the numerator with that of $V_L(x)$ in the denominator.

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#### 14.2.3 2.3. Evaluation Inside a Spherically Symmetric Neutron Star ($h_{rr} = e^{2\lambda(r)} = B(r)^{-1}$) and Connection to Formulation II (Yukawa)

In a spherically symmetric metric ($ds^2_\Sigma = e^{2\lambda(r)}dr^2 + r^2 d\Omega^2$, where $e^{-2\lambda(r)} = B(r) = 1 - 2m(r)/r$), the radial metric variation $\delta h_{rr}(r) = 2e^{2\lambda(r)}\delta\lambda(r)$ in the $(rr)$ Einstein equation ($G^r_{\ r} = 8\pi T^r_{\ r,\mathrm{tot}}$) adds the following nonlocal radial window pressure ($P_{r,\mathrm{window}}(r)$) to the `bilateral_lapse_weighted` equations:

$$P_{r,\mathrm{window}}(r) \;=\; -\frac{1}{4\pi r^2 N(r) e^{\lambda(r)}}\,\frac{\delta S_{\mathrm{I},X}}{\delta\lambda(r)}.$$

This expression illuminates two fundamental statements of the V23 manuscript:

1. **Why does `bilateral_lapse_weighted` computed with a flat-space kernel differ from the EL solution of action (2)?** Because the flat-space kernel ($d_{\mathrm{flat}}(x,y) = |\mathbf{x}-\mathbf{y}|$) omits the curved volume-element factor $e^{\lambda(r)} = (1-2m/r)^{-1/2}$ and the geodesic boundary line integral $\delta d_\Sigma(x,y)$.
2. **Why does Formulation II (Helmholtz–Yukawa, `yukawa_action_el`) eliminate the distributional surface boundary stress $\delta_D(L-d_\Sigma)$?** When the sharp top-hat window $\Theta(L-d_\Sigma)$ is replaced by the smooth 3D Helmholtz–Yukawa resolvent ($(-\Box_\Sigma + L^{-2})^{-1}$), the nonlocal boundary stress arising from geodesic line integrals transforms into the **local, continuously differentiable scalar pressure** derived in Step 7 (`step7_neutron_star_tov_stability.md`):

$$p_{r,\phi}(r) \;=\; \frac{1}{2\alpha_Y}\left(e^{-2\lambda(r)}\left[\phi_L'(r)\right]^2 - \frac{\phi_L(r)^2}{L^2}\right),$$

in which the variation of the factor $g^{rr} = e^{-2\lambda(r)} = B(r)$ ($\delta g^{rr} = -2e^{-2\lambda}\delta\lambda$) automatically and algebraically incorporates all metric back-reactions of the curved-space distance measure!

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### 14.3 3. Checks

1. **Homogeneous-Density Limit ($\varepsilon(y) \equiv \varepsilon_0 = $ const.):**
   - If $\varepsilon(y) = \varepsilon_0$ is constant inside the ball $d_\Sigma(x,y) \le L$, then $\bar\rho_L(x) = \varepsilon_0$, so both the volume-element variation ($\varepsilon(y) - \bar\rho_L(x) = 0$) and the $\delta_D(L-d_\Sigma)$ surface boundary stress ($\varepsilon(y)\big|_{d_\Sigma = L} - \bar\rho_L(x) = 0$) **vanish identically**: $\delta\bar\rho_L(x) \equiv 0$ ✓.
2. **Flat-Spacetime ($\lambda(r) \to 0$, $N(r) \to 1$) Limit:**
   - $h_{ij} \to \delta_{ij} \implies d_\Sigma(x,y) \to |\mathbf{x}-\mathbf{y}|$, $V_L(x) \to \frac{4}{3}\pi L^3$, and the variational equation recovers the `bilateral_eom` / `bilateral_lapse_weighted` expressions derived in Step 10 ✓.

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# 15 Step 14: Conformal vs. Disformal Gravitational Lensing (Null-Geodesic Light Deflection Angle $\hat{\alpha}_{\mathbf{def}}(b)$), NICER Ray-Tracing, and Cosmological Background Embedding ($L_{\mathbf{hor}}^{(m)} \approx 22.94$ Gpc vs. $L_{\mathbf{hor}}^{(b)} \approx 57.98$ Gpc)

Source file: step14_gravitational_lensing_null_geodesics_cosmological_background_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step14_gravitational_lensing_null_geodesics_cosmological_background_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step13_curved_window_metric_variation_synge_world_function_en.md (step13_curved_window_metric_variation_synge_world_function.md) **License:** CC BY-NC 4.0
**Local Archival Sources:**

- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex (Chapter 2, line 199), hu_p2.tex (Chapter 4, line 31; Chapter 5, lines 52–55), and hu_p3.tex (Chapter 9, lines 59–60; Chapter 11, line 104; DOI: 10.5281/zenodo.23604423)
- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Section 3, lines 161–186; DOI: 10.5281/zenodo.23639084)

## 15.1 1. Verbatim Citations from the V22–V23 Manuscripts

In Step 5 (step5_coupling_dictionary_ppn_screening.md), we derived the weak-field PPN parameter $\gamma_{\rm PPN}$, and in Step 9 (step9_vhe_photon_propagation_hess_dispersion.md) we analyzed straight line-of-sight VHE gamma-ray time-of-flight dispersion. In this step, we derive three closely connected results: **(A)** what difference the purely temporal disformal clock metric ($\tilde{g}_{00} = f_L^2 g_{00}$, $\tilde{g}_{ij} = g_{ij}$, **Branch II-Disf**) induces in the **light deflection angle (gravitational lensing,** $\hat{\alpha}_{\mathbf{def}}(b)$**)** compared to the conformal case $\tilde{g}_{\mu\nu}^{\rm conf} = f_L^2 g_{\mu\nu}$ (**Branch II-Conf**); **(B)** how **cosmological background embedding** ($\gamma_L \rho_{\rm bg} = 1$) enters the exponential factor, computing separately the horizon scales associated with the baryonic density ($\Omega_b = 0.0493 \implies L_{\rm hor}^{(b)} \approx 57.98$ Gpc) and the total cosmic matter density ($\Omega_m \approx 0.315 \implies L_{\rm hor}^{(m)} \approx 22.94$ Gpc); and **(C)** what this implies for **NICER pulsar ray-tracing likelihoods** (PSR J0030+0451, PSR J0740+6620, PSR J0437-4715).

1. **Temporal Refractive Index and Cosmological Background Embedding in the V22 Monograph (hu_p1.tex line 199 and hu_p2.tex line 54):**
   - hu_p1.tex, line 199:

     *"I interpret the flow of time as possessing an effective local carrier density (analogous to an optical medium with refractive index $n_t(\mathbf{x}) = f_L(\mathbf{x})^{-1} = \exp(+\gamma_L \bar{\rho}_L(\mathbf{x}))$ or a compressible time fluid)."*

   - hu_p2.tex, Chapter 5, line 54:

     *"**Cosmological Background Embedding ($L \approx 22.94$ Gpc):** For $\bar{\rho}_L$ on megaparsecs scales to reach $\gamma_L \bar{\rho}_L \sim 1$, the ball $B(\mathbf{x}, L)$ must include the ambient intergalactic and cosmological background density $\rho_{bg} \approx \Omega_b \rho_{crit}$. At uniform cosmic baryon density, the condition $\gamma_L \bar{\rho}_{bg} = 1$ occurs at the formal horizon scale $L \approx 22.94$ Gpc ($7.08 \times 10^{26}$ m)."*

2. **NICER Pulsar Likelihoods in the V22 Monograph (hu_p3.tex, lines 59–60 and 104):**
   - hu_p3.tex, Chapter 9, line 60:

     *"The joint log-likelihood computed for the three NICER pulsars (PSR J0030+0451, PSR J0740+6620, PSR J0437-4715) under causal endpoint ($M_{max}(causal)$) mass normalization shows that at $L = 1$ km the shift relative to GR is at most $4.8 \times 10^{-4}$ **dex** ($\Delta \ln \mathcal{L} \approx 1.1 \times 10^{-3}$). Consequently, **current NICER mass–radius data cannot statistically distinguish the scale $L$ from GR under either hypothesis**."*

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## 15.2 2. Conformal vs. Purely Temporal Disformal Gravitational Lensing ($\hat{\alpha}_{\mathbf{def}}(b)$)

Consider a static, weak gravitational field in which the line element of the Einstein-frame gravitational metric ($g_{\mu\nu}$) in isotropic coordinates is:

$$ds^2 \;=\; g_{\mu\nu}dx^\mu dx^\nu \;=\; -\left(1+\frac{2\Phi_N(\mathbf{x})}{c^2}\right)c^2dt^2 \;+\; \left(1-\frac{2\Psi_N(\mathbf{x})}{c^2}\right)d\mathbf{x}^2, \qquad \Phi_N(\mathbf{x})=\Psi_N(\mathbf{x})=-\frac{GM(<r)}{r}.$$

### 15.2.1 2.1. Light Deflection under Purely Conformal Jordan Coupling (Branch II-Conf: $\tilde{g}^{\mathbf{conf}}_{\mu\nu}=f_L^2 g_{\mu\nu}$)

If the Maxwell field couples to the conformally rescaled metric $\tilde{g}^{\rm conf}_{\mu\nu}=f_L(\mathbf{x})^2 g_{\mu\nu}=e^{-2\phi_L(\mathbf{x})}g_{\mu\nu}$, then in $D=4$ spacetime dimensions the conformal invariance $\sqrt{-\tilde{g}^{\rm conf}}\tilde{g}^{\mu\alpha}_{\rm conf}\tilde{g}^{\nu\beta}_{\rm conf}F_{\mu\nu}F_{\alpha\beta}=\sqrt{-g}g^{\mu\alpha}g^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}$ ensures that null geodesics of light rays ($\tilde{g}^{\rm conf}_{\mu\nu}dx^\mu dx^\nu=0 \iff g_{\mu\nu}dx^\mu dx^\nu=0$) are **independent of the scalar field** $\phi_L(\mathbf{x})$. The effective optical refractive index entering Fermat's principle ($\delta\int n_{\rm conf}(\mathbf{x})\,dl=0$) is:

$$n_{\rm conf}(\mathbf{x}) \;=\; \sqrt{\frac{g_{rr}}{-g_{00}}} \;\simeq\; 1 \;-\; \frac{\Phi_N(\mathbf{x})+\Psi_N(\mathbf{x})}{c^2} \;=\; 1 \;-\; \frac{2\Phi_N(\mathbf{x})}{c^2}.$$

Consequently, the deflection angle of a light ray with impact parameter $b$ ($\hat{\alpha}_{\rm def}(b)=-\int_{-\infty}^{+\infty}\nabla_\perp \ln n\,dz$) under purely conformal coupling **senses only the baryonic Weyl potential** ($\Phi^{\mathbf{conf}}_{\mathbf{lens}}=\frac{1}{2}(\Phi_N+\Psi_N)=\Phi_N$):

$$\boxed{\hat{\alpha}^{\rm conf}_{\rm def}(b) \;=\; \frac{2}{c^2}\int_{-\infty}^{+\infty}\nabla_\perp\Phi_N\big(\sqrt{b^2+z^2}\big)\,dz \;=\; \frac{4GM_{\rm bary}(<b)}{c^2 b}.}$$

That is, in purely conformal coupling the scalar field ($\phi_L$) enhances the circular orbital velocity of non-relativistic stars and gas ($V^2_{\rm rot}(r)=V^2_{\rm bar}(r)-c^2 r\phi_L'(r)$), **but does not contribute to gravitational lensing**!

### 15.2.2 2.2. Light Deflection in the V23 Purely Temporal Disformal Clock Metric (Branch II-Disf: $\tilde{g}_{00}=f_L^2 g_{00}$, $\tilde{g}_{ij}=g_{ij}$)

By contrast, if electromagnetic waves propagate on the V23 disformal physical metric ($\tilde{g}^{\rm disf}_{\mu\nu}=g_{\mu\nu}+\frac{1-f_L^2}{c^2}u_\mu u_\nu$), then the null-cone equation ($d\tilde{s}^2_{\rm disf}=0$) is:

$$0 \;=\; \tilde{g}^{\rm disf}_{00}\,c^2dt^2 \;+\; \tilde{g}^{\rm disf}_{ij}\,dx^i dx^j \;=\; -f_L(\mathbf{x})^2\left(1+\frac{2\Phi_N(\mathbf{x})}{c^2}\right)c^2dt^2 \;+\; \left(1-\frac{2\Psi_N(\mathbf{x})}{c^2}\right)d\mathbf{x}^2.$$

From this, the effective optical refractive index associated with Fermat's principle (`hu_p1.tex`, line 199: $n_t(\mathbf{x})=f_L(\mathbf{x})^{-1}=e^{+\phi_L(\mathbf{x})}$) is:

$$n_{\rm disf}(\mathbf{x}) \;=\; \frac{c}{v_{\rm phase}(\mathbf{x})} \;=\; f_L(\mathbf{x})^{-1}\sqrt{\frac{1-2\Psi_N/c^2}{1+2\Phi_N/c^2}} \;\simeq\; 1 \;-\; \frac{\Phi_N(\mathbf{x})+\Psi_N(\mathbf{x})}{c^2} \;+\; \phi_L(\mathbf{x}).$$

Substituting the transverse gradient $\nabla_\perp \ln n_{\rm disf}\simeq -\frac{1}{c^2}\nabla_\perp(\Phi_N+\Psi_N-c^2\phi_L)$ into the Fermat integral, the disformal lensing potential and light deflection angle are:

$$\boxed{\Phi^{\rm disf}_{\rm lens}(\mathbf{x}) \;=\; \Phi_N(\mathbf{x}) \;-\; \frac{1}{2}c^2\,\phi_L(\mathbf{x}), \qquad \hat{\alpha}^{\rm disf}_{\rm def}(b) \;=\; \frac{4GM_{\rm bary}(<b)}{c^2 b} \;-\; \int_{-\infty}^{+\infty}\frac{b}{r}\frac{d\phi_L(r)}{dr}\,dz.}$$

Since $\frac{d\phi_L}{dr}<0$ for an attractive time potential (density decreases outward), the second term produces a **positive (inward-directed) excess light deflection**: the disformal temporal refractive index ($n_t=e^{+\phi_L}$) around galaxies and galaxy clusters amplifies not only rotation curves, but gravitational lensing as well!

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## 15.3 3. Cosmological Background Embedding: Exact Arithmetic for $\Omega_b=0.0493$ (57.98 Gpc) and $\Omega_m=0.315$ (22.94 Gpc)

Let us decompose the cosmic mass–energy density into a spatially homogeneous FLRW background density ($\rho_{\rm bg}(t)=\Omega_{\rm bg}\,\rho_{\rm crit,0}$) and the local overdensity ($\delta\rho(\mathbf{x})$):

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$$\rho(\mathbf{x}) \;=\; \rho_{\rm bg} \;+\; \delta\rho(\mathbf{x}) \quad\Longrightarrow\quad \bar\rho_L(\mathbf{x}) \;=\; \frac{3}{4\pi L^3}\int_{|\mathbf{y}-\mathbf{x}|\le L}\big[\rho_{\rm bg}+\delta\rho(\mathbf{y})\big]d^3y \;=\; \rho_{\rm bg} \;+\; \delta\bar\rho_L(\mathbf{x}).$$

### 15.3.1 3.1. General Formula and Exact Separation of the Two Cosmological Density Parameters ($\Omega_b$ vs. $\Omega_m$)

Using the Planck 2018 (TT,TE,EE+lowE+lensing) parameters:

$$H_0 \;=\; 67.4\,\mathrm{km\,s^{-1}Mpc^{-1}} \;=\; 2.184285\times10^{-18}\,\mathrm{s^{-1}}, \qquad R_H \;\equiv\; \frac{c}{H_0} \;=\; 1.37250\times10^{26}\,\mathrm{m} \;=\; 4.4480\,\mathrm{Gpc},$$

$$\rho_{\rm crit,0} \;=\; \frac{3H_0^2}{8\pi G} \;=\; 8.53287\times10^{-27}\,\mathrm{kg/m^3}.$$

The exact general solution of the unit-background-exponent condition ($\gamma_L\rho_{\rm bg}=1$, where $\gamma_L=\frac{GL_{\rm hor}^2}{c^2}$) for any density fraction $\Omega_{\rm bg}\equiv\rho_{\rm bg}/\rho_{\rm crit,0}$ is:

$$\gamma_L\,\rho_{\rm bg} \;=\; \frac{GL_{\rm hor}^2}{c^2}\,\Omega_{\rm bg}\,\frac{3H_0^2}{8\pi G} \;=\; \frac{3\,\Omega_{\rm bg}}{8\pi}\left(\frac{L_{\rm hor}}{c/H_0}\right)^2 \;=\; 1 \quad\Longleftrightarrow\quad \boxed{L_{\rm hor}(\Omega_{\rm bg}) \;=\; \sqrt{\frac{8\pi}{3\,\Omega_{\rm bg}}}\,\frac{c}{H_0} \;=\; \frac{c}{\sqrt{G\,\rho_{\rm bg}}}.}$$

Observe that Newton's constant $G$ cancels identically from the product $\gamma_L\rho_{\rm bg}$. To clarify the numerical values cited in line 54 of V22 `hu_p2.tex`, it is essential to separate the purely baryonic ($\Omega_b$) and combined cosmic matter ($\Omega_m$) cases:

| Background Density Model | Density Parameter $\Omega_{\rm bg}$ | Background Density $\rho_{\rm bg}=\Omega_{\rm bg}\rho_{\rm crit,0}$ (kg/m$^3$) | Dimensionless Factor $\sqrt{\frac{8\pi}{3\Omega_{\rm bg}}}$ | Horizon Scale $L_{\rm hor}$ (m) | Horizon Scale $L_{\rm hor}$ (Gpc) | Note / Source |
|---|---|---|---|---|---|---|
| **Purely Baryonic Cosmic Background ($\Omega_b$)** | $\Omega_b=0.0493$ | $4.2067\times10^{-28}$ | 13.0306 | $1.7885\times10^{27}$ | 57.96–57.98 **Gpc** | Exact $\gamma_L\rho_{b,0}=1$ solution for $\Omega_b=0.0493$ |
| **Cosmic Matter Density Background ($\Omega_m$)** | $\Omega_m=0.3147$–$0.3150$ | $2.6855$–$2.6879\times10^{-27}$ | 5.1583–5.1605 | $7.080$–$7.083\times10^{26}$ | 22.94–22.95 **Gpc** | Value $7.08\times10^{26}$ m $\approx$ 22.94 Gpc in V22 `hu_p2.tex` line 54 (corresponds to $\Omega_m\approx0.315$!) |

*(Note on line 54 of V22 `hu_p2.tex`: the manuscript prose verbally mentions $\Omega_b\rho_{crit}$, but the reported numerical value $7.08\times10^{26}$ $m$ = 22.94 Gpc arises arithmetically from the cosmic matter density $\Omega_m\approx0.315$, $\rho_{bg}=\frac{1}{\gamma_L(L_{hor})}=2.686\times10^{-27}$ kg/m$^3$ and the factor $\sqrt{8\pi/(3\Omega_m)}=5.160$, whereas for $\Omega_b=0.0493$ one obtains $\sqrt{8\pi/(3\Omega_b)}=13.031$ and $L_{hor}^{(b)}=1.789\times10^{27}$ $m\approx57.98$ Gpc.)*

### 15.3.2 3.2. Factorization of the Homogeneous Background Factor from Local Dynamical and Lensing Gradients

By Cauchy's multiplicative property (`hu_p1.tex`, Eq. 2: $f(\rho_1+\rho_2)=f(\rho_1)f(\rho_2)$), the exponential time factor factorizes as a product:

$$f_L(\mathbf{x}) \;=\; \exp\!\big(-\gamma_L[\rho_{\rm bg}+\delta\bar\rho_L(\mathbf{x})]\big) \;=\; \underbrace{e^{-\gamma_L\rho_{\rm bg}}}_{f_{L,\rm bg}\ \text{(global constant)}} \cdot \underbrace{e^{-\gamma_L\,\delta\bar\rho_L(\mathbf{x})}}_{f_{L,\rm loc}(\mathbf{x})}.$$

Since $f_{L,\rm bg}$ is spatially constant ($\nabla f_{L,\rm bg}=0$), rescaling the global time coordinate $d\tilde t=f_{L,\rm bg}\,dt$ absorbs the homogeneous background factor into the cosmological background clock, while the logarithmic gradient appearing in local dynamical acceleration ($\mathbf{a}_{\rm eff}=-\nabla\Phi_N+c^2\nabla\phi_L$) and disformal light deflection ($\hat\alpha_{\rm def}^{\rm disf}$) is **independent of the homogeneous background $\rho_{\rm bg}$**:

$$\nabla\ln f_L(\mathbf{x}) \;=\; -\nabla\phi_L(\mathbf{x}) \;=\; -\gamma_L\,\nabla\delta\bar\rho_L(\mathbf{x}).$$

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## 15.4 4. Implication for NICER Pulsar Ray-Tracing (PSR J0030+0451, PSR J0740+6620, PSR J0437-4715)

In ray-tracing of X-ray photons emitted from the surface ($r \geq R_*$) of a neutron star of radius $R_*$ (Beloborodov / Pechenick–Ftaclas–Cohen formula) in the exterior vacuum ($\rho(r > R_*) = 0$):

- **For a 3D top-hat kernel ($L \leq 1$ km $\ll R_*$):** Outside the stellar surface ($r > R_* + L$), the top-hat sphere does not intersect the star, so $\bar{\rho}_L(r > R_* + L) = 0 \implies f_L = 1$; exterior light deflection follows exact Schwarzschild geodesics, and at $L = 1$ km the relative shift of the mass–radius curve is only $4.8 \times 10^{-4}$ dex ($\Delta \ln \mathcal{L} \approx 1.1 \times 10^{-3}$, V22 `hu_p3.tex` line 60).
- **For a Yukawa kernel (`yukawa_action_el`):** The exterior scalar profile in curved exterior spacetime decays exponentially as $r \to \infty$ ($\phi_L(r) \propto r^{-(1+M_{\rm ADM}/L)} e^{-r/L}$), and the asymptotic gravitational mass of the star is absorbed into the ADM mass $M_{\rm ADM} = M_{\rm surf} + \Delta M_{\phi,\rm ext}^{\rm curved}$ derived in Steps 7 and 22 (`step7_neutron_star_tov_stability.md`, `step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability.md`).

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## 15.5 5. Checks

1. **Vacuum and GR Limit ($\phi_L \to 0 \implies f_L \to 1$):**
   - As $\phi_L(\mathbf{x}) \to 0$, $\frac{d\phi_L}{dr} \to 0$, so both the conformal ($\hat{\alpha}_{\rm def}^{\rm conf}$) and disformal ($\hat{\alpha}_{\rm def}^{\rm disf}$) light deflection angles recover the Einstein value $\hat{\alpha}_{\rm GR}(b) = \frac{4GM}{c^2 b}$ ✓.
2. **Numerical Verification of $L_{\rm hor}^{(m)} \approx 22.94$ Gpc and $L_{\rm hor}^{(b)} \approx 57.98$ Gpc:**
   - $\Omega_m = 0.3147 \implies \rho_{m,0} = 2.6855 \times 10^{-27}$ kg/m$^3$ $\implies \sqrt{8\pi/(3\Omega_m)} = 5.1605 \implies L_{\rm hor}^{(m)} = 7.083 \times 10^{26}$ m $= 22.95$ Gpc (and for $\rho_{\rm bg} = 2.686 \times 10^{-27}$ kg/m$^3$, $7.08 \times 10^{26}$ m $= 22.94$ Gpc) ✓.
   - $\Omega_b = 0.0493 \implies \rho_{b,0} = 4.2067 \times 10^{-28}$ kg/m$^3$ $\implies \sqrt{8\pi/(3\Omega_b)} = 13.0306 \implies L_{\rm hor}^{(b)} = 1.7885 \times 10^{27}$ m $=$ 57.96–57.98 Gpc ✓.

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# 16 Step 15: Electromagnetic Trace-Freeness ($T^{\mu}_{\mu,\mathbf{EM}} = 0$) vs. Disformal Magnetic Source Coupling ($\rho_B = |\mathbf{B}|^2/(2\mu_0 c^2)$), V20 Polarity-Even Cross-Term Cancellation, and the Methodological Boundary ($\Phi_t \equiv 0$)

Source file: step15_electromagnetic_trace_anomaly_disformal_magnetic_source_v20_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step15_electromagnetic_trace_anomaly_disformal_magnetic_source_v20_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step14_gravitational_lensing_null_geodesics_cosmological_background_en.md (step14_gravitational_lensing_null_geodesics_cosmological_background.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p1.tex (Chapter 2, points 3–4, lines 208–221) and hu_p3.tex (Chapter 10, point 2, lines 75–85; DOI: 10.5281/zenodo.23604423)
- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Sections 1 and 3, lines 33–35, 58–66, and 161–166; DOI: 10.5281/zenodo.23639084)
- README.md (scalar-tensor-yukawa-fit/README.md, Section 3, point 5, lines 60–61)

## 16.1 1. Verbatim Citations from the V22 (hu_p1.tex, hu_p3.tex) and V23 Manuscripts

In Step 8 (step8_microscopic_clock_coupling_siriusb_lab_v20.md), we derived how a given scalar profile $\phi_L$ shifts the frequencies of atomic clocks and quartz oscillators (H0 vs. H1), and evaluated the numerical value for the V20 8-coil solenoid ($\Delta y_{\text{peak}} = -1.2017108 \times 10^{-48}$ at $L = 0.630957$ m). Within the scalar–tensor framework, however, there is a fundamental theoretical question highlighted in hu_p1.tex (lines 208–221), hu_p3.tex (lines 75–85), and README.md (line 61): **how can pure magnetic energy density ($\rho_B = \frac{|\mathbf{B}|^2}{2\mu_0 c^2}$) act as a source for the scalar field $\phi_L$ in a covariant action principle when the trace of the 4-dimensional Maxwell stress–energy tensor vanishes identically ($T^{\mu}_{\mu,\mathbf{EM}} = g^{\mu\nu}T^{\mathbf{EM}}_{\mu\nu} = 0$)?**

1. **Methodological Boundary Between the Audited Physical Core ($\rho = T_{00}/c^2$, $\Phi_t \equiv 0$) and the V13–V16 Perspective Layer (hu_p1.tex, Chapter 2, points 3–4, lines 209–221):**

$$\rho_{\text{eff}} \;=\; \rho \;+\; \Xi\,\varepsilon_0\,\Phi_t \;+\; \beta\,\Phi_B \;+\; \gamma\,\kappa, \qquad T^{\text{total}}_{\mu\nu} \;=\; T^{\text{matter}}_{\mu\nu} \;+\; T^{\text{flux}}_{\mu\nu} \;+\; T^{\text{thought}}_{\mu\nu}, \tag{V22 Eq. 2}$$

   *"When presenting this book to the academic community, I draw a clear boundary between the two layers:*

   - ***Audited Physical Core (V19–V21):*** *Every numerical result in the SPARC galactic rotation analysis, the H.E.S.S. timing bounds, the neutron-star TOV and pulsation sequences, and the V20 granite-chamber and 8-coil quartz-oscillator laboratory protocol is built exclusively on the standard baryonic and electromagnetic mass–energy density $\rho = T_{00}/c^2$, with $\Phi_t \equiv 0$.*
   - ***Perspective Bio-Coherence Layer (V13–V16):*** *The conscious-coherence terms ($\Phi_t$, $T^{thought}_{\mu\nu}$, EEG/HRV bio-coherence) are retained as an open, not-yet-audited theoretical perspective representing part of the intellectual history of the Time Attraction Law."*

2. **V20 Eight-Coil Magnetic Energy-Density Source and Background Cross-Term Cancellation (hu_p3.tex, Chapter 10, point 2, lines 76 and 84):**

   - *"To test whether pure magnetic energy density $u_B(\mathbf{x}) = \frac{|\mathbf{B}(\mathbf{x})|^2}{2\mu_0}$ slows the flow of time via $\rho_B(\mathbf{x}) = u_B(\mathbf{x})/c^2$, I designed a symmetric eight-coil arrangement..."*
   - *"**Cancellation of the Background Cross-Term via Polarity Reversal:** Since $|\mathbf{B}_{tot}|^2 = |\mathbf{B}_{bg}|^2 + 2\mathbf{B}_{bg} \cdot \mathbf{B}_{coil} + |\mathbf{B}_{coil}|^2$, reversing the current direction ($+I \leftrightarrow -I$) reverses $\mathbf{B}_{coil}$, so both the linear Zeeman sensitivity of the quartz oscillator and the cross-term $2\mathbf{B}_{bg} \cdot \mathbf{B}_{coil}$ cancel out of the polarity-even average $\Delta y_{even}$."*

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## 16.2 2. Why Does a Magnetic Field Not Source a Purely Conformal Brans–Dicke / Dilaton Scalar, and Why Does It Source the V23 Disformal Coupling?

### 16.2.1 2.1. Electromagnetic Blindness of Purely Conformal Coupling ($T^{\mu}_{\mu,\mathbf{EM}} = 0$ in $D = 4$)

Consider the standard Maxwell action of the free electromagnetic field ($F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$) on a general Jordan metric $\tilde{g}_{\mu\nu}$:

$$S_{\mathrm{EM}}\big[\tilde{g}_{\mu\nu}, A_\mu\big] \;=\; -\frac{1}{4\mu_0}\int d^4x\,\sqrt{-\tilde{g}}\,\tilde{g}^{\mu\alpha}\tilde{g}^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}.$$

If the metric depends on the scalar field $\phi_L$ in a purely **conformal** manner ($\tilde{g}^{\mathrm{conf}}_{\mu\nu} = \Omega^2(\phi_L)\,g_{\mu\nu} = f_L^2\,g_{\mu\nu}$, so that $\tilde{g}^{\mu\alpha}_{\mathrm{conf}} = f_L^{-2}g^{\mu\alpha}$ and in $D = 4$ dimensions $\sqrt{-\tilde{g}^{\mathrm{conf}}} = f_L^4\sqrt{-g}$), then the factors of $f_L$ cancel identically:

$$\sqrt{-\tilde{g}^{\mathrm{conf}}}\,\tilde{g}^{\mu\alpha}_{\mathrm{conf}}\tilde{g}^{\nu\beta}_{\mathrm{conf}}F_{\mu\nu}F_{\alpha\beta} \;=\; \big(f_L^4\sqrt{-g}\big)\big(f_L^{-2}g^{\mu\alpha}\big)\big(f_L^{-2}g^{\nu\beta}\big)F_{\mu\nu}F_{\alpha\beta} \;=\; \sqrt{-g}\,g^{\mu\alpha}g^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}.$$

Accordingly, the variation of the Maxwell action with respect to $\phi_L$ vanishes identically in purely conformal coupling (which is equivalent to the vanishing of the trace of the Maxwell stress–energy tensor in $D = 4$, $T^{\mu}_{\mu,\mathrm{EM}} = g^{\mu\nu}T^{\mathrm{EM}}_{\mu\nu} = -u_E - u_B + 3\cdot\frac{1}{3}(u_E + u_B) = 0$):

$$\boxed{\frac{\delta S_{\mathrm{EM}}\big[\tilde{g}^{\mathrm{conf}}_{\mu\nu}, A_\mu\big]}{\delta\phi_L(x)} \;=\; \frac{\partial\tilde{g}^{\mathrm{conf}}_{\mu\nu}}{\partial\phi_L}\,\frac{\delta S_{\mathrm{EM}}}{\delta\tilde{g}^{\mathrm{conf}}_{\mu\nu}} \;=\; \sqrt{-g}\,f_L^2\,\frac{\partial\ln f_L}{\partial\phi_L}\,g^{\mu\nu}T^{\mathrm{EM}}_{\mu\nu} \;=\; 0.}$$

**Physical consequence:** A purely conformal Jordan–Brans–Dicke scalar field is not sourced at tree level by a static magnetic field **B** at all!

### 16.2.2 2.2. Exact Magnetic Source Term of the V23 Disformal Clock Metric ($\tilde{g}^{\mathbf{disf}}_{\mu\nu} = g_{\mu\nu} + \frac{1-f_L^2}{c^2}u_\mu u_\nu$)

Now let us substitute the V23 **temporal disformal metric** (`tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, line 165):

$$\tilde{g}^{\mathrm{disf}}_{\mu\nu} \;=\; g_{\mu\nu} \;+\; \frac{1-f_L^2(\phi_L)}{c^2}\,u_\mu u_\nu, \qquad \tilde{g}^{\mu\nu}_{\mathrm{disf}} \;=\; g^{\mu\nu} \;-\; \frac{1-f_L^{-2}(\phi_L)}{c^2}\,u^\mu u^\nu, \qquad \sqrt{-\tilde{g}^{\mathrm{disf}}} \;=\; f_L(\phi_L)\,\sqrt{-g}.$$

Decomposing the field-strength tensor $F_{\mu\nu}$ into electric ($E_\mu = F_{\mu\nu}u^\nu/c$) and magnetic ($B_\mu = \frac{1}{2}\epsilon_{\mu\nu\alpha\beta}u^\nu F^{\alpha\beta}/c$) field four-vectors comoving with the laboratory four-velocity $u^\mu$:

$$\frac{1}{4}\tilde{g}^{\mu\alpha}_{\mathrm{disf}}\tilde{g}^{\nu\beta}_{\mathrm{disf}}F_{\mu\nu}F_{\alpha\beta} \;=\; \frac{1}{2}\Big[\tilde{g}^{ij}_{\mathrm{disf}}\tilde{g}^{kl}_{\mathrm{disf}}F_{ik}F_{jl} \;+\; 2\,\tilde{g}^{00}_{\mathrm{disf}}\tilde{g}^{ij}_{\mathrm{disf}}F_{0i}F_{0j}\Big] \;=\; \frac{|\mathbf{B}|^2}{2\mu_0}\cdot\mu_0 \;-\; f_L^{-2}(\phi_L)\,\frac{\varepsilon_0|\mathbf{E}|^2}{2}\cdot\mu_0.$$

Substituting this together with the volume element $\sqrt{-\tilde{g}^{\mathrm{disf}}} = f_L(\phi_L)\sqrt{-g}$ into the Maxwell action gives:

$$\boxed{S_{\mathrm{EM}}\big[\tilde{g}^{\mathrm{disf}}_{\mu\nu}, A_\mu\big] \;=\; \int d^4x\,\sqrt{-g}\left[-\,f_L(\phi_L)\,\frac{|\mathbf{B}(\mathbf{x})|^2}{2\mu_0} \;+\; f_L(\phi_L)^{-1}\,\frac{\varepsilon_0|\mathbf{E}(\mathbf{x})|^2}{2}\right].}$$

In a static magnetic field ($\mathbf{E} = \mathbf{0}$, as in the V20 8-coil solenoid), the Lagrangian density $\mathcal{L}^{\mathrm{disf}}_{\mathrm{EM}}(\mathbf{E} = \mathbf{0}) = -f_L(\phi_L)\,u_B(\mathbf{x})$ takes the exact same $-f_L(\phi_L)\varepsilon$ form as a static baryonic fluid (`step4_covariant_conservation_hydrostatic.md`: $\mathcal{L}_{\mathrm{matter}} = -f_L(\phi_L)\varepsilon_{\mathrm{baryon}}$)! Varying the V23 Formulation II (Helmholtz–Yukawa) or Formulation I (bilateral top-hat) action with respect to $\phi_L$ ($f_L = e^{-\phi_L}$, $\partial f_L/\partial\phi_L = -f_L$) yields:

$$\frac{\partial\big(\mathcal{L}_{\mathrm{matter}} + \mathcal{L}^{\mathrm{disf}}_{\mathrm{EM}}\big)}{\partial\phi_L} \;=\; f_L(\phi_L)\,\Big(\varepsilon_{\mathrm{baryon}}(\mathbf{x}) + u_B(\mathbf{x})\Big) \;=\; f_L(\phi_L)\,c^2\Big(\rho_{\mathrm{baryon}}(\mathbf{x}) + \rho_B(\mathbf{x})\Big), \qquad \rho_B(\mathbf{x}) \;\equiv\; \frac{|\mathbf{B}(\mathbf{x})|^2}{2\mu_0 c^2}.$$

We have thus derived from a covariant action principle that the V23 disformal clock metric ($\sqrt{-\tilde{g}} = f_L\sqrt{-g}$, $\tilde{g}_{ij} = g_{ij}$) automatically selects the magnetic mass–energy density $T^{\mathrm{EM}}_{00}/c^2 = \rho_B = \frac{|\mathbf{B}|^2}{2\mu_0 c^2}$ as the source of the scalar field $\phi_L$!

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## 16.3 3. Covariant Derivation of V20 Polarity-Even ($\Delta y_{\mathrm{even}}$) Background Cross-Term Cancellation

In the laboratory, the net magnetic field is the superposition of the static background field $\mathbf{B}_{\mathrm{bg}}(\mathbf{x})$ of the Earth (and laboratory environment) and the 8-coil field $\mathbf{B}_{\mathrm{coil}}(\mathbf{x};I) = I\,\mathbf{b}_{\mathrm{coil}}(\mathbf{x})$ proportional to the current $I$ (`hu_p3.tex`, line 84):

$$\rho_B(\mathbf{x};I) \;=\; \frac{|\mathbf{B}_{\mathrm{bg}}(\mathbf{x}) + I\,\mathbf{b}_{\mathrm{coil}}(\mathbf{x})|^2}{2\mu_0 c^2} \;=\; \underbrace{\frac{|\mathbf{B}_{\mathrm{bg}}(\mathbf{x})|^2}{2\mu_0 c^2}}_{\rho_{B,\mathrm{bg}}(\mathbf{x})\ (I^0)} + \underbrace{I\,\frac{\mathbf{B}_{\mathrm{bg}}(\mathbf{x})\cdot\mathbf{b}_{\mathrm{coil}}(\mathbf{x})}{\mu_0 c^2}}_{\rho_{B,\mathrm{cross}}(\mathbf{x};I)\ (I^1,\ \mathrm{odd})} + \underbrace{I^2\,\frac{|\mathbf{b}_{\mathrm{coil}}(\mathbf{x})|^2}{2\mu_0 c^2}}_{\rho_{B,\mathrm{coil}}(\mathbf{x};I)\ (I^2,\ \mathrm{even})}\,.$$

Because both the 3D top-hat ($\mathcal{K}_L^{\mathrm{TH}}$) and 3D Yukawa ($\mathcal{G}_L^{\mathrm{Yuk}}$) operators are linear in the source density, the ON ($I = \pm I_0$) minus OFF ($I = 0$) difference of the scalar potential at the oscillator position $\mathbf{x}_1 = \mathbf{0}$ is:

$$\Delta\phi_L(\pm I_0) \;\equiv\; \phi_L(\mathbf{0};\pm I_0) - \phi_L(\mathbf{0};0) \;=\; \pm I_0\,\gamma_L\bar{\rho}_{B,\mathrm{cross}}(\mathbf{0};1) \;+\; I_0^2\,\gamma_L\bar{\rho}_{B,\mathrm{coil}}(\mathbf{0};1).$$

Even if the oscillator possesses a parasitic linear Zeeman sensitivity ($\kappa_Z\,\mathbf{B}_{\mathrm{coil}}(\mathbf{0};\pm I_0) = \pm\kappa_Z I_0 \mathbf{b}_{\mathrm{coil}}(\mathbf{0})$), the **polarity-even average** of the H1 fractional frequency shift ($\Delta y(\pm I_0) \simeq -\Delta\phi_L(\pm I_0) \pm \kappa_Z I_0 b_{\mathrm{coil}}$) (`hu_p3.tex`, lines 79 and 84) is:

$$\boxed{\Delta y_{\mathrm{even}}(L) \;\equiv\; \frac{\Delta y(+I_0) + \Delta y(-I_0)}{2} \;=\; -I_0^2\,\gamma_L\bar{\rho}_{B,\mathrm{coil}}(\mathbf{0};1) \;=\; -\frac{GL^2}{c^2}\int d^3y\,K_L(\mathbf{0},\mathbf{y})\,\frac{|\mathbf{B}_{\mathrm{coil}}(\mathbf{y};I_0)|^2}{2\mu_0 c^2},}$$

from which both the linear Zeeman term $\pm\kappa_Z I_0$ and the background cross-term $\pm 2\mathbf{B}_{\mathrm{bg}}\cdot\mathbf{B}_{\mathrm{coil}}$ **cancel identically**!

---

## 16.4 4. Checks

1. **Conformal vs. Disformal Maxwell-Trace Check:**
   - Conformal metric ($\tilde{g}_{\mu\nu} = f_L^2 g_{\mu\nu}$): $\sqrt{-\tilde{g}}\tilde{g}^{\mu\alpha}\tilde{g}^{\nu\beta} = f_L^4 f_L^{-2} f_L^{-2}\sqrt{-g}g^{\mu\alpha}g^{\nu\beta} = \sqrt{-g}g^{\mu\alpha}g^{\nu\beta}$, independent of $\phi_L$ $\Longrightarrow$ $\delta S_{\mathrm{EM}}^{\mathrm{conf}}/\delta\phi_L \equiv 0$ ✓.
   - Disformal metric ($\tilde{g}_{00} = f_L^2 g_{00}$, $\tilde{g}_{ij} = g_{ij}$): $\sqrt{-\tilde{g}}\tilde{g}^{ik}\tilde{g}^{jl} = f_L\cdot 1\cdot 1\,\sqrt{-g}g^{ik}g^{jl}$ $\Longrightarrow$ $-\partial\mathcal{L}_{\mathrm{EM}}^{\mathrm{disf}}/\partial\phi_L\big|_{\mathbf{E}=\mathbf{0}} = -f_L u_B$ ✓.
2. **Polarity-Reversal ($\pm I_0$) Parity Check:**
   - $\Delta\phi_L(+I_0) + \Delta\phi_L(-I_0) = (+I_0 - I_0)\gamma_L\bar{\rho}_{B,\mathrm{cross}} + 2I_0^2\gamma_L\bar{\rho}_{B,\mathrm{coil}} = 2\,\gamma_L\bar{\rho}_{B,\mathrm{coil}}(I_0)$, which at $L = 0.630957$ m and $I_0 = 14.0$ mA reproduces the audited V20 value $\Delta y_{\mathrm{peak}} = -1.2017108\times 10^{-48}$ ✓.

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# 17 Step 16: Derivation of the V21/V23 Algebraic X-Sector Anisotropic Stress Reconstruction (Mass and Gradient Modes, V23 Eqs. 4–5) from the Bianchi Identity and Comparison with the Dynamical Yukawa Scalar Stress Tensor ($T^{\mu}_{\nu}(\phi)$)

Source file: step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step15_electromagnetic_trace_anomaly_disformal_magnetic_source_v20_en.md (step15_electromagnetic_trace_anomaly_disformal_magnetic_source_v20.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Section 1.1, lines 47–53, Eqs. 4–5 and Fig. 1(a–b); DOI: 10.5281/zenodo.23639084)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/tal_book_HU_2026-10-24.tex (Chapter 8, point 1, lines 400–408; DOI: 10.5281/zenodo.23604423) and hu_p3.tex (Chapter 8, point 1, lines 6–12)

## 17.1 1. Verbatim Citations from the V22 and V23 Manuscripts

In Steps 4 and 7 (step4_covariant_conservation_hydrostatic.md, step7_neutron_star_tov_stability.md), we derived the local covariant stress–energy tensor ($T^{(\phi)}_{\mu\nu}$) of Formulation II (yukawa_action_el), but we had not yet derived step by step **Equations (4) and (5)** of the V23 manuscript (eq:mass_X and eq:grad_X), the non-zero central limit ($p_{tX}(0) = p_{rX}(0) = -(1-f_c)(\varepsilon_c/3 + P_c)$) in Chapter 8 of the V22 monograph, and the three-tier covariant divergence residual hierarchy (conservation_Residuals.json):

1. **Equations (4)–(5) of V23 and the Three-Tier Bianchi Residual (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, Section 1.1, lines 47–53):**

   *"In a static spherically symmetric interior, if an auxiliary sector $T^{\mu}_{\nu}(X) = diag(-\varepsilon_X, P_{r,X}, P_{t,X}, P_{t,X})$ absorbs the window stress on the right-hand side of $G^{\mu}_{\nu} = \frac{8\pi G}{c^4}(T^{\mu}_{\nu,matter} + T^{\mu}_{\nu,X})$ while fluid elements satisfy $\nabla_\mu T^{\mu\nu}_{matter} = 0$ ($P' = -(\varepsilon + P)\nu'$ in geometrized units, $G = c = 1$, $\nu = \Phi_{eff}/c^2$, $g_{tt} = -e^{2\nu} = -e^{2\Phi_{eff}/c^2}$), then the Bianchi identity $\nabla_\mu G^{\mu}_{r} \equiv 0$ uniquely fixes $T^{\mu}_{\nu}(X)$ in both V21 modes:"*

$$\textbf{Mass mode } (m' = 4\pi r^2 f_L \varepsilon,\ \nu' = A_{\rm geo}): \quad \varepsilon_X = -(1-f_L)\varepsilon, \quad P_{r,X} = 0, \quad P_{t,X} = -\frac{r}{2}(1-f_L)\varepsilon\nu',$$
$$\textbf{Gradient mode } (m' = 4\pi r^2 \varepsilon,\ \nu' = f_L A_{\rm geo}): \quad \varepsilon_X = 0, \quad P_{r,X} = -(1-f_L)\Big(P + \frac{m}{4\pi r^3}\Big),\ P_{t,X} = P_{r,X} + \frac{r}{2}(P'_{r,X} + P_{r,X}\nu'),$$

(V23 Eq. 4, V23 Eq. 5)

   *"where $A_{geo}(r) = (m + 4\pi r^3 P)/[r(r-2m)]$. Symbolic verification in SymPy (conservation_Residuals.json) confirms that $\nabla_\mu T^{\mu}_{r}(matter) \equiv 0$ and $\nabla_\mu T^{\mu}_{r}(X) \equiv 0$ hold identically. In Table 1 and Fig. 1(b), evaluating relations (4)–(5) on numerical TOV grids ($L = 10\,km$, $\log_{10}\rho_{0c} = 15.0$) yields three distinct tiers (conservation_Residuals.json): (i) symbolic identities, residual exactly 0; (ii) algebraic X-sector quadrature residual $\max|\nabla_\mu T^{\mu}_{r}(X)|/|\rho_X\nu'| = 4.44\times10^{-16}$ (mass) and $/|P'_{r,X}| = 4.39\times10^{-14}$ (gradient); (iii) divergence of the numerically (spline-) differentiated background $\max|\nabla_\mu T^{\mu}_{r}|/|P'| = 2.70\times10^{-5}$ (mass) and $2.36\times10^{-5}$ (gradient), governed by the finite-difference derivative of $h(r)$, not by the action."*

2. **Central Limit of the Gradient Mode in the V22 Monograph (tal_book_HU_2026-10-24.tex, Chapter 8, point 1, lines 404–408):**

   *"The background X-sector differs between the two insertion modes:*

$$\textbf{Gradient mode:} \quad \rho_X = 0, \quad p_{rX} = -(1-f)\Big(P + \frac{m}{4\pi r^3}\Big), \quad p_{tX} = p_{rX} + \frac{r}{2}(p'_{rX} + p_{rX}\nu'),$$

   *whose central limit $p_{tX}(0) = p_{rX}(0) = -(1-f_c)(\varepsilon_c/3 + P_c)$ is generally non-zero. In the **mass-source mode**, by contrast, $\rho_X = -(1-f)\varepsilon$, $p_{rX} = 0$, and $p_{tX} = -\frac{r}{2}(1-f)\varepsilon\nu'$ ($p_{tX}(0) = 0$). The value 0.140 in the 36-configuration*

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*stress-diagnostics table is a background bookkeeping diagnostic of that configuration, which does not replace a covariant action principle (`50aa2db0` task)."*

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## 17.2 2. Exact Derivation of V23 Equations (4) and (5) from the Einstein Equations and the Bianchi Identity

In a static, spherically symmetric spacetime ($ds^2 = -e^{2\nu(r)}dt^2 + e^{2\lambda(r)}dr^2 + r^2 d\Omega^2$, in geometrized units $G = c = 1$, where $e^{-2\lambda(r)} \equiv 1 - \frac{2m(r)}{r}$), the non-vanishing mixed components of the Einstein tensor are:

$$G^t_{\ t} \;=\; -\frac{2m'(r)}{r^2}, \qquad G^r_{\ r} \;=\; -\frac{2m(r)}{r^3} \;+\; \frac{2}{r}\left(1 - \frac{2m(r)}{r}\right)\nu'(r).$$

Let us write the Einstein equations for the sum of the baryonic perfect fluid ($T^{\mu}_{\nu,\mathrm{matter}} = \mathrm{diag}(-\varepsilon, P, P, P)$) and the anisotropic $X$-sector ($T^{\mu}_{\nu,X} = \mathrm{diag}(-\varepsilon_X, P_{r,X}, P_{t,X}, P_{t,X})$):

$$G^t_{\ t} \;=\; -8\pi\big(\varepsilon + \varepsilon_X\big), \qquad G^r_{\ r} \;=\; 8\pi\big(P + P_{r,X}\big), \qquad G^{\theta}_{\ \theta} \;=\; G^{\phi}_{\ \phi} \;=\; 8\pi\big(P + P_{t,X}\big).$$

If the fluid satisfies $P'(r) = -(\varepsilon + P)\nu'(r)$, then $\nabla_\mu T^{\mu}_{\ r}(\mathrm{matter}) = P' + (\varepsilon + P)\nu' \equiv 0$. Because $\nabla_\mu G^{\mu}_{\ r} \equiv 0$ by the contracted Bianchi identity, the $X$-sector by itself must also satisfy the radial covariant conservation equation:

$$\boxed{\nabla_\mu T^{\mu}_{\ r}(X) \;=\; P'_{r,X}(r) \;+\; \big[\varepsilon_X(r) + P_{r,X}(r)\big]\nu'(r) \;+\; \frac{2}{r}\big[P_{r,X}(r) - P_{t,X}(r)\big] \;=\; 0.}$$

### 17.2.1 2.1. Derivation in the V21/V23 `mass` (Mass-Source) Mode (V23 Eq. 4)

In the `mass` mode, the mass function and lapse gradient are defined by:

$$m'(r) \;=\; 4\pi r^2 f_L(r)\,\varepsilon(r), \qquad \nu'(r) \;=\; A_{\mathrm{geo}}(r) \;\equiv\; \frac{m(r) + 4\pi r^3 P(r)}{r\big(r - 2m(r)\big)}.$$

1. **From the $(tt)$ Einstein Equation ($\varepsilon_X$):**

$$-\frac{2m'}{r^2} = -8\pi f_L \varepsilon = -8\pi(\varepsilon + \varepsilon_X) \quad\Longrightarrow\quad \varepsilon_X(r) \;=\; -\big(1 - f_L(r)\big)\,\varepsilon(r).$$

2. **From the $(rr)$ Einstein Equation ($P_{r,X}$):** Since $\nu' = A_{\mathrm{geo}}$, the expression for $G^r_{\ r}$ yields $8\pi P(r)$ directly, hence:

$$8\pi P(r) = 8\pi\big(P(r) + P_{r,X}(r)\big) \quad\Longrightarrow\quad P_{r,X}(r) \;\equiv\; 0.$$

3. **From the Radial Bianchi Identity ($P_{t,X}$):** Substituting $P_{r,X} = 0$ and $P'_{r,X} = 0$ into $\nabla_\mu T^{\mu}_{\ r}(X) = 0$ gives:

$$\varepsilon_X(r)\,\nu'(r) \;-\; \frac{2}{r}P_{t,X}(r) \;=\; 0 \quad\Longrightarrow\quad \boxed{P_{t,X}(r) \;=\; \frac{r}{2}\,\varepsilon_X(r)\,\nu'(r) \;=\; -\frac{r}{2}\big(1 - f_L(r)\big)\,\varepsilon(r)\,\nu'(r),}$$

which is precisely **V23 Equation (4)** (`eq:mass_X`), and since $\nu'(r) = \mathcal{O}(r)$ at the origin, $P_{t,X}(r) = \mathcal{O}(r^2) \xrightarrow{r\to 0} 0$.

### 17.2.2 2.2. Derivation in the V21/V23 `gradient` Mode (V23 Eq. 5) and Proof of Central Isotropy

In the `gradient` mode, the mass function and lapse gradient are defined by:

$$m'(r) \;=\; 4\pi r^2\,\varepsilon(r), \qquad \nu'(r) \;=\; f_L(r)\,A_{\mathrm{geo}}(r) \;=\; f_L(r)\,\frac{m(r) + 4\pi r^3 P(r)}{r\big(r - 2m(r)\big)}.$$

1. **From the $(tt)$ Einstein Equation ($\varepsilon_X$):**

$$-\frac{2m'}{r^2} = -8\pi\varepsilon = -8\pi(\varepsilon + \varepsilon_X) \quad\Longrightarrow\quad \varepsilon_X(r) \;\equiv\; 0.$$

2. **From the $(rr)$ Einstein Equation ($P_{r,X}$):** Substituting $\nu'(r) = f_L(r)\frac{m + 4\pi r^3 P}{r(r-2m)}$ into $G^r_{\ r} = 8\pi(P + P_{r,X})$ gives:

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$$P(r)+P_{r,X}(r) \;=\; \frac{G^{r}_{\ r}}{8\pi} \;=\; -\frac{m(r)}{4\pi r^3} \;+\; f_L(r)\left(\frac{m(r)}{4\pi r^3}+P(r)\right) \quad\Longrightarrow\quad \boxed{P_{r,X}(r) \;=\; -\big(1-f_L(r)\big)\left(P(r)+\frac{m(r)}{4\pi r^3}\right).}$$

3. **From the Radial Bianchi Identity ($P_{t,X}$):** Since $\varepsilon_X = 0$, solving $\nabla_\mu T^\mu_r(X) = P'_{r,X} + P_{r,X}\nu' + \frac{2}{r}(P_{r,X} - P_{t,X}) = 0$ for $P_{t,X}(r)$ yields:

$$\boxed{P_{t,X}(r) \;=\; P_{r,X}(r) \;+\; \frac{r}{2}\Big[P'_{r,X}(r) \;+\; P_{r,X}(r)\,\nu'(r)\Big],}$$

which is precisely **V23 Equation (5)** (`eq:grad_X`).

4. **Central Limit and Isotropy ($r \to 0$):** Near the stellar center ($r \to 0$), $m(r) = \frac{4\pi}{3}\varepsilon_c r^3 + \mathcal{O}(r^5)$, so $\lim_{r\to 0}\frac{m(r)}{4\pi r^3} = \frac{\varepsilon_c}{3}$. Because the derivative of every regular scalar profile vanishes at the origin ($P'(0) = 0$, $f'_L(0) = 0$, $\nu'(0) = 0$), we have $P'_{r,X}(r) = \mathcal{O}(r)$, so the anisotropy term $\frac{r}{2}[P'_{r,X} + P_{r,X}\nu'] = \mathcal{O}(r^2)$ vanishes at the origin, and:

$$\boxed{\lim_{r\to 0} P_{t,X}(r) \;=\; \lim_{r\to 0} P_{r,X}(r) \;=\; -\big(1-f_c\big)\left(\frac{\varepsilon_c}{3}+P_c\right) \;\neq\; 0, \qquad \lim_{r\to 0}\big[P_{t,X}(r) - P_{r,X}(r)\big] \;=\; 0.}$$

Hence no geometric conical singularity arises at the origin: although the central pressure is non-zero, the stress of the $X$-sector at the origin is **isotropic** ($P_{r,X}(0) = P_{t,X}(0)$)!

---

## 17.3 3. Comparison with the Dynamical Scalar Stress Tensor of Formulation II (`yukawa_action_el`)

In Formulation II of V23 (`yukawa_action_el`, `step4` and `step7`), the $X$-sector is determined not by algebraic reconstruction, but directly from the covariant action ($S = \int d^4x\sqrt{-g}\big[\frac{R}{16\pi} - \frac{(\nabla\phi_L)^2+\phi_L^2/L^2}{2\alpha_Y} - f_L(\phi_L)\varepsilon(n)\big]$). Relative to the standard GR source $(\varepsilon, P)$, the resulting effective $X$-sector consists of two dynamical parts:

$$T^\mu_{\nu,X,\mathrm{dyn}} \;=\; \big(T^\mu_{\nu,\mathrm{matter}} - T^\mu_{\nu,\mathrm{GR}}\big) \;+\; T^\mu_\nu(\phi),$$

that is, component by component ($B(r) = 1 - 2m(r)/r$):

$$\begin{aligned}
\varepsilon_{X,\mathrm{dyn}}(r) \;&=\; -\big(1-f_L(r)\big)\,\varepsilon(r) \;+\; \frac{1}{2\alpha_Y}\left(B(r)\big[\phi'_L(r)\big]^2 + \frac{\phi_L(r)^2}{L^2}\right),\\
P_{r,X,\mathrm{dyn}}(r) \;&=\; -\big(1-f_L(r)\big)\,P(r) \;+\; \frac{1}{2\alpha_Y}\left(B(r)\big[\phi'_L(r)\big]^2 - \frac{\phi_L(r)^2}{L^2}\right),\\
P_{t,X,\mathrm{dyn}}(r) \;&=\; -\big(1-f_L(r)\big)\,P(r) \;-\; \frac{1}{2\alpha_Y}\left(B(r)\big[\phi'_L(r)\big]^2 + \frac{\phi_L(r)^2}{L^2}\right).
\end{aligned}$$

Three deep structural relationships between Formulations I and II follow immediately:

1. **Origin of the $-(1-f_L)\varepsilon$ Term in the Mass Mode:** The first term of $\varepsilon_{X,\mathrm{dyn}}(r)$ is the exact dilution term $\varepsilon_X = -(1-f_L)\varepsilon$ of the V21/V23 `mass` mode, supplemented by the positive self-energy density $\rho_\phi = \frac{1}{2\alpha_Y}(B[\phi'_L]^2 + \phi_L^2/L^2)$ of the scalar field.

2. **Purely Gradient Origin of the Anisotropy ($\Pi_\phi \equiv P_{r,X,\mathbf{dyn}} - P_{t,X,\mathbf{dyn}}$):**

$$\boxed{\Pi_\phi(r) \;\equiv\; P_{r,X,\mathrm{dyn}}(r) - P_{t,X,\mathrm{dyn}}(r) \;=\; p_{r,\phi}(r) - p_{t,\phi}(r) \;=\; \frac{B(r)}{\alpha_Y}\big[\phi'_L(r)\big]^2 \;\geq\; 0.}$$

Because $\phi'_L(r) = \mathcal{O}(r)$ near the origin, the dynamical anisotropy also obeys $\Pi_\phi(r) = \mathcal{O}(r^2) \xrightarrow{r\to 0} 0$, so at the origin the pressure is strictly isotropic: $P_{r,X,\mathrm{dyn}}(0) = P_{t,X,\mathrm{dyn}}(0) = -(1-f_c)P_c - \frac{\phi_L(0)^2}{2\alpha_Y L^2} < 0$.

3. **Why the Euler–Lagrange Equation for $\phi_L$ Replaces Separate Bianchi Reconstruction:** Whereas in the algebraic V21 modes $P_{t,X}(r)$ had to be reconstructed a posteriori from $\nabla_\mu T^\mu_r(X) = 0$, in the `yukawa_action_el` system the Noether identity $\nabla_\mu\big(T^\mu_{r,\mathrm{matter}} + T^\mu_r(\phi)\big) \equiv 0$ proved in Step 4 (`step4_covariant_conservation_hydrostatic.md`) is satisfied automatically by virtue of the field equation for $\phi_L$!

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## 17.4 4. Checks

1. **Symbolic Bianchi Identity Check for Both V23 Algebraic Modes:**
   - **Mass mode:** $P'_{r,X} + (\varepsilon_X + P_{r,X})\nu' + \frac{2}{r}(P_{r,X} - P_{t,X}) = 0 + \varepsilon_X\nu' + \frac{2}{r}\left(0 - \frac{r}{2}\varepsilon_X\nu'\right) = \varepsilon_X\nu' - \varepsilon_X\nu' \equiv 0$ ✓.
   - **Gradient mode:** $P'_{r,X} + (0 + P_{r,X})\nu' + \frac{2}{r}\Big(P_{r,X} - \big[P_{r,X} + \frac{r}{2}(P'_{r,X} + P_{r,X}\nu')\big]\Big) = P'_{r,X} + P_{r,X}\nu' - (P'_{r,X} + P_{r,X}\nu') \equiv 0$ ✓.
2. $f_L \to 1$ **($\phi_L \to 0$) GR Limit:**
   - In the `mass` mode ($\varepsilon_X, P_{r,X}, P_{t,X} \to 0$), in the `gradient` mode ($\varepsilon_X, P_{r,X}, P_{t,X} \to 0$), and in the dynamical `yukawa_action_el` case ($\varepsilon_{X,\mathrm{dyn}}, P_{r,X,\mathrm{dyn}}, P_{t,X,\mathrm{dyn}} \to 0$), the $X$-sector vanishes identically, recovering the isotropic GR TOV equations ✓.

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# 18 Step 17: Exact Analytical Derivation of the V23 Eq. (8) Factorized Chandrasekhar–Hydrostatic Difference Identity (eq:factored_diff) and the gradient-Mode metric Inertia-Weight Indefiniteness ($\mathcal{W}_{\mathbf{metric}}(r) \leq 0$, $L \geq 367$ m)

Source file: step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefiniteness_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefiniteness_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress_en.md (step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Section 2, lines 121–126, Eq. 8 and Table 2 footnote, line 152; DOI: 10.5281/zenodo.23639084)
- V23 verification scripts: /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/covariant_closure_50aa2db0.py (lines 100–120) and radial_modes.py (lines 20–39)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/tal_book_HU_2026-10-24.tex (Chapter 8, point 2, lines 409–419; DOI: 10.5281/zenodo.23604423)

## 18.1 1. Verbatim Citations from the V22 and V23 Manuscripts

In Step 16 (step16_anisotropic_x_sector_bianchi_reconstruction_vs_yukawa_stress.md), we derived the background stress tensor $T^{\mu}_{\nu}(X)$ appearing in V23 Equations (4)–(5). Section 2 of the V23 manuscript (lines 121–126, Eq. 8, and line 152) and Chapter 8 of the V22 monograph (lines 409–418) show how this same background modification causes both frozen-$f_L$ ($\delta f_L = 0$) Sturm–Liouville operators (matter and metric) to depart from the static turning point $dM/d\rho_c = 0$:

1. **V23 Eq. (8) Factorized Chandrasekhar–Hydrostatic Difference Identity (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, Section 2, lines 121–125, Eq. 8):**

   *"To understand why both frozen-$f_L$ operators depart from the static mass peak $dM/d\rho_c = 0$, I first determined in SymPy the exact difference between the frozen-$f_L$ Chandrasekhar operator $\mathcal{L}_{SL}^{matter}[\xi]$ and the linearized hydrostatic operator $\mathcal{H}_{lin}[\xi] = (\delta P)' + (\delta\varepsilon + \delta P)\nu' + (\varepsilon + P)\delta\nu'\big|_{\delta f_L=0}$ in mass mode:"*

$$\mathcal{L}_{\mathrm{SL}}^{\mathrm{matter}}[\xi] - \sqrt{\frac{r}{r-2m}}\, e^{2\nu}\, \mathcal{H}_{\mathrm{lin}}[\xi]\Big|_{\delta f_L=0} = \frac{4\pi r^{5/2}(f_L-1)\Gamma P\varepsilon e^{2\nu}}{(r-2m)^{5/2}}\Big[(4\pi r^3 P - 3r + 7m)\xi - r(r-2m)\xi'\Big]. \qquad \text{(V23 Eq. 8)}$$

   *"Expression (8) vanishes identically for $f_L = 1$ (GR), because Chandrasekhar's integration by parts substitutes the GR Einstein identity $\lambda' + 2\nu' = 8\pi r e^{\lambda}(\varepsilon + P)$. When $f_L \neq 1$, however, $\lambda' + 2\nu' = 8\pi r e^{\lambda}(f_L\varepsilon + P)$, which generates the $(f_L - 1)$ residual term in (8) AND ALSO omits the nonlocal perturbation $\delta f_L[\delta\varepsilon, \xi]$."*

2. $\mathcal{W}(r) \leq 0$ **Inertia-Weight Indefiniteness of the metric Operator Above $L \geq 367$ m (tal_book_HU_2026-10-24.tex, line 417 and V23 Table 2 footnote, line 152):**

   – tal_book_HU_2026-10-24.tex, line 417:

   *"**Approximate metric operator:** Above $L \geq 367$ m across 12 sequences (609 models), the inertia weight $\mathcal{W}(r)$ becomes indefinite ($\mathcal{W} \leq 0$) and yields negative $\omega_0^2$ values along the selected positive-norm branch; determining physical stability remains open (50aa2db0 task)."*

   – tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, line 152:

   *"* In gradient mode at $L = 10$ km, the weight of the approximate frozen-$f_L$ metric operator becomes indefinite near the surface ($\mathcal{W}(r) \leq 0$); the static tangent model (9) uses a positive Gram weight (this is not a proof of kinetic positivity)."*

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## 18.2 2. Step-by-Step Derivation of the V23 Eq. (8) Factorized Difference Identity (eq:factored_diff)

### 18.2.1 2.1. Definition of the Two Operators in the Convention of V23 covariant_closure_50aa2db0.py (Lines 101–120)

In the notation of `covariant_closure_50aa2db0.py` (lines 103–118) and `radial_modes.py` (lines 3–7) ($G = c = 1$, $g_{tt} = -e^{\nu_C}$ where $\nu_C \equiv 2\nu$, and $e^{\lambda} \equiv \left(1 - \frac{2m}{r}\right)^{-1} = \frac{r}{r-2m}$, so that $e^{\lambda/2} = \sqrt{\frac{r}{r-2m}}$):

- The background equations in `mass` mode (`sub_mass_full`):

$$m'(r) \;=\; 4\pi r^2 f_L(r)\,\varepsilon(r), \qquad \nu'(r) \;=\; A(r) \;\equiv\; \frac{m + 4\pi r^3 P}{r(r-2m)}, \qquad P'(r) \;=\; -(\varepsilon + P)A(r), \qquad \varepsilon'(r) \;=\; -\frac{(\varepsilon+P)^2}{\Gamma P}A(r).$$

- The **frozen-$f_L$ Chandrasekhar `matter` Sturm–Liouville operator** ($\zeta(r) \equiv r^3 e^{-\nu(r)}\xi(r)$, where $\xi(r) = \Delta r/r$):

$$\mathcal{L}_{\rm SL}^{\rm matter}[\xi] \;\equiv\; -\frac{d}{dr}\left(\Pi(r)\frac{d\zeta}{dr}\right) \;-\; Q_{\rm matter}(r)\,\zeta(r),$$

where the coefficients (`covariant_closure_50aa2db0.py`, lines 106–108) are:

$$\Pi(r) \;=\; \sqrt{\frac{r}{r-2m}}\,e^{3\nu} r^{-2}\,\Gamma P, \qquad Q_{\rm matter}(r) \;=\; \sqrt{\frac{r}{r-2m}}\,e^{3\nu} r^{-2}\,(\varepsilon+P)\left(\nu'^2 + \frac{4\nu'}{r} - \frac{8\pi r\,P}{r-2m}\right).$$

- The **linearized hydrostatic operator** at $\delta f_L = 0$ (`covariant_closure_50aa2db0.py`, lines 111–117):

$$\mathcal{H}_{\rm lin}[\xi]\Big|_{\delta f_L=0} \;\equiv\; \frac{d(\delta P)}{dr} \;+\; (\delta\varepsilon + \delta P)\,A(r) \;+\; (\varepsilon+P)\,\delta A(r)\Big|_{\delta f_L=0},$$

where the Eulerian pressure, energy-density, and mass perturbations are:

$$\Delta P \;=\; -\Gamma P\,e^{\nu} r^{-2}\zeta'(r), \qquad \delta P \;=\; \Delta P - r\xi P'(r), \qquad \delta\varepsilon \;=\; \frac{\varepsilon+P}{\Gamma P}\delta P, \qquad \delta m\Big|_{\delta f_L=0} \;=\; -4\pi r^3\big(f_L\varepsilon + P\big)\xi,$$

and the linearized variation of the geometric factor is:

$$\delta A\Big|_{\delta f_L=0} \;=\; \frac{\partial A}{\partial m}\,\delta m\Big|_{\delta f_L=0} \;+\; \frac{\partial A}{\partial P}\,\delta P \;=\; \frac{r + 8\pi r^3 P}{r(r-2m)^2}\,\delta m\Big|_{\delta f_L=0} \;+\; \frac{4\pi r^2}{r-2m}\,\delta P.$$

### 18.2.2 2.2. Why Does a Residual Proportional to $(f_L - 1)$ Arise Under Integration by Parts?

Let us expand the difference $\mathcal{L}_{\rm SL}^{\rm matter}[\xi] - \sqrt{\frac{r}{r-2m}}e^{2\nu}\mathcal{H}_{\rm lin}[\xi]\big|_{\delta f_L=0}$:

1. Because $\delta P = -e^{-\nu}\sqrt{\frac{r-2m}{r}}\,\Pi(r)\zeta'(r) + r(\varepsilon+P)A\xi$, the leading second-order derivative term $-\frac{d}{dr}(\Pi\zeta')$ ($\propto \xi''(r)$) **cancels identically** against the leading term of $-\sqrt{\frac{r}{r-2m}}e^{2\nu}\frac{d(\delta P)}{dr}$!
2. When collecting the remaining first-order ($\propto \xi'(r)$) and zeroth-order ($\propto \xi(r)$) terms, the key step is the radial derivative of the metric prefactor $\sqrt{\frac{r}{r-2m}}e^{2\nu}$:

$$\frac{d}{dr}\left(\ln\left[\sqrt{\frac{r}{r-2m}}\,e^{2\nu}\right]\right) \;=\; \frac{1}{2}\lambda'(r) \;+\; 2\nu'(r) \;=\; \frac{rm'(r) - m(r)}{r(r-2m)} \;+\; \frac{2(m + 4\pi r^3 P)}{r(r-2m)} \;=\; \frac{m + 4\pi r^3\big[f_L(r)\varepsilon + 2P\big]}{r(r-2m)}.$$

In general relativity ($f_L = 1$), the numerator has $f_L\varepsilon = \varepsilon$, and likewise $(\varepsilon + P)$ appears in the mass perturbation $\delta m_{\rm GR} = -4\pi r^3(\varepsilon+P)\xi$, so the difference between the two expressions vanishes identically.

3. When $f_L(r) \neq 1$, however, a shift proportional to $(f_L - 1)\varepsilon$ survives from both the background derivative $m'(r) = 4\pi r^2 f_L\varepsilon$ and the perturbation $\delta m\big|_{\delta f_L=0} = -4\pi r^3(f_L\varepsilon + P)\xi$:

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– The coefficient of the $\xi'(r)$ term (arising from the product of the $r\xi'$ part of $\Delta P = -\Gamma P(3\xi - r\nu'\xi + r\xi')$ and the factor $\frac{1}{2}\lambda'\big|_{f_L} - \frac{1}{2}\lambda'\big|_{f_L=1} = \frac{4\pi r^2(f_L-1)\varepsilon}{r-2m}$) is:

$$\text{Coeff}(\xi') \;=\; -\sqrt{\frac{r}{r-2m}}\, e^{2\nu}\,\Gamma P\, r\cdot\frac{4\pi r^2(f_L-1)\varepsilon}{r-2m}\cdot\frac{r}{r-2m} \;=\; -\frac{4\pi r^{7/2}(f_L-1)\Gamma P\varepsilon e^{2\nu}}{(r-2m)^{3/2}}.$$

– Combining the contributions from $(3-r\nu')\Delta P$ and $\frac{\partial A}{\partial m}\big(\delta m|_{f_L} - \delta m|_{f_L=1}\big)$, the coefficient of the $\xi(r)$ term is $\frac{4\pi r^{5/2}(f_L-1)\Gamma P\varepsilon e^{2\nu}}{(r-2m)^{5/2}}(4\pi r^3 P - 3r + 7m)$. Factoring out the common prefactor $\frac{4\pi r^{5/2}(f_L-1)\Gamma P\varepsilon e^{2\nu}}{(r-2m)^{5/2}}$ yields **V23 Equation (8)** identically:

$$\boxed{\mathcal{L}_{\text{SL}}^{\text{matter}}[\xi] - \sqrt{\frac{r}{r-2m}}\, e^{2\nu}\,\mathcal{H}_{\text{lin}}[\xi]\Big|_{\delta f_L=0} \;=\; \frac{4\pi r^{5/2}(f_L-1)\Gamma P\varepsilon e^{2\nu}}{(r-2m)^{5/2}}\Big[(4\pi r^3 P - 3r + 7m)\xi - r(r-2m)\xi'\Big].}$$

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## 18.3 3. Why Does the Weight $\mathcal{W}_{\text{metric}}(r)$ of the `metric` Operator Turn Negative Near the Surface ($L \geq 367$ m, `gradient` Mode)?

By the definition in `radial_modes.py` (lines 11–12 and 30–38), the `metric` operator variant replaces the matter quantities $(\varepsilon, P)$ in the Sturm–Liouville coefficients $Q(r)$ and $\mathcal{W}(r)$ with the effective Einstein-tensor sources of the modified background ($\varepsilon_{\text{eff}} = \varepsilon + \varepsilon_X$, $P_{r,\text{eff}} = P + P_{r,X}$):

$$\mathcal{W}_{\text{metric}}(r) \;=\; e^{(3\lambda-\nu_C)/2}\, r^4\,\big[\varepsilon_{\text{eff}}(r) \;+\; P_{r,\text{eff}}(r)\big].$$

1. **In `mass` Mode (`radial_modes.py`, lines 31–32):** By V23 Eq. (4) derived in Step 16, $\varepsilon_{\text{eff}} = \varepsilon + \varepsilon_X = f_L(r)\varepsilon(r) > 0$ and $P_{r,\text{eff}} = P(r) > 0$, hence:

$$\varepsilon_{\text{eff}}^{\text{mass}}(r) + P_{r,\text{eff}}^{\text{mass}}(r) \;=\; f_L(r)\,\varepsilon(r) \;+\; P(r) \;>\; 0 \qquad \forall\, r \in [0, R_*),$$

so in `mass` mode the inertia weight remains positive everywhere.

2. **In `gradient` Mode (`radial_modes.py`, lines 33–34):** By V23 Eq. (5) derived in Step 16, $\varepsilon_{\text{eff}} = \varepsilon(r)$, whereas $P_{r,\text{eff}}(r) = P(r) + P_{r,X}(r) = f_L(r)P(r) - \big(1 - f_L(r)\big)\frac{m(r)}{4\pi r^3}$. Therefore, the effective inertial sum is:

$$\boxed{\varepsilon_{\text{eff}}^{\text{grad}}(r) + P_{r,\text{eff}}^{\text{grad}}(r) \;=\; \varepsilon(r) \;+\; f_L(r)\,P(r) \;-\; \big(1 - f_L(r)\big)\,\frac{m(r)}{4\pi r^3}.}$$

Let us examine the behavior of this expression near the stellar surface ($r \to R_*$):

– At the surface, matter pressure and energy density vanish: $\lim_{r\to R_*} P(r) = 0$ and $\lim_{r\to R_*} \varepsilon(r) = 0$.

– By contrast, the mean enclosed density ($\frac{m(R_*)}{4\pi R_*^3} = \frac{M}{4\pi R_*^3} > 0$) is strictly positive, and for any nonlocal smoothing radius $L > 0$ the surface ball reaches into the stellar interior, so $\bar{\rho}_L(R_*) > 0 \implies 1 - f_L(R_*) > 0$!

– Consequently, near the surface ($r \to R_*$) the limit of the expression is **strictly negative**:

$$\boxed{\lim_{r\to R_*}\Big[\varepsilon_{\text{eff}}^{\text{grad}}(r) + P_{r,\text{eff}}^{\text{grad}}(r)\Big] \;=\; -\big(1 - f_L(R_*)\big)\,\frac{M}{4\pi R_*^3} \;<\; 0 \qquad (\forall\, L > 0)!}$$

When $L \geq 367$ m, the factor $1 - f_L(R_*)$ and the crustal density profile are wide enough that the outermost Gauss–Legendre points of the $N = 1200$ finite-element quadrature grid (`radial_modes.py`, lines 45–54) fall inside the crustal zone where $\varepsilon(r) + f_L(r)P(r) < (1 - f_L(r))\frac{m(r)}{4\pi r^3}$, making $\mathcal{W}_{\text{metric}}(r) \leq 0$ (across 12 sequences, 609 models in `item2_radial_grid.csv`). This proves analytically that the `metric` weight indefiniteness is not a numerical artifact, but a consequence of the `metric` ansatz substituting the static background stress ($P_{r,X} < 0$) as inertial mass, whereas the genuine kinetic weights ($\mathcal{W}_\xi \propto f_L(\varepsilon + P) > 0$ and $\mathcal{W}_\phi \propto \alpha_Y^{-1} > 0$) derived in Step 12 (`step12_second_variation_coupled_pulsation_hessian.md`) from the second variation $\delta^2 S$ of the covariant action are positive-definite for all $r \in (0, R_*)$ and all $L > 0$!

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## 18.4 4. Checks

1. $f_L \to 1$ **(GR) Limit in V23 Equation (8):**
   - If $f_L(r) \equiv 1$, the factor $(f_L - 1)$ makes the right-hand side of V23 Eq. (8) vanish identically (`diff_gr_limit = 0` in line 120 of `covariant_closure_50aa2db0.py`), so $\mathcal{L}_{\rm SL}^{\rm matter}[\xi] \equiv \sqrt{\frac{r}{r-2m}} e^{2\nu} \mathcal{H}_{\rm lin}[\xi]$, and the eigenvalue crossing $\omega_0^2 = 0$ coincides with the static turning point $dM/d\rho_c = 0$ ✓.
2. **Sign-Reversal Radius ($r_{\rm sign}$) of $\mathcal{W}_{\rm metric}(r)$ in `gradient` Mode:**
   - The sign reversal occurs in the crust precisely where $\varepsilon(r_{\rm sign}) + f_L(r_{\rm sign}) P(r_{\rm sign}) = \big(1 - f_L(r_{\rm sign})\big) \frac{m(r_{\rm sign})}{4\pi r_{\rm sign}^3}$; as $L \to 0$, $1 - f_L(R_*) \to 0$ so $r_{\rm sign} \to R_*$ (pushed outside the quadrature points), whereas for $L \geq 367$ m we have $r_{\rm sign} < r_{\rm quad,max} < R_*$ ✓.

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# 19 Step 18: Exact Derivation of the V23 Eq. (9) Nonlocal Static Tangent Operator (eq:coupled_bvp), the 3D Spherical-Shell Intersection Kernel $K_{ij}$, and the $W \to \kappa W$ Inertia-Scaling Invariance

Source file: step18_nonlocal_tangent_bvp_shell_kernel_and_inertia_scaling_invariance_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step18_nonlocal_tangent_bvp_shell_kernel_and_inertia_scaling_invariance_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefiniteness_en.md (step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefiniteness.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Section 2, lines 127–153, Eq. 9 and Table 2; DOI: 10.5281/zenodo.23639084)
- V23 verification script and results: /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/covariant_closure_50aa2db0.py (lines 179–191 and 583–695) and coupled_stability_results.csv
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/tal_book_HU_2026-10-24.tex (Chapter 7, point 1, line 374 and Chapter 8, point 2, lines 409–419; DOI: 10.5281/zenodo.23604423)

## 19.1 1. Verbatim Citations from the V22 and V23 Manuscripts

In Step 17 (step17_chandrasekhar_factored_difference_eq8_and_metric_weight_indefiniteness.md), we derived V23 Equation (8) (eq:factored_diff) and the surface sign reversal $\mathcal{W}_{\rm metric}(r) \le 0$ of the gradient-mode metric operator. Immediately following this in Section 2 of the V23 manuscript comes **V23 Equation (9) (eq:coupled_bvp)**, the nonlocal kernel perturbation $K_{ij}$ aligned to the piecewise-polytrope transition radii $r_{\rm trans}$, the $W \to 4W$ inertia-scaling invariance, the first-order tangent estimate $\omega^2 \simeq -F(0)/F'(0)$, and the symmetrized Galerkin matrix $\mathbf{M}_{\rm sym} = \mathbf{D}^T \mathrm{diag}(w\,\Delta r)\mathbf{D}$:

1. **V23 Eq. (9) Coupled Static Nonlocal Tangent Model and Quasi-Dynamical BVP Test (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, Section 2, lines 127–136, Eq. 9):**

   *"I linearize the coupled system (around the V21 dense_profile background, not around the new branches) in terms of the enthalpy perturbation $\delta h(r) = \delta P/(\varepsilon + P)$ and mass perturbation $\delta m(r)$ on a grid aligned with the exact piecewise-polytrope transition radii $r_{trans}$, incorporating the nonlocal kernel perturbation operator:"*

$$\delta f_L(r_i) = -\gamma_L f_L(r_i) \sum_j K_{ij}\, \frac{\varepsilon(r_j) + P(r_j)}{c_s^2(r_j)}\, \delta h(r_j), \qquad \begin{cases} \dfrac{d(\delta h)}{dr} = -\delta\nu'[\delta h, \delta m, \delta f_L[\delta h]] + \omega_0^2\, e^{\lambda - 2\nu}\xi, \\ \dfrac{d(\delta m)}{dr} = 4\pi r^2 \left[ f_{\rm mass} \frac{\varepsilon + P}{c_s^2}\delta h + \varepsilon\, \delta f_{\rm mass}[\delta h] \right]. \end{cases} \tag{V23 Eq. 9}$$

   *"At $\omega_0^2 = 0$, system (9) is the exact tangent operator of the static equilibrium equations with respect to $\log_{10}\rho_{0c}$. As shown in Table 2 and Fig. 3, solving the coupled eigenvalue problem (9) along the SLy $L = 10$ km sequences yields a zero crossing within $\sim 7\times10^{-5}$ dex of $dM/d\rho_c = 0$ (the pure GR discretization baseline is $7.12\times10^{-5}$ dex). Because at $\omega_0^2 = 0$ the problem reduces to $\mathcal{H}_{lin}[\delta h] = 0$, this agreement verifies static nonlocal linearization rather than dynamical inertia: scaling $W \to 4W$ at $\log_{10}\rho_{0c} = 15.30427$ yields a ratio $\omega^2_{4W}/\omega^2_W = 0.2500$ ($\omega^2_W = 4.897\times10^6\ s^{-2}$) with the identical shooting function $F(0) = 0.023252$, so the zero crossing is inertia-invariant. The raw BVP $W$ matrix is not symmetric; the symmetrized Galerkin form $\mathbf{M}_{sym} = \mathbf{D}^T\mathrm{diag}(w\,\Delta r)\mathbf{D}$ is symmetric with $\min \mathrm{diag}\,\mathbf{M}_{sym} = 5.68\times10^{-9} > 0$; this is a Gram-matrix property, not a proof of positive-definiteness of the physical kinetic Hessian. The omega0_sq_sym_tangent_s2 column gives the first-order tangent estimate ($\omega^2 \simeq -F(0)/F'(0)$)."*

2. **3D Spherical-Shell Intersection Integral (tal_book_HU_2026-10-24.tex, Chapter 7, point 1, line 374):**

   *"Inside a spherically symmetric star of radius $R_*$, the exact 3D top-hat average $\bar\rho_L(r)$ at radial coordinate $r$ can be evaluated analytically over the intersection of the ball $B(\mathbf{x}, L)$ with spherical shells $r' \in [|r - L|, r + L]$, yielding $f_L(r) = \exp(-\gamma_L \bar\rho_L(r))$."*

## 19.2 2. Exact Derivation of the 3D Spherical-Shell Intersection Quadrature Kernel $K_{ij}$ and the $r_{\mathrm{trans}}$ Transition Grid

### 19.2.1 2.1. Solid-Angle Fraction of the 3D Spherical Top-Hat Shell Intersection (`build_tophat_matrix`)

Let $\mathbf{x}_i = (0, 0, r_i)$ be the query point ($r_i > 0$), and consider a concentric spherical shell of radius $r_j > 0$ and thickness $\Delta r_j$ around the stellar center. By the law of cosines, the squared Euclidean distance of a point $\mathbf{y} = (r_j \sin\theta\cos\varphi,\, r_j \sin\theta\sin\varphi,\, r_j\cos\theta)$ on the shell from $\mathbf{x}_i$ is:

$$|\mathbf{y} - \mathbf{x}_i|^2 \;=\; r_i^2 \;+\; r_j^2 \;-\; 2\, r_i\, r_j \cos\theta.$$

The condition $|\mathbf{y} - \mathbf{x}_i| \le L$ is equivalent to:

$$\cos\theta \;\ge\; \mu_{\min}(r_i, r_j; L) \;\equiv\; \frac{r_i^2 + r_j^2 - L^2}{2\, r_i\, r_j}.$$

The surface fraction of the spherical shell of radius $r_j$ falling inside the ball $B(\mathbf{x}_i, L)$ ($\mathrm{frac}(r_i, r_j; L) = \frac{1}{4\pi}\int_{|\mathbf{y}-\mathbf{x}_i|\le L} \sin\theta\, d\theta\, d\varphi = \frac{1}{2}\int_{\max(-1,\mu_{\min})}^{1} d(\cos\theta)$) splits into three geometric regimes:

1. **Fully Enclosed Shell ($r_i + r_j \le L$):** Here $r_i^2 + r_j^2 - L^2 \le -2r_ir_j \implies \mu_{\min} \le -1$, so the entire spherical shell lies inside $B(\mathbf{x}_i, L)$: $\mathrm{frac}(r_i, r_j; L) = 1$.
2. **Partially Intersected Shell ($|r_i - r_j| < L < r_i + r_j$):** Here $-1 < \mu_{\min} < +1$, so the solid-angle fraction of the spherical cap is:

$$\mathrm{frac}(r_i, r_j; L) \;=\; \frac{1 - \mu_{\min}}{2} \;=\; \frac{2r_ir_j - (r_i^2 + r_j^2 - L^2)}{4\, r_i\, r_j} \;=\; \frac{L^2 - (r_i - r_j)^2}{4\, r_i\, r_j}.$$

3. **Exterior Shell ($|r_i - r_j| \ge L$):** Here $\mu_{\min} \ge +1$, so the intersection is empty: $\mathrm{frac}(r_i, r_j; L) = 0$.

Because the volume of the $j$-th spherical shell is $4\pi r_j^2\, \Delta r_j$ and the volume of the ball $B(\mathbf{x}_i, L)$ is $V_L = \frac{4}{3}\pi L^3$, the entry of the discrete 3D top-hat kernel matrix is (`covariant_closure_50aa2db0.py`, lines 188–190):

$$K_{ij} \;=\; \frac{4\pi r_j^2\, \mathrm{frac}(r_i, r_j; L)\, \Delta r_j}{\frac{4}{3}\pi L^3} \;=\; \frac{3\, r_j^2\, \Delta r_j}{L^3} \begin{cases} 1, & r_i + r_j \le L, \\ \dfrac{L^2 - (r_i - r_j)^2}{4\, r_i\, r_j}, & |r_i - r_j| < L < r_i + r_j, \\ 0, & |r_i - r_j| \ge L. \end{cases}$$

Consequently, the first variation of the nonlocal average $\bar\rho_L(r_i) = \sum_j K_{ij}\, \varepsilon(r_j)$ and of the factor $f_L(r_i) = \exp(-\gamma_L \bar\rho_L(r_i))$ with respect to the enthalpy perturbation $\delta\varepsilon(r_j) = \frac{d\varepsilon}{dh}\big|_{r_j} \delta h(r_j) = \frac{\varepsilon(r_j) + P(r_j)}{c_s^2(r_j)} \delta h(r_j)$ gives the first part of V23 Equation (9) identically:

$$\delta f_L(r_i) \;=\; -\gamma_L\, f_L(r_i) \sum_{j=1}^{N_{\mathrm{tot}}} K_{ij}\, \frac{\varepsilon(r_j) + P(r_j)}{c_s^2(r_j)}\, \delta h(r_j).$$

### 19.2.2 2.2. Why Must Cell Boundaries Be Aligned with the Transition Radii $r_{\mathrm{trans}}$?

In piecewise-polytrope equations of state (SLy, APR4, H4), the pressure $P(h)$ and energy density $\varepsilon(h)$ are continuous across the polytrope boundaries $h_{\mathrm{tr}} \in \mathrm{EOS}.h_b$, but the adiabatic index $\Gamma_k$ and hence the squared sound speed $c_s^2(h) = \frac{dP}{d\varepsilon}$ undergo a **discontinuous jump** at the transition radii $h(r_{\mathrm{trans}}) = h_{\mathrm{tr}}$! If a quadrature cell straddled a transition radius $r_{\mathrm{trans}}$, the jump in the integrand $\frac{\varepsilon+P}{c_s^2}\delta h$ would introduce a first-order $\mathcal{O}(\Delta r)$ quadrature error, spoiling the turning-point agreement at $dM/d\rho_c = 0$. Therefore, `solve_coupled_radial_omega2` (`covariant_closure_50aa2db0.py`, lines 604–616) uses Brent's method (`brentq(lambda x: p["h"](x) - h_tr, 0.0, R)`) to locate all transition radii $0 < r_{\mathrm{trans}} < R_*$ to machine precision and embeds them as fixed cell boundaries in the grid.

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## 19.3 3. Matrix Form of the V23 Eq. (9) Coupled BVP, the $W \to \kappa W$ Scaling Invariance, and Galerkin Symmetrization

### 19.3.1 3.1. Operator Matrices (S, $\mathbf{D}_M$, $\mathbf{D}_A$, W) and the Fredholm Shooting Function $F(\omega^2)$

In geometrized units ($G = c = 1$, where $\gamma_L = L^2$), the two first-order equations of system V23 (9) for the vector of cell-center enthalpy perturbations $\delta\mathbf{h} = (\delta h(r_1), \ldots, \delta h(r_{N_{\rm tot}}))^T$ (`covariant_closure_50aa2db0.py`, lines 629–660) take the form:

1. **Linearized Mass-Source Matrix (S) and Integrated Mass Perturbation ($\delta\mathbf{m} = \mathbf{D}_M\,\delta\mathbf{h}$):**

$$\frac{d(\delta m)}{dr} \;=\; (\mathbf{S}\,\delta\mathbf{h})_i, \qquad \mathbf{S} \;=\; \begin{cases} \mathrm{diag}\big(4\pi r^2 f_L\big)\,\big[\mathbf{I} - L^2\,\mathrm{diag}(\varepsilon)\,\mathbf{K}\big]\,\mathrm{diag}\left(\dfrac{\varepsilon+P}{c_s^2}\right), & (\texttt{mass mode}), \\ \mathrm{diag}\left(4\pi r^2\dfrac{\varepsilon+P}{c_s^2}\right), & (\texttt{gradient mode}). \end{cases}$$

With the trapezoidal lower-triangular integration matrix $\big(\mathbf{M}_{\rm int}\big)_{ij} = \Delta r_j\,\Theta(i-j) + \frac{1}{2}\Delta r_i\,\delta_{ij}$, the mass-perturbation operator is $\mathbf{D}_M \equiv \mathbf{M}_{\rm int}\mathbf{S}$, i.e., $\delta\mathbf{m} = \mathbf{D}_M\,\delta\mathbf{h}$ (automatically satisfying the central boundary condition $\delta m(0) = 0$).

2. **Linearized Hydrostatic Gradient Matrix ($\mathbf{D}_A$):** Since $A(r) = \frac{m+4\pi r^3 P}{r(r-2m)}$, the partial derivatives are $c_{dm}(r) \equiv \frac{\partial A}{\partial m} = \frac{1+2rA}{r(r-2m)}$ and $c_{dh}(r) \equiv (\varepsilon+P)\frac{\partial A}{\partial P} = \frac{4\pi r^3(\varepsilon+P)}{r(r-2m)}$, so the base perturbation is $\mathbf{D}_{A,\rm base} = \mathrm{diag}(c_{dm})\,\mathbf{D}_M + \mathrm{diag}(c_{dh})$, and:

$$\delta\boldsymbol{\nu}' \;=\; \mathbf{D}_A\,\delta\mathbf{h}, \qquad \mathbf{D}_A \;=\; \begin{cases} \mathbf{D}_{A,\rm base}, & (\texttt{mass mode}), \\ \mathrm{diag}(f_L)\,\mathbf{D}_{A,\rm base} \;-\; L^2\,\mathrm{diag}(f_L A)\,\mathbf{K}\,\mathrm{diag}\left(\dfrac{\varepsilon+P}{c_s^2}\right), & (\texttt{gradient mode}). \end{cases}$$

3. **Adiabatic Mass-Conservation Relation and the Inertia Matrix (W):** For an adiabatic radial displacement, Lagrangian mass conservation $\Delta m = 0 \implies \delta m(r) = -4\pi r^2 f_{\rm mass}(\varepsilon+P)\,r\xi(r)$ ($f_{\rm mass} = f_L$ in `mass` mode and $f_{\rm mass} = 1$ in `gradient` mode), allowing the displacement to be expressed in terms of the mass perturbation:

$$-\omega^2 e^{2(\lambda-\nu)} r\xi(r) \;=\; +\omega^2\,w_{\rm inert}(r)\,\delta m(r), \qquad w_{\rm inert}(r) \;\equiv\; \frac{e^{2(\lambda-\nu)}}{4\pi r^2 f_{\rm mass}(r)\big(\varepsilon(r)+P(r)\big)} \;>\; 0.$$

Therefore, the inertia matrix is $\mathbf{W}(\kappa) \equiv \kappa\,\mathrm{diag}(w_{\rm inert})\,\mathbf{D}_M$ (where $\kappa \equiv W_{\rm scale} > 0$ is the scaling factor), and integrating the V23 Eq. (9) enthalpy equation with central normalization $\delta h(0) = 1$ yields the discrete Fredholm system (`covariant_closure_50aa2db0.py`, lines 664–665):

$$\boxed{\Big[\mathbf{I} \;+\; \mathbf{M}_{\rm int}\big(\mathbf{D}_A \;+\; \omega^2\,\mathbf{W}(\kappa)\big)\Big]\,\delta\mathbf{h}(\omega^2;\kappa) \;=\; \mathbf{1}, \qquad F(\omega^2;\kappa) \;\equiv\; \delta m(R_*;\omega^2,\kappa) \;=\; \boldsymbol{\Delta r}^T\mathbf{S}\,\delta\mathbf{h}(\omega^2;\kappa) \;=\; 0.}$$

The surface Birkhoff boundary condition $F(\omega^2;\kappa) = \delta m(R_*;\omega^2,\kappa) = 0$ is equivalent to the free-surface boundary condition $\delta m(R_*) = -4\pi R_*^3 f_{\rm mass}(\varepsilon+P)\xi(R_*)$ and $\Delta P(R_*) = 0$.

### 19.3.2 3.2. Proof of the $W \to \kappa W$ Inertia-Scaling Invariance and the First-Order Tangent Theorem

Three exact algebraic consequences follow from the Fredholm equation above:

1. **Exact Scaling Identity ($\forall\,\kappa > 0$):** Because $\mathbf{W}(\kappa) = \kappa\,\mathbf{W}(1)$, the combination $\mathbf{D}_A + \omega^2\mathbf{W}(\kappa) = \mathbf{D}_A + (\kappa\omega^2)\mathbf{W}(1)$ depends on the eigenvalue $\omega^2$ and inertia scale $\kappa$ solely through the product $\tilde{\omega}^2 \equiv \kappa\,\omega^2$:

$$\delta\mathbf{h}(\omega^2;\kappa) \;\equiv\; \delta\mathbf{h}(\kappa\omega^2;1) \quad\Longrightarrow\quad F(\omega^2;\kappa) \;\equiv\; F(\kappa\omega^2;1).$$

Consequently, if $\omega_W^2$ is the eigenvalue at $\kappa = 1$ ($F(\omega_W^2;1) = 0$), then under $\kappa$-fold inertia scaling the exact eigenvalue is:

$$\boxed{\omega_{\kappa W}^2 \;=\; \frac{\omega_W^2}{\kappa}, \qquad \frac{\omega_{4W}^2}{\omega_W^2} \;=\; \frac{1}{4} \;=\; 0.2500 \quad \text{identically!}}$$

Specifically, at $\log_{10}\rho_{0c} = 15.30427$ (`tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` / `HU`, line 135), $F(0;\kappa) \equiv F(0;1) = 0.023252$ (independent of $\kappa$), $\omega_W^2 = 4.897 \times 10^6\ \mathrm{s}^{-2}$, and $\omega_{4W}^2/\omega_W^2 = 0.2500$.

2. **Inertia Invariance of the Static Turning Point ($\omega_0^2 = 0$):** At $\omega^2 = 0$, the term $\omega^2\mathbf{W}(\kappa)$ drops out, leaving $(\mathbf{I} + \mathbf{M}_{\rm int}\mathbf{D}_A)\delta\mathbf{h}(0) = \mathbf{1}$. Because $\mathbf{D}_A$ and $\mathbf{S}$ are the exact Fréchet derivatives of the nonlocal static TOV equations

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with respect to the central enthalpy perturbation $\delta h(0)$, the solution $\delta\mathbf{h}(0)$ is proportional to the static sequence derivative $\frac{\partial h(r;\rho_{0c})}{\partial \log_{10}\rho_{0c}}$, and the shooting function is:

$$F(0) \;=\; \delta m(R_*;0) \;=\; \left(\frac{\partial h_c}{\partial \log_{10}\rho_{0c}}\right)^{-1} \frac{dM}{d\log_{10}\rho_{0c}}.$$

Therefore $F(0)=0 \iff \frac{dM}{d\rho_c}=0$, explaining why the turning-point offset drops to $\Delta\log_{10}\rho_{0c} = +6.97\times10^{-5}$ dex (`mass`, a 19.2×/58.3× improvement) and $+7.39\times10^{-5}$ dex (`gradient`, a 48.3×/59.3× improvement), matching the pure GR discretization baseline ($+7.12\times10^{-5}$ dex, V23 Table 2).

3. **First-Order Tangent Estimate (`omega0_sq_sym_tangent_s2`) and the Galerkin Gram Matrix ($\mathbf{M_{sym}}$):** First-order Taylor expansion of the root $F(\omega^2)=0$ around $\omega^2=0$ gives $F(0)+F'(0)\,\omega^2+\mathcal{O}(\omega^4)=0$, i.e. (`covariant_closure_50aa2db0.py`, lines 680–690):

$$\omega^2_{\text{tangent}} \;\simeq\; -\frac{F(0)}{F'(0)}.$$

Although the raw BVP matrix $\mathbf{W}=\mathrm{diag}(w_{\text{inert}})\mathbf{D}_M$ is not symmetric (since $\mathbf{D}_M=\mathbf{M}_{\text{int}}\mathbf{S}$ is multiplied by a lower-triangular integrator), the symmetrized Galerkin matrix of the quadratic kinetic form expressed in terms of the mass perturbations $\delta\mathbf{m}=\mathbf{D}_M\delta\mathbf{h}$ (`covariant_closure_50aa2db0.py`, lines 685–686) is:

$$\boxed{\mathbf{M}_{\text{sym}} \;\equiv\; \mathbf{D}_M^T\,\mathrm{diag}\big(w_{\text{inert}}\,\Delta r\big)\,\mathbf{D}_M \;=\; \mathbf{M}_{\text{sym}}^T.}$$

Because $w_{\text{inert}}(r_i)\,\Delta r_i>0$ in every cell $i=1,\dots,N_{\text{tot}}$ (in both `mass` and `gradient` modes, since the denominator contains the matter sum $\varepsilon+P>0$!), for any vector $\mathbf{v}\neq\mathbf{0}$ we have $\mathbf{v}^T\mathbf{M}_{\text{sym}}\mathbf{v}=\sum_{i=1}^{N_{\text{tot}}} w_{\text{inert}}(r_i)\Delta r_i\,(\mathbf{D}_M\mathbf{v})_i^2\geq 0$, with minimum diagonal entry $\min\mathrm{diag}\,\mathbf{M}_{\text{sym}}=5.68\times10^{-9}>0$. As emphasized in the V23 manuscript (lines 135 and 152), this is by construction a **Gram-matrix property** ($\mathbf{B}^T\mathbf{B}$ form with $\mathbf{B}=\mathrm{diag}(\sqrt{w_{\text{inert}}\Delta r})\mathbf{D}_M$), whereas the positive-definiteness of the genuine dynamical kinetic Hessian was derived in Step 12 (`step12_second_variation_coupled_pulsation_hessian.md`) from the second variation $\delta^2 S$ of the action.

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## 19.4 4. Checks

1. **Normalization of the 3D Spherical-Shell Intersection Kernel $K_{ij}$ ($\sum_j K_{ij}=1$):**
   - When the stellar radius satisfies $R_*>r_i+L$ and the density is homogeneous ($\varepsilon(r_j)\equiv1$), in the continuum limit:

$$\sum_j K_{ij} \;\longrightarrow\; \frac{3}{L^3}\left[\int_0^{L-r_i} r'^2\,dr' \;+\; \int_{|L-r_i|}^{L+r_i} r'^2\,\frac{L^2-(r_i-r')^2}{4\,r_i\,r'}\,dr'\right] \;=\; 1 \qquad \forall\, r_i\geq 0\ \checkmark.$$

2. **Agreement Between $\omega^2_{\text{tangent}}\simeq -F(0)/F'(0)$ and $\omega^2_{\text{BVP}}$ Near the Turning Point (`coupled_stability_results.csv`):**
   - In row 23 of `coupled_stability_results.csv` (SLy GR, $\log_{10}\rho_{0c}=15.302445$), $\omega^2_{\text{BVP}}=1.04011\times10^5$ s$^{-2}$ and $\omega^2_{\text{tangent}}=1.03992\times10^5$ s$^{-2}$ (0.018% difference); in row 56 (SLy `mass`, $L=10$ km, $\log_{10}\rho_{0c}=15.308269$), $\omega^2_{\text{BVP}}=9.84022\times10^4$ s$^{-2}$ and $\omega^2_{\text{tangent}}=9.83845\times10^4$ s$^{-2}$ (0.018% difference), confirming the accuracy of first-order tangent linearization in the neighborhood of the $\omega^2_0=0$ turning point ✓.

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## 19.5 5. Explicit Separation Between the V21 Nonlocal 3D Top-Hat Tangent Model (`step18`) and the V23 Local Curved `yukawa_action_el` Coupled Stability (`step12`, `step22`)

To distinguish the model branches unambiguously, we record the exact scope of the two stability calculations:

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| Property | Branch I-A: V21 Nonlocal 3D Top-Hat (step18, covariant_closure_50aa2db0.py) | Branch II-Fluid: V23 Curved Exterior yukawa_action_el (step12, step22, curved_yukawa_adm_and_stability.py) |
|---|---|---|
| **Equilibrium Background** | V21 tophat_mass and tophat_gradient (tp.dense_profile) | Coupled interior + curved exterior ($r \in [0, R_* + 12L]$) yukawa_action_el |
| **Perturbation Variables** | $(\delta h, \delta m)$ with nonlocal shell kernel $\delta f_L = -\gamma_L f_L K_{ij} \delta\varepsilon_j$ | Coupled $(\delta h, \delta\phi_L, \delta m)$ in the interior and curved exterior vacuum tail |
| **Mass Turning Point (SLy, $L = 10$ km)** | $\log_{10}\rho_{0c}(M_{\max}) = 15.29924$ (mass), 15.30106 (gradient) | $\log_{10}\rho_{0c}(M_{\mathrm{ADM,max}}) = 15.316404$ ($\alpha_Y = 1$), 15.307166 ($\alpha_Y = 1/3$) |
| $\omega_0^2 = 0$ **Zero Crossing** | 15.29931 (mass), 15.30113 (gradient) | 15.316410 ($\alpha_Y = 1$), 15.307170 ($\alpha_Y = 1/3$) |
| **Turning-Point Offset** $\Delta\log_{10}\rho_{0c}$ | $+6.97 \times 10^{-5}$ dex (mass), $+7.39 \times 10^{-5}$ dex (gradient) | $+5.80 \times 10^{-6}$ dex ($\alpha_Y = 1$, 2097× reduction), $+4.20 \times 10^{-6}$ dex ($\alpha_Y = 1/3$, 995× reduction; $L = 1$ km: $-6.86 \times 10^{-6}$ dex, 165×) |

**V25 Update (curved yukawa_action_el branch):** 4-component exterior BVP ($\delta\nu_R = 1.63 \times 10^{-14}$), finite-difference verified exterior ($\partial\phi_L'(R_*)/\partial m(R_*) = -1.68365 \times 10^{-4}$ km$^{-2}$) and interior ($r = 6$ km $\partial\phi_L''/\partial m = -1.72833 \times 10^{-4}$ km$^{-3}$, scalar_m_sign = -1.0 main operator) derivatives, dynamical scalar wave, GL2 root convergence ($p_{\mathrm{eff,root}} = 4.248$). SLy $L = 10$ km, $\alpha_Y = 1$: $M_{\mathrm{ADM}}$ turning point 15.316404, main dynamical $\omega_0^2 = 0$ root 15.316410 ($+5.80 \times 10^{-6}$ dex). Detailed tables: step22, Sections 4 and 7.

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# 20 Step 19: Exact Derivation of the V20 8558 kg Granite-Chamber 3D IN–OUT Differential Quadrature (Three Geometric $L$-Regimes, Top-Hat vs. Yukawa) and the Sirius B–Sirius A Common Effective-Mass Proxy (`common_background` vs. `fixedM`)

Source file: step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b_common_proxy_en.md

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**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b_common_proxy_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step18_nonlocal_tangent_bvp_shell_kernel_and_inertia_scaling_invariance_en.md (step18_nonlocal_tangent_bvp_shell_kernel_and_inertia_scaling_invariance.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Section 3 and Table 3 + footnote, lines 161–189; DOI: 10.5281/zenodo.23639084)
- V23 verification script and dataset: /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/covariant_closure_50aa2db0.py (Part 4, lines 700–885) and clock_coupling_H0_H1.csv (rows 1–13)
- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/tal_book_HU_2026-10-24.tex (Chapter 9, lines 426–455 and Chapter 10, point 1, lines 463–468; DOI: 10.5281/zenodo.23604423)

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## 20.1 1. Verbatim Citations from the V22 and V23 Manuscripts

In Step 8 (step8_microscopic_clock_coupling_siriusb_lab_v20.md), we derived the Dirac–Maxwell atomic-clock action ($S_{\text{atomic}}[\tilde{g}_{\mu\nu}, \psi_e, A_\mu]$) under H0 ($\tilde{g}_{\mu\nu} = g_{\mu\nu}$) and conditional H1 ($\tilde{g}_{00} = f_L^2 g_{00}$). Table 3 of Section 3 of the V23 manuscript (lines 170–189) and Part 4 of covariant_closure_50aa2db0.py (lines 700–885), however, contain two additional exact mathematical structures that we now derive step by step:

1. **Footnote to V23 Table 3 on the Granite-Chamber IN–OUT Differential Signal and the Sirius B Common Effective-Mass Proxy (tal_covariant_closure_50aa2db0_EN_2026-10-24.tex / HU, line 188):**

   *"V20 rows: H0 column = TAL excess over Newtonian GR at the IN ($\mathbf{r} = \mathbf{0}$) position; H1 columns = IN shift (IN−OUT differential signal, $D_{out} = 2.0\,m$). Newtonian GR: $\Phi_N(IN)/c^2 = -9.269 \times 10^{-24}$, IN−OUT $= -6.094 \times 10^{-24}$; $\sigma_y = 10^{-18}$ is an illustrative reference value. On the Sirius B common effective-mass proxy (fixed $n = 3$ Lane–Emden profile with the H0 effective-mass shift, not a newly solved equilibrium) at $L = 10^3$ km: H0 80.69839, H1 top-hat 80.69878 km/s."*

2. **Geometry and Mass Budget of the 8558 kg Granite Chamber (tal_book_HU_2026-10-24.tex, Chapter 10, point 1, line 466):**

   *"**Exact Geometry and Mass Budget:** The open cylindrical barrel of dimensions $R_{in} = 0.25\,m$, $R_{out} = 0.85\,m$, $H = 1.20\,m$ has volume $V_{cyl} = \pi(0.85^2 - 0.25^2) \times 1.20 = 2.48814\,m^3$ and mass $M_{cyl} = 6842.39\,kg$; the nominal $V_{tot} = 3.1102\,m^3$ ($M_{tot} = 8558\,kg$) includes an additional $0.6220\,m^3$ (1710.6 kg) of endcap/support granite elements (or an outer radius $R_{out} = 0.942\,m$)."*

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## 20.2 2. 3D IN–OUT Quadrature of the V20 8558 kg Granite Chamber and Derivation of the Three Geometric $L$-Regimes

### 20.2.1 2.1. Domain of the Cylindrical Barrel + Endcaps ($\Omega_{\text{granite}}$) and the Off-Axis Distance Function

In covariant_closure_50aa2db0.py (lines 802–822), the granite chamber of density $\rho_{\text{granite}} = 2750\ \text{kg/m}^3$ and mass $M_{\text{tot}} = 8558$ kg ($V_{\text{tot}} = M_{\text{tot}}/\rho_{\text{granite}} = 8558/2750 = 3.112000\ \text{m}^3$) is constructed in cylindrical coordinates ($\varrho \in [0, R_{\text{out}}]$, $z \in [-z_{\text{out}}, +z_{\text{out}}]$, $\varphi \in [0, 2\pi)$), exploiting $z \leftrightarrow -z$ mirror symmetry, as the union of:

- **Open Cylindrical Barrel:** $R_{\rm in} = 0.25\ \text{m} \le \varrho \le R_{\rm out} = 0.85\ \text{m}$, $|z| \le z_{\rm in} = 0.60\ \text{m}$ ($H_{\rm in} = 2z_{\rm in} = 1.20\ \text{m}$), whose volume and mass are:

$$V_{\rm cyl} \;=\; \pi\big(R_{\rm out}^2 - R_{\rm in}^2\big)\,(2z_{\rm in}) \;=\; 2.488141\ \text{m}^3, \qquad M_{\rm cyl} \;=\; \rho_{\rm granite}\,V_{\rm cyl} \;=\; 6842.39\ \text{kg}.$$

- **Two Solid Endcaps:** $0 \le \varrho \le R_{\rm out} = 0.85\ \text{m}$, $z_{\rm in} \le |z| \le z_{\rm out}$, where the endcap thickness $\Delta z_{\rm cap} = z_{\rm out} - z_{\rm in}$ (`covariant_closure_50aa2db0.py`, line 808) is:

$$\Delta z_{\rm cap} \;=\; \frac{V_{\rm tot} - V_{\rm cyl}}{2\pi R_{\rm out}^2} \;=\; \frac{3.112000 - 2.488141}{2\pi \times 0.85^2} \;=\; 0.137426\ \text{m} \quad\Longrightarrow\quad z_{\rm out} \;=\; 0.60 + 0.137426 \;=\; 0.737426\ \text{m}.$$

Thus, the characteristic domain $\Omega_{\rm granite}$ occupied by granite on the half-meridian plane ($\varrho \in [0, R_{\rm out}]$, $z \in [0, z_{\rm out}]$) is:

$$\Omega_{\rm granite} \;=\; \big\{(\varrho, z) \in [0, R_{\rm out}] \times [0, z_{\rm out}] \;:\; (\varrho \ge R_{\rm in}) \;\vee\; (z \ge z_{\rm in})\big\}, \qquad dV \;=\; 2 \cdot \big(2\pi\varrho\, d\varrho\, dz\big),$$

where the factor of 2 accounts for the sum over the $\pm z$ half-spaces (giving $M_{\rm granite,num} = 8558.23$ kg on an $N_\varrho \times N_z = 500 \times 500$ grid).

The Euclidean distance of a source point $(\varrho, \varphi, \pm z) \in \Omega_{\rm granite}$ from the two clock positions (`covariant_closure_50aa2db0.py`, lines 820 and 834) is:

1. **IN Position ($\mathbf{r_{in}} = (0,0,0)$, Geometric Center of the Chamber):** independent of azimuth:

$$d_{\rm in}(\varrho, z) \;=\; \sqrt{\varrho^2 + z^2}.$$

2. **OUT Position ($\mathbf{r_{out}} = (D_{\rm out}, 0, 0)$, $D_{\rm out} = 2.0$ m in the Midplane):** by the law of cosines:

$$d_{\rm out}(\varrho, z, \varphi) \;=\; \sqrt{\varrho^2 + D_{\rm out}^2 - 2\varrho D_{\rm out}\cos\varphi + z^2},$$

integrated over $N_\varphi = 256$ uniform azimuthal points ($\varphi_k = \frac{2\pi(k+1/2)}{256}$, $dm_p = dm_g / N_\varphi$). From this, the Newtonian gravitational potential and standard GR redshift difference (`clock_coupling_H0_H1.csv`, rows 8–13) are:

$$\frac{\Phi_N(\text{IN})}{c^2} \;=\; -\frac{G}{c^2}\int_{\Omega_{\rm granite}} \frac{dm_g}{d_{\rm in}} \;=\; -9.268639 \times 10^{-24}, \qquad \frac{\Phi_N(\text{OUT})}{c^2} \;=\; -\frac{G}{c^2}\int_{\Omega_{\rm granite}} \frac{dm_p}{d_{\rm out}} \;=\; -3.174686 \times 10^{-24},$$

$$\Delta y_{\rm GR}(\text{IN} - \text{OUT}) \;=\; \frac{\Phi_N(\text{IN}) - \Phi_N(\text{OUT})}{c^2} \;=\; -6.093953 \times 10^{-24}.$$

### 20.2.2 2.2. Exact Proof of the Four Geometric Threshold Distances and Top-Hat vs. Yukawa IN–OUT Behavior

The extremal Euclidean distances of the chamber domain $\Omega_{\rm granite}$ from the IN and OUT points are:

- **At the IN Position ($\mathbf{r_{in}} = 0$):**

$$d_{\rm min}^{\rm IN} \;=\; \min_{(\varrho,z)\in\Omega_{\rm granite}} \sqrt{\varrho^2 + z^2} \;=\; \min(R_{\rm in}, z_{\rm in}) \;=\; \min(0.25,\, 0.60) \;=\; 0.25\ \text{m},$$

$$d_{\rm max}^{\rm IN} \;=\; \max_{(\varrho,z)\in\Omega_{\rm granite}} \sqrt{\varrho^2 + z^2} \;=\; \sqrt{R_{\rm out}^2 + z_{\rm out}^2} \;=\; \sqrt{0.85^2 + 0.737426^2} \;=\; 1.12530\ \text{m}.$$

- **At the OUT Position ($D_{\rm out} = 2.00$ m):**

$$d_{\rm min}^{\rm OUT} \;=\; \min_{\Omega_{\rm granite}} d_{\rm out} \;=\; D_{\rm out} - R_{\rm out} \;=\; 2.00 - 0.85 \;=\; 1.15\ \text{m} \quad (\varphi = 0,\ \varrho = R_{\rm out},\ z = 0),$$

$$d_{\rm max}^{\rm OUT} \;=\; \max_{\Omega_{\rm granite}} d_{\rm out} \;=\; \sqrt{(D_{\rm out} + R_{\rm out})^2 + z_{\rm out}^2} \;=\; \sqrt{2.85^2 + 0.737426^2} \;=\; 2.94385\ \text{m} \quad (\varphi = \pi,\ \varrho = R_{\rm out},\ z = z_{\rm out}).$$

From these four threshold distances ($0.25\ \text{m} < 1.1253\ \text{m} < 1.15\ \text{m} < 2.9439\ \text{m}$), the sharp physical contrast between the 3D spherical top-hat ($\phi_L^{\text{top-hat}}(\mathbf{r}) = \frac{3G}{4\pi c^2 L}\int_{d\le L} dm$) and the local Yukawa kernel ($\phi_L^{\rm Yuk}(\mathbf{r}) = \frac{G}{4\pi c^2}\int \frac{e^{-d/L}}{d}\, dm$ at $\alpha_Y = 1$) follows immediately (`clock_coupling_H0_H1.csv`, rows 8–13):

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1. **Cavity Regime ($L < d_{\min}^{\mathbf{IN}} = 0.25$ m, e.g. $L = 0.20$ m):** Both balls $B(\mathbf{0}, L)$ and $B(\mathbf{r}_{\text{out}}, L)$ remain outside the granite walls (inside the inner cavity and outside the chamber, respectively), so $\phi_L^{\text{top-hat}}(\text{IN}) = \phi_L^{\text{top-hat}}(\text{OUT}) = 0$, giving $\Delta y_{\text{H1}}^{\text{top-hat}} = 0$. By contrast, the Yukawa Green's function $e^{-d/L}/d$ senses the walls across the cavity: $\Delta y_{\text{H1}}^{\text{Yuk}}(\text{IN}) = -3.690 \times 10^{-26}$ and $\Delta y_{\text{H1}}^{\text{Yuk}}(\text{IN} - \text{OUT}) = -3.683 \times 10^{-26}$.

2. **Purely Interior Top-Hat Regime ($d_{\min}^{\mathbf{IN}} < L < d_{\min}^{\mathbf{OUT}} = 1.15$ m, e.g. $L = 0.50,\ 0.85,\ 1.00$ m in V23 Table 3):** Because $L < 1.15$ m, the OUT ball $B(\mathbf{r}_{\text{out}}, L)$ **does not intersect the granite chamber** ($d_{\text{out}} \geq 1.15\ \text{m} > L$), so:

$$\boxed{\phi_L^{\text{top-hat}}(\text{OUT}) \equiv 0 \quad \Longrightarrow \quad \Delta y_{\text{H1}}^{\text{top-hat}}(\text{IN} - \text{OUT}) \equiv \Delta y_{\text{H1}}^{\text{top-hat}}(\text{IN}) \qquad \forall\, L < 1.15\ \text{m}!}$$

This is why V23 Table 3 (lines 183–185) has the exact same number before and inside the parentheses in the Top-Hat column: $-3.318 \times 10^{-25}$ ($-3.318 \times 10^{-25}$) ($L = 0.50$ m), $-1.303 \times 10^{-24}$ ($-1.303 \times 10^{-24}$) ($L = 0.85$ m), and $-1.436 \times 10^{-24}$ ($-1.436 \times 10^{-24}$) ($L = 1.00$ m). By contrast, the exponential tail of the Yukawa kernel is non-zero even at $D_{\text{out}} = 2.0$ m ($\phi_L^{\text{Yuk}}(\text{OUT}) = +3.735 \times 10^{-26}$ at $L = 1.00$ m), so the Yukawa differential signal is smaller in magnitude than the absolute IN signal: $-3.781 \times 10^{-25}$ ($-3.407 \times 10^{-25}$) at $L = 1.00$ m!

3. **Partial Exterior Overlap ($d_{\min}^{\mathbf{OUT}} < L < d_{\max}^{\mathbf{OUT}}$, e.g. $L = 1.50$ m):** Here the IN ball already encloses the entire chamber ($L = 1.50\ \text{m} > d_{\max}^{\text{IN}} = 1.1253$ m), while the OUT ball partially intersects the chamber: $\Delta y_{\text{H1}}^{\text{top-hat}}(\text{IN}) = -1.0115 \times 10^{-24}$, $\Delta y_{\text{H1}}^{\text{top-hat}}(\text{OUT}) = -1.0064 \times 10^{-25}$, yielding $\Delta y_{\text{H1}}^{\text{top-hat}}(\text{IN} - \text{OUT}) = -9.1086 \times 10^{-25}$.

4. **Common Enclosure Regime ($L \geq d_{\max}^{\mathbf{OUT}} = 2.94385$ m, e.g. $L = 10.0$ m):** When $L \geq 2.944$ m, both balls $B(\mathbf{0}, L)$ and $B(\mathbf{r}_{\text{out}}, L)$ enclose the entire $M_{\text{tot}} = 8558$ kg granite chamber! Consequently, the 3D spherical top-hat average at both positions is $\bar{\rho}_L^{\text{top-hat}}(\text{IN}) = \bar{\rho}_L^{\text{top-hat}}(\text{OUT}) = \frac{3M_{\text{tot}}}{4\pi L^3}$, so the **Top-Hat IN−OUT differential signal vanishes identically**:

$$\boxed{\Delta y_{\text{H1}}^{\text{top-hat}}(\text{IN} - \text{OUT})\Big|_{L \geq 2.944\ \text{m}} = 0 \quad (\text{numerically } -5.28 \times 10^{-40} \text{ rounding noise at } L = 10\ \text{m}),}$$

whereas the Yukawa kernel, owing to its $1/d$ weighting, still produces a distinct differential signal at $L = 10.0$ m: $\Delta y_{\text{H1}}^{\text{Yuk}}(\text{IN} - \text{OUT}) = -4.818 \times 10^{-25}$!

5. **Second-Order H0 TAL Excess in the Cavity (`dy_H0_cavity`):** Under H0 there is no direct $f_L$ clock prefactor, so the excess over Newtonian GR arises from the $1 - f_L \simeq \phi_L$ modification of the effective gravitating mass of the chamber (`covariant_closure_50aa2db0.py`, line 852):

$$\Delta y_{\text{H0}}^{\text{excess}}(\text{IN}) = \left|\frac{\Phi_N(\text{IN})}{c^2}\right| \phi_L^{\text{top-hat}}(\text{IN}) = 9.268639 \times 10^{-24} \times 1.436124 \times 10^{-24} = +1.331091 \times 10^{-47} \quad (L = 1.00\ \text{m}).$$

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## 20.3 3. Derivation of the Sirius B–Sirius A Differential Redshift and the `common_background` vs. `fixedM` Proxy

In `covariant_closure_50aa2db0.py` (lines 710–799), the white dwarf Sirius B ($M_B = 1.018 \pm 0.011\, M_\odot$, $R_B = 0.00803 \pm 0.00011\, R_\odot = 5586.471$ km) and the main-sequence reference star Sirius A ($M_A = 2.042\, M_\odot$, $R_A = 1.711\, R_\odot = 1.19034 \times 10^6$ km) are modeled as $n = 3$ Lane–Emden polytropes:

1. **GR Baseline and Combined Uncertainty (`covariant_closure_50aa2db0.py`, lines 719–723):** The surface compactness is $\mathcal{C}_B = \frac{GM_B}{R_B c^2} = 2.690878 \times 10^{-4}$, giving the exact Schwarzschild redshift and velocity equivalent:

$$z_{\text{GR},B} = \left(1 - 2\mathcal{C}_B\right)^{-1/2} - 1 = 2.6918805 \times 10^{-4}, \qquad v_{\text{GR}} = c\, z_{\text{GR},B} = 80.700550\ \text{km/s}.$$

The parameter uncertainty is $\sigma_{\text{GR}} = v_{\text{GR}} \sqrt{(\sigma_M/M_B)^2 + (\sigma_R/R_B)^2} = 1.4080$ km/s, which combined in quadrature with the observational uncertainty $\sigma_{\text{obs}} = 0.77$ km/s yields $\sigma_{\text{comb}} = \sqrt{\sigma_{\text{obs}}^2 + \sigma_{\text{GR}}^2} = 1.604806$ km/s (`clock_coupling_H0_H1.csv`, row 2).

2. **Sirius B–Sirius A Differential H1 Clock Shift (`covariant_closure_50aa2db0.py`, lines 767–773):** Because the astrometric gravitational redshift is measured from the spectral-line difference between Sirius B and Sirius A, and under H1 the surface redshift is $1 + z_{\text{H1}} = (1 + z_{\text{metric}})\, f_L(R_*)^{-1} = (1 + z_{\text{metric}})\, e^{+x_*}$ (where $x_* \equiv \gamma_L \bar{\rho}_L(R_*)$), the direct H1 excess velocity between the two stars is:

$$\boxed{\Delta v_{\text{H1,extra}}(L) = \Delta v_B(L) - \Delta v_A(L) = c\,(1 + z_{\text{GR},B})\left(e^{x_B(L)} - 1\right) - c\left(e^{x_A(L)} - 1\right).}$$

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- At $L = 1000$ km $= 10^6$ m: $\Delta v_{\mathrm{H1,extra}}^{\text{top-hat}} = +0.000392$ km/s and $\Delta v_{\mathrm{H1,extra}}^{\mathrm{Yuk}} = +0.044215$ km/s.
- At $L = 5000$ km $= 5 \times 10^6$ m: $\Delta v_{\mathrm{H1,extra}}^{\text{top-hat}} = +4.688557$ km/s ($v_{\mathrm{H1}}^{\text{top-hat}} = 85.38911$ km/s, $2.95\sigma$) and $\Delta v_{\mathrm{H1,extra}}^{\mathrm{Yuk}} = +2.150821$ km/s ($v_{\mathrm{H1}}^{\mathrm{Yuk}} = 82.85137$ km/s, $1.37\sigma$).

3. **Difference Between Fixed-Mass (`fixedM`) and the Common Effective-Mass Proxy (`common_background`) (`covariant_closure_50aa2db0.py`, lines 758–766 and 794–798; V23 line 188):**
   - If the gravitating mass $M_B = 1.018\, M_\odot$ of the star is held fixed from orbital astrometry (`fixedM`), the exterior Schwarzschild metric is unchanged, so $v_{\mathrm{H0,fixedM}} \equiv v_{\mathrm{GR}} = 80.70055$ km/s, and $v_{\mathrm{H1,fixedM}}(L) = v_{\mathrm{GR}} + \Delta v_{\mathrm{H1,extra}}(L)$ (80.70094 km/s at $L = 10^3$ km).
   - If instead one evaluates the `mass`-mode effective mass reduction at a fixed $n = 3$ Lane–Emden baryonic profile $\rho_B(r)$ (`common_background` proxy):

$$\frac{\Delta M_{\mathrm{eff}}(L)}{M_B} \;=\; \frac{\int_0^{R_B} 4\pi r^2 \rho_B(r)\,\big[f_L(r) - 1\big]\,dr}{\int_0^{R_B} 4\pi r^2 \rho_B(r)\,dr} \;<\; 0,$$

then the H0 prediction is $v_{\mathrm{H0,mass}}(L) = v_{\mathrm{GR}}\big(1 + \frac{\Delta M_{\mathrm{eff}}(L)}{M_B}\big)$ (80.69839 km/s at $L = 10^3$ km and 80.69487 km/s at $L = 5 \times 10^3$ km), while the H1 value built on that same common effective-mass proxy is:

$$\boxed{v_{\mathrm{H1,common}}(L) \;=\; v_{\mathrm{H0,mass}}(L) \;+\; \Delta v_{\mathrm{H1,extra}}(L) \;=\; 80.698385 + 0.000392 \;=\; 80.69878 \text{ km/s} \quad (\text{at } L = 10^3 \text{ km, Top-Hat}).}$$

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## 20.4 4. Checks

1. **Ratio of Top-Hat and Yukawa $\phi_L$(IN) at Large $L$ ($L = 10.0$ m $\gg d_{\max}^{\mathrm{IN}} = 1.125$ m):**
   - When $L \gg d_{\max}^{\mathrm{IN}}$, the entire mass $M_{\mathrm{tot}}$ lies inside the ball of volume $\frac{4}{3}\pi L^3$ in the 3D spherical top-hat average, so $\phi_L^{\text{top-hat}}(\mathrm{IN}) = \frac{GL^2}{c^2}\frac{3M_{\mathrm{tot}}}{4\pi L^3} = \frac{3GM_{\mathrm{tot}}}{4\pi c^2 L} = 1.517259 \times 10^{-25}$ (at $L = 10$ m), whereas in the Yukawa integral $e^{-d_{\mathrm{in}}/L} \approx 1 - d_{\mathrm{in}}/L$, yielding:

$$\phi_L^{\mathrm{Yuk}}(\mathrm{IN})\Big|_{L \gg d_{\max}^{\mathrm{IN}}} \;=\; \frac{G}{4\pi c^2}\int_{\Omega_{\mathrm{granite}}} \frac{e^{-d_{\mathrm{in}}/L}}{d_{\mathrm{in}}}\,dm_g \;\approx\; \frac{1}{4\pi}\left|\frac{\Phi_N(\mathrm{IN})}{c^2}\right| - \frac{GM_{\mathrm{tot}}}{4\pi c^2 L} \;=\; 7.3757 \times 10^{-25} - 0.5058 \times 10^{-25} \;\approx\; 6.87 \times 10^{-25},$$

in close agreement with the exact quadrature value ($6.888137 \times 10^{-25}$) in row 13 of `clock_coupling_H0_H1.csv` ✓.

2. **Verification of the Sirius B `common_background` vs. `fixedM` Difference at $L = 10^3$ km:**
   - $v_{\mathrm{H1,fixedM}}^{\text{top-hat}} - v_{\mathrm{H1,common}}^{\text{top-hat}} = 80.70094147 - 80.69877704 = +0.00216443$ km/s $\equiv v_{\mathrm{GR}} - v_{\mathrm{H0,mass}}$ ($80.70054975 - 80.69838532 = +0.00216443$ km/s) to machine precision ✓.

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# 21 Step 20: Exact Derivation of the V20 Eight-Coil Elliptic Biot–Savart Loop Integral, the $r = L/t$ Compactified Exterior Quadrature ($3.7\times10^{-16}$ / $7.4 \times 10^{-16}$ Agreement), and the $L = 0.95$ m Switching Identity

Source file: step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_quadrature_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-26 **File Path:** scalar-tensor-yukawa-fit/step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_quadrature_en.md **Previous Steps:** step1_dilaton_form_en.md (step1_dilaton_form.md) – step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b_common_proxy_en.md (step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b_common_proxy.md) **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/tal_book_HU_2026-10-24.tex (Chapter 10, point 2, lines 470–485 and Appendix, lines 557–618; DOI: 10.5281/zenodo.23604423)
- V20 (10.5281/zenodo.23540741) verification script and dataset: /workspace/ta-v18/V20/make_revision3_checks.py (lines 1–77) and /workspace/ta-v18/V20/revision3_checks.json (lines 123–197)
- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (line 28; DOI: 10.5281/zenodo.23639084)

## 21.1 1. Verbatim Citations from the V22 and V20 Manuscripts

In Step 15 (step15_electromagnetic_trace_anomaly_disformal_magnetic_source_v20.md), we derived why the trace-free Maxwell tensor generates a source $\rho_B(\mathbf{x}) = \frac{|\mathbf{B}(\mathbf{x})|^2}{2\mu_0 c^2}$ in the V23 disformal metric, and in Step 19 we closed the 3D IN–OUT quadrature of V20 Branch "A" (8558 kg granite chamber). Point 2 of Chapter 10 (lines 470–485) and the Appendix (lines 557–618) of the V22 monograph, together with V20 make_revision3_checks.py (lines 25–75), contain the exact elliptic-integral magnetic-field evaluation of V20 Branch "B" (the eight-coil magnetic-energy experiment), the two exterior quadrature paths (Path A and Path B), and the $L = 0.95$ m interior/exterior switching identity:

1. **Parameters of the 8-Coil Arrangement and the $L = 0.630957$ m Grid Maximum (tal_book_HU_2026-10-24.tex, Chapter 10, point 2, lines 473–478):**

   *"**Audited 8-Coil Hardware Parameters:** 8 identical solenoids, each of length $l_{coil} = 400\,mm$, outer structural ⌀300 **mm flange diameter**, effective ⌀291.7 **mm copper winding diameter** ($R_{coil} = 0.14585\,m$), $N = 2000$ turns of 0.15 mm enameled copper wire, DC resistance $R_{DC} = 1789\,\Omega$ per coil, driven by a current $I = 14.0\,mA$ (0.351 W/coil, 2.805 W in all)."*

   *"**Maximum on the Sampled 25-Point $L$-Grid ($L = 0.630957$ m):** Evaluating the magnetic energy density of the 8 coils on the 25-point logarithmic grid $L \in [0.1, 3.0]\,m$, the conditional polarity-even shift $\Delta y(L) = \frac{1}{2}[\Delta y(+I) + \Delta y(-I)]$ attains its discrete maximum on the grid at $L = 0.630957$ **m** (not a continuous extremum fit):*

   $$\Delta y_{\text{grid max}}(L = 0.630957\,\text{m}) \;=\; -1.2017108 \times 10^{-48}. \qquad \text{(V22 Eq. 14)}$$

   *Here I separate: (i) the $3.7 \times 10^{-16}$ agreement of the two deterministic integration routines at $L = 3.0\,m$, (ii) the $-0.0722\%$ Sobol quadrature cross-check, (iii) the $1.23 \times 10^{-3}$ matching residual at $L = 0.95\,m$, and (iv) the actual measurement sensitivity."*

2. **The $r = L/t$ Compactified Exterior Integral (Paths A and B) and the $L = 0.95$ m Interior/Exterior Switch (tal_book_HU_2026-10-24.tex, Appendix, lines 581 and 615):**

   – Line 581:

   *"The new Path A uses panel integration plus an analytic tail to infinity, while Path B uses the direct substitution $r = L/t$. In the 2.6–3.01 m range, the targeted rerun shows the two paths to agree to a **relative precision of about** $10^{-15}$."*

   – Line 615:

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*"**B. The new, smaller matching residual is reported and genuine.** The report already notes that at $L = 0.95\,m$ the interior/exterior switch is not strictly continuous. The archived values are 2.17738947 **and** 2.17597607 $\boldsymbol{J/A^2}$, with difference 0.00141340 $\boldsymbol{J/A^2}$, i.e. 0.0649% of the interior value. This agrees algebraically with the difference of the total-energy estimates; it is not a persistence of the old large 3 m error."*

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## 21.2 2. Exact Elliptic Biot–Savart Field of Circular Current Loops and the 8-Coil Superposition

### 21.2.1 2.1. Off-Axis Magnetic Field of a Single Circular Current Loop of Radius $a = D_{\mathrm{turn}}/2$ (`make_revision3_checks.py`, Lines 30–35)

Consider a circular current loop of radius $a = D_{\mathrm{turn}}/2 = 0.2917/2 = 0.14585$ m lying in the $xy$-plane and carrying current $I$. In cylindrical coordinates ($\varrho = \sqrt{x^2+y^2}$, $z$), the azimuthal vector potential $A_\varphi(\varrho, z)$ from the Biot–Savart law is:

$$A_\varphi(\varrho, z) \;=\; \frac{\mu_0 I\, a}{4\pi} \int_0^{2\pi} \frac{\cos\phi'\, d\phi'}{\sqrt{a^2+\varrho^2+z^2-2a\varrho\cos\phi'}}.$$

With the substitution $\phi' = \pi - 2\theta$ ($\cos\phi' = 2\sin^2\theta - 1$) and the elliptic parameter (`make_revision3_checks.py`, lines 31–32):

$$m(\varrho, z; a) \;\equiv\; \frac{4a\varrho}{(a+\varrho)^2+z^2} \;\in\; [0,1), \qquad d(\varrho, z; a) \;\equiv\; \sqrt{(a+\varrho)^2+z^2}, \qquad q(\varrho, z; a) \;\equiv\; (a-\varrho)^2+z^2,$$

the components of $\mathbf{B} = \nabla\times(A_\varphi\hat{\boldsymbol{\varphi}}) = -\frac{\partial A_\varphi}{\partial z}\hat{\boldsymbol{\varrho}} + \frac{1}{\varrho}\frac{\partial(\varrho A_\varphi)}{\partial \varrho}\hat{\mathbf{z}}$ can be expressed in closed form in terms of the elliptic integrals of the first and second kind, $K(m) = \int_0^{\pi/2}(1-m\sin^2\theta)^{-1/2}d\theta$ and $E(m) = \int_0^{\pi/2}(1-m\sin^2\theta)^{1/2}d\theta$ (`make_revision3_checks.py`, lines 33–34):

$$\boxed{\begin{aligned} B_z(\varrho, z; a, I) \;&=\; \frac{\mu_0 I}{2\pi\, d(\varrho, z; a)}\left[K(m) \;+\; \frac{a^2-\varrho^2-z^2}{q(\varrho, z; a)}\,E(m)\right], \\ B_\varrho(\varrho, z; a, I) \;&=\; \frac{\mu_0 I\, z}{2\pi\,\varrho\, d(\varrho, z; a)}\left[-K(m) \;+\; \frac{a^2+\varrho^2+z^2}{q(\varrho, z; a)}\,E(m)\right] \qquad (\varrho>0,\;\; B_\varrho(0,z)=0). \end{aligned}}$$

### 21.2.2 2.2. Superposition of the $N_c = 8\times N = 2000$ Loops and the Derby–Olbert Current-Sheet Limit (`make_revision3_checks.py`, Lines 25–51)

In the V20 layout, the centers of the $N_c = 8$ solenoids lie on a circle of radius $R_{\mathrm{ax}} = 0.45$ m at vertical coordinate $z_c = 0.200 - 0.300 = -0.100$ m:

$$\mathbf{c}_k \;=\; \left(R_{\mathrm{ax}}\cos\frac{2\pi k}{8},\; R_{\mathrm{ax}}\sin\frac{2\pi k}{8},\; z_c\right), \qquad k = 0, 1, \ldots, 7.$$

Each coil has length $\ell = 0.400$ m and contains $N = 2000$ turns at midpoint positions $z_n = \left(\frac{n+1/2}{N} - \frac{1}{2}\right)\ell$ ($n = 0, \ldots, N-1$). At an arbitrary point $\mathbf{P} = (x, y, z)$, the net magnetic field is:

$$\mathbf{B}_{\mathrm{loops}}(\mathbf{P}; I_0) \;=\; \sum_{k=0}^{7}\sum_{n=0}^{N-1}\left[B_\varrho(\varrho_k, z-z_c-z_n; a, I_0)\,\frac{(x-c_{k,x})\hat{\mathbf{x}} + (y-c_{k,y})\hat{\mathbf{y}}}{\varrho_k} \;+\; B_z(\varrho_k, z-z_c-z_n; a, I_0)\,\hat{\mathbf{z}}\right],$$

where $\varrho_k = \sqrt{(x-c_{k,x})^2+(y-c_{k,y})^2}$. As $N\to\infty$ at fixed surface current density $NI_0/\ell$, the midpoint sum $\frac{1}{N}\sum_{n=0}^{N-1} f(z_n) = \frac{1}{\ell}\int_{-\ell/2}^{+\ell/2} f(z')\,dz' + \mathcal{O}(N^{-2})$ converges to second order to the continuous Derby–Olbert cylindrical current-sheet integral (`fields.coils_B`). Accordingly, at $I_0 = 14.0$ mA, the two independent evaluations (`revision3_checks.json`, lines 124–136) give:

- **At the Main Clock ($\mathbf{P_{main}} = (0,0,0)$):** $|\mathbf{B}_{\mathrm{loops}}| = 11.63290030\ \mu$T, $|\mathbf{B}_{\mathrm{sheet}}| = 11.63289980\ \mu$T (relative difference $-4.28\times10^{-8}$).
- **At the Reference Clock ($\mathbf{P_{ref}} = (1.60, 0, 0)$ m):** $|\mathbf{B}_{\mathrm{loops}}| = 0.430539231\ \mu$T, $|\mathbf{B}_{\mathrm{sheet}}| = 0.430539228\ \mu$T (relative difference $-7.56\times10^{-9}$).

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## 21.3 3. Differential Magnetic Energy ($\Delta W(L)$), the $r = L/t$ Compactified Exterior Quadrature, and the $L = 0.95$ m Identity

### 21.3.1 3.1. Expression of the Polarity-Even Fractional Frequency Shift in Terms of $\Delta W(L)$ (`make_revision3_checks.py`, Lines 53–58)

By definition, the magnetic energy normalized to unit current ($I = 1$ A) inside a ball $B(\mathbf{P}, L)$ of radius $L$ centered at $\mathbf{P}$ is:

$$W_{\rm in}(\mathbf{P}; L) \;\equiv\; \frac{1}{2\mu_0 I_0^2} \int_{|\mathbf{x}-\mathbf{P}|\le L} |\mathbf{B}(\mathbf{x}; I_0)|^2 \, d^3x \qquad [\mathrm{J/A^2}],$$

and the differential enclosed energy between the main clock ($\mathbf{P}_{\rm main} = \mathbf{0}$) and the reference clock ($\mathbf{P}_{\rm ref}$) is $\Delta W(L) \equiv W_{\rm in}(\mathbf{P}_{\rm main}; L) - W_{\rm in}(\mathbf{P}_{\rm ref}; L)$. Because the 3D top-hat averaged magnetic mass-density difference is $\Delta\bar{\rho}_{B,L} = \frac{I_0^2\,\Delta W(L)}{c^2\,V_L}$ (where $V_L = \frac{4}{3}\pi L^3$), the conditional H1 clock-shift sensitivity coefficient ($k_g(L)$, [$\mathrm{A^{-2}}$]) and the signal at $I_0 = 14$ mA (`make_revision3_checks.py`, lines 55–57; `revision3_checks.json`, lines 139–145) are:

$$\boxed{k_g(L) \;=\; -\frac{GL^2}{c^2}\frac{\Delta W(L)}{c^2\,\frac{4}{3}\pi L^3} \;=\; -\frac{3G\,\Delta W(L)}{4\pi c^4 L}, \qquad \Delta y(L) \;=\; k_g(L)\, I_0^2.}$$

At the grid peak on the 25-point logarithmic grid ($L_p = 10^{-0.2}$ m $= 0.6309573444801932$ m):

$$\Delta W(L_p) \;=\; 1.9611445207162488\ \mathrm{J/A^2} \quad\Longrightarrow\quad k_g(L_p) \;=\; -6.131177672606294\times 10^{-45}\ \mathrm{A^{-2}},$$

$$\Delta y(L_p) \;=\; k_g(L_p)\times (0.014\ \mathrm{A})^2 \;=\; -1.2017108238308338\times 10^{-48}.$$

### 21.3.2 3.2. Why Is the Exterior Complementary Integral Advantageous at Large $L$, and How Does the $r = L/t$ Compactification Work?

When $L > L_{\rm jump} = 0.95$ m is large enough that both the main-clock ball $B(\mathbf{P}_{\rm main}, L)$ and the reference-clock ball $B(\mathbf{P}_{\rm ref}, L)$ enclose the intense near-coil zone, directly subtracting the two interior integrals ($W_{\rm in,main}(L) - W_{\rm in,ref}(L)$) would subtract two nearly equal numbers. In Euclidean space, however, the magnetic energy integrated over all of $\mathbb{R}^3$ is **translation-invariant**:

$$W_{\rm tot} \;\equiv\; \frac{1}{2\mu_0 I_0^2}\int_{\mathbb{R}^3} |\mathbf{B}(\mathbf{x}; I_0)|^2\, d^3x \;=\; W_{\rm in}(\mathbf{P}_{\rm main}; L) + W_{\rm ext}(\mathbf{P}_{\rm main}; L) \;=\; W_{\rm in}(\mathbf{P}_{\rm ref}; L) + W_{\rm ext}(\mathbf{P}_{\rm ref}; L),$$

where $W_{\rm ext}(\mathbf{P}; L) \equiv \frac{1}{2\mu_0 I_0^2}\int_{|\mathbf{x}-\mathbf{P}|\ge L} |\mathbf{B}(\mathbf{x}; I_0)|^2\, d^3x$ is the exterior (complementary) energy outside the ball. Because the exact total energy cancels ($W^{\rm exact}_{\rm tot,main} - W^{\rm exact}_{\rm tot,ref} \equiv 0$), the interior energy difference equals the negative difference of the smooth, singularity-free exterior integrals:

$$\boxed{\Delta W_{\rm out}(L) \;\equiv\; -\Big[W_{\rm ext}(\mathbf{P}_{\rm main}; L) \;-\; W_{\rm ext}(\mathbf{P}_{\rm ref}; L)\Big] \;=\; \frac{1}{2\mu_0 I_0^2}\int_{r\ge L} r^2\, dr \int_{S^2} d\Omega \Big[|\mathbf{B}(\mathbf{P}_{\rm ref} + r\hat{\mathbf{n}})|^2 - |\mathbf{B}(\mathbf{P}_{\rm main} + r\hat{\mathbf{n}})|^2\Big].}$$

V20 `energy_integrator.py` evaluates this exterior integral over the half-line $[L, \infty)$ along two independent paths (`make_revision3_checks.py`, lines 62–68; `tal_book_HU_2026-10-24.tex`, line 581):

- **Path "B" (`dW_B`): Exact $r = L/t$ Compactification Without Finite Cutoff.** Introducing the dimensionless variable $t \equiv L/r \in (0, 1]$, the differential is $dr = -\frac{L}{t^2}\, dt$ and the volume measure is $r^2\, dr = -\frac{L^3}{t^4}\, dt$. Thus the infinite domain $[L, \infty)$ maps onto the compact interval $(0, 1]$:

$$\boxed{\Delta W_{\rm B}(L) \;=\; \frac{L^3}{2\mu_0 I_0^2}\int_0^1 \frac{dt}{t^4}\int_{S^2} d\Omega \Big[\big|\mathbf{B}(\mathbf{P}_{\rm ref} + \tfrac{L}{t}\hat{\mathbf{n}})\big|^2 \;-\; \big|\mathbf{B}(\mathbf{P}_{\rm main} + \tfrac{L}{t}\hat{\mathbf{n}})\big|^2\Big].}$$

Let us inspect the behavior of the integrand as $t \to 0^+$ ($r = L/t \to \infty$): because the net charge of the 8 closed current loops is zero, the far magnetic field is dipolar at leading order: $|\mathbf{B}(r\hat{\mathbf{n}})| = \mathcal{O}(r^{-3}) = \mathcal{O}(t^3/L^3)$, so the energy density is

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$|\mathbf{B}|^2 = \mathcal{O}(t^6/L^6)$, and the difference between the two shifted balls is $|\mathbf{B}(\mathbf{P}_{\rm ref} + \frac{L}{t}\hat{\mathbf{n}})|^2 - |\mathbf{B}(\mathbf{P}_{\rm main} + \frac{L}{t}\hat{\mathbf{n}})|^2 = \mathcal{O}(t^7/L^7)$. Multiplied by the Jacobian factor $\frac{L^3}{t^4}$, the integrand vanishes at $t \to 0^+$ **as a regular polynomial ($\mathcal{O}(t^2)$–$\mathcal{O}(t^3)$)!** Consequently, $n_t = 32$ (or 64) point Gauss–Legendre quadrature on $(0, 1]$ achieves machine precision while simultaneously proving the far-field scaling:

$$\Delta W(L) \;\propto\; L^{-3} \text{ (or } L^{-5} \text{ under quadrupole/shift symmetry)} \quad\Longrightarrow\quad |\Delta y(L)| \;\propto\; \frac{\Delta W(L)}{L} \;\propto\; L^{-4}\text{–}L^{-6}.$$

- **Path "A" (`dW_A`): Panel Integral + Analytic Asymptotic Multipole Tail.** At $L = 3.0$ m, the relative difference between Path "A" (`dW_A = 0.0023231314942858453 8 J/A^2`) and Path "B" (`dW_B = 0.0023231314942858455 5 J/A^2`) is $-7.44 \times 10^{-16}$ (`revision3_checks.json`, lines 164–165; the $3.7 \times 10^{-16}$ agreement of the two deterministic routines in `tal_book_HU_2026-10-24.tex`, line 478).

### 21.3.3 3.3. Exact Algebraic Identity of the $L_{\rm jump} = 0.95$ m Interior/Exterior Switching Jump (`make_revision3_checks.py`, Lines 70–74)

When the interior and exterior quadratures evaluated at finite angular resolution $n_{\rm hi} = 24000$ are added together, the numerical total energies centered on the two clock positions (`revision3_checks.json`, lines 186–195) are:

$$W^{\rm num}_{\rm tot,main} \;=\; W^{\rm num}_{\rm in,main}(L) + W^{\rm num}_{\rm ext,main}(L) \;=\; 2.237861258956951 \text{ J/A}^2,$$

$$W^{\rm num}_{\rm tot,ref} \;=\; W^{\rm num}_{\rm in,ref}(L) + W^{\rm num}_{\rm ext,ref}(L) \;=\; 2.2364478629240048 \text{ J/A}^2.$$

Subtracting the two equations yields an exact algebraic identity valid for every $L$ (including the switching point $L_{\rm jump} = 0.95$ m):

$$\boxed{\Delta W^{\rm num}_{\rm in}(0.95\text{ m}) \;-\; \Delta W^{\rm num}_{\rm out}(0.95\text{ m}) \;\equiv\; W^{\rm num}_{\rm tot,main} \;-\; W^{\rm num}_{\rm tot,ref} \;=\; 0.001413396032946 \text{ J/A}^2}$$

($2.17738947 - 2.17597607 = 0.00141340$ J/A$^2$, i.e., 0.06491% of the interior value; the algebraic difference check residual is `identity_check_abs` $= 8.88 \times 10^{-16}$ J/A$^2$).

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## 21.4 4. Checks

1. **Consistency of the DC Coil Resistance with the Measured Winding Diameter $D_{\rm turn} = 291.7$ mm (`tal_book_HU_2026-10-24.tex`, Line 577):**
   - For $N = 2000$ turns of copper wire of diameter $d_{\rm wire} = 0.15$ mm ($A_{\rm wire} = \frac{\pi}{4} d^2_{\rm wire} = 1.767146 \times 10^{-8}$ m$^2$, $\rho_{\rm Cu}(20°\text{C}) = 1.7241 \times 10^{-8}\ \Omega\,$m) and mean winding diameter $D_{\rm turn} = 0.2917$ m: $L_{\rm wire} = \pi D_{\rm turn} N = 1832.805$ m, giving $R_{\rm DC,calc} = 1788.160\ \Omega$, which differs by only $-0.0470\%$ from the nominal $1789\ \Omega$ ✓.
2. **Verification of the Peak Signal via Formula (1) at $L_p = 0.63095734448$ m:**
   - $k_g(L_p) = -\frac{3\times 6.67430\times 10^{-11}\times 1.961144520716}{4\pi\times(299792458)^4\times 0.63095734448} = -6.1311776726 \times 10^{-45}$ A$^{-2}$, and $\Delta y(L_p) = k_g(L_p) \times 0.014^2 = -1.2017108238 \times 10^{-48}$, reproducing the main V20/V22 result ✓.

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# 22 Step 21: Exact Derivation of the 7-Segment Piecewise-Polytrope Log-Enthalpy ($h$) TOV Transformation, the $s = r^2$ Even-Parity Central Regularization (`Audit Fix 01`), and the `gradient`-Mode Exterior Vacuum-Halo Lapse Integral ($\Delta\nu_{\mathbf{ext}}$) over $r \in [R_*, R_* + L]$

Source file: step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_lapse_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-27 **File Path:** scalar-tensor-yukawa-fit/step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_lapse_en.md **Hungarian Original:** step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_lapse.md **Previous Steps:** step1_dilaton_form_en.md – step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_quadrature_en.md **License:** CC BY-NC 4.0 **Local Archival Sources:**

- V22 Book Edition: /workspace/artifacts/tal_booklet_2026-10-24/hu_p2.tex (Chapter 7, lines 102–120; DOI: 10.5281/zenodo.23604423)
- V23 (50aa2db0 task): /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_HU_2026-10-24.tex (Section 1.3 and Table 1, lines 109–156; DOI: 10.5281/zenodo.23639084)
- V21/V23 profile and TOV integrator: /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_profiles.py (lines 14–108) and /workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/stellar_tal_v21.py (lines 33–136; DOI: 10.5281/zenodo.23602501)

## 22.1 1. Verbatim Citations from the V22 and V23 Manuscripts and the `tal_profiles.py` Module

In Steps 7, 16, 17, and 18 we used the V21/V23 neutron-star equilibrium profiles (`dense_profile`, SLy, APR4, and H4 equations of state, continuum Bianchi residuals $1.82 \times 10^{-11}$ and $1.59 \times 10^{-11}$), without yet deriving the three analytical transformations in `stellar_tal_v21.py` and `tal_profiles.py` that guarantee this accuracy and central regularity:

1. **The 3 Equations of State (SLy, APR4, H4), the Causal $c_s = c$ Bound, and the Two Conditional TOV Modes (`hu_p2.tex`, Chapter 7, lines 108–115):**

$$\textbf{(A) mass mode:}\quad \frac{dm_{\rm eff}}{dr} = 4\pi r^2 \rho(r)\, f_L(r), \qquad \textbf{(B) gradient mode:}\quad \frac{d\Phi_{\rm eff}}{dr} = f_L(r)\, \frac{G\big(m(r) + 4\pi r^3 P(r)/c^2\big)}{c^2 r\big(r - 2Gm(r)/c^2\big)}, \qquad \text{(V22 Ch. 7)}$$

*"I integrated 33 equilibrium sequences across three nuclear equations of state — **SLy** (nucleonic), **APR4** (variational nucleonic; reaches the causal sound-speed threshold $c_s = c$ at rest-mass density $\rho_0 \approx 1.4458 \times 10^{15}$ **g/cm$^3$**), and **H4** (stiff hyperonic) — across 11 logarithmically spaced smoothing scales $L \in [1\,m, 100\,km]$. The V21 archival baseline table contains **182 equilibrium models**, whereas the dense diagnostic stability grid comprises **769 stellar models** in total (including all **33 exact causal endpoints** where $c_s/c = 1$)."*

2. **The $s = r^2$ Even-Parity Central Mapping and the `gradient`-Mode Exterior Lapse Integral (`tal_profiles.py`, lines 53–56 and 91–93):**
   - Lines 53–56:

```
"# Profile mapping, regular at the centre (audit fix 01, run 2):  h and mu = m/r^3 are even,
analytic functions of r, so they are interpolated in s = r^2 with their exact centre values
h(0)=hc, mu(0)=(4pi/3) eps_c f_m.  Then h'(r)=2r dh/ds -> 0, m=r^3 mu, A=r(mu+4pi P)/(1-2
mu r^2) are regular.  (Run 1 interpolated h(r), m(r) in r with a single node at r=0 -> wrong
slopes near the centre.)"
```

   - Lines 91–93:

```
"def exterior_lapse(cur, L, M, R, nq=160):  \"\"\"nu(R) = -int_R^inf f(r) M/(r(r-2M)) dr with
f=exp(-L^2 rhobar_L) from the converged profile; f=1 beyond R+L. Returns nu(R) and diagnostic
dict.\"\"\""
```

3. **Continuum Bianchi Conservation Residuals in V23 Table 1 (`tal_covariant_closure_50aa2db0_HU_2026-10-24.tex`, lines 144–147):**
   - `tophat_mass`: $1.82 \times 10^{-11}$, `tophat_gradient`: $1.59 \times 10^{-11}$ (exact symbolic residual `0.0`).

---

## 22.2 2. Read et al. (2009) 7-Segment Piecewise-Polytrope Log-Enthalpy ($h$) Closure and the $h$-Transformation of the TOV System

### 22.2.1 2.1. Thermodynamic Consistency, $a_i$ Continuity Recursion, and the $c_s^2/c^2 = 1$ Causal Threshold (`stellar_tal_v21.py`, lines 33–69)

The Read et al. (2009) parameterization consists of 4 crust segments $(\Gamma_0, \ldots, \Gamma_3)$ and 3 core segments $(\Gamma_4, \Gamma_5, \Gamma_6 = g_1, g_2, g_3$, with dividing densities $\rho_{0,4} = \rho_{\rm match}$, $\rho_{0,5} = 10^{14.7}$ g/cm$^3$, $\rho_{0,6} = 10^{15}$ g/cm$^3$). In the $i$-th segment $(\rho_0 \in [\rho_{0,i-1}, \rho_{0,i}])$, the pressure (in $c^2$-scaled CGS units) is $P(\rho_0) = K_i \rho_0^{\Gamma_i}$. Pressure continuity at the segment boundaries enforces:

$$K_{i+1} \;=\; K_i\, \rho_{0,i}^{\Gamma_i - \Gamma_{i+1}}, \qquad \rho_{\rm match} \;=\; \left(\frac{K_3}{K_4}\right)^{\frac{1}{\Gamma_4 - \Gamma_3}}.$$

Integrating the first law of thermodynamics at constant entropy per baryon ($d(\varepsilon/\rho_0) = -P\, d(1/\rho_0) = \frac{P}{\rho_0^2} d\rho_0$) yields the total mass-energy density (`stellar_tal_v21.py`, lines 47–49 and 67):

$$\varepsilon(\rho_0) \;=\; (1 + a_i)\,\rho_0 \;+\; \frac{P(\rho_0)}{\Gamma_i - 1}, \qquad a_{i+1} \;=\; a_i \;+\; \frac{K_i}{\Gamma_i - 1}\rho_{0,i}^{\Gamma_i - 1} \;-\; \frac{K_{i+1}}{\Gamma_{i+1} - 1}\rho_{0,i}^{\Gamma_{i+1} - 1} \quad (a_0 = 0).$$

The specific relativistic log-enthalpy ($dh \equiv \frac{dP}{\varepsilon + P}$) can then be integrated and inverted analytically (`stellar_tal_v21.py`, lines 50 and 60–68):

$$\boxed{h(\rho_0) \;=\; \ln\!\left(\frac{\varepsilon + P}{\rho_0}\right) \;=\; \ln\!\left(1 + a_i + \frac{\Gamma_i}{\Gamma_i - 1} K_i\, \rho_0^{\Gamma_i - 1}\right) \iff \rho_0(h) \;=\; \left[\frac{\left(e^h - 1 - a_i\right)(\Gamma_i - 1)}{\Gamma_i K_i}\right]^{\frac{1}{\Gamma_i - 1}}.}$$

The squared adiabatic sound speed in each segment is:

$$\frac{c_s^2(h)}{c^2} \;=\; \left(\frac{\partial P}{\partial \varepsilon}\right)_s \;=\; \frac{\Gamma_i\, P(h)}{\varepsilon(h) + P(h)} \;=\; (\Gamma_i - 1)\,\frac{e^h - 1 - a_i}{e^h}.$$

Because $\Gamma_i > 2$ in the high-density core (for APR4, $\Gamma_5 = 3.445$, $\Gamma_6 = 3.348$), $c_s^2(h)/c^2$ increases strictly monotonically with $h$, and the causal threshold ($c_s^2/c^2 = 1$) is located by `brentq` to $10^{-15}$ accuracy at $h_{\rm causal}$ (for APR4, $\rho_0(h_{\rm causal}) \approx 1.4458 \times 10^{15}$ g/cm$^3$, `hu_p2.tex`, line 115).

### 22.2.2 2.2. Rewriting the TOV Equations with $h$ as Independent Variable and Central Taylor Startup (`tal_profiles.py`, lines 38–50)

Because the hydrostatic balance equation is $P'(r) = (\varepsilon + P)h'(r) = -(\varepsilon + P) f_{\rm grad}(r) A_{\rm GR}(r)$, the $(\varepsilon + P)$ factor cancels identically:

$$\frac{dh}{dr} \;=\; -f_{\rm grad}(r)\, A_{\rm GR}(r) \;=\; -f_{\rm grad}(r)\,\frac{m(r) + 4\pi r^3 P(h)}{r\big(r - 2m(r)\big)},$$

where $f_{\rm grad}(r) = f_L(r)$ in `gradient` mode and 1 in `mass` (as well as GR) mode, while $f_{\rm mass}(r) = f_L(r)$ in `mass` mode and 1 in `gradient` mode. Since $h(r)$ decreases strictly monotonically inside the star from central $h_c$ down to surface $h(R_*) = 0$, the ODE system for $(r(h), m(h))$ reads (`tal_profiles.py`, lines 43–48):

$$\boxed{\frac{dr}{dh} \;=\; -\frac{r(r - 2m)}{\big(m + 4\pi r^3 P(h)\big)\, f_{\rm grad}(r)}, \qquad \frac{dm}{dh} \;=\; 4\pi r^2 \varepsilon(h)\, f_{\rm mass}(r)\,\frac{dr}{dh}.}$$

**Why is integrating in $h$ advantageous?** When $r$ is the independent variable, the stellar surface $P(R_*) = 0$ must be found by event detection where $\varepsilon(P) \to 0$ varies steeply. With $h$ as the independent variable, the surface is a fixed

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integration endpoint ($h_{\rm surf} = 10^{-10}$) at which $R_* = r(h_{\rm surf})$ and $M = m(h_{\rm surf})$ are read out directly without root-finding overshoot! To avoid the $r = 0$ coordinate singularity, the integration starts at $r_0 = 10^{-4}$ km using the second-order central Taylor expansion (`tal_profiles.py`, lines 41–42):

$$m(r_0) \;=\; \frac{4\pi}{3}\varepsilon_c f_m\, r_0^3 + \mathcal{O}(r_0^5), \qquad h(r_0) \;=\; h_c - \delta h, \qquad \delta h \;=\; \frac{2\pi}{3}(\varepsilon_c f_m + 3P_c)\, f_g\, r_0^2 + \mathcal{O}(r_0^4).$$

## 22.3 3. Even-Parity $s = r^2$ Central Regularization (`Audit Fix O1`, `tal_profiles.py`, lines 53–79)

### 22.3.1 3.1. Why Direct Spline Interpolation in $r$ Causes Central Singularities, and How $s = r^2$ Resolves It

Any spherically symmetric scalar field that is smooth at the origin ($C^\infty(\mathbb{R}^3)$) — such as the log-enthalpy $h(\mathbf{x})$, the mean interior mass density $\mu(\mathbf{x}) \equiv \frac{m(r)}{r^3}$, or the 3D top-hat exponent $x(\mathbf{x}) \equiv \gamma_L \bar{\rho}_L(r)$ — can contain only powers of $|\mathbf{x}|^2 = x^2 + y^2 + z^2 = r^2 \equiv s$ in its Taylor expansion, i.e., it is an **even function of** $r$:

$$h(r) \;=\; h_s(r^2), \qquad \mu(r) \;\equiv\; \frac{m(r)}{r^3} \;=\; \mu_s(r^2), \qquad x(r) \;=\; x_s(r^2),$$

with $h_s(0) = h_c$ and $\mu_s(0) = \frac{4\pi}{3}\varepsilon_c f_m$. If $h(r)$ and $m(r)$ are interpolated directly in $r$ via PCHIP splines (`Run 1`), the cubic Hermite polynomial at the single node $r = 0$ enforces neither the exact value of $h''(0)$ nor $m'(0) = m''(0) = 0$, injecting numerical noise and $\mathcal{O}(1/r)$ errors into $\frac{m(r)}{r^3}$ and $\frac{h'(r)}{r}$ near the center!

By contrast, `tal_profiles.py` (lines 64–79, `Audit Fix O1`) constructs `PchipInterpolator` objects on the grid $s = r^2 \in [0, R_*^2]$:

$$h_s(s), \qquad \mu_s(s) \;=\; \frac{m(\sqrt{s})}{s^{3/2}} \quad (\mu_s(0) = \mu_c), \qquad x_s(s) \;=\; x(\sqrt{s}).$$

By the chain rule ($\frac{d}{dr} = 2r\frac{d}{ds}$), every central derivative and curvature factor is **analytically regular** as $r \to 0^+$ (`tal_profiles.py`, lines 66–79):

$$\begin{aligned}
h'(r) \;&=\; 2r\,\frac{dh_s}{ds}(r^2) \;=\; \mathcal{O}(r), & m(r) \;&=\; r^3\,\mu_s(r^2) \;=\; \mathcal{O}(r^3),\\
\frac{d\mu}{dr}(r) \;&=\; 2r\,\frac{d\mu_s}{ds}(r^2) \;=\; \mathcal{O}(r), & \frac{d\ln f_L}{dr}(r) \;&=\; -2r\,\frac{dx_s}{ds}(r^2) \;=\; \mathcal{O}(r),\\
A_{\rm GR}(r) \;&=\; \frac{m(r) + 4\pi r^3 P(r)}{r\big(r - 2m(r)\big)} \;=\; \frac{r\big(\mu_s(r^2) + 4\pi P(r)\big)}{1 - 2r^2\,\mu_s(r^2)} \;=\; \mathcal{O}(r).
\end{aligned}$$

Consequently, in the ratios $\frac{h'(r)}{r} = 2\frac{dh_s}{ds}(r^2)$ and $\frac{A_{\rm GR}(r)}{r} = \frac{\mu_s(r^2)+4\pi P(r)}{1-2r^2\mu_s(r^2)}$ entering the Chandrasekhar operator and the Bianchi identity, the factor $r$ **cancels algebraically** at $r = 0$! This regularization reduces the continuum Bianchi conservation residual in V23 Table 1 to $1.82 \times 10^{-11}$ (`tophat_mass`) and $1.59 \times 10^{-11}$ (`tophat_gradient`).

## 22.4 4. `gradient`-Mode Exterior Vacuum-Halo Lapse Integral ($\Delta\nu_{\rm ext}$) over $r \in [R_*, R_* + L]$ and the Interior Bernoulli Integral (`tal_profiles.py`, lines 80–106)

### 22.4.1 4.1. Why the $f_L(r) < 1$ Domain Extends Beyond the Stellar Surface $R_*$ out to $R_* + L$

Outside the star ($r > R_*$), the local matter density vanishes identically ($\varepsilon(r) = 0$, $P(r) = 0$), so $m(r) \equiv M$ and $A_{\rm GR}(r) = \frac{M}{r(r-2M)}$ for all $r \geq R_*$.

- **In `mass` mode (and in GR):** the exterior lapse gradient $\nu'(r) = A_{\rm GR}(r) = \frac{M}{r(r-2M)}$ carries no $f_L(r)$ factor. Thus the exterior spacetime is exact Schwarzschild geometry, and the surface lapse is (`tal_profiles.py`, line 81):

$$\nu_{\rm mass}(R_*) \;=\; \nu_{\rm GR}(R_*) \;=\; -\int_{R_*}^{\infty} \frac{M}{r(r-2M)}\,dr \;=\; \frac{1}{2}\ln\!\left(1 - \frac{2M}{R_*}\right).$$

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- **In `gradient` mode:** the lapse gradient is $\nu'(r) = f_L(r)\, A_{\rm GR}(r)$ everywhere. Although $\varepsilon(r) = 0$ outside the star, a 3D top-hat ball $B(\mathbf{x}, L)$ of radius $L$ centered at an exterior point $r \in (R_*, R_* + L)$ **still intersects the outer layer** $[r - L, R_*]$ **of the star** (via the spherical-cap intersection formula derived in Step 18)! Consequently, a **nonlocal vacuum-halo shell** of thickness $L$ ($r \in [R_*, R_* + L]$) forms around the star, within which:

$$x(r) \;=\; L^2 \bar{\rho}_L(r) \;=\; \frac{3}{4r\,L} \int_{r-L}^{R_*} s\,\varepsilon(s)\,\left[L^2 - (r-s)^2\right]\,ds \;>\; 0 \quad\Longrightarrow\quad f_L(r) \;=\; e^{-x(r)} \;<\; 1 \qquad (R_* \le r < R_* + L),$$

and only for $r \ge R_* + L$ does $x(r) \equiv 0 \implies f_L(r) \equiv 1$!

### 22.4.2 4.2. Derivation of the Finite-Range Integral $\Delta\nu_{\rm ext}$ and the Interior $\nu(r)$ Profile (`tal_profiles.py`, lines 86–105)

Integrating the `gradient`-mode equation $\nu'(r) = f_L(r)\, A_{\rm GR}(r)$ from asymptotically flat infinity ($\nu(\infty) = 0$) inward to the stellar surface $r = R_*$, and using $f_L(r) = 1 - (1 - e^{-x(r)})$ where $1 - e^{-x(r)} \equiv 0$ for all $r \ge R_* + L$, splits the improper integral into the analytical Schwarzschild value plus a finite correction over the compact interval $[R_*, R_* + L]$ (`tal_profiles.py`, lines 91–105):

$$\begin{aligned}\nu_{\rm grad}(R_*) \;&=\; -\int_{R_*}^{\infty} e^{-x(r)}\,\frac{M}{r(r-2M)}\,dr \;=\; -\int_{R_*}^{\infty} \frac{M}{r(r-2M)}\,dr \;+\; \int_{R_*}^{R_*+L} \left(1 - e^{-x(r)}\right)\frac{M}{r(r-2M)}\,dr \\ &=\; \frac{1}{2}\ln\!\left(1 - \frac{2M}{R_*}\right) \;+\; \Delta\nu_{\rm ext}, \qquad \Delta\nu_{\rm ext} \;\equiv\; \int_{R_*}^{R_*+L} \left(-\,\mathrm{expm1}(-x(r))\right)\frac{M}{r(r-2M)}\,dr \;>\; 0,\end{aligned}$$

which `exterior_lapse()` evaluates via 200-point Gauss–Legendre quadrature on $[R_*, R_* + L]$. In the stellar interior ($0 \le r \le R_*$), both `mass` and `gradient` modes satisfy the relativistic Bernoulli identity $\nu'(r) = -h'(r)$, yielding the interior lapse profile without any additional numerical quadrature (`tal_profiles.py`, lines 86–87):

$$\nu(r) \;=\; \nu(R_*) \;-\; \big[h(r) - h(R_*)\big], \qquad \forall\, r \in [0, R_*].$$

---

## 22.5 5. Checks

1. **Physical Verification of the Sign and Limits of $\Delta\nu_{\rm ext} > 0$:**
   - Because $x(r) > 0$ on $r \in [R_*, R_* + L)$ and $r > 2M$, the integrand $(1 - e^{-x(r)})\frac{M}{r(r-2M)} > 0$, so $\Delta\nu_{\rm ext} > 0$, meaning `gradient` mode softens the surface gravitational potential depth ($|\nu_{\rm grad}(R_*)| < |\nu_{\rm GR}(R_*)|$). As $L \to 0^+$, the integration interval length $L \to 0$ and $\Delta\nu_{\rm ext} \to 0$, continuously recovering the Schwarzschild limit ✓.
2. **Verification of the Derivative of the Interior Bernoulli Integral ($\nu(r) = \nu(R_*) - [h(r) - h(R_*)]$):**
   - In both modes, $\frac{d\nu}{dr} = -\frac{dh}{dr} = -2r\frac{dh_s}{ds}(r^2) = f_{\rm grad}(r) A_{\rm GR}(r)$, which gives $\nu'(0) = 0$ at $r = 0$ and $\nu(R_*)$ at $r = R_*$ ✓.

---

## 22.6 6. Explicit Comparison with the V23 Local Yukawa (`yukawa_action_el`, Branch II-Fluid) Branch (Point 5)

For audit clarity, we strictly separate the properties of the V21 phenomenological 3D top-hat branch (`tophat_mass`, `tophat_gradient` — **Branch I-A**) derived above from the V23 local Yukawa-screened Einstein–scalar–fluid branch (`yukawa_action_el` — **Branch II-Fluid**, see `step4`, `step7`, and `step22`):

| Physical Property | V21 `tophat_mass` / `tophat_gradient` (Branch I-A, Step 21) | V23 `yukawa_action_el` (Branch II-Fluid, Steps 4, 7, and 22) |
|---|---|---|
| **Interior Hydrostatic / Bernoulli Equation ($0 \le r \le R_*$)** | $h'(r) = -\nu'(r) \implies \nu(r) = \nu(R_*) - [h(r) - h(R_*)]$ ($f_L$ enters directly into $\nu' = f_L A_{\rm GR}$ or $m' = 4\pi r^2 f_L \varepsilon$) | $h'(r) = -\nu'(r) + \phi_L'(r) \implies \nu(r) - \phi_L(r) = \nu(R_*) - \phi_L(R_*) - [h(r) - h(R_*)]$ (distinct scalar fifth force $\phi_L'$) |

| Physical Property | V21 `tophat_mass` / `tophat_gradient` (Branch I-A, Step 21) | V23 `yukawa_action_el` (Branch II-Fluid, Steps 4, 7, and 22) |
|---|---|---|
| **Structure of Exterior Domain ($r > R_*$)** | Compact-support 3D top-hat halo on $r \in [R_*, R_* + L]$; for $r \geq R_* + L$, $x(r) \equiv 0$, $f_L(r) \equiv 1$, exact Schwarzschild vacuum | Exponentially decaying gravitating scalar tail over $r \in [R_*, \infty)$: $\rho_\phi = \frac{1}{2\alpha_Y}[B(\phi_L')^2 + \phi_L^2/L^2] > 0$ |
| **Surface Mass ($M_{\mathrm{surf}}$) vs. ADM Mass ($M_{\mathrm{ADM}}$)** | $m'(r > R_*) = 0 \implies M_{\mathrm{ADM}} \equiv M_{\mathrm{surf}} = m(R_*)$ | $m'(r > R_*) = 4\pi r^2 \rho_\phi(r) > 0 \implies M_{\mathrm{ADM}} = M_{\mathrm{surf}} + \Delta M_{\phi,\mathrm{ext}}^{\mathrm{curved}} > M_{\mathrm{surf}}$ (see `step22`) |

**V25 Update (Curved `yukawa_action_el` Branch):** 4-component exterior BVP ($\delta\nu_R = 1.63 \times 10^{-14}$), exact exterior linearization (`a_m = (2.0 * psi0 / (r * r) + 2.0 * dpsi0 / r) / B0`) and interior left-hand side scalar coupling (`scalar_m_sign = -1.0`: `a_m_sc = -(2.0 * ddphi_n / r_nodes + 2.0 * dphi_n / (r_nodes ** 2)) / B_n`), dynamical scalar wave, and GL2 root convergence ($p_{\mathrm{eff,root}} = 4.248$). For SLy $L = 10$ km, $\alpha_Y = 1$: $M_{\mathrm{ADM}}$ turning point 15.316404, adaptive Gauss–Kronrod $M_B$ turning point 15.316409, main GL2 operator root (`scalar_m_sign=-1.0`) 15.316410 (within $+5.80 \times 10^{-6}$ dex of the $M_{\mathrm{ADM}}$ peak; with diagnostic `scalar_m_sign=+1.0`, 15.319129, $+2.725 \times 10^{-3}$ dex). Detailed tables and diagnostics: `step22`, Sections 4 and 7.

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# 23 Step 22 (V25): Curved Exterior Vacuum Boundary-Value Problem of `yukawa_action_el`, Exact ADM Mass ($M_{\mathrm{ADM}}$), Recomputed Mass–Radius ($M_{\mathrm{ADM}}$–$R_*$) Sequences, and Coupled $(\delta h, \delta\phi_L, \delta m)$ Radial Pulsation Stability

Source file: step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability_en.md

**Author:** Dávid Szabolcsi (ORCID: 0009-0005-6300-074X) **Date:** 2026-10-27 **File Path:** `scalar-tensor-yukawa-fit/step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability_en.md` **Previous Steps:** `step1_dilaton_form_en.md` – `step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_lapse_en.md` **License:** CC BY-NC 4.0 **Local Sources and Runnable Code:**

- V23 (`50aa2db0` task): `/workspace/artifacts/tal_covariant_closure_50aa2db0_2026-10-24/tal_covariant_closure_50aa2db0_EN_2026-10-24.tex` (Sections 1.2 and 2, lines 65–153; DOI: `10.5281/zenodo.23639084`)
- Runnable solver code: `scalar-tensor-yukawa-fit/curved_yukawa_adm_and_stability.py` (and integrated into `covariant_closure_50aa2db0.py`)
- Generated output files: `curved_yukawa_adm_sequences.csv`, `curved_yukawa_coupled_stability.csv`, `curved_yukawa_audit_summary.json`, `closure_tov_comparison.csv`, `conservation_Residuals.json`

## 23.1 1. Exact Exterior Coupled Einstein–Scalar Vacuum Equations ($r \geq R_*$) and Curved Robin Matching (Point 3)

In the earlier V23 (`50aa2db0`) `yukawa_action_el` computation (`covariant_closure_50aa2db0.py`, `solve_yukawa_to_surface`), the interior boundary-value problem was closed at the stellar surface $r = R_*$ using the flat-space Yukawa decay relation $\phi'_{L,\mathrm{flat}}(R_*) = -\left(\frac{1}{R_*} + \frac{1}{L}\right)\phi_L(R_*)$, and the mass–radius sequences were optimized for the surface mass $M_{\mathrm{surf}} \equiv m(R_*)$.

In reality, in the exterior vacuum outside the star ($r \geq R_*$, where $\varepsilon(r) = P(r) = 0$), the gravitational field is strongly curved ($B(R_*) \equiv e^{-2\lambda(R_*)} = 1 - \frac{2M_{\mathrm{surf}}}{R_*} \approx 0.50$–$0.62$), and the energy density and radial pressure of the scalar field $\phi_L(r)$ continue to source the metric components $(m(r), \nu(r))$:

$$\boxed{\rho_\phi(r) \;=\; \frac{1}{2\alpha_Y}\left[B(r)\big(\phi'_L(r)\big)^2 + \frac{\phi_L(r)^2}{L^2}\right], \qquad p_{r,\phi}(r) \;=\; \frac{1}{2\alpha_Y}\left[B(r)\big(\phi'_L(r)\big)^2 - \frac{\phi_L(r)^2}{L^2}\right], \qquad B(r) \;=\; 1 - \frac{2m(r)}{r}.}$$

Substituting the exact algebraic identity of the vacuum Einstein equations ($\varepsilon = P = 0$),

$$B(r)\nu'(r) + \frac{B'(r)}{2} \;=\; \frac{2m(r)}{r^2} + 4\pi r\big(p_{r,\phi}(r) - \rho_\phi(r)\big) \;=\; \frac{2m(r)}{r^2} - \frac{4\pi r}{\alpha_Y L^2}\,\phi_L(r)^2,$$

into the radial Klein–Gordon equation $B\phi''_L + \left(\frac{2B}{r} + B\nu' + \frac{B'}{2}\right)\phi'_L - \frac{\phi_L}{L^2} = 0$, we obtain the **exact coupled exterior Einstein–scalar ODE system on** $r \in [R_*, \infty)$:

$$\boxed{\begin{aligned}
&\left(1 - \frac{2m(r)}{r}\right)\phi''_L(r) \;+\; \left[\frac{2}{r}\left(1 - \frac{m(r)}{r}\right) - \frac{4\pi r}{\alpha_Y L^2}\,\phi_L(r)^2\right]\phi'_L(r) \;-\; \frac{\phi_L(r)}{L^2} \;=\; 0, \\
&\qquad\qquad \frac{dm}{dr} \;=\; 4\pi r^2\,\rho_\phi(r) \;=\; \frac{2\pi r^2}{\alpha_Y}\left[\left(1 - \frac{2m(r)}{r}\right)\big(\phi'_L(r)\big)^2 + \frac{\phi_L(r)^2}{L^2}\right], \\
&\qquad\qquad \frac{d\nu}{dr} \;=\; \frac{m(r) + 4\pi r^3\,p_{r,\phi}(r)}{r\big(r - 2m(r)\big)}, \qquad \lim_{r\to\infty}\nu(r) \;=\; 0.
\end{aligned}}$$

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#### 23.1.1 Numerical Verification (`curved_yukawa_adm_and_stability.py`): Curved vs. Flat Exterior Robin Slope and $R_{\max}$-Convergence

The script `curved_yukawa_adm_and_stability.py` solves the interior + exterior two-domain boundary-value problem ($s = r^2$ regular core up to $r = R_*$, coupled continuously in $C^1$ to the 4-component exterior curved vacuum $y = [\phi_L, \phi'_L, m, \nu]$ on $r \in [R_*, R_{\max}]$ with asymptotic WKB boundary condition $\phi'_L(R_{\max}) + \left(\frac{1}{L} + \frac{1+m(R_{\max})/L}{R_{\max}}\right)\phi_L(R_{\max}) = 0$ and $\nu(R_{\max}) = \frac{1}{2}\ln(1 - 2m(R_{\max})/R_{\max})$). For the SLy benchmark configuration ($\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$, $R_* = 11.46572423$ km, $M_{\rm surf} = 1.50766122\,M_\odot$, $\phi_L(R_*) = 0.0054086823$, $B(R_*) = 0.6119$; `curved_yukawa_adm_sequences.csv` and `curved_yukawa_audit_summary.json`):

- **$R_{\max}$-independence across $N_{\rm mult} \in \{10, 12, 14, 16\}$ ($R_{\max} = R_* + N_{\rm mult}L$):**
  - Relative spread of the curved Dirichlet–Neumann slope: $\delta k_{\rm curved}/k_{\rm curved} = \mathbf{6.17 \times 10^{-12}}$,
  - Relative spread of the ADM mass: $\delta M_{\rm ADM}/M_{\rm ADM} = \mathbf{5.54 \times 10^{-13}}$,
  - Spread of the surface lapse $\nu(R_*)$: $\delta\nu_R = \mathbf{1.63 \times 10^{-14}}$.
- **Curved vs. flat surface slope ($\log_{10}\rho_{0c} = 15.0$, $R_* = 11.46572423$ km):**
  - Exact curved exterior Dirichlet–Neumann slope: $k_{\rm curved} = -\phi'_{L,\rm curved}(R_*)/\phi_L(R_*) = \mathbf{0.23354233}$ km$^{-1}$ ($\phi'_{L,\rm curved}(R_*) = -1.26315627 \times 10^{-3}$ km$^{-1}$),
  - Flat Robin approximation: $k_{\rm flat} = \frac{1}{R_*} + \frac{1}{L} = \mathbf{0.18721647}$ km$^{-1}$ ($\phi'_{L,\rm flat}(R_*) = -1.01259441 \times 10^{-3}$ km$^{-1}$),
  - Relative difference in the surface derivative: $+\mathbf{24.74}\%$ (spacetime curvature $B(R_*)^{-1/2} \approx 1.278$ steepens the exterior Yukawa decay rate).
  - Consequently, the exact curved exterior scalar halo mass is $\Delta M^{\rm curved}_{\phi,\rm ext} = \mathbf{0.00233692\,M_\odot}$ at $\log_{10}\rho_{0c} = 15.0$ (giving $M_{\rm ADM}(15.0) = \mathbf{1.50999814\,M_\odot}$, whereas the flat Robin estimate gave $\Delta M^{\rm flat}_{\phi,\rm ext} = 0.003416\,M_\odot$), and $\Delta M^{\rm curved}_{\phi,\rm ext} = \mathbf{0.00410774\,M_\odot}$ at the ADM mass maximum (whereas the flat estimate gave $0.00898\,M_\odot$).

| $N_{\rm mult}$ | $R_{\max}$ (km) | $k_{\rm curved}$ (km$^{-1}$) | $\phi'_{L,\rm curved}(R_*)$ (km$^{-1}$) | $\Delta M^{\rm curved}_{\phi,\rm ext}$ ($M_\odot$) | $M_{\rm ADM}$ ($M_\odot$) | $\nu_R \equiv \nu_{\rm exact}(R_*)$ |
|---|---|---|---|---|---|---|
| 10 | 111.4657 | 0.233542329280 | -0.001263156266 | 0.002336923550 | 1.509998138706 | -0.246306687228 |
| 12 | 131.4657 | 0.233542329278 | -0.001263156266 | 0.002336923550 | 1.509998138707 | -0.246306687228 |
| 14 | 151.4657 | 0.233542329278 | -0.001263156266 | 0.002336923550 | 1.509998138707 | -0.246306687228 |
| 16 | 171.4657 | 0.233542329278 | -0.001263156266 | 0.002336923550 | 1.509998138707 | -0.246306687228 |

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## 23.2 2. Exact Exterior Lapse ($\nu(r)$) and Asymptotic Redshift Synchronization at $r = R_*$ (Point 5)

In the `yukawa_action_el` model (**Branch II-Fluid**), the exterior lapse $\nu(r)$ is **not** equal to the vacuum Schwarzschild formula $\frac{1}{2}\ln\!\left(1-2M_{\rm surf}/r\right)$, because in the exterior domain $r \in [R_*, \infty)$ both $m(r)$ and $p_{r,\phi}(r)$ vary with radius. Integrating the 4-component exterior BVP $y = [\phi_L, \phi'_L, m, \nu]$ with outer boundary condition $\nu(R_{\max}) = \frac{1}{2}\ln\!\left(1 - 2m(R_{\max})/R_{\max}\right)$ from $R_{\max} = R_* + N_{\rm mult}L$ down to the stellar surface $r = R_*$ (defined by $h(R_*) = P(R_*) = 0$), the exact surface lapse is:

$$\boxed{\nu_{\rm exact}(R_*) \;\equiv\; \nu_R \;=\; \frac{1}{2}\ln\!\left(1 - \frac{2m(R_{\max})}{R_{\max}}\right) \;-\; \int_{R_*}^{R_{\max}} \frac{m(r) + 4\pi r^3\, p_{r,\phi}(r)}{r\big(r - 2m(r)\big)}\, dr.}$$

For the SLy ($\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$) star at the exact surface boundary point $R_* = 11.46572423$ km ($M_{\rm surf} = 1.50766122\,M_\odot$, $M_{\rm ADM} = 1.50999814\,M_\odot$; `curved_yukawa_adm_sequences.csv` column `nu_R_15p0` and `curved_yukawa_audit_summary.json` block `rmax_independence_checks_SLy_15p0_L10km`), the three definitions at $r = R_*$ yield:

- $\nu_{\rm Schw}(M_{\rm surf}) = \frac{1}{2}\ln\!\left(1 - 2M_{\rm surf}/R_*\right) = -0.2457823130$,
- $\nu_{\rm Schw}(M_{\rm ADM}) = \frac{1}{2}\ln\!\left(1 - 2M_{\rm ADM}/R_*\right) = -0.2462745915$,
- $\nu_{\rm exact}(R_*) \equiv \nu_R(15.0) = \mathbf{-0.2463066872}$ (the curved exterior scalar tail shifts the surface lapse at $r = R_*$ by $-5.2437 \times 10^{-4}$ relative to $\nu_{\rm Schw}(M_{\rm surf})$ and by $-3.2096 \times 10^{-5}$ relative to $\nu_{\rm Schw}(M_{\rm ADM})$, and the 4-component exterior BVP matches backward quadrature to $\delta\nu_R = 1.63 \times 10^{-14}$).

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Inside the star ($0 \le r \le R_*$), the modified Euler equation $h'(r) = -\nu'(r) + \phi_L'(r)$ integrates directly to the exact modified Bernoulli first integral:

$$\boxed{h(r) + \nu(r) - \phi_L(r) \;=\; \nu_{\rm exact}(R_*) - \phi_L(R_*) \;=\; \text{const.}}$$

## 23.3 3. Recomputing the $M_{\rm ADM}$–$R_*$ Sequences with the Asymptotic ADM Mass ($M_{\rm ADM}$) and Baryonic Rest Mass ($M_B$)

The table below (`curved_yukawa_adm_sequences.csv` and `closure_tov_comparison.csv`) reports the **ADM-mass ($M_{\rm ADM}$) optimized** quantities across all 8 `yukawa_action_el` sequences, together with the surface mass ($M_{\rm surf}$) and the maximum and turning point of the conserved baryonic rest mass ($M_B^{\rm quad} = \int_0^{R_*} 4\pi r^2 B(r)^{-1/2} \rho_0(r)\, dr$, evaluated via PCHIP + adaptive Gauss–Kronrod quadrature in $s = r^2$ with `epsrel=1e-12`):

| EOS | $L$ (km) | $\alpha_Y$ | $M_{\rm ADM,max}$ ($M_\odot$) | $M_{\rm surf,max}$ ($M_\odot$) | $M_{B,\rm max}^{\rm quad}$ ($M_\odot$) | $\Delta M_{\phi,\rm ext}^{\rm curved}(M_{\rm max})$ ($M_\odot$) | $\log_{10}\rho_{0c}(M_{\rm ADM,max})$ | $\log_{10}\rho_{0c}(M_{B,\rm max}^{\rm quad})$ | $R(M_{\rm ADM,max})$ (km) | $R(M_{\rm ADM} = 1.4\,M_\odot)$ (km) | $R(M_{\rm surf} = 1.4\,M_\odot)$ (km) | $M_{\rm ADM}(15.0)$ ($M_\odot$) | $\Omega(0; 15.0)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **SLy** | 1.0 | 1.000 ($1/4\pi$) | 2.04684 | 2.04684 | 2.42634 | 0.000002 | 15.303086 | 15.303082 | 9.9689 | 11.6933 | 11.6933 | 1.56171 | 0. 999198 |
| **SLy** | 1.0 | 0.333 ($1/12\pi$) | 2.04789 | 2.04789 | 2.42744 | 0.000001 | 15.302656 | 15.302660 | 9.9733 | 11.7018 | 11.7018 | 1.56386 | 0. 999733 |
| **SLy** | 10.0 | 1.000 ($1/4\pi$) | 2.03321 | 2.02911 | 2.42339 | 0.004108 | 15.316404 | 15.316409 | 9.8853 | 11.5359 | 11.5348 | 1.51000 | 0. 982378 |
| **SLy** | 10.0 | 0.333 ($1/12\pi$) | 2.04321 | 2.04183 | 2.42630 | 0.001381 | 15.307166 | 15.307169 | 9.9446 | 11.6475 | 11.6471 | 1.54638 | 0. 993967 |
| **APR4** | 1.0 | 1.000 ($1/4\pi$) | 2.18693 | 2.18693 | 2.65730 | 0.000003 | 15.282808 | 15.282805 | 9.8947 | 11.3085 | 11.3085 | 1.56316 | 0. 999201 |
| **APR4** | 10.0 | 1.000 ($1/4\pi$) | 2.18068 | 2.17624 | 2.66733 | 0.004449 | 15.293022 | 15.293025 | 9.8462 | 11.1830 | 11.1828 | 1.51271 | 0. 982098 |
| **H4** | 1.0 | 1.000 ($1/4\pi$) | 2.01139 | 2.01138 | 2.29535 | 0.000001 | 15.209751 | 15.209739 | 11.6917 | 13.9388 | 13.9388 | 1.88120 | 0. 999166 |
| **H4** | 10.0 | 1.000 ($1/4\pi$) | 1.97809 | 1.97502 | 2.26306 | 0.003077 | 15.227785 | 15.227787 | 11.5289 | 13.7179 | 13.7169 | 1.82871 | 0. 981076 |

**Baryonic Mass Definition and Turning-Point Precision (`baryonic_mass_and_first_law_audit`):** In `curved_yukawa_adm_sequences.csv` and `closure_tov_comparison.csv`, the columns `M_baryon_max_Msun` and `log10_rho0c_at_M_baryon_max` are defined via adaptive Gauss–Kronrod quadrature on $s = r^2$ PCHIP-interpolated interior profiles ($M_B^{\rm quad}$, `epsrel=1e-12`) with a 7-point quadratic fit over $\pm 0.002$ dex. For SLy $L = 10$ km, $\alpha_Y = 1$, $M_{B,\rm max}^{\rm quad} = \mathbf{2.42338503\,M_\odot}$ at $\log_{10}\rho_{0c}(M_{B,\rm max}^{\rm quad}) = 15.31640854$ (converging across $N_{\rm grid} = 301 \to 2401$ within $1.2 \times 10^{-6}$ dex), refined trapezoidal quadrature at $N_{\rm grid} = 2401$ gives $M_B^{\rm trapz} = 2.42338667\,M_\odot$ with a 7-point peak at 15.31641814 ($+9.6 \times 10^{-6}$ dex), whereas single-pass integration of $M_B^{\rm ODE}$ as a 5th ODE state variable gives $2.42203137\,M_\odot$ and a 3-point peak at 15.31652192 (shifted by $-0.001354\,M_\odot$ and $+1.13 \times 10^{-4}$ dex due to step-size sensitivity of the steep $\sqrt{s}\,\rho_0(h)$ integrand near the crust edge $h \to 0$).

*(Comparison with the earlier flat-Robin `yukawa_action_el` values for SLy $L = 10$ km, $\alpha_Y = 1$: under the flat-Robin approximation, $M_{surf,\rm max}^{flat} = 2.03583\,M_\odot$ and $M_{surf,\rm max}^{flat} + \Delta M_{\phi,ext}^{flat} = 2.04481\,M_\odot$ at $\log_{10}\rho_{0c} = 15.31307$; with the exact curved exterior BVP solution, the surface mass peak is $M_{surf,\rm max}^{curved} = 2.02911\,M_\odot$ and the exact ADM mass peak is $M_{ADM,\rm max}^{curved} = \mathbf{2.03321\,M_\odot}$ at $\log_{10}\rho_{0c} = \mathbf{15.316404}$.)*

## 23.4 4. Coupled $(\delta h, \delta\phi_L, \delta m)$ Radial Pulsation Eigenvalue Problem on the Curved `yukawa_action_el` Background (Point 4)

To ensure that the radial stability analysis covers not only the V21 nonlocal 3D top-hat background (`step18`), we solved the coupled $(\delta h, \delta\phi_L, \delta m)$ linear perturbation system derived in Step 12 directly on the curved-exterior `yukawa_action_el` backgrounds (`curved_yukawa_adm_and_stability.py`, `solve_yukawa_coupled_omega2_gl2`):

- **Interior coupled equations ($0 \le r \le R_*$):** $(\delta h, \delta m, \delta\phi_L, \delta\phi_L')$ integrated with regular center expansion ($\delta h(r) = \delta h_0 + \mathcal{O}(r^2)$, $\delta\phi_L(r) = \delta\phi_c + \mathcal{O}(r^2)$, $\delta m(r) = \mathcal{O}(r^3)$), where the fluid inertia term $\omega^2 e^{2(\lambda-\nu)}\xi(r)$ and the dynamical scalar wave term $+\omega^2 e^{-2\nu}\delta\phi_L$ are closed via the exact adiabatic relation $(\varepsilon + P)r\big(\delta\nu' - \delta\phi_L' + \delta h'\big) = \big[\text{static background derivative}\big]\,\xi(r)$.

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- **Moving surface and curved exterior tail ($R_* \le r \le R_{\max}$):** at $r = R_*^-$, because $\varepsilon(R_*^-) = P(R_*^-) = 0$ on the polytropic crust, the jump in every perturbation variable vanishes ($[\![\delta\phi_L]\!] = [\![\delta\phi_L']\!] = [\![\delta m]\!] = 0$). In the exterior domain $r \in [R_*, R_{\max}]$, the linearized exterior $(\delta\phi_L, \delta m)$ system is integrated out to $R_{\max} = R_* + 14L$, and the eigenvalue $\omega_0^2$ is determined by the vanishing of the asymptotic ADM mass perturbation $\delta M_{\rm ADM}(\omega^2) = \delta m(R_{\max}; \omega^2) = 0$.

### 23.4.1 Numerical Results (`curved_yukawa_coupled_stability.csv` and `curved_yukawa_audit_summary.json`): Dynamical vs. Quasi-Static vs. Frozen-$\phi_L$ ($\delta\phi_L = 0$) Zero Crossings

**Methodological Summary of `curved_yukawa_adm_and_stability.py`:**

- **4-component exterior BVP (`solve_curved_yukawa_star`, lines 234–364):** $y = [\phi_L, \phi_L', m, \nu]$ integrated simultaneously on $r \in [R_*, R_{\max}]$ with boundary conditions $\phi_L(R_*) = \phi_{\rm surf}$, $m(R_*) = M_{\rm surf}$, $\phi_L'(R_{\max}) + \big(1/R_{\max} + 1/(L\sqrt{B(R_{\max})})\big)\phi_L(R_{\max}) = 0$, and $\nu(R_{\max}) = \frac{1}{2}\ln\big(1 - 2m(R_{\max})/R_{\max}\big)$. Consistency between the backward quadrature and the 4th BVP component at $r = R_*$ is $\delta\nu_R = 1.63 \times 10^{-14}$.
- **Exterior linearization and finite-difference derivative tests (SLy, $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$):** `linearized_exterior_response` (line 593: `a_m = (2.0 * psi0 / (r * r) + 2.0 * dpsi0 / r) / B0`) and the left-hand side ($L_u u = R_h\,\delta h$) interior scalar operator coefficient `a_m_sc` (`scalar_m_sign = -1.0`: `a_m_sc = -(2.0 * ddphi_n / r_nodes + 2.0 * dphi_n / (r_nodes ** 2)) / B_n`, obtained by moving the right-hand side derivative $\partial(\phi_L'')/\partial m = +(2\phi_L''/r + 2\phi_L'/r^2)/B$ to the left-hand side) verified against central finite differences:
  - Exterior response $\partial\phi_L'(R_*)/\partial m(R_*)$: linearized BVP `-1.68365059e-04` km$^{-2}$ vs. 4-component exterior BVP central finite difference `-1.68365059e-04` km$^{-2}$ (relative difference: $3.98 \times 10^{-10}$; $K_\phi = 0.23358323$ km$^{-1}$, rel. diff. $1.42 \times 10^{-10}$; $\partial M_{\rm ADM}/\partial m(R_*) = 0.99947576$, rel. diff. $4.26 \times 10^{-10}$).
  - Interior $r = 6.0$ km scalar-mass derivative ($\partial\phi_L''/\partial m$): analytic `-1.72833429e-04` km$^{-3}$ vs. interior scalar RHS central finite difference `-1.72833428e-04` km$^{-3}$ (relative difference: $6.08 \times 10^{-9}$).
- **Dynamical scalar block:** the scalar block includes the $-\omega^2 e^{-2\nu}\delta\phi_L$ term and the exterior response is obtained from an $\omega$-dependent BVP (`dynamical_scalar_wave=True`); solver: GL2 piecewise segment Volterra discretization, $N = 480$, with the exact left-hand side sign `scalar_m_sign = -1.0`. The quasi-static and dynamical roots agree to all 6 reported decimal places (since at $\omega^2 = 0$ the two operators are identical).
- **Turning points and zero crossings:** `minimize_scalar` + 7-point quadratic fit over $\pm 0.002$ dex; zero crossings from a 3-point quadratic fit over $\pm 0.0005$ dex around the root.

| Sequence | $\log_{10}\rho_{0c}(dM_{\rm ADM}=0)$ | $\log_{10}\rho_{0c}(dM_B=0)$ | Dynamical $\omega_0^2=0$ | $\Delta_{\rm ADM}$ Dynamical (dex) | $\Delta_{M_B}$ Dynamical (dex) | Quasi-Static $\omega_0^2=0$ | Frozen $\delta\phi_L=0$ | $\Delta_{\rm ADM}$ Frozen (dex) |
|---|---|---|---|---|---|---|---|---|
| **SLy,** $L=10$ **km,** $\alpha_Y=1$ | 15.316404 | 15.316409 | 15.316410 | +5.80e-06 | +1.33e-06 | 15.316410 | 15.328565 | +1.22e-02 |
| **SLy,** $L=10$ **km,** $\alpha_Y=1/3$ | 15.307166 | 15.307169 | 15.307170 | +4.20e-06 | +6.41e-07 | 15.307170 | 15.311340 | +4.17e-03 |
| **SLy,** $L=1$ **km,** $\alpha_Y=1$ | 15.303086 | 15.303082 | 15.303080 | -6.86e-06 | -2.61e-06 | 15.303080 | 15.304220 | +1.13e-03 |
| **APR4,** $L=10$ **km,** $\alpha_Y=1$ | 15.293022 | 15.293025 | 15.293025 | +3.57e-06 | -1.40e-07 | 15.293025 | 15.301564 | +8.54e-03 |
| **H4,** $L=10$ **km,** $\alpha_Y=1$ | 15.227785 | 15.227787 | 15.227783 | -1.34e-06 | -4.05e-06 | 15.227783 | 15.251384 | +2.36e-02 |

**Convergence Study on the $\omega_0^2 = 0$ Zero Crossing Itself (with respect to $N$ and $R_{\max}$)** (SLy, $L = 10$ km, $\alpha_Y = 1$; together with the convergence of $\omega_0^2$ at $\log_{10}\rho_{0c} = 15.311404$):

- **Zero-crossing $N$-convergence ($N_{\rm mult} = 14$):** $p_{\rm eff,root} = 4.248$ (from the three finest levels), Richardson-extrapolated root $\log_{10}\rho_{0c,\rm Rich}(\omega_0^2 = 0) =$ @@TOK0@@; at $N = 480$, the discretization error relative to the Richardson root is $6.72 \times 10^{-8}$ dex (and between $N = 240 \to 800$ across all 5 sequences $\le 2.02 \times 10^{-5}$ dex).
- **Zero-crossing $R_{\max}$-convergence ($N = 480$, $R_{\max} = R_* + N_{\rm mult}L$, $N_{\rm mult} \in \{10, 12, 14, 16, 20\}$, $R_{\max} \in [109.89,\ 209.89]$ km):** the spread of the root is $2.42 \times 10^{-13}$ dex (across all 5 sequences $\le 2.72 \times 10^{-13}$ dex).
- **Fixed-point ($\log_{10}\rho_{0c} = 15.311404$) $\omega_0^2$ convergence:** $p_{\rm eff} = 5.223$, Richardson value $\omega_0^2 = 6369417.611$ s$^{-2}$ (at $N = 480$, relative error $1.29 \times 10^{-5}$).

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| $N$ | $\log_{10}\rho_{0c}(\omega_0^2=0)$ | $ | \Delta\log_{10}\rho_{0c} | $ from Richardson root (dex) | $\Delta_{\rm ADM}$ (dex) | $\Delta_{M_B}$ (dex) | $\omega_0^2(15.311404)$ ($s^{-2}$) | Rel. error ($\omega_0^2$) |
|---|---|---|---|---|---|---|---|---|
| 80 | 15.31645808 | 4.83e-05 | +5.40e-05 | +4.95e-05 | 6432300.503336 | 9.87e-03 | | |
| 120 | 15.31642223 | 1.24e-05 | +1.82e-05 | +1.37e-05 | 6385729.603235 | 2.56e-03 | | |
| 160 | 15.31641426 | 4.46e-06 | +1.02e-05 | +5.72e-06 | 6375276.486537 | 9.20e-04 | | |
| 240 | 15.31641079 | 9.80e-07 | +6.71e-06 | +2.24e-06 | 6370703.105233 | 2.02e-04 | | |
| 360 | 15.31641001 | 2.06e-07 | +5.94e-06 | +1.47e-06 | 6369680.451299 | 4.13e-05 | | |
| 480 | 15.31640987 | 6.72e-08 | +5.80e-06 | +1.33e-06 | 6369499.539632 | 1.29e-05 | | |
| 600 | 15.31640983 | 2.56e-08 | +5.76e-06 | +1.29e-06 | 6369446.532399 | 4.54e-06 | | |
| 800 | 15.31640982 | 7.55e-09 | +5.74e-06 | +1.27e-06 | 6369424.048161 | 1.01e-06 | | |
| 1200 | 15.31640981 | 1.35e-09 | +5.73e-06 | +1.26e-06 | 6369418.385568 | 1.22e-07 | | |

**Interpretive Note:** With the exact left-hand side sign (`scalar_m_sign = -1.0`), the root of the GL2 operator converges with respect to $N$ and $R_{\max}$ to better than $< 10^{-7}$ dex (`15.31640981`), the $\omega_0^2 = 0$ zero crossing lies within $\leq 6.9 \times 10^{-6}$ dex of the ADM mass maximum across all 5 sequences (for SLy $L = 10$ km, $\alpha_Y = 1$: 15.316410 vs. 15.316404, $\Delta_{\rm ADM} = +5.80 \times 10^{-6}$ dex, $\Delta_{M_B^{\rm quad}} = +1.33 \times 10^{-6}$ dex), and the interior relative $L^2$ differences between the eigenfunction and the static equilibrium tangent for SLy $L = 10$ km, $\alpha_Y = 1$ are $\delta h = 5.04 \times 10^{-4}$, $\delta m = 1.77 \times 10^{-4}$, $\delta\phi_L = 2.27 \times 10^{-4}$, $\delta\phi_L' = 3.09 \times 10^{-4}$ (while the relative difference of the free-surface displacements is $\Delta\xi(R_*)/\xi = 3.54\%$; see Section 7.4). Because the 3.54% free-surface displacement difference and the ODE vs. quadrature baryonic-mass turning-point spread show that the minus-sign root and grid convergence do not by themselves constitute an independent proof of the full radial fundamental spectrum and free-surface variation, **the stability closing statement is formulated conditionally until the independent verification of the coupled second-variation derivation and interior–exterior boundary-condition matching is concluded.**

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## 23.5 5. Checks

1. **Comparison of the ADM Mass ($M_{\rm ADM}$), Baryonic Mass ($M_B$), and Coupled $\omega_0^2 = 0$ Zero Crossing:**
   - For SLy, $L = 10$ km, $\alpha_Y = 1$:

$$\log_{10}\rho_{0c}\left(\frac{dM_{\rm ADM}}{d\rho_c} = 0\right) = 15.316404, \quad \log_{10}\rho_{0c}\left(\frac{dM_B}{d\rho_c} = 0\right) = 15.316409, \quad \log_{10}\rho_{0c}\left(\omega_0^2 = 0\right) = 15.316410,$$

meaning the coupled dynamical stability zero crossing coincides with the ADM mass peak within $+5.80 \times 10^{-6}$ dex (and with the baryonic mass peak within $+1.33 \times 10^{-6}$ dex), while across the remaining 4 sequences the offset is $+4.20 \times 10^{-6}$ dex ($\alpha_Y = 1/3$), $-6.86 \times 10^{-6}$ dex ($L = 1$ km), $+3.57 \times 10^{-6}$ dex (APR4), and $-1.34 \times 10^{-6}$ dex (H4) ✓.

2. **Physical Effect of Omitting the Scalar-Field Perturbation ($\delta\phi_L$):**
   - If the scalar field is frozen on the `yukawa_action_el` background ($\delta\phi_L \equiv 0$), the zero crossing for SLy $L = 10$ km, $\alpha_Y = 1$ shifts to $\log_{10}\rho_{0c} = 15.328565$ ($+1.22 \times 10^{-2}$ dex above the ADM peak), because without the compressional perturbation of the attractive scalar self-energy ($\delta\phi_L > 0$) the star appears artificially stiffer. Including the coupled $(\delta h, \delta\phi_L, \delta m)$ system reduces this offset by a factor of 2097× (to $+5.80 \times 10^{-6}$ dex, and by factors of 165 × –17593× across all 5 sequences) ✓.

---

## 23.6 6. Reproducibility and Generated Files

All results of Step 22 can be reproduced with a single command:

```
python3 curved_yukawa_adm_and_stability.py
```

Running the script generates and updates the following files (in the root directory and in the `csv/` and `data/v23/` subdirectories):

- `curved_yukawa_adm_sequences.csv`: summary table of the 8 mass–radius sequences recomputed for the ADM mass ($M_{\rm ADM}$), surface mass ($M_{\rm surf}$), and three baryonic rest-mass definitions ($M_B^{\rm quad}$, $M_B^{\rm ODE}$, $M_B^{\rm trapz}$).

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- `curved_yukawa_coupled_stability.csv`: coupled $(\delta h, \delta\phi_L, \delta m)$ dynamical, quasi-static, and frozen-$\phi_L$ ($\delta\phi_L = 0$) radial eigenvalue scans and zero crossings across the 5 `yukawa_action_el` sequences (generated with the `scalar_m_sign = -1.0` main operator).
- `curved_yukawa_audit_summary.json`: machine-readable audit log of $R_{\max}$-convergence, curved vs. flat Robin slope, finite-difference derivative tests, the 108-point $4 \times 4$ Jacobian verification, the baryonic mass / First-Law audit, and the eigenmode vs. static tangent comparison.
- `closure_tov_comparison.csv` and `conservation_Residuals.json`: V23 reference files updated with the `yukawa_action_el_curved_adm` rows and the `curved_exterior_yukawa_adm_and_coupled_stability` block.

---

## 23.7 7. V25 Diagnostic Analysis: Sign Fixes, the Three New Diagnostics, and the Conditional Stability Closing Statement (`curved_yukawa_audit_summary.json`)

**Publication Status and Version Delineation (V24 vs. V25):** The earlier Zenodo publication (**V24**, DOI: `10.5281/zenodo.23680715`) refers exclusively to the earlier V24 release. The sign fixes in `curved_yukawa_adm_and_stability.py` (`a_m = (2.0 * psi0 / (r * r) + 2.0 * dpsi0 / r) / B0` in `linearized_exterior_response` and making the exact left-hand side sign `scalar_m_sign = -1.0`, i.e., `a_m_sc = -(2.0 * ddphi_n / r_nodes + 2.0 * dphi_n / (r_nodes ** 2)) / B_n`, the default in the $L_u u = R_h\, \delta h$ operator of `solve_yukawa_coupled_omega2_gl2`), the three new diagnostics in `curved_yukawa_audit_summary.json` (`full_jacobian_multi_eos_and_points`, `baryonic_mass_and_first_law_audit`, `eigenmode_xi_nodes_and_static_tangent_audit`), and the closure of the theoretical verification belong exclusively to the new **V25** version — the earlier V24 DOI must never be cited as if these fixes and closures had already been included in it.

### 23.7.1 7.1. Finite-Difference Derivative Verification (`derivative_verification_tests_SLy_15p0_L10km`)

Finite-difference verification of the exterior `linearized_exterior_response` and interior scalar-mass coupling (SLy, $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$):

| Quantity | Linearized BVP / Analytic | Central Finite Difference (FD) | Relative Difference |
|---|---|---|---|
| `exterior_dphi_prime_dm_R` ($\partial\phi'_L(R_*)/\partial m(R_*)$, km$^{-2}$) | -1.68365059e-04 | -1.68365059e-04 | 3.98e-10 |
| `exterior_K_phi_R` ($-\partial\phi'_L(R_*)/\partial\phi_L(R_*)$, km$^{-1}$) | 2.33583231e-01 | 2.33583231e-01 | 1.42e-10 |
| `exterior_M_m_R` ($\partial M_{\rm ADM}/\partial m(R_*)$) | 9.99475763e-01 | 9.99475764e-01 | 4.26e-10 |
| `exterior_M_phi_R` ($\partial M_{\rm ADM}/\partial\phi_L(R_*)$, km) | 1.27572889e+00 | 1.27572919e+00 | 2.31e-07 |
| `interior_r6km_a_m_sc` ($\partial\phi''_L/\partial m$ at $r = 6$ km, km$^{-3}$) | -1.72833429e-04 | -1.72833428e-04 | 6.08e-09 |

### 23.7.2 7.2. Multi-EOS, Multi-Density, and Multi-Radius Full $4 \times 4$ Jacobian Verification (`full_jacobian_multi_eos_and_points`)

The function `verify_full_jacobian_multi_eos_and_points()` verifies the full interior $4 \times 4$ Jacobian matrix ($\delta h'$, $\delta\phi''_L$, $\delta m'$, $\delta\nu'$, and the coefficients of the adiabatic displacement $\xi$) against central finite differences across 4 configurations (SLy at $\log_{10}\rho_{0c} = 15.0$ and the turning point 15.316404; APR4 at $\log_{10}\rho_{0c} = 15.0$ and the turning point 15.293022), at 3 interior radii per configuration ($r \in \{3.0,\ 6.0,\ 8.5\}$ km) plus surface matching. Across all **108 checks**, the maximum relative error is `max_relative_error_across_all_108_checks = 2.74e-05` (`all_checks_pass_tol_1e_4 = true`):

| EOS | $\log_{10}\rho_{0c}$ | $K_m$ rel. err | $K_\phi$ rel. err | $M_m$ rel. err | $M_\phi$ rel. err | Interior radii $r$ (km) | Max interior rel. err |
|---|---|---|---|---|---|---|---|
| SLy | 15.000000 | 3.98e-10 | 1.42e-10 | 4.26e-10 | 2.31e-07 | 3.0, 6.0, 8.5 | 2.74e-05 |
| SLy | 15.316404 | 7.12e-10 | 4.46e-10 | 4.64e-10 | 7.40e-08 | 3.0, 6.0, 8.5 | 1.88e-05 |
| APR4 | 15.000000 | 1.42e-09 | 6.91e-10 | 5.42e-10 | 8.70e-08 | 3.0, 6.0, 8.5 | 1.94e-05 |
| APR4 | 15.293022 | 5.79e-10 | 4.84e-10 | 6.21e-10 | 1.12e-07 | 3.0, 6.0, 8.5 | 1.59e-05 |

### 23.7.3 7.3. Baryonic Rest Mass ($M_B$) and First-Law Audit (`baryonic_mass_and_first_law_audit`)

The function `verify_baryonic_mass_and_first_law()` computes the baryonic rest mass ($M_B = \int_0^{R_*} 4\pi r^2 B(r)^{-1/2}\rho_0(r)\, dr$) via three independent methods: (i) post-facto trapezoidal rule (`M_B_trapz`), (ii) as the 5th state variable of the ($s = r^2$) TOV ODE system (`dM_B/ds = 2*pi*sqrt(s)*rho0/sqrt(B)`, `rtol=1e-12`,

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`atol=1e-14`, `M_B_ode`), and (iii) adaptive Gauss–Kronrod quadrature on $s = r^2$ PCHIP-interpolated profiles (`quad`, `epsrel=1e-12`, `M_B_quad`). Grid convergence (SLy, $L = 10$ km, $\alpha_Y = 1$):

| ngrid | 7-point peak $M_B^{\rm trapz}$ | 7-point peak $M_B^{\rm quad}$ | 7-point peak $M_{\rm ADM}$ | 3-point peak $M_B^{\rm ODE}$ |
|---|---|---|---|---|
| 301 | 15.316389 | 15.316409 | 15.316404 | 15.316522 |
| 601 | 15.316471 | 15.316410 | 15.316414 | 15.316522 |
| 1201 | 15.316433 | 15.316410 | 15.316406 | 15.316523 |
| 2401 | 15.316418 | 15.316410 | 15.316415 | 15.316522 |

Across all 8 sequences, the turning points of $M_B^{\rm quad}$ (computed with adaptive Gauss–Kronrod quadrature in $s = r^2$) and $M_{\rm ADM}$ agree to within $\leq 8.43 \times 10^{-6}$ dex (`delta_dex_ADM_vs_baryon_turning` between $-4.58 \times 10^{-6}$ and $+8.43 \times 10^{-6}$ dex). The 9-point First-Law test around the turning point (`dM_ADM/dx - e^{\nu(R_*) - \phi_L(R_*)} dM_B/dx`) quantifies the exact residual arising from the non-conformal ($\mathcal{L}_m = -e^{-\phi_L}\varepsilon$, `USE_TRACE_SOURCE=False`) coupling:

$$\frac{dM_{\rm ADM}}{dx} - e^{\nu(R_*)-\phi_L(R_*)}\frac{dM_B}{dx} \;=\; \int_0^{R_*} \mathcal{O}_{\rm nc}(r)\,\frac{\partial \phi_L}{\partial x}\,dr \;\approx\; +8.23\times 10^{-5}\,M_\odot/\text{dex}.$$

### 23.7.4 7.4. Eigenmode Reconstruction ($\xi(r)$, Node Count) and Comparison with the Static Equilibrium Tangent (`eigenmode_xi_nodes_and_static_tangent_audit`)

The function `verify_eigenmode_xi_nodes_and_static_tangent()` reconstructs the coupled fundamental radial displacement profile $\xi(r)$, verifies the interior node count (`n_nodes = 0`) and the free-surface Lagrangian pressure condition ($\Delta P(R_*) = -(\varepsilon + P)r\,\delta h(R_*) = 0$), and compares the $\omega_0^2 = 0$ eigenmode against the static equilibrium tangent vector $\partial_{\log_{10}\rho_{0c}}(h, m, \phi_L, \phi_L')$ for both the main (`scalar_m_sign = -1.0`, exact LHS) and plus-sign (`scalar_m_sign = +1.0`) operator conventions:

| Sequence | Main Zero Crossing (`scalar_m_sign = -1.0`, exact LHS) | $\Delta_{\rm ADM}$ (dex) | Diagnostic Zero Crossing (`scalar_m_sign = +1.0`) | $\Delta_{\rm ADM}$ (dex) | $\xi(r)$ nodes (-) | $\xi(R_*)$ free surface vs. static tangent (km) | Surface $\xi(R_*)$ rel. diff. | $\xi/r$ central reg. |
|---|---|---|---|---|---|---|---|---|
| SLy_L10km_alphaY_1.0000 | 15.316410 | +5.813e-06 | 15.319129 | +2.725e-03 | 0 | -2.8994 vs. -2.8003 | 3.541e-02 (3.54%) | 1.08e-03 |
| SLy_L10km_alphaY_0.3333 | 15.307170 | +4.089e-06 | 15.308114 | +9.479e-04 | 0 | -2.9237 vs. -2.8986 | 8.666e-03 (0.87%) | 1.03e-03 |
| SLy_L1km_alphaY_1.0000 | 15.303080 | -6.412e-06 | 15.303088 | +2.212e-06 | 0 | -2.9406 vs. -2.9411 | 1.767e-04 (0.018%) | 1.01e-03 |
| APR4_L10km_alphaY_1.0000 | 15.293025 | +3.065e-06 | 15.294175 | +1.153e-03 | 0 | -2.1155 vs. -2.0681 | 2.289e-02 (2.29%) | 1.29e-03 |
| H4_L10km_alphaY_1.0000 | 15.227783 | -1.658e-06 | 15.234109 | +6.324e-03 | 0 | -5.5394 vs. -5.5623 | 4.116e-03 (0.41%) | 5.30e-04 |

Relative $L^2$-norm difference between the static equilibrium tangent ($\partial_{\log_{10}\rho_{0c}}$) and the $\omega_0^2 = 0$ eigenfunction across the entire stellar volume:

| Sequence | Operator Variant | $\ | \delta h - \partial h\ | _2$ | $\ | \delta m - \partial m\ | _2$ | $\ | \delta\phi_2$ L - \partial\phi_ L\ | $\ | \delta\phi_2$ L' - \partial\phi_ L'\ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| SLy_L10km_alphaY_1.0000 | exact_minus_a_m_sc (scalar_m_sign=-1.0, main result) | 5.043e-04 | 1.767e-04 | 2.273e-04 | 3.093e-04 | | | | | | |
| SLy_L10km_alphaY_1.0000 | code_plus_a_m_sc (scalar_m_sign=+1.0, diagnostic) | 5.026e-03 | 4.099e-03 | 9.347e-01 | 5.974e-01 | | | | | | |
| SLy_L10km_alphaY_0.3333 | exact_minus_a_m_sc (scalar_m_sign=-1.0, main result) | 1.319e-04 | 1.588e-04 | 5.441e-04 | 3.635e-04 | | | | | | |

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| Sequence | Operator Variant | $\ | \delta h - \partial h\ | _2$ | $\ | \delta m - \partial m\ | _2$ | $\ | \delta\phi_2$ L - \partial\phi_ L\ | $\ | \delta\phi_2$ L' - \partial\phi_ L'\ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| SLy_L10km_alphaY_0.3333 | code_plus_a_m_sc (scalar_m_sign=+1.0, diagnostic) | 1.525e-03 | 1.341e-03 | 9.397e-01 | 5.967e-01 | | | | | | |
| SLy_L1km_alphaY_1.0000 | exact_minus_a_m_sc (scalar_m_sign=-1.0, main result) | 5.177e-05 | 1.638e-04 | 1.709e-04 | 2.478e-03 | | | | | | |
| SLy_L1km_alphaY_1.0000 | code_plus_a_m_sc (scalar_m_sign=+1.0, diagnostic) | 4.052e-05 | 1.468e-04 | 2.243e-02 | 2.143e-02 | | | | | | |
| APR4_L10km_alphaY_1.0000 | exact_minus_a_m_sc (scalar_m_sign=-1.0, main result) | 3.048e-04 | 2.108e-04 | 1.239e-03 | 6.328e-04 | | | | | | |
| APR4_L10km_alphaY_1.0000 | code_plus_a_m_sc (scalar_m_sign=+1.0, diagnostic) | 4.077e-02 | 6.161e-03 | 1.112e+00 | 7.202e-01 | | | | | | |
| H4_L10km_alphaY_1.0000 | exact_minus_a_m_sc (scalar_m_sign=-1.0, main result) | 7.340e-05 | 1.885e-04 | 6.043e-05 | 1.400e-04 | | | | | | |
| H4_L10km_alphaY_1.0000 | code_plus_a_m_sc (scalar_m_sign=+1.0, diagnostic) | 1.059e-01 | 1.964e-03 | 5.713e-01 | 3.420e-01 | | | | | | |

**Summary and Boundary of Theoretical Closure (`mathematical_origin_of_0p002725_dex_offset`):** In the nonlinear interior scalar equation ($B\phi_L'' + C_0\phi_L' = \phi_L/L^2 - \alpha_Y e^{-\phi_L}\varepsilon$), the partial derivative of the right-hand side (RHS) with respect to $m$ at fixed $(h, \phi_L, \phi_L')$ is $\partial(\phi_L'')/\partial m = +(2\phi_L''/r + 2\phi_L'/r^2)/B$ (at $r = 6$ km, $-1.728334 \times 10^{-4}$ km$^{-3}$). When rearranged onto the left-hand side ($L_u u = R_h\,\delta h$) of the Volterra system in `solve_yukawa_coupled_omega2_gl2`, the exact left-hand side sign is `scalar_m_sign = -1.0` (`a_m_sc = -\partial(\phi_L")/\partial m`), with which all V25 main results (`curved_yukawa_coupled_stability.csv`, `curved_yukawa_audit_summary.json`) are generated: across all 5 sequences it yields a nodeless ($n = 0$) $\xi(r)$ fundamental mode and an $\omega_0^2 = 0$ zero crossing (`15.316410`) lying within $\leq 6.9 \times 10^{-6}$ dex of the ADM mass peak (whereas `scalar_m_sign = +1.0` shifts the root by +0.002725 dex to `15.319129`). For SLy $L = 10$ km, $\alpha_Y = 1$, the relative $L^2$ differences between the eigenfunction and the static equilibrium tangent are $\delta h = 5.04 \times 10^{-4}$, $\delta m = 1.77 \times 10^{-4}$, $\delta\phi_L = 2.27 \times 10^{-4}$, $\delta\phi_L' = 3.09 \times 10^{-4}$, while the relative difference of the free-surface displacements is 3.54% (`rel_diff_xi_R_exact_minus_vs_static_tangent = 0.035406`). While the minus-sign root and grid convergence represent a major step forward, they do **not by themselves constitute an independent proof of the full radial fundamental spectrum and free-surface variation**, and therefore **the stability closing statement is formulated conditionally until the independent verification of the coupled variational derivation and boundary-condition matching is concluded.**

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# 24 Reproducibility Manifest and SHA-256 Checksums (MANIFEST_en.txt)

Source file: MANIFEST_en.txt

```
# MANIFEST --- Scalar-Tensor and Yukawa/Disformal Action Mapping (Steps 1--22 Derivation and Audit Reproduction Package, V25)
# Total files tracked in this manifest: 336 (including the 22 Hungarian chapters, 22 English translation chapters, README.md, and
↪  README_en.md)
# Publication status: The earlier Zenodo DOI (10.5281/zenodo.23680715) is cited exclusively as a reference to the prior V24 release; the
↪  two sign fixes and the three new diagnostics belong to this V25 package, and the stability closing statement is formulated
↪  conditionally until the variational and boundary-condition verification is independently concluded.
# Format: <SHA-256>  <size_in_bytes>  <relative_path>

f0f52a32a094aea127d713f29d70e90726086db3a257716453a5a9b1384d0fbd      47093  README.md
622886ef5be8bce89303979e7cc42836782a41ea000659b456288232ce4fc931      44466  README_en.md
1a8d58f9a15adba4cb805522e52c91275eaceb2a4fcddefc3f1ac317582c6d2a       5720  SPARC_mode.py
ad054b61c758080335f2d9d5827ab5a2fe2dcff4183b6e788119fd1f15cd19ab      14002  closure_tov_comparison.csv
1a8d58f9a15adba4cb805522e52c91275eaceb2a4fcddefc3f1ac317582c6d2a       5720  code/SPARC_mode.py
fbe82c9dfc89501de589935a223b950013ad455525f4440ce81bc5c48c2046f7        883  code/build_pdf.py
54699ce07f594b952f3e233b49d99a43d72365f6e27672279e8c0d8236c6ae4b      71437  code/covariant_closure_50aa2db0.py
dee2cf8b0a345d8131ef5f2cb168bb8566f8a3be712e26c98e23e209e5f39cd6      94362  code/curved_yukawa_adm_and_stability.py
acf334f7e5594b9af1ec1f2f9ecb6c39eba34e9abbeca864fae5d62a3227526c       6133  code/curved_yukawa_adm_sequences.csv
a97669f24dade5a83a26e476946313c161cf692db7f845b368e5d1e51aebbaed      97385  code/curved_yukawa_audit_summary.json
f4fe6d056841cfa9c0a072388624a5ffe51a732fd8da8655e1fbacb3c6b8747e       7033  code/curved_yukawa_coupled_stability.csv
20e5084ee072c4d9fff7d588ab9fe83b6c6ecee27e2f9f9bcaa56c5e90a09c65       7500  code/diagnostics.py
665d533feeb7a6091830f8516aeeaf54fda55fcef6dc8bb9b91b19e4b9055dca       4039  code/energy_integrator.py
d55cff98421e34d15010efe5dfb499ab3d9ee582fa72f458691e14b297c61599       3467  code/fields.py
93ca2b3955333869c5f29eadc381d26c1932b5a4560a01db1a3025b2aaed9533       5405  code/make_revision3_checks.py
8d4ff9681fead3323e1b5b0e4d3dbdc8e640180c3217ea7fecdef915f64de96d       7128  code/radial_modes.py
eb40036842162dc9a42216bd7a4d35c7f2ce8addcf7e3ed8ffd3626427b22313       6259  code/sparc_full_runner.py
a92ec4bf10668f566d9a587445bb62052650ea5bdebdb29e2141ea7f320c4b1f      29815  code/sparc_tophat.py
637f892bb25468fc0c058fd1ecdd60ca53d51059675b94e61b885398ad162bdb        876  code/sparkto_b_raw.py
cb93c90fb261009c987ebc98a516fc15d66c7f8c34ae8098b5617fd7ddd21906      27472  code/stellar_tal.py
cb93c90fb261009c987ebc98a516fc15d66c7f8c34ae8098b5617fd7ddd21906      27472  code/stellar_tal_v21.py
e07e8c6e06bd7bc002b3728c98ec94ce28bd6a67dc9513e736acc7fdca1f901d      16463  code/ta_foundation.py
e96e57443dc1c9a691111a82aae51a2c10644a6b336ef530ec12226d729beafe       7988  code/ta_gammaL.py
70cad9a97511e3b0d37a15a58eb22b656e323615d1cce5df1ab2c778250fb3a2      18181  code/ta_pipeline.py
a624ee69abec04f274b351518e47f9f3f227dd10a7fc854f8c7c9c5adfd8c80d       6112  code/ta_run.py
2b9fe3d788aa1bc2699cf090374fdfb3136416aa6daba06c0b8a44a4f1b5c7ba       6387  code/ta_run_l0.py
c158ee17dd26be2186c6b4650775e261717d56f2bf4b01661a38fa9230e32023       7271  code/ta_run_l0_v2.py
73962aed3380230f91c898db2d1876cb7c4984c4fe01523bdcf6ef1727f572ef      13086  code/tal_factor_7th_model.py
96eb449555c5b89fcb59a5648aa9dca37aec91ec92032e977c23b4ddf3c5df3c       6277  code/tal_profiles.py
53a30947b77d5ee9e9a73a292631f1cf1dfbf0e4224a7e83fc5449ca32403322      14351  code/tophat_finite_source.py
c2d52e451ba042790e469f8bf00b8019bb59779ec137121c2a72397904aeced7      14767  code/v20_fixed_case_runner.py
059ac76b1b8fc4dcbf1537a2b3fa0e95dcca0e88415e51a400f4f1f88360a226     115783  conservation_Residuals.json
54699ce07f594b952f3e233b49d99a43d72365f6e27672279e8c0d8236c6ae4b      71437  covariant_closure_50aa2db0.py
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↪  step18_nonlocal_tangent_bvp_shell_kernel_and_inertia_scaling_invariance_en.md
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↪  step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b_common_proxy.md
9fab3349db68588b0864ca86443e11d8690f71422d29b4bb62c898f1de7699ec       18067
↪  step19_in_out_differential_clock_quadrature_granite_chamber_and_sirius_b_common_proxy_en.md
ecd894436341cbdcab40378de8de26f06a28f82a1a29039769d9532dcb4c4d12       11205  step1_dilaton_form.md
83a2e8e88808373947d5b078ee3f20f616a9c23c865f3a75d384697bf11e05bf       10890  step1_dilaton_form_en.md
bedb9d8511f3c2872e936672c86e8c083f35d6473e141191e7f47e564995a146       17367
↪  step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_quadrature.md
800ff0bb696640791b0baedf6a95ae5349b560dc3d4bd036097a3fd3037d151b       16820
↪  step20_eight_coil_elliptic_biot_savart_and_exterior_compactified_quadrature_en.md
29eb46950ed70c57ad076158afb23d3f5361f4c14040e1ba672235baa24e6cf3       18431
↪  step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_lapse.md
106b9ee27d97f4a47d69e6ddc3adbbeb48b30a5d076d84e307e314c197385760       17542
↪  step21_enthalpy_tov_even_parity_s_r2_regularity_and_exterior_halo_lapse_en.md
0c09a01ca8973d53ce7ad1733cbec73ba3d73fb10febc6dcadb8225e555cd565       35771
↪  step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability.md
dc84080449d2fcba870df8d7273c9495cafe153624e27f8a8867f89df020686c       36896
↪  step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability_en.md
7a3761a4adcca6fd38443be0666b49577b49ca18582ffe395fce4038623e2a3d       18402  step2_conformal_disformal.md
abd9db8d104e3bcf157194a5796e75d9695c20d7ef0b47403fb5fc8a1028bfc9       18014  step2_conformal_disformal_en.md
77fa5ad7646bad0bc2b5c3e6987f779f45460ceec016f9ee4fa1330b57ce0d29       12647  step3_yukawa_helmholtz_eom.md
8ca1db6f110e2617d2a00cd9cd066a54c4d3926ef4f4d3b135f2fd9e0c852ac9       12258  step3_yukawa_helmholtz_eom_en.md
52b1cab653da99d60994267b39b1799c91a9401bc37a21a08b5012e17b879442       12356  step4_covariant_conservation_hydrostatic.md
1deaba07182e7cfa7c65db0462577f37ea3652269982afc0673906015da67baa       11986  step4_covariant_conservation_hydrostatic_en.md
d9c9d69ff93e6625f489b98a9761f43625e2c79a1a57ee2a12012b27449ccf73       14949  step5_coupling_dictionary_ppn_screening.md
e57d506cdd3ba02b569695d30d68d8b2812ef52b46bad1f6fb7d1980655646d4       14701  step5_coupling_dictionary_ppn_screening_en.md
8fbdd2157d0a4aa9fbd78dca87444c4689f7ecb35f660424bfa8a252a1b32594       14615  step6_sparc_galactic_ngc3198_consistency.md
6f75c1f0ccba91dee96325413d81e3ba7983b1bead0fa440f44a98e8e0b8fc4f       13870  step6_sparc_galactic_ngc3198_consistency_en.md
d656ac9249a495ae66a4909afb4eafc8e5ab094a7d0e2002d8a4894c362b60f6       21215  step7_neutron_star_tov_stability.md
e219a8cfaf4c81a28e91d7cbfb2842c1023bac0034cc6b1f82f73276c9101109       20546  step7_neutron_star_tov_stability_en.md
f4d790f47b5ab8d47a1a4fbe1b7b8465fb76eafe1db2a9ce492e88f97a516125       16306  step8_microscopic_clock_coupling_siriusb_lab_v20.md
b6caf38ba27692eab8ef1298ab9fefde937a4204b95c54aa9498d5e71b766773       15620  step8_microscopic_clock_coupling_siriusb_lab_v20_en.md
af5b8d0ab66ff02c831ac4774b84446dc589a7d43e4c485d2819410a427af197       12164  step9_vhe_photon_propagation_hess_dispersion.md
dabd6f01ef17584b44c0fa63e8fe871a2d33a021d42894e4d76330d6a3291e22       11454  step9_vhe_photon_propagation_hess_dispersion_en.md
e07e8c6e06bd7bc002b3728c98ec94ce28bd6a67dc9513e736acc7fdca1f901d       16463  ta_foundation.py
e96e57443dc1c9a691111a82aae51a2c10644a6b336ef530ec12226d729beafe        7988  ta_gammaL.py
70cad9a97511e3b0d37a15a58eb22b656e323615d1cce5df1ab2c778250fb3a2       18181  ta_pipeline.py
a624ee69abec04f274b351518e47f9f3f227dd10a7fc854f8c7c9c5adfd8c80d        6112  ta_run.py
2b9fe3d788aa1bc2699cf090374fdfb3136416aa6daba06c0b8a44a4f1b5c7ba        6387  ta_run_l0.py
c158ee17dd26be2186c6b4650775e261717d56f2bf4b01661a38fa9230e32023        7271  ta_run_l0_v2.py
73962aed3380230f91c898db2d1876cb7c4984c4fe01523bdcf6ef1727f572ef       13086  tal_factor_7th_model.py
96eb449555c5b89fcb59a5648aa9dca37aec91ec92032e977c23b4ddf3c5df3c        6277  tal_profiles.py
3637d2f5bda6b2b97e645c5fba5cf39d310a1826bee388286c2c74b9171aa283        1039  v17_package/github/data/blind_123.txt
a7405768522cc7a4cdaba1232a20e0c6eea75a45eee32654c73646526c1457ae       30859  v21_radial_summary_ref.csv
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